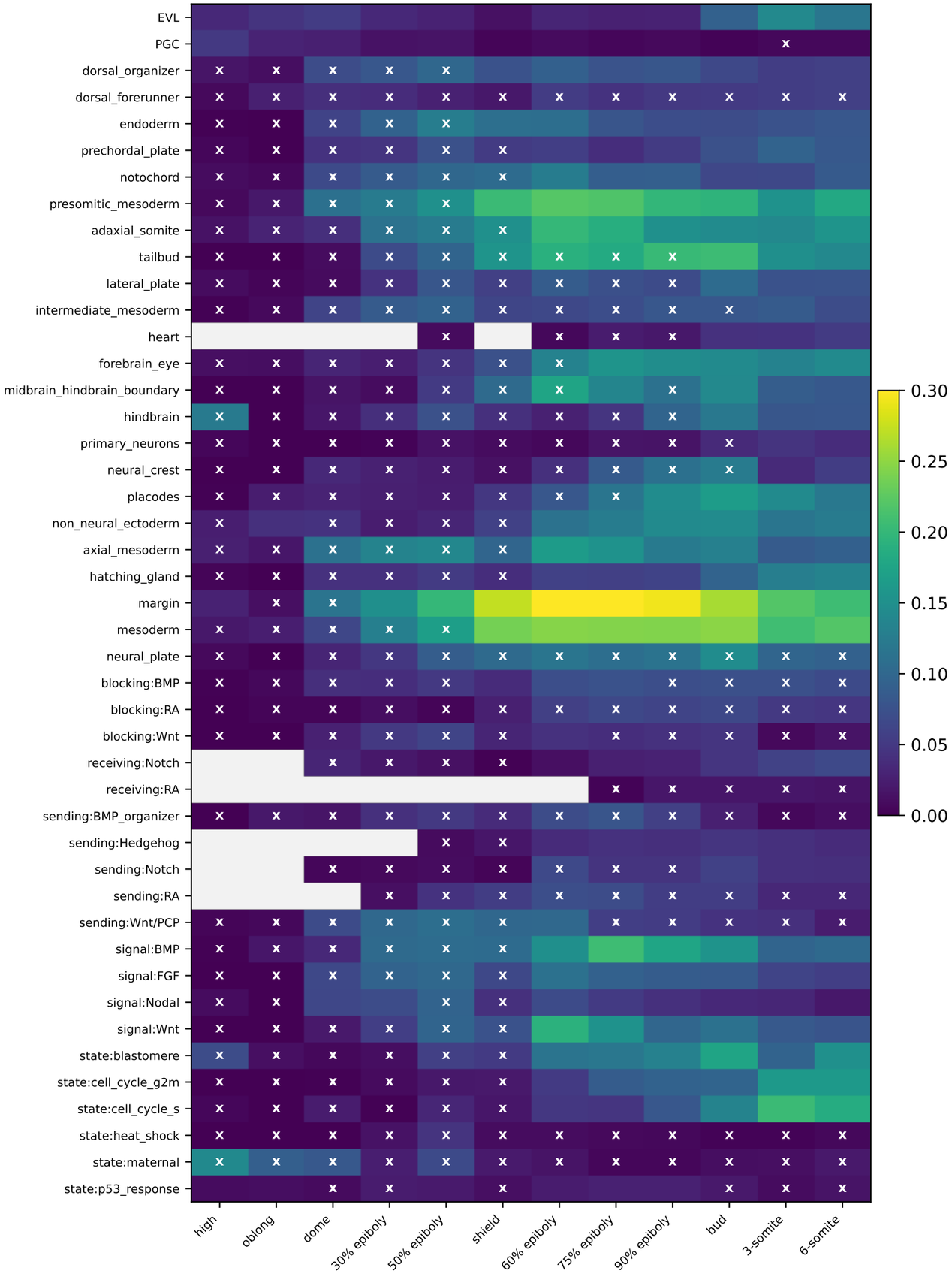
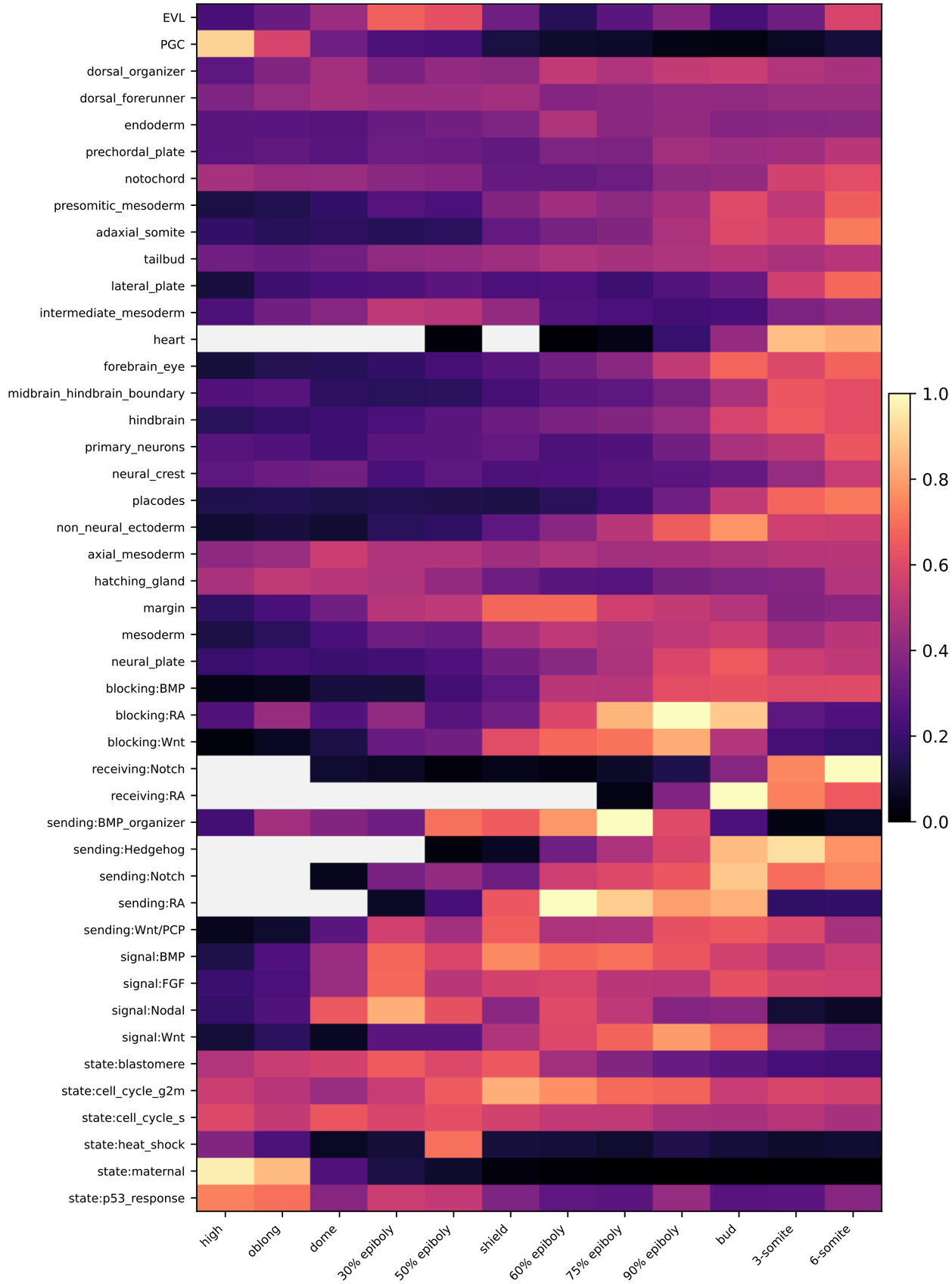
share of cells with the program active (median library); x: not supported



pseudobulk strength (script 25)

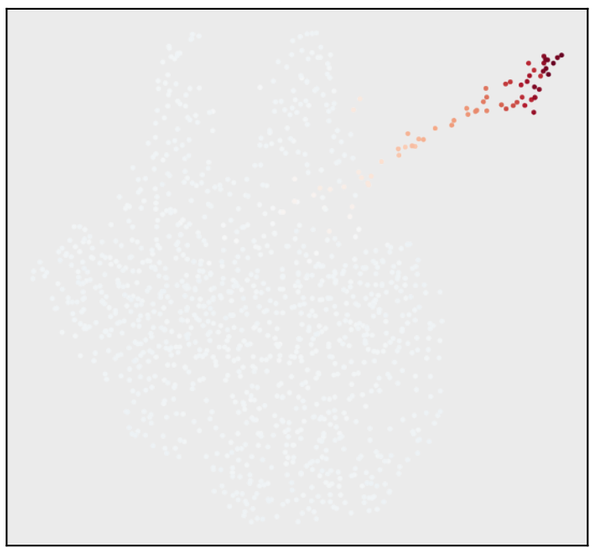


EVL

high: 17 genes, supported
active in 4% of cells, pseudobulk strength 0.23

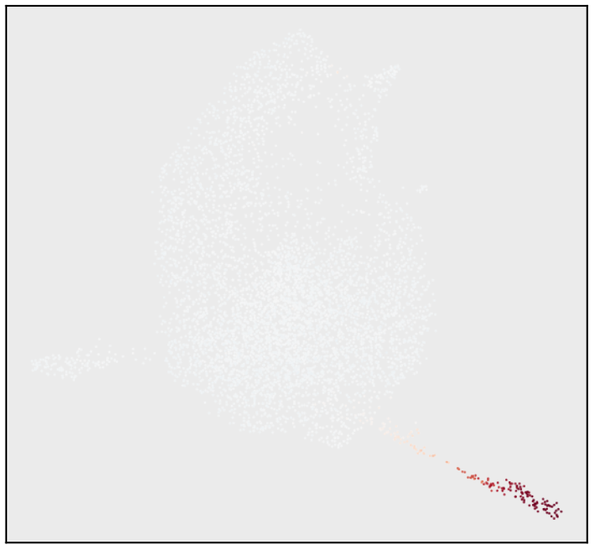
oblong: 18 genes, supported
active in 5% of cells, pseudobulk strength 0.31

dome: 28 genes, supported
active in 5% of cells, pseudobulk strength 0.44

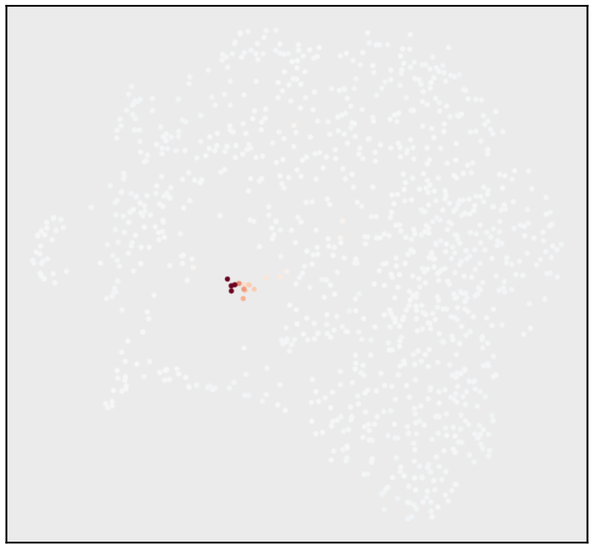


30% epiboly: 30 genes, supported
active in 4% of cells, pseudobulk strength 0.67

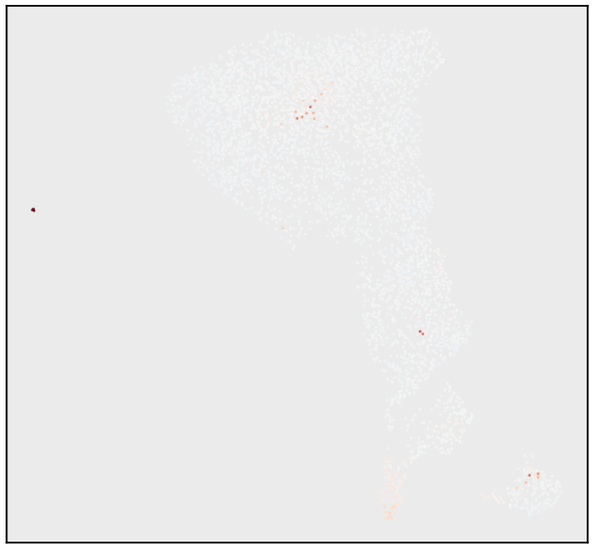
50% epiboly: 30 genes, supported
active in 3% of cells, pseudobulk strength 0.62



shield: 30 genes, supported
active in 1% of cells, pseudobulk strength 0.33

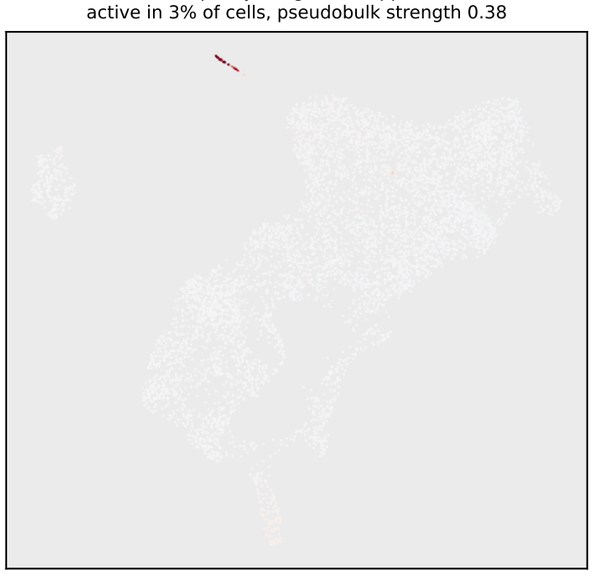


60% epiboly: 30 genes, supported
active in 3% of cells, pseudobulk strength 0.15

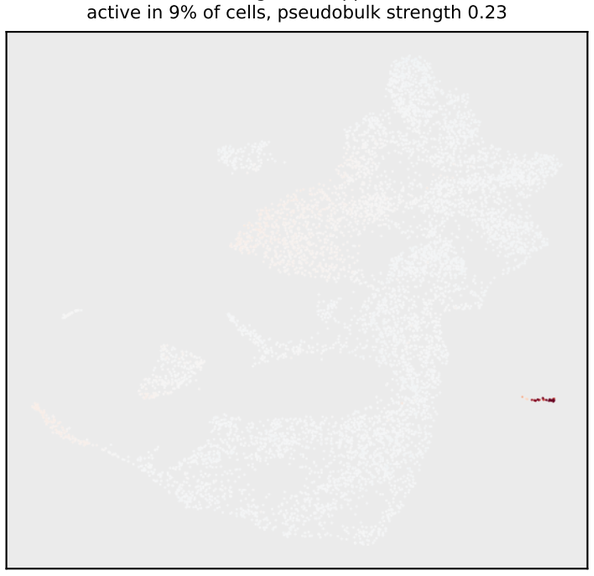


75% epiboly: 30 genes, supported
active in 3% of cells, pseudobulk strength 0.27

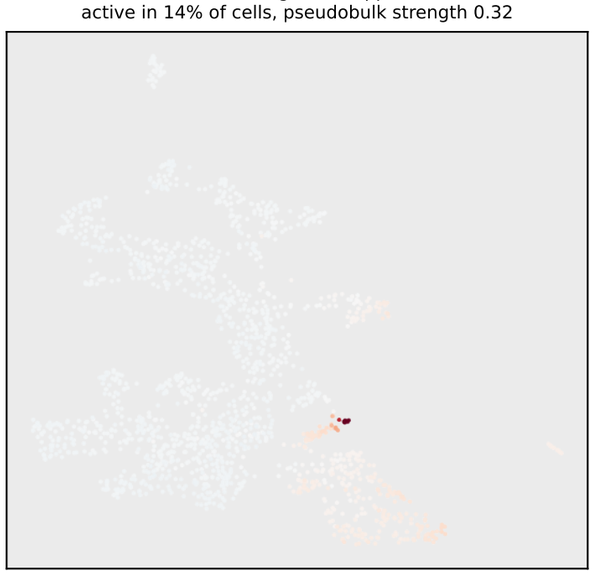
90% epiboly: 30 genes, supported
active in 3% of cells, pseudobulk strength 0.38



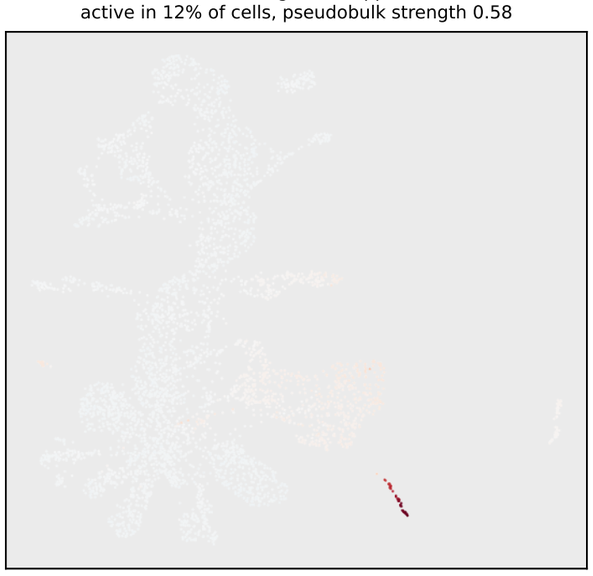
bud: 30 genes, supported
active in 9% of cells, pseudobulk strength 0.23



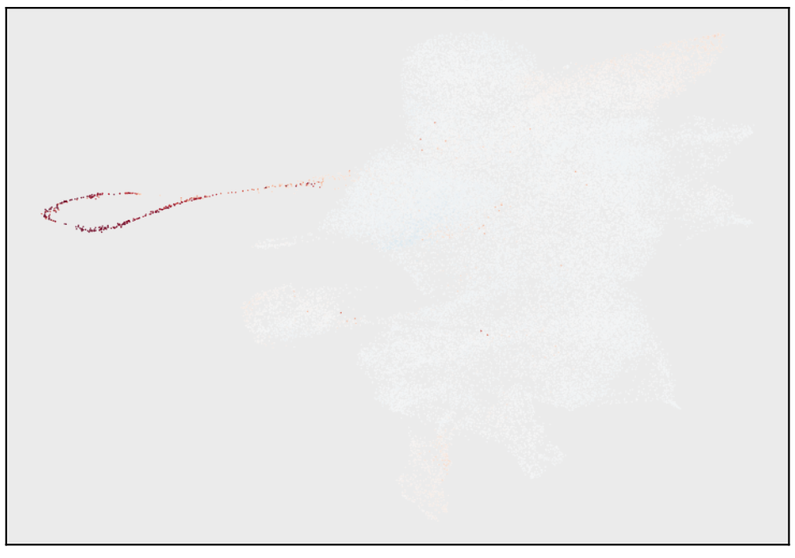
3-somite: 27 genes, supported
active in 14% of cells, pseudobulk strength 0.32



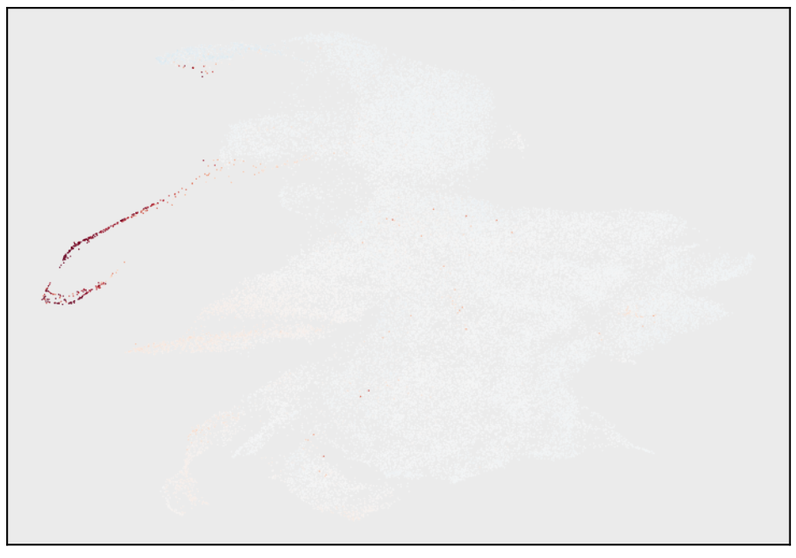
6-somite: 30 genes, supported
active in 12% of cells, pseudobulk strength 0.58



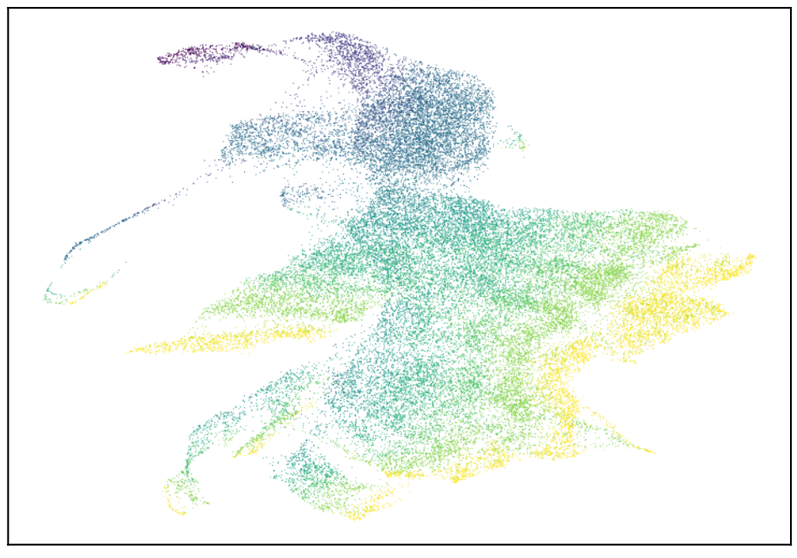
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



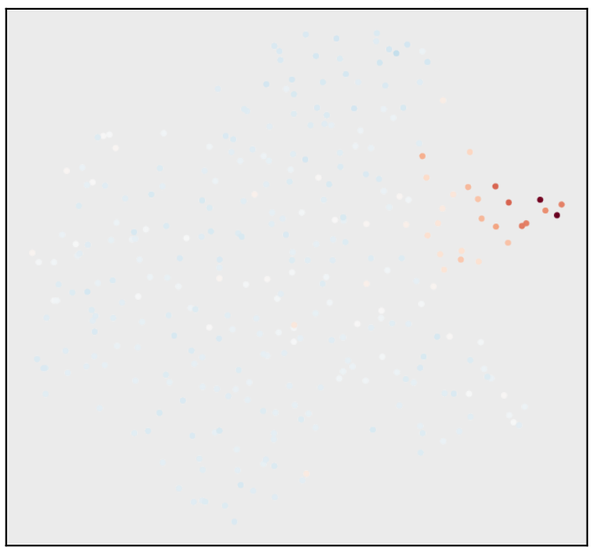
atlas, unaligned: stages



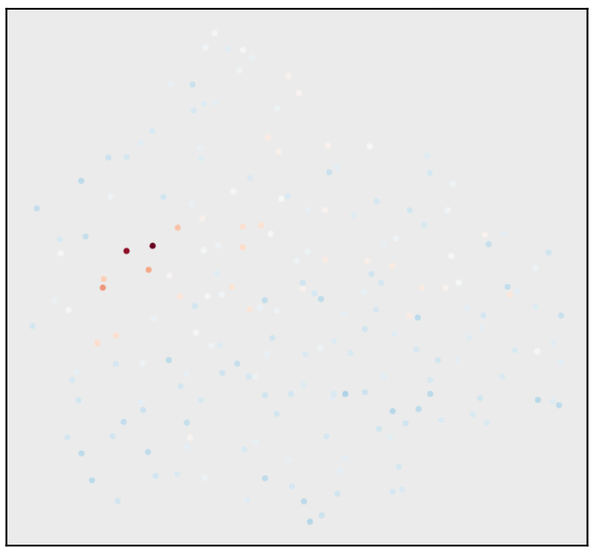
Genes (number of stages): krt4 (12), krt18 (12), cldne (12), cldnb (12), lye (12), krt8 (12), mgll (12), cebpb (12), epcam (12), ppl (12), gbgt1l4 (12), gmds (12), f11r.1 (12), hsd17b12a (12), krt5 (11), grhl3 (11), alox5b.3 (11), elovl7b (11), cyt1 (10), capn9 (10), evpla (10), eppk1 (10), krt97 (10), rab25a (10), zgc:101000 (10), zgc:110333 (10), znf185 (10), dspa (10), krt17 (9), uox (9)

PGC

high: 9 genes, supported
active in 5% of cells, pseudobulk strength 0.91



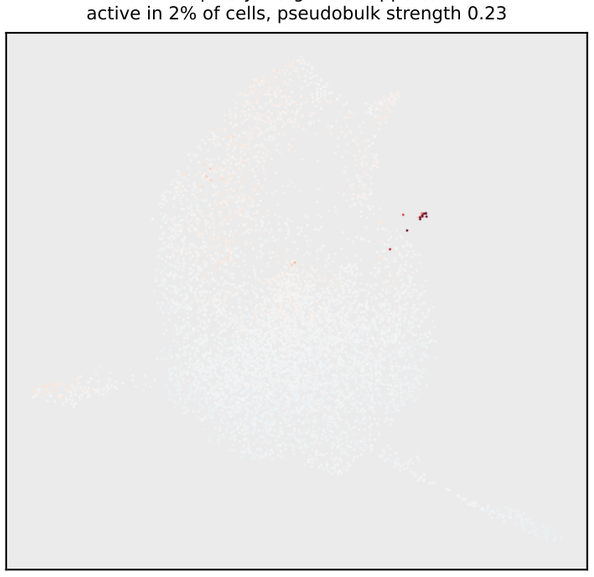
oblong: 9 genes, supported
active in 3% of cells, pseudobulk strength 0.58



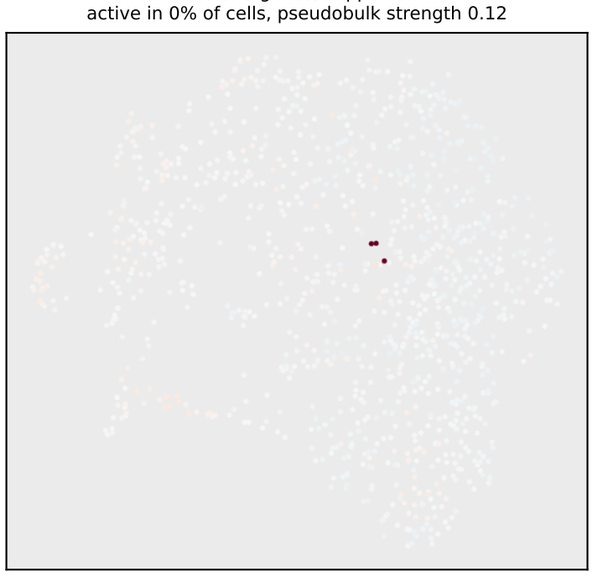
dome: 9 genes, supported
active in 3% of cells, pseudobulk strength 0.33

30% epiboly: 9 genes, supported
active in 1% of cells, pseudobulk strength 0.25

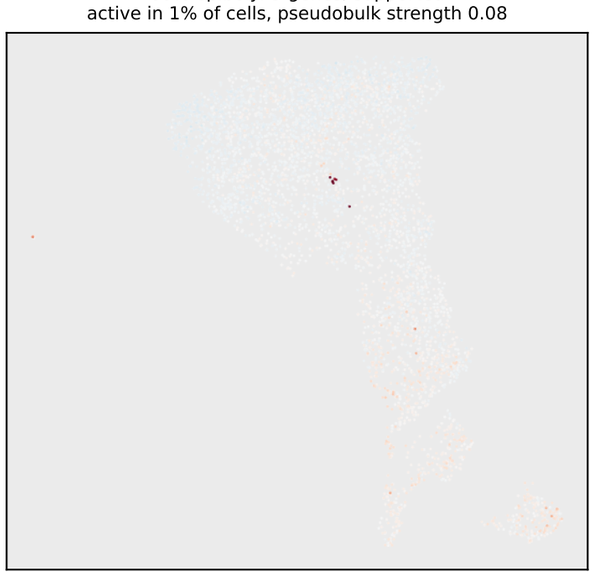
50% epiboly: 10 genes, supported
active in 2% of cells, pseudobulk strength 0.23



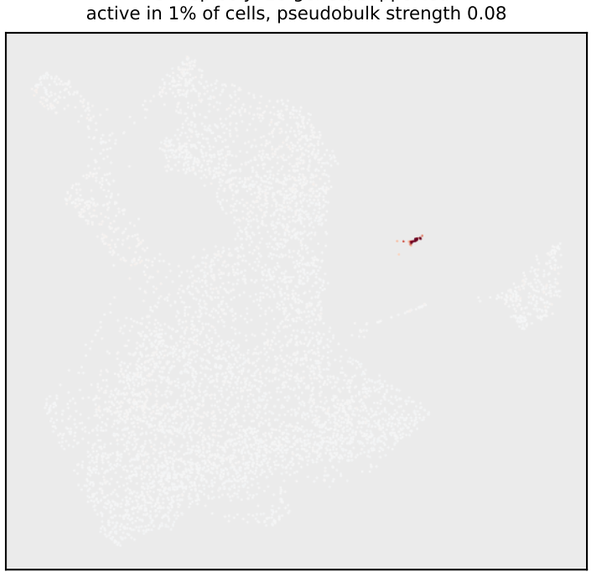
shield: 8 genes, supported
active in 0% of cells, pseudobulk strength 0.12



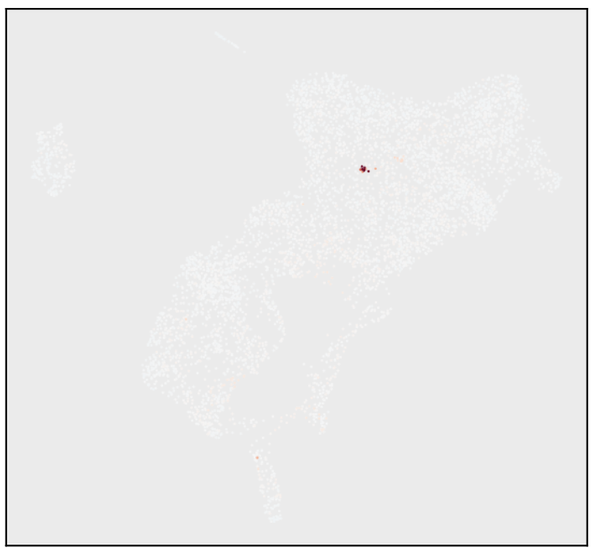
60% epiboly: 9 genes, supported
active in 1% of cells, pseudobulk strength 0.08



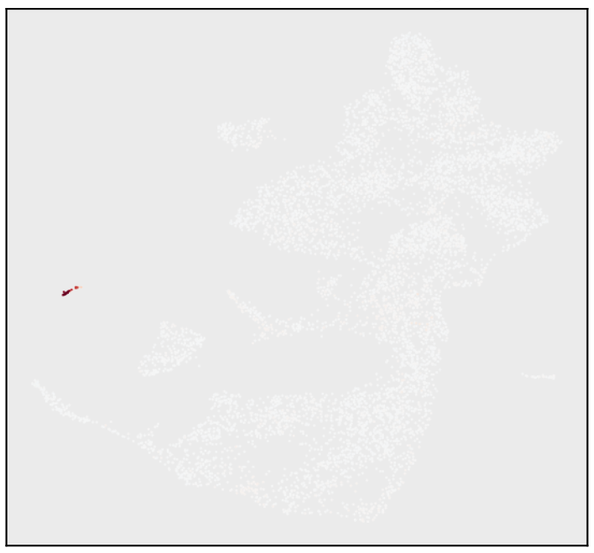
75% epiboly: 10 genes, supported
active in 1% of cells, pseudobulk strength 0.08



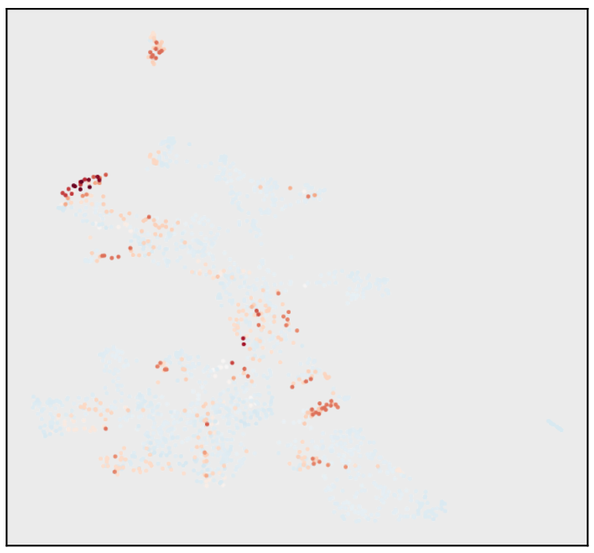
90% epiboly: 9 genes, supported
active in 1% of cells, pseudobulk strength 0.04



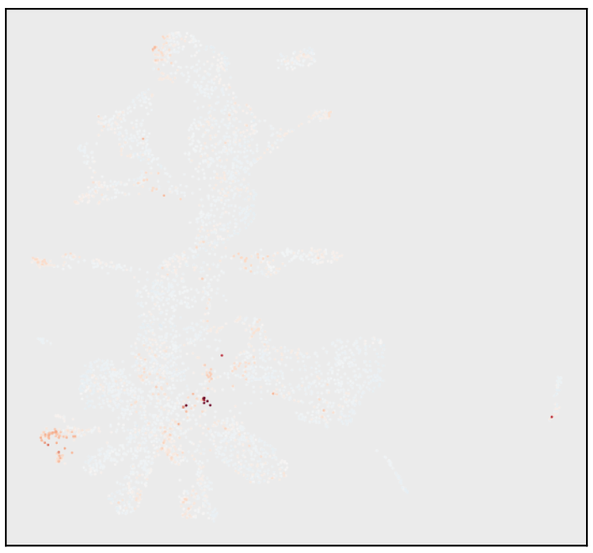
bud: 9 genes, supported
active in 0% of cells, pseudobulk strength 0.03



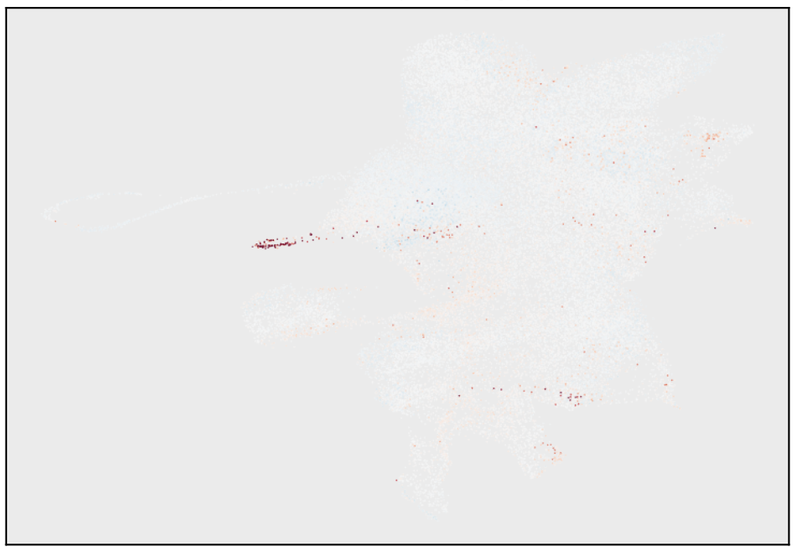
3-somite: 5 genes, not supported
active in 1% of cells, pseudobulk strength 0.07



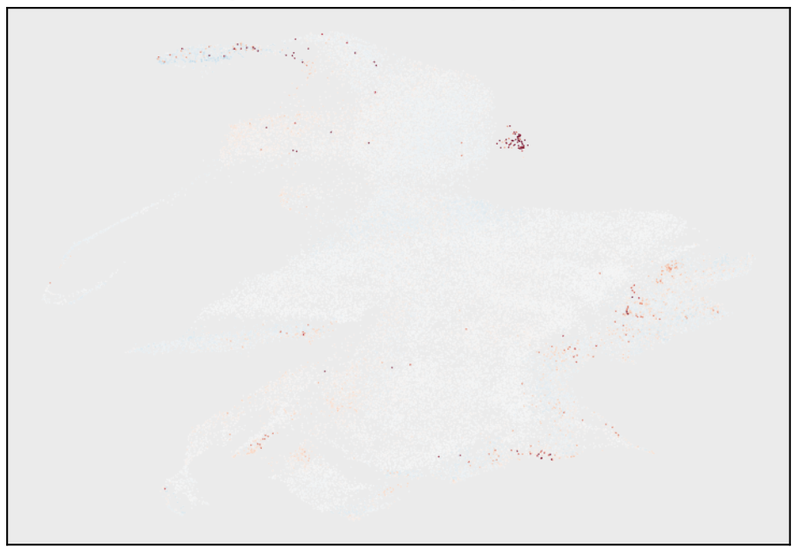
6-somite: 9 genes, supported
active in 1% of cells, pseudobulk strength 0.11



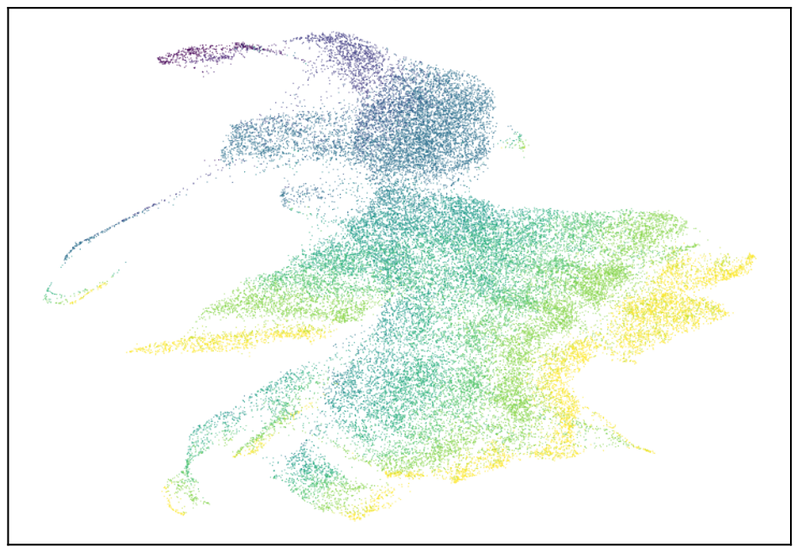
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



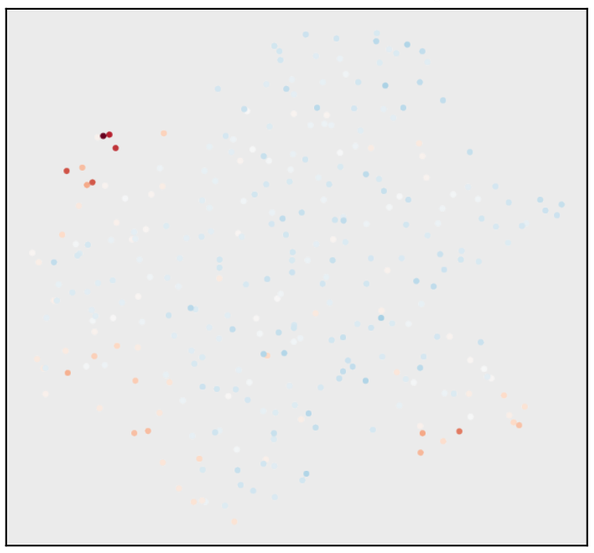
atlas, unaligned: stages



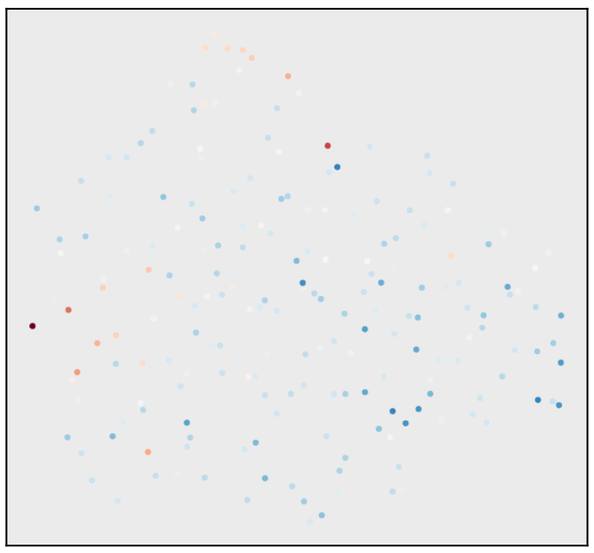
Genes (number of stages): ddx4 (12), gra (12), slc16a3 (12), org (12), nanos3 (11), dnd1 (11), ca15b (11), tdrd7a (11), rgs14a (7), nanos1 (6)

dorsal_organizer

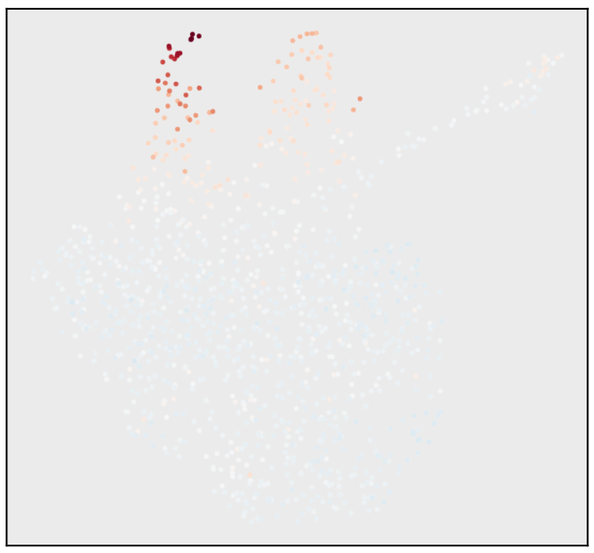
high: 13 genes, not supported
active in 2% of cells, pseudobulk strength 0.29



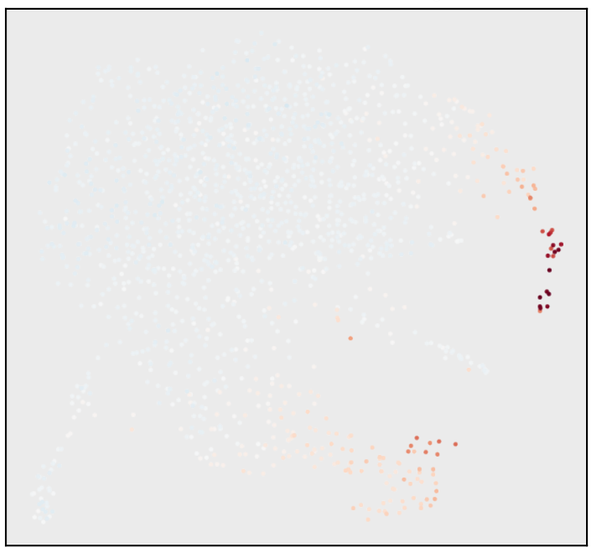
oblong: 12 genes, not supported
active in 1% of cells, pseudobulk strength 0.37



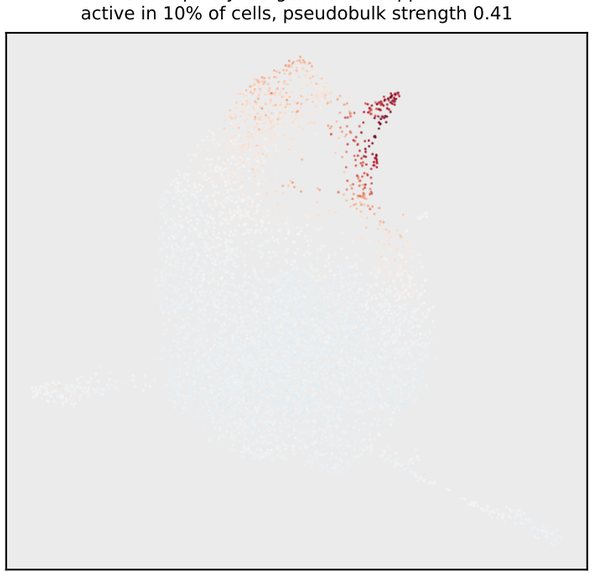
dome: 16 genes, not supported
active in 7% of cells, pseudobulk strength 0.45



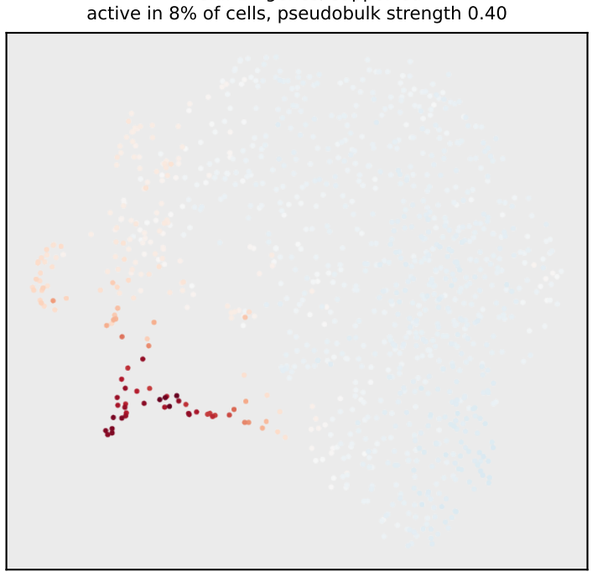
30% epiboly: 16 genes, not supported
active in 8% of cells, pseudobulk strength 0.36



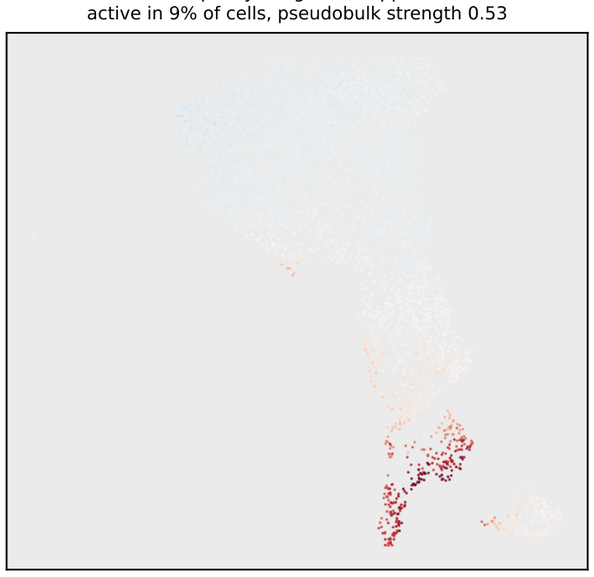
50% epiboly: 20 genes, not supported
active in 10% of cells, pseudobulk strength 0.41



shield: 14 genes, supported
active in 8% of cells, pseudobulk strength 0.40

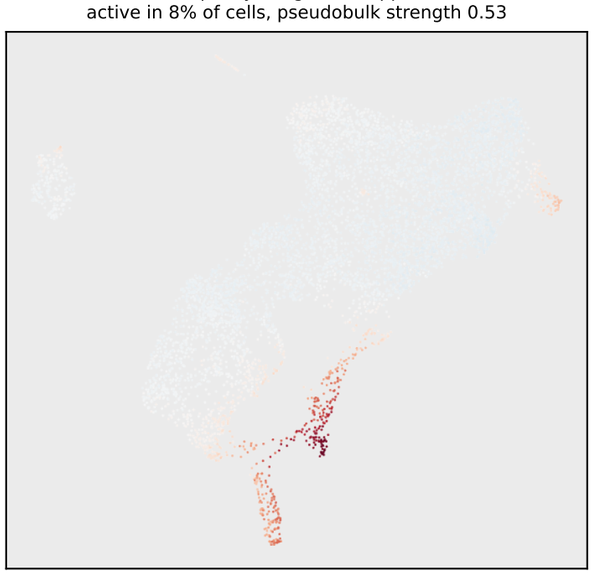


60% epiboly: 20 genes, supported
active in 9% of cells, pseudobulk strength 0.53

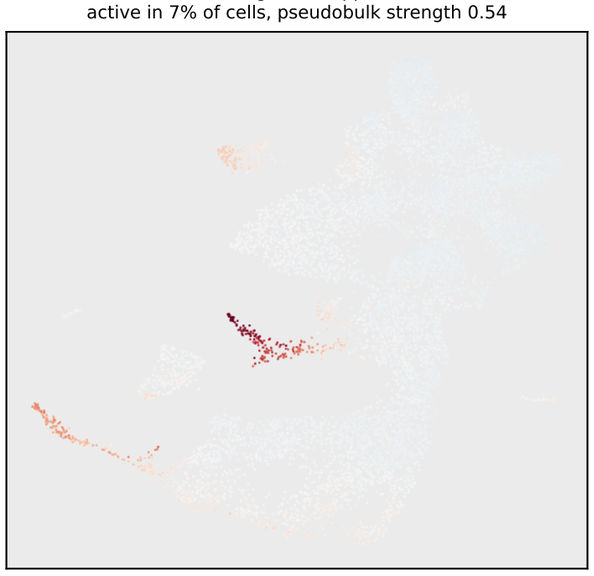


75% epiboly: 21 genes, supported
active in 8% of cells, pseudobulk strength 0.49

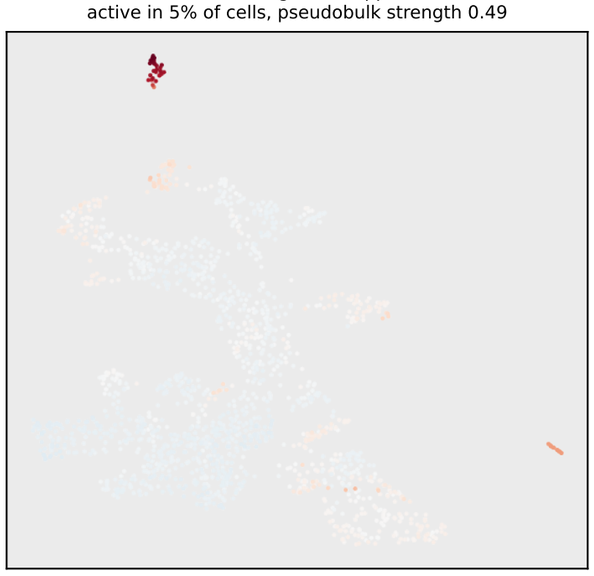
90% epiboly: 21 genes, supported
active in 8% of cells, pseudobulk strength 0.53



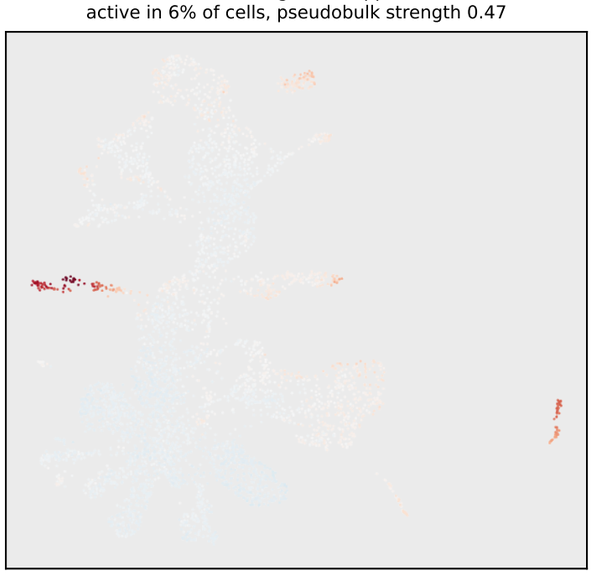
bud: 21 genes, supported
active in 7% of cells, pseudobulk strength 0.54



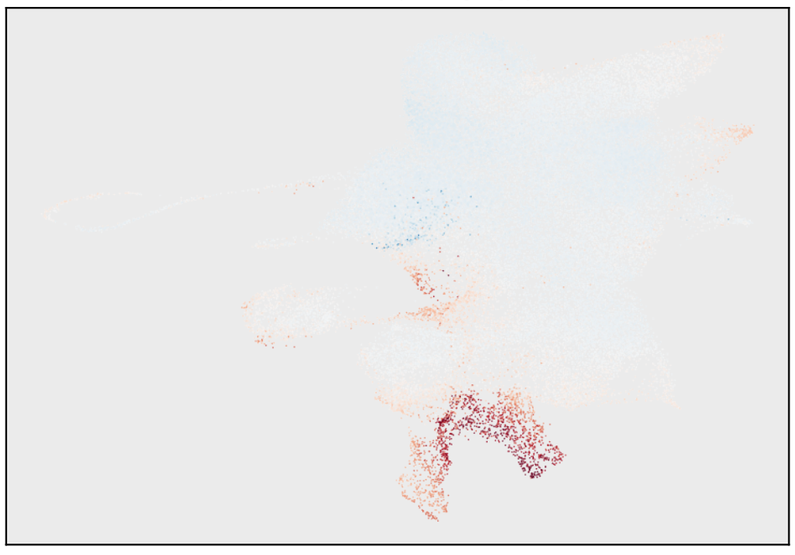
3-somite: 21 genes, supported
active in 5% of cells, pseudobulk strength 0.49



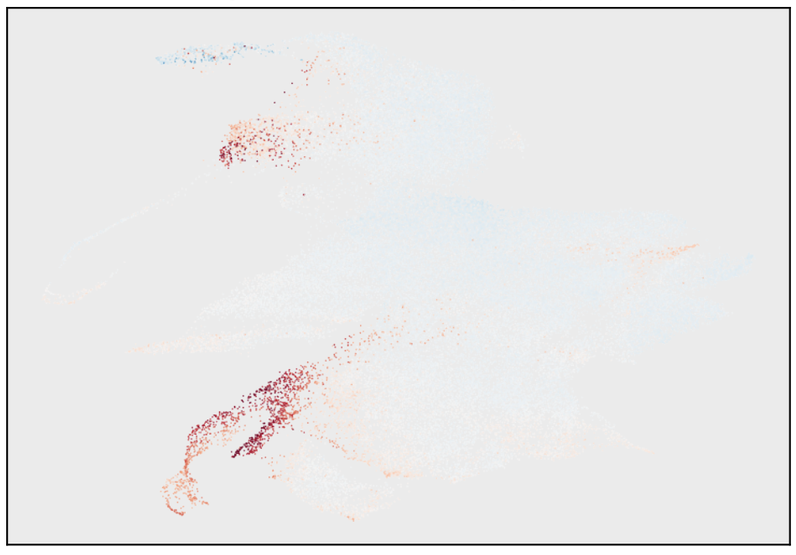
6-somite: 21 genes, supported
active in 6% of cells, pseudobulk strength 0.47



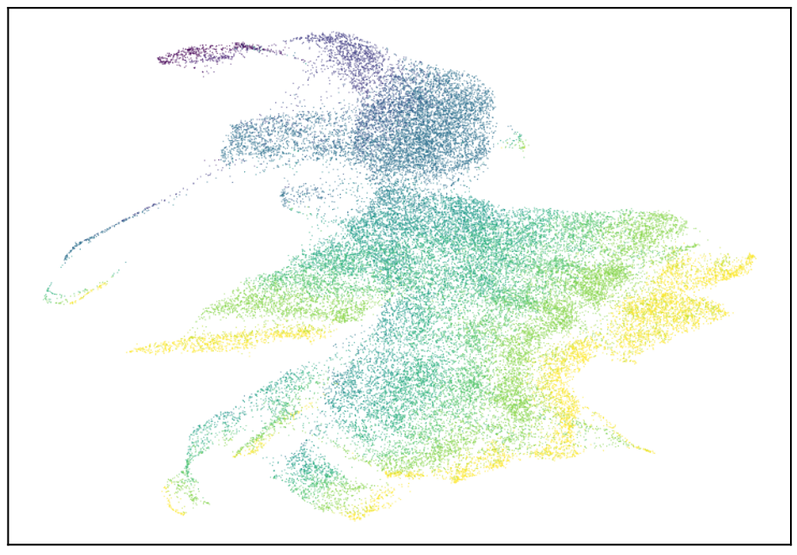
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



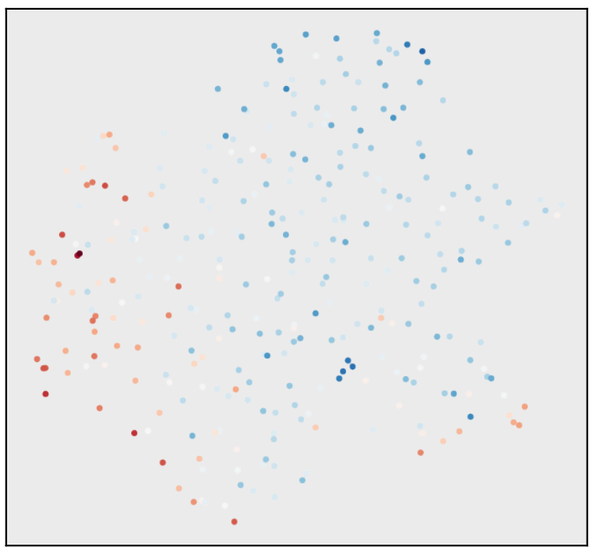
atlas, unaligned: stages



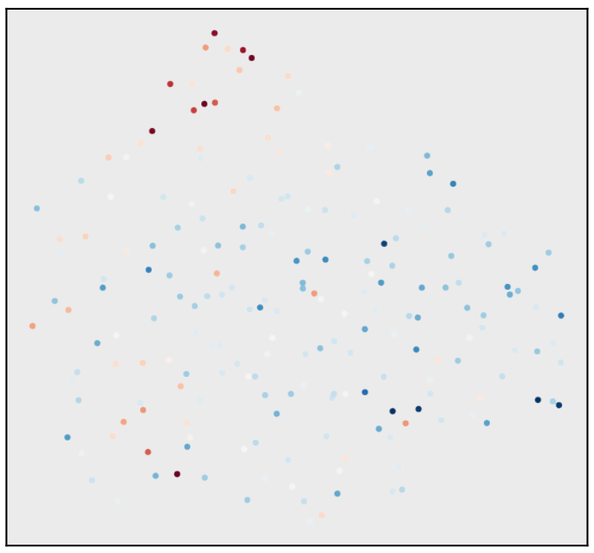
Genes (number of stages): gsc (12), noto (12), xbp1 (12), dact2 (12), net1 (12), fstl1b (12), mycb (12), cbsb (12), acvr1ba (12), rab32a (12), smad7 (12), insb (11), pitx2 (10), kirrel3l (10), plod1a (9), twist2 (7), creb3l1 (7), plod2 (6), cthrc1a (6), capn2b (5), igf2b (5), dharma (5), tbr1b (3)

dorsal_forerunner

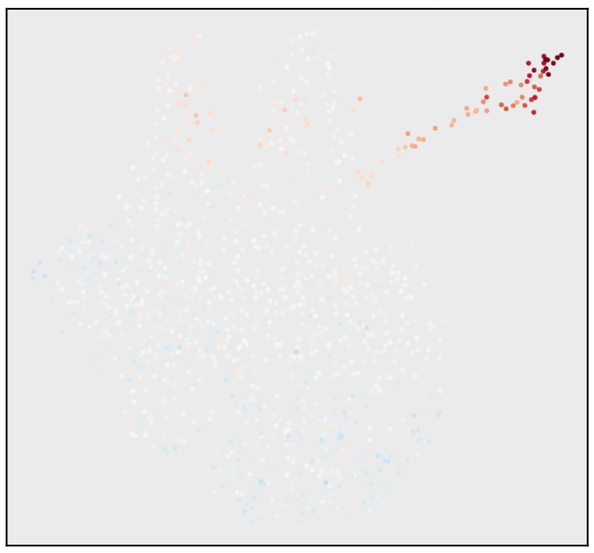
high: 22 genes, not supported
active in 1% of cells, pseudobulk strength 0.36



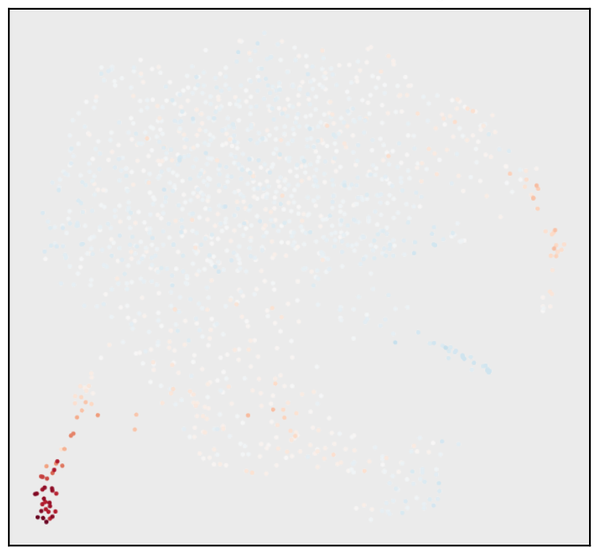
oblong: 22 genes, not supported
active in 3% of cells, pseudobulk strength 0.42



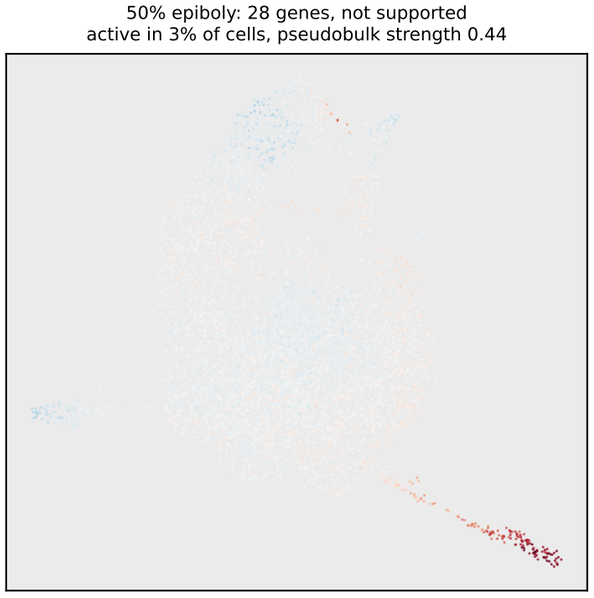
dome: 26 genes, not supported
active in 4% of cells, pseudobulk strength 0.46



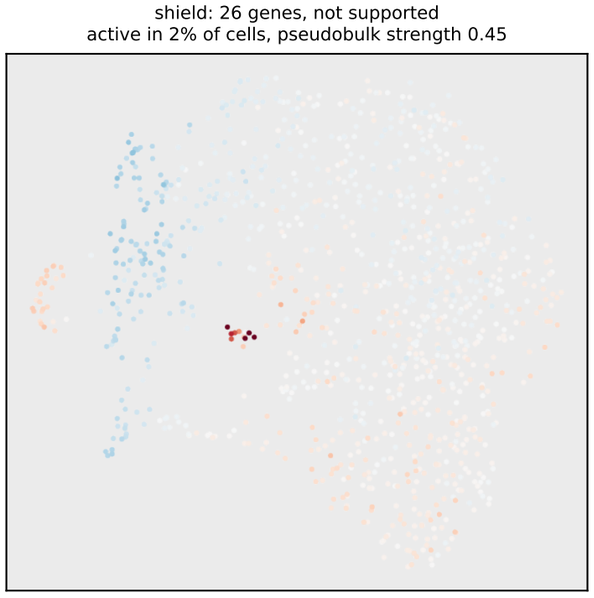
30% epiboly: 25 genes, not supported
active in 4% of cells, pseudobulk strength 0.44



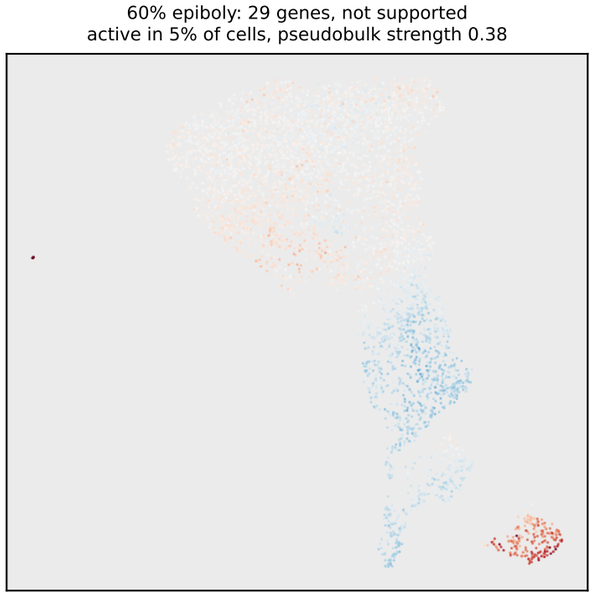
50% epiboly: 28 genes, not supported
active in 3% of cells, pseudobulk strength 0.44



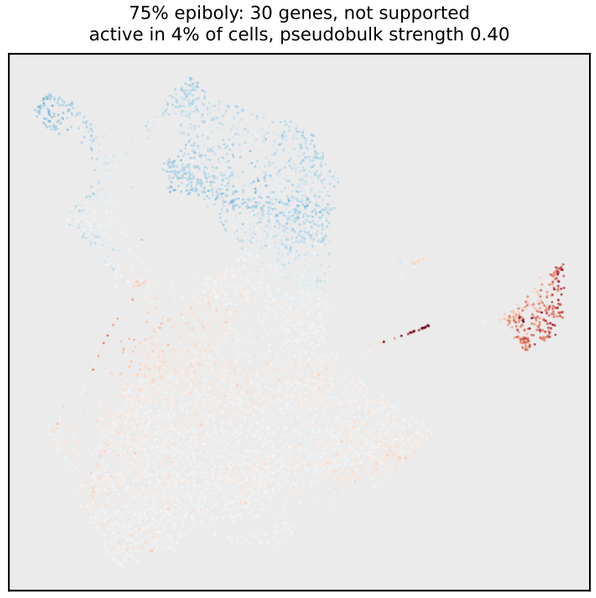
shield: 26 genes, not supported
active in 2% of cells, pseudobulk strength 0.45



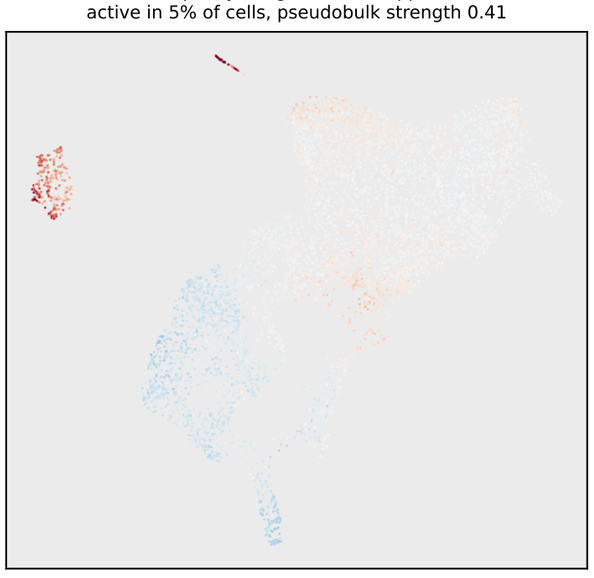
60% epiboly: 29 genes, not supported
active in 5% of cells, pseudobulk strength 0.38



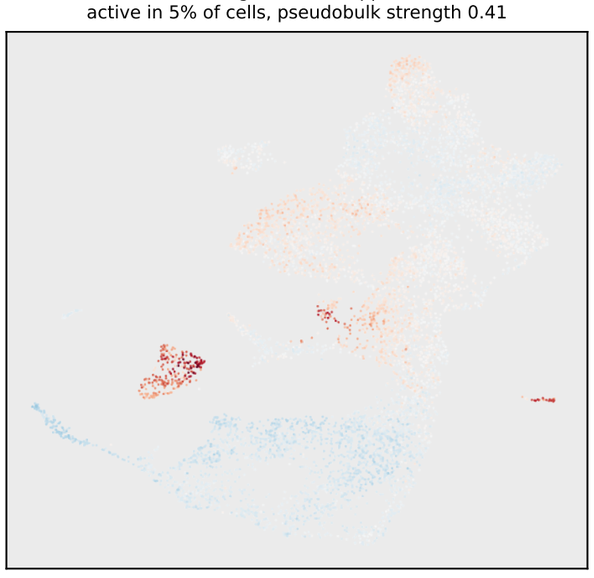
75% epiboly: 30 genes, not supported
active in 4% of cells, pseudobulk strength 0.40



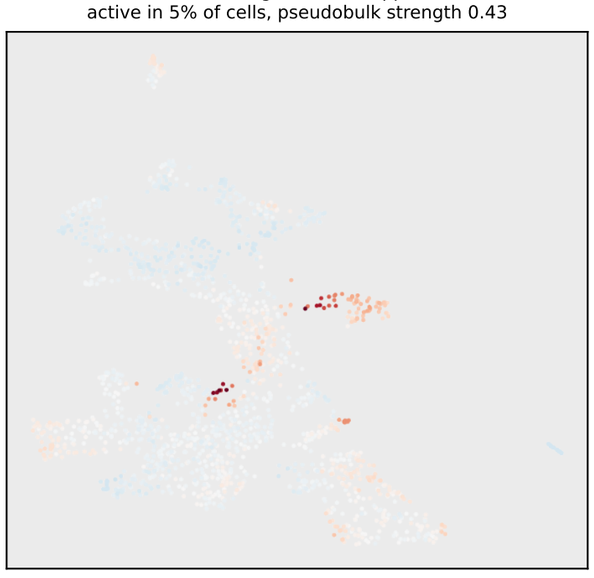
90% epiboly: 30 genes, not supported
active in 5% of cells, pseudobulk strength 0.41



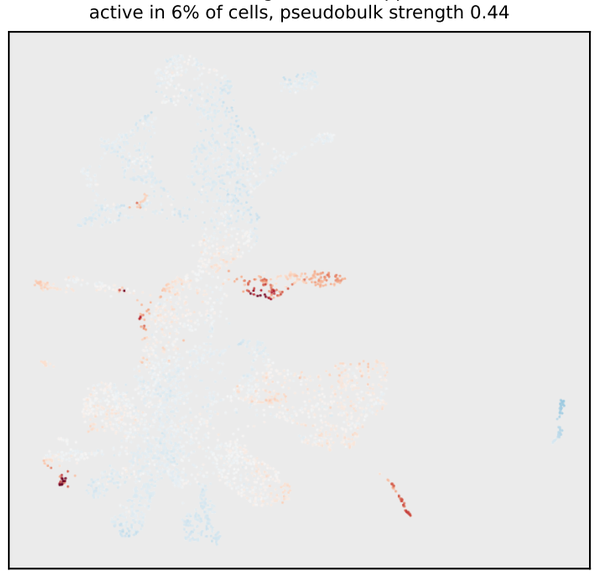
bud: 30 genes, not supported
active in 5% of cells, pseudobulk strength 0.41



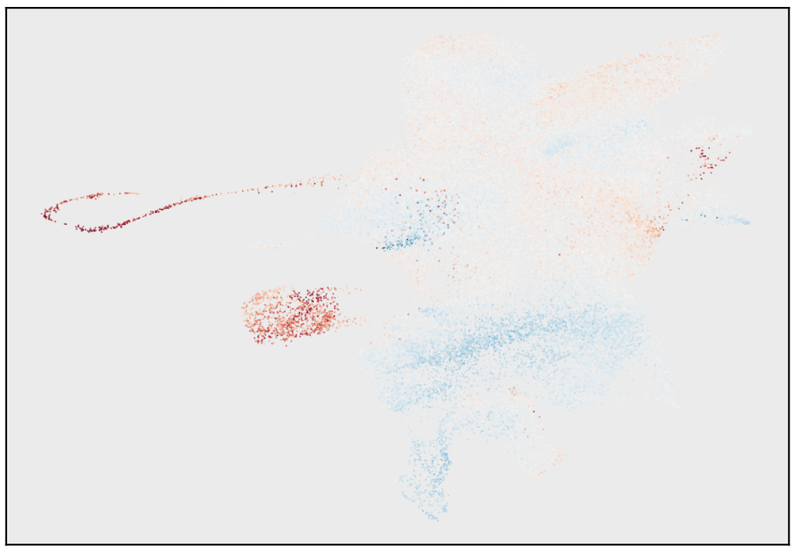
3-somite: 29 genes, not supported
active in 5% of cells, pseudobulk strength 0.43



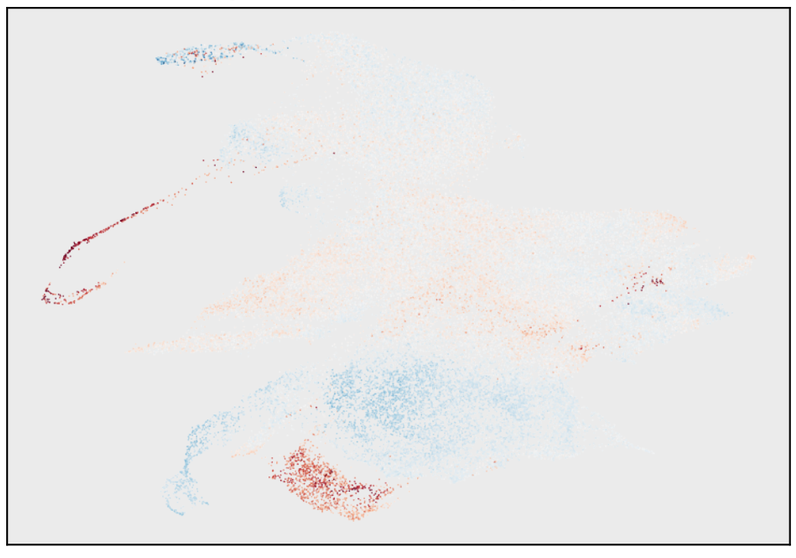
6-somite: 30 genes, not supported
active in 6% of cells, pseudobulk strength 0.44



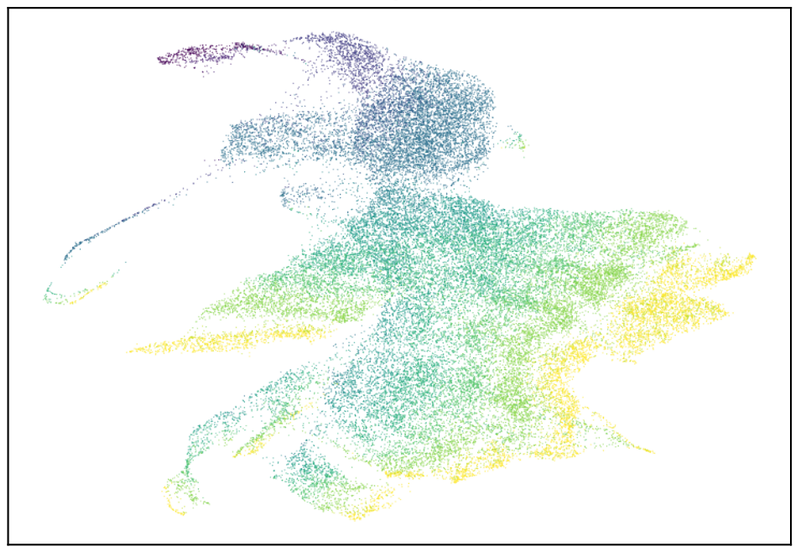
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



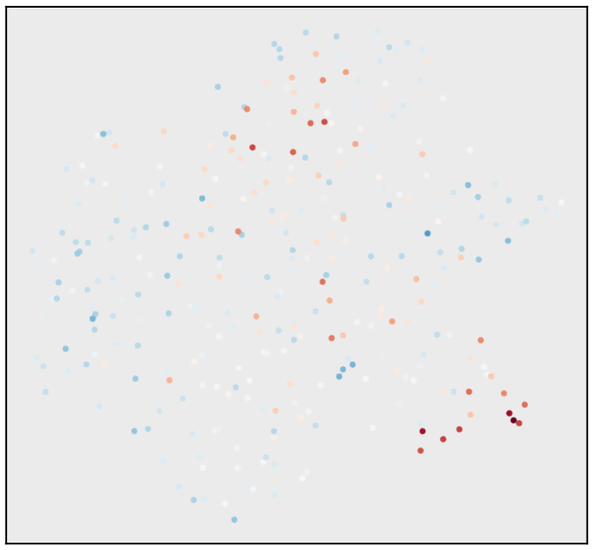
atlas, unaligned: stages



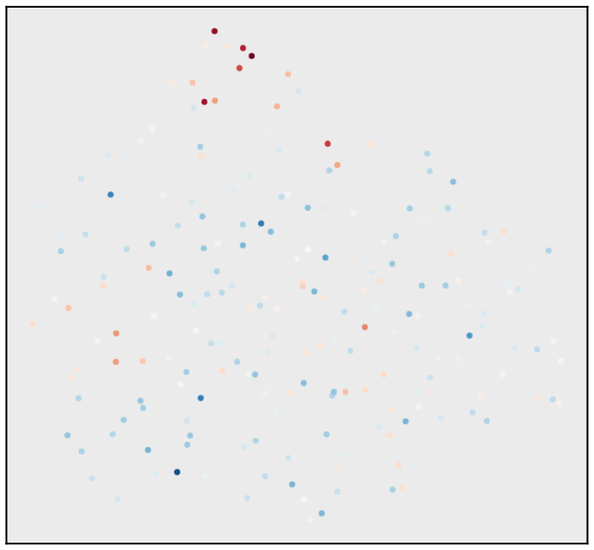
Genes (number of stages): sult6b1 (12), cxcr4a (12), rassf7b (12), vgll4l (12), si:ch211-152c2.3 (12), tjp3 (12), apela (12), rasgef1ba (12), jupa (12), mns1 (12), arg2 (12), dnmt3bb.1 (12), rock2b (12), pard6gb (12), tmem106ba (12), zgc:55461 (12), prickle1a (12), pfkfb4b (12), ca9 (11), cfap53 (11), cfap45 (11), cdh2 (11), nnr (11), si:ch73-234b20.5 (10), relt (10), cldn5a (10), foxj1a (8), ribc2 (7), tagln3b (6), ccdc151 (5)

endoderm

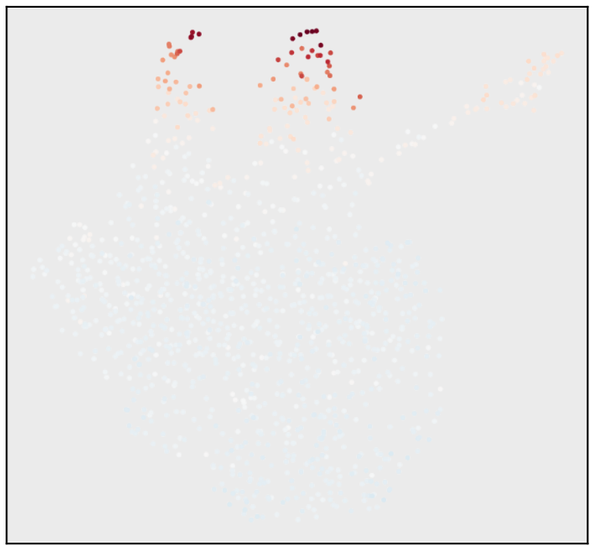
high: 9 genes, not supported
active in 0% of cells, pseudobulk strength 0.28



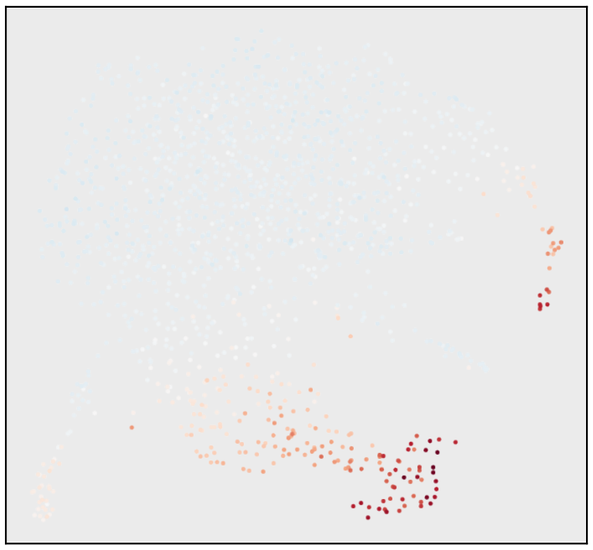
oblong: 8 genes, not supported
active in 0% of cells, pseudobulk strength 0.28



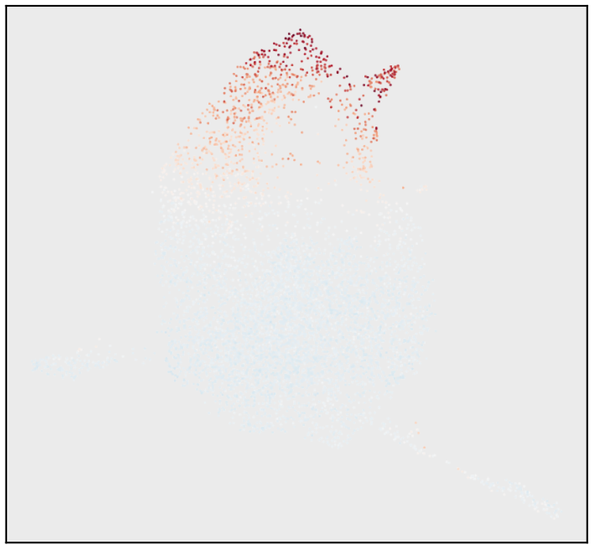
dome: 14 genes, not supported
active in 6% of cells, pseudobulk strength 0.27



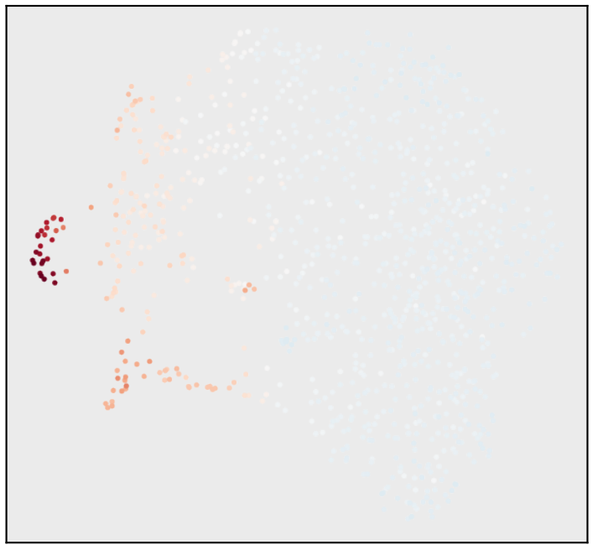
30% epiboly: 14 genes, not supported
active in 9% of cells, pseudobulk strength 0.32



50% epiboly: 17 genes, not supported
active in 13% of cells, pseudobulk strength 0.34

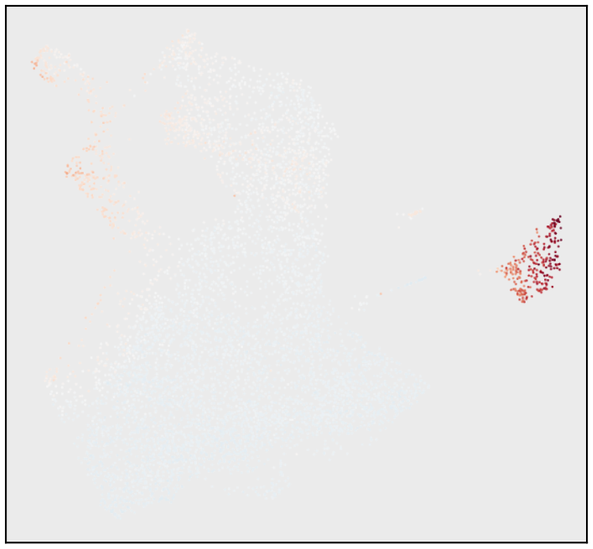


shield: 13 genes, supported
active in 11% of cells, pseudobulk strength 0.36

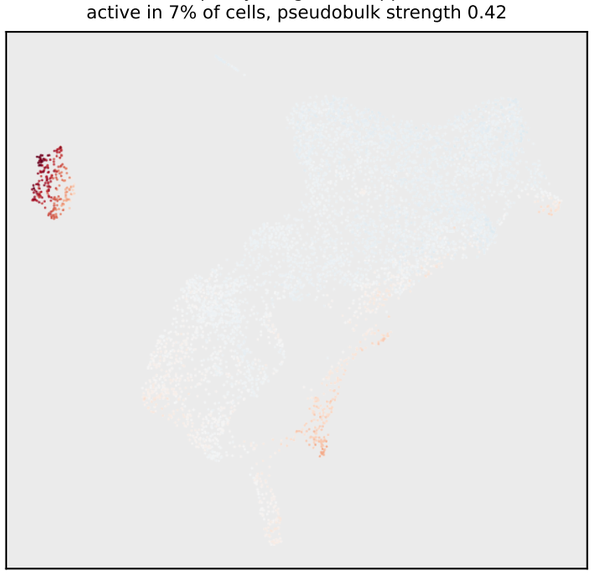


60% epiboly: 19 genes, supported
active in 11% of cells, pseudobulk strength 0.48

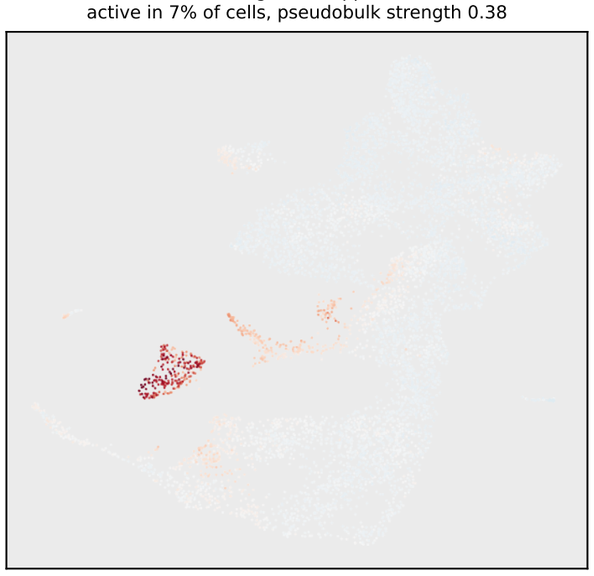
75% epiboly: 21 genes, supported
active in 8% of cells, pseudobulk strength 0.40



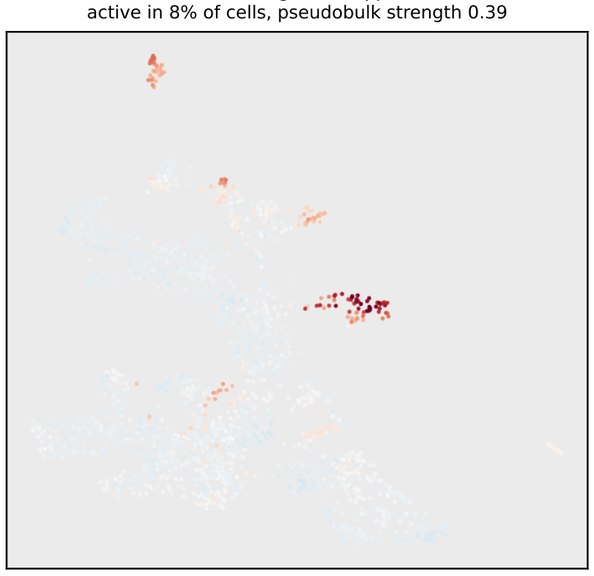
90% epiboly: 23 genes, supported
active in 7% of cells, pseudobulk strength 0.42



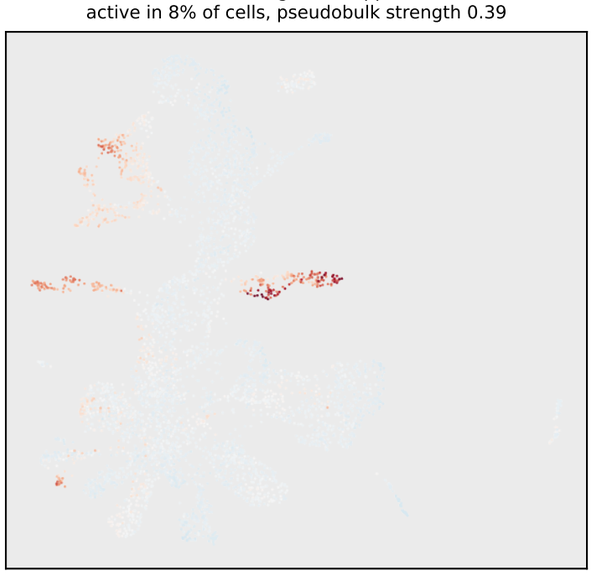
bud: 23 genes, supported
active in 7% of cells, pseudobulk strength 0.38



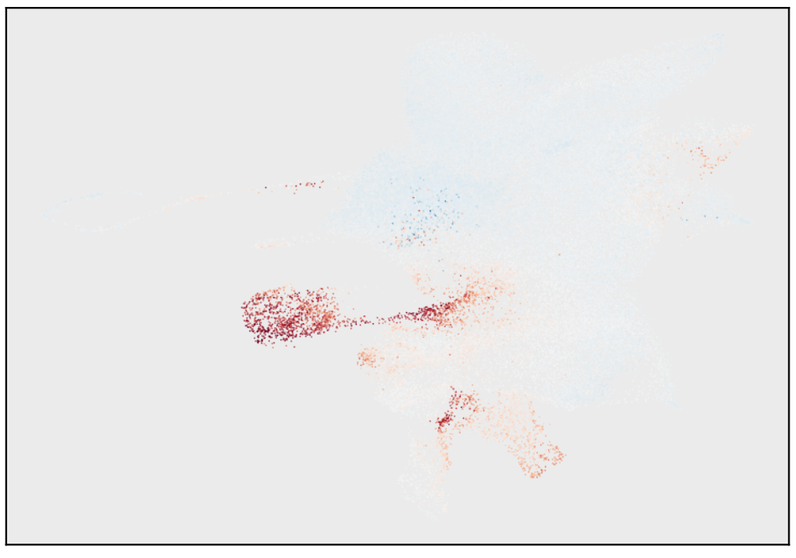
3-somite: 18 genes, supported
active in 8% of cells, pseudobulk strength 0.39



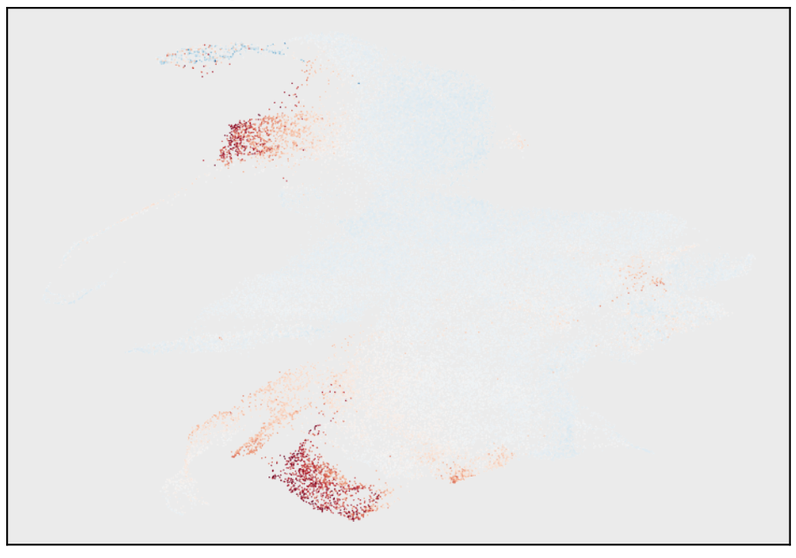
6-somite: 20 genes, supported
active in 8% of cells, pseudobulk strength 0.39



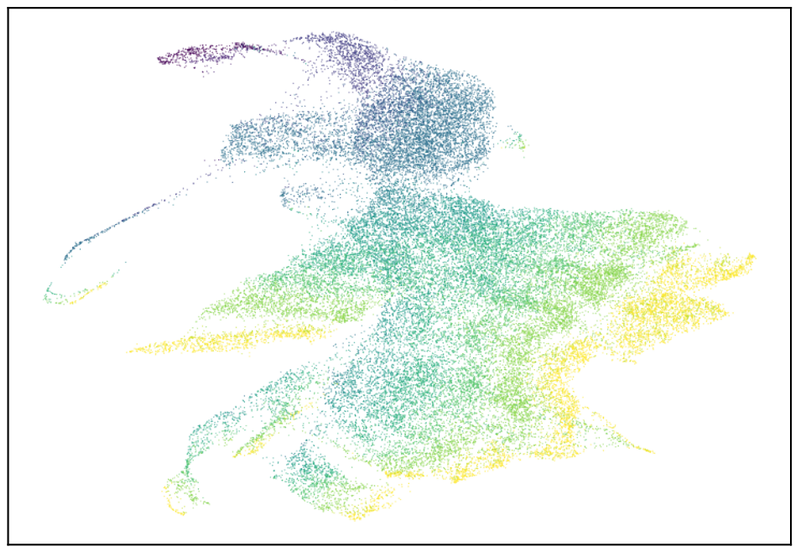
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



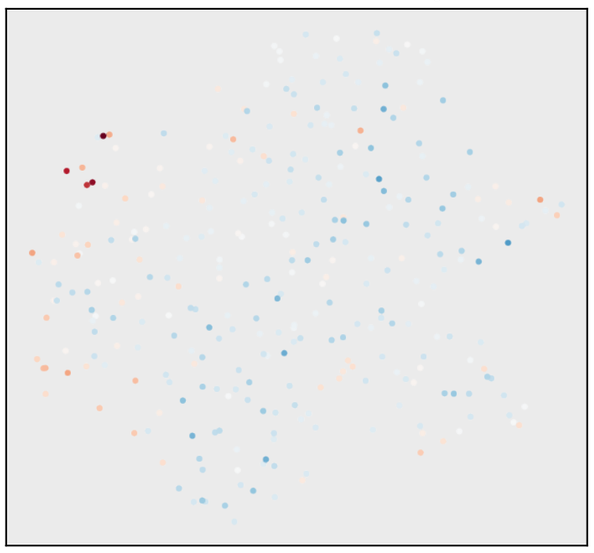
atlas, unaligned: stages



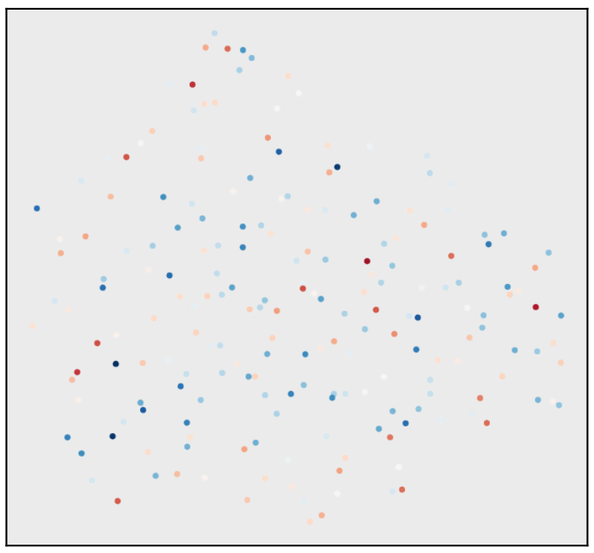
Genes (number of stages): ackr3b (12), gid8b (12), sept9a (12), etv5a (12), flrt3 (11), nme4 (11), gata5 (10), foxa2 (10), foxa3 (10), mixl1 (10), nipsnap1 (10), nkx6.1 (9), sox17 (8), met (8), sox32 (8), sebox (8), foxa1 (6), nkx6.2 (6), fgfrl1b (6), bmp6 (5), mnx1 (5), col2a1a (5), mnx2b (5)

prechordal_plate

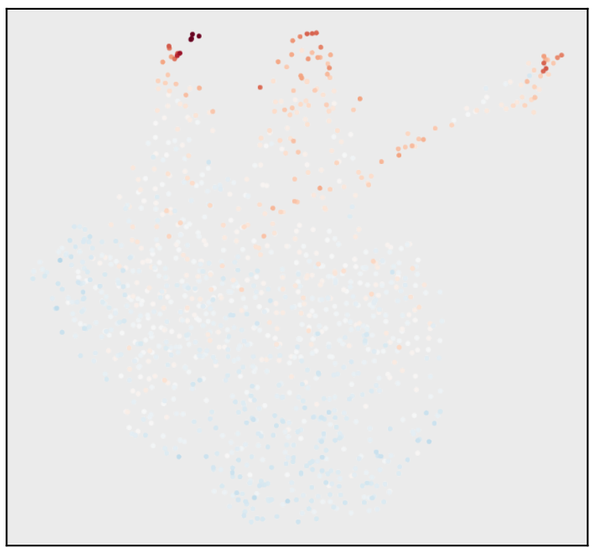
high: 14 genes, not supported
active in 1% of cells, pseudobulk strength 0.28



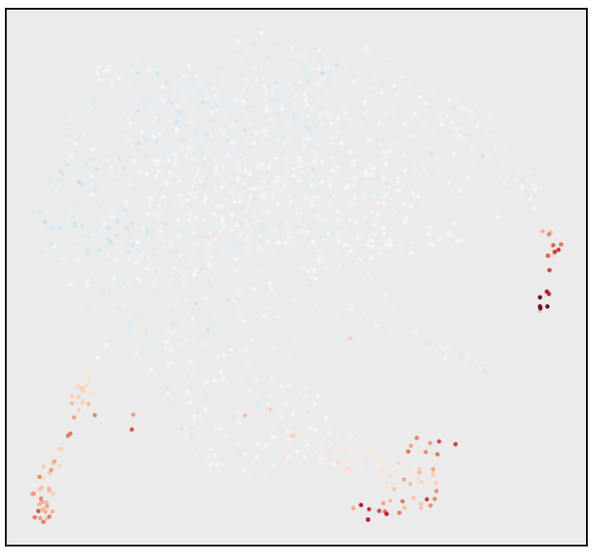
oblong: 14 genes, not supported
active in 0% of cells, pseudobulk strength 0.29



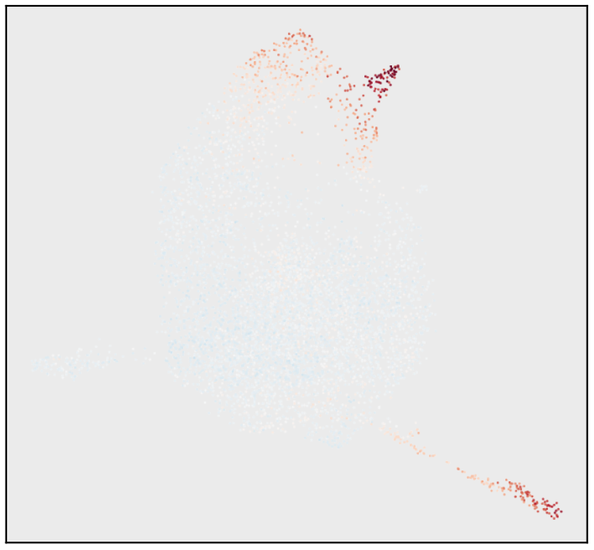
dome: 18 genes, not supported
active in 4% of cells, pseudobulk strength 0.27



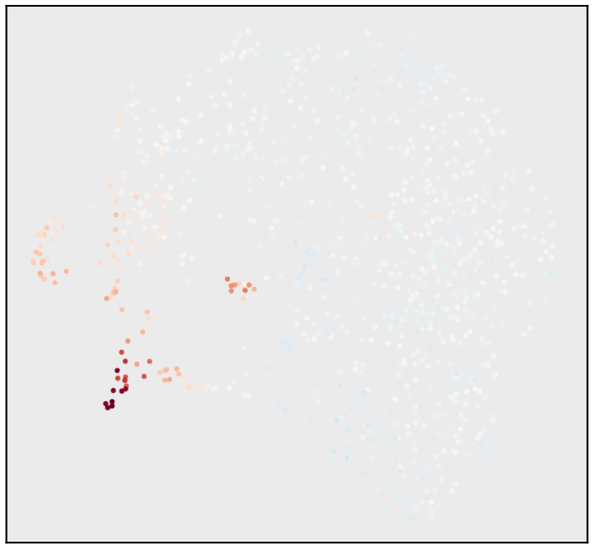
30% epiboly: 18 genes, not supported
active in 5% of cells, pseudobulk strength 0.32



50% epiboly: 26 genes, not supported
active in 7% of cells, pseudobulk strength 0.32



shield: 20 genes, not supported
active in 5% of cells, pseudobulk strength 0.30

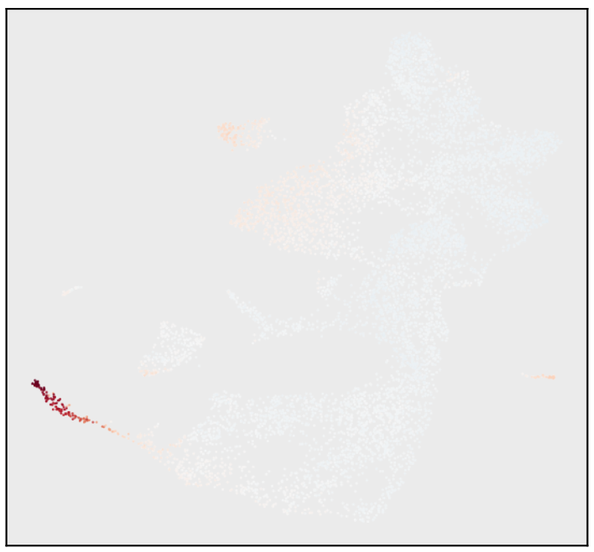


60% epiboly: 26 genes, supported
active in 6% of cells, pseudobulk strength 0.37

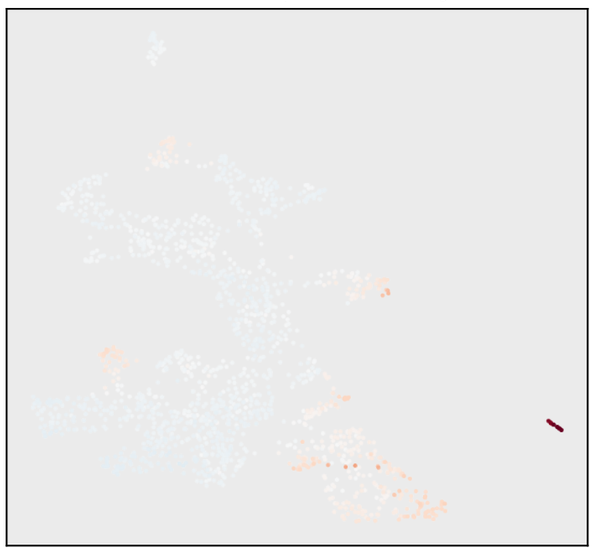
75% epiboly: 30 genes, supported
active in 4% of cells, pseudobulk strength 0.36

90% epiboly: 30 genes, supported
active in 5% of cells, pseudobulk strength 0.45

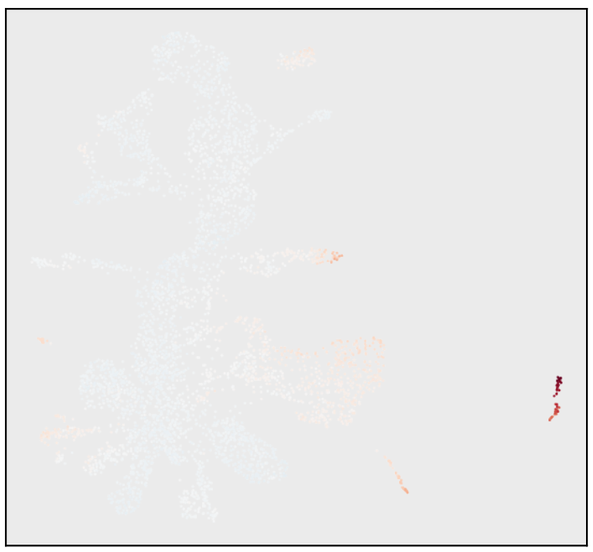
bud: 30 genes, supported
active in 7% of cells, pseudobulk strength 0.44



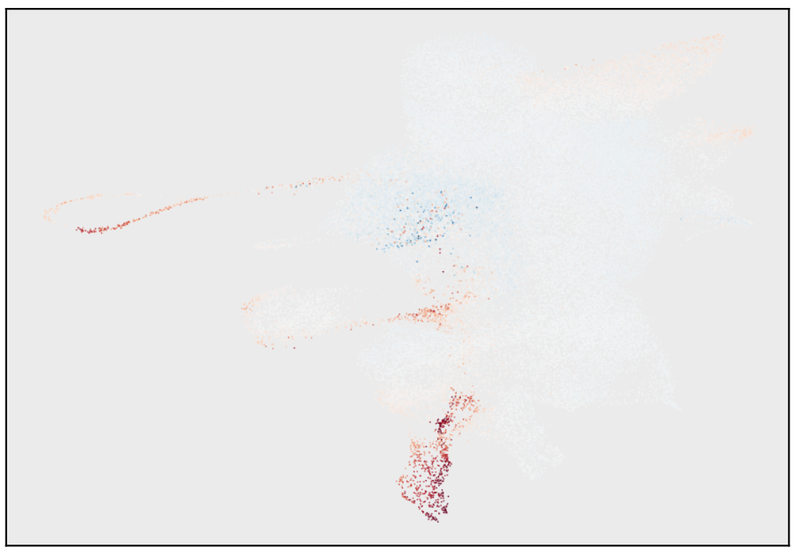
3-somite: 27 genes, supported
active in 10% of cells, pseudobulk strength 0.45



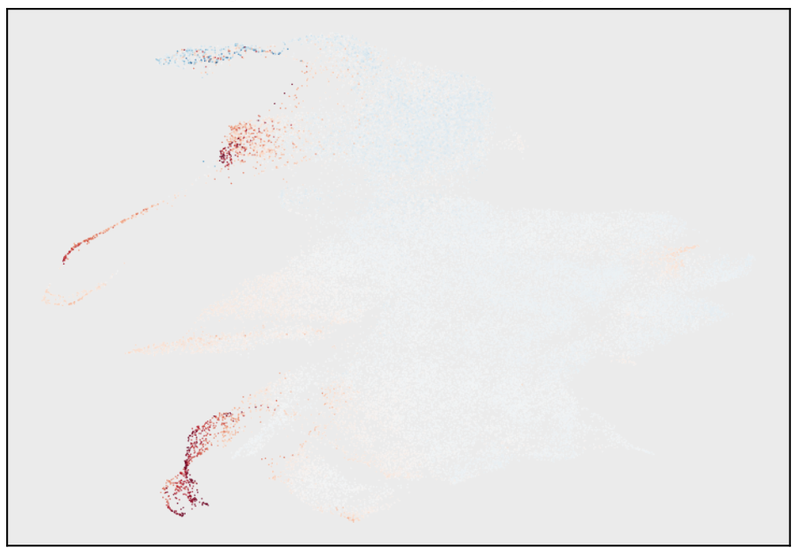
6-somite: 30 genes, supported
active in 8% of cells, pseudobulk strength 0.51



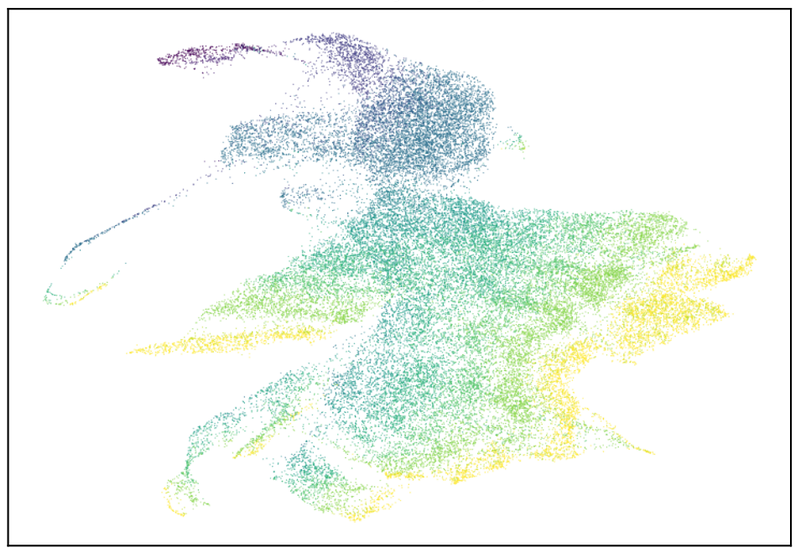
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



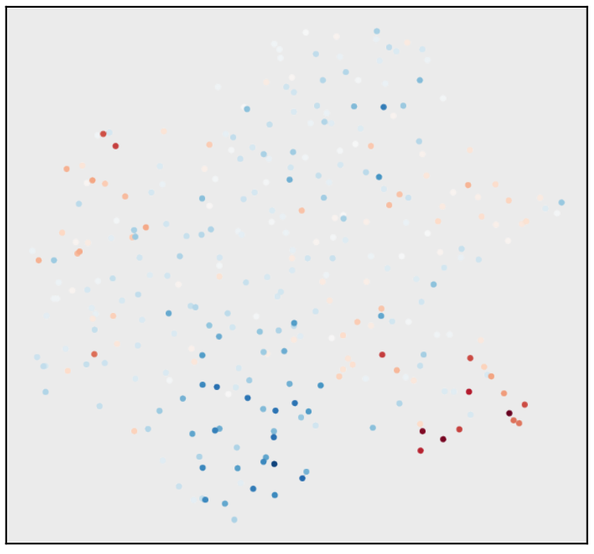
atlas, unaligned: stages



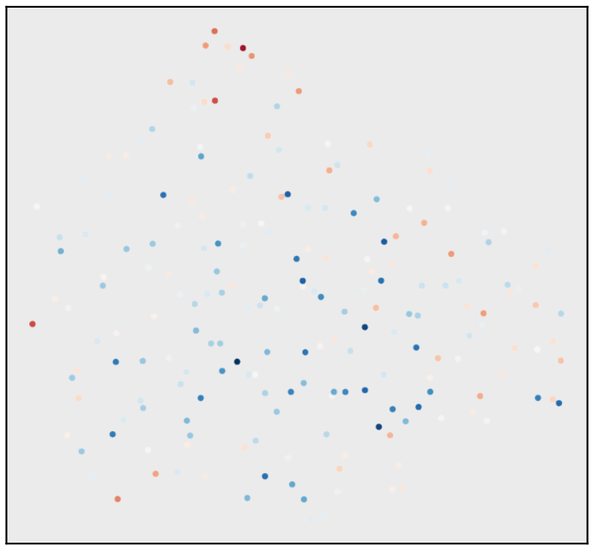
Genes (number of stages): gsc (12), gadd45ga (12), ddit4 (12), cd63 (12), asb11 (12), cdh1 (12), b3gnt5b (12), fzd8a (12), hyou1 (12), fam84a (12), nav2b (12), ezrb (12), si:dkey-27i16.2 (12), tagln2 (11), mllt1b (11), pitx2 (10), elovl6 (10), atp1b3a (10), icn (7), rdh8a (7), pcdh10b (7), nr2f6b (7), tmem120a (7), lmo1 (7), ctslb (6), nkx2.2b (6), si:ch211-163l21.8 (6), inhbaa (5), nmt1b (5), capn2b (5)

notochord

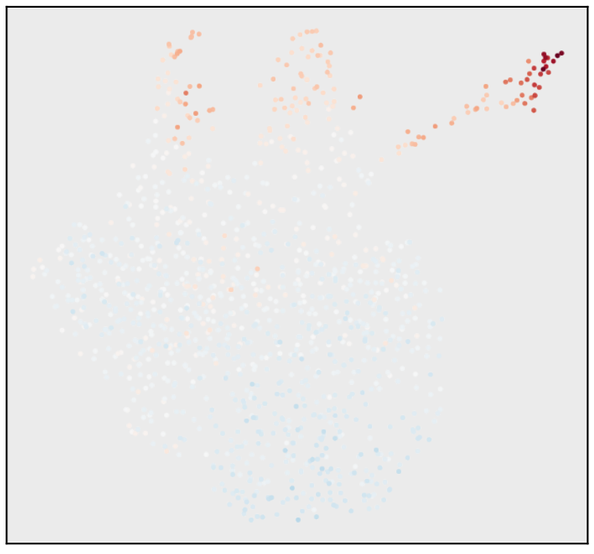
high: 20 genes, not supported
active in 1% of cells, pseudobulk strength 0.46



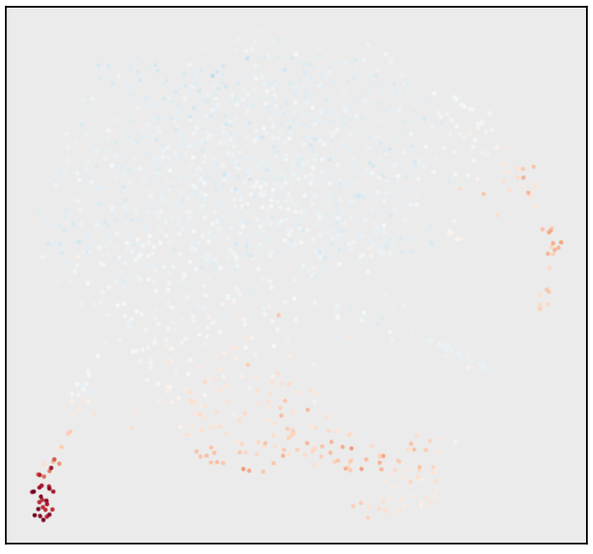
oblong: 17 genes, not supported
active in 1% of cells, pseudobulk strength 0.43



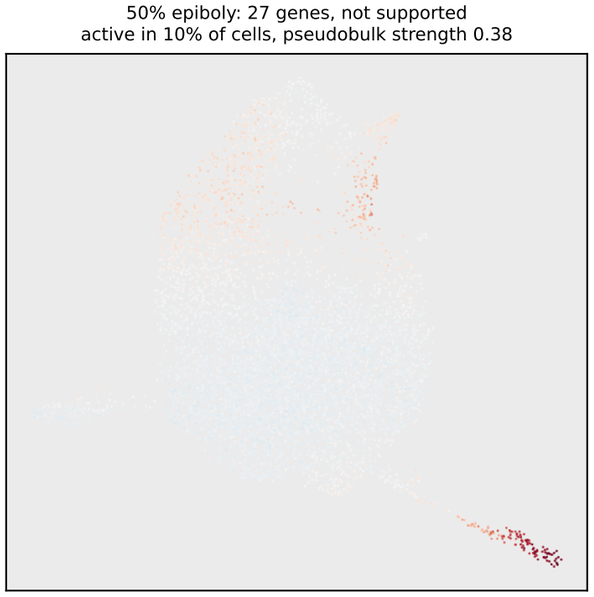
dome: 21 genes, not supported
active in 7% of cells, pseudobulk strength 0.43



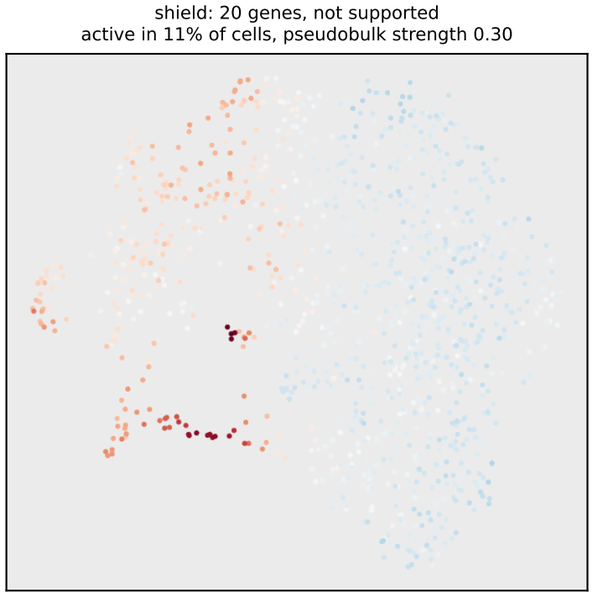
30% epiboly: 21 genes, not supported
active in 9% of cells, pseudobulk strength 0.39



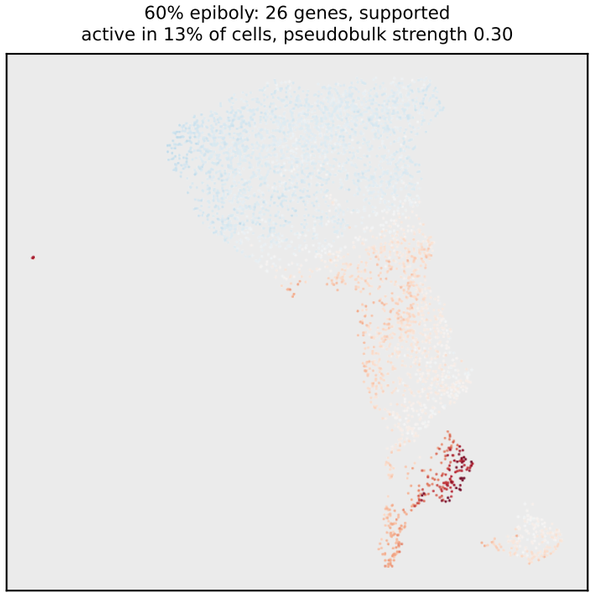
50% epiboly: 27 genes, not supported
active in 10% of cells, pseudobulk strength 0.38



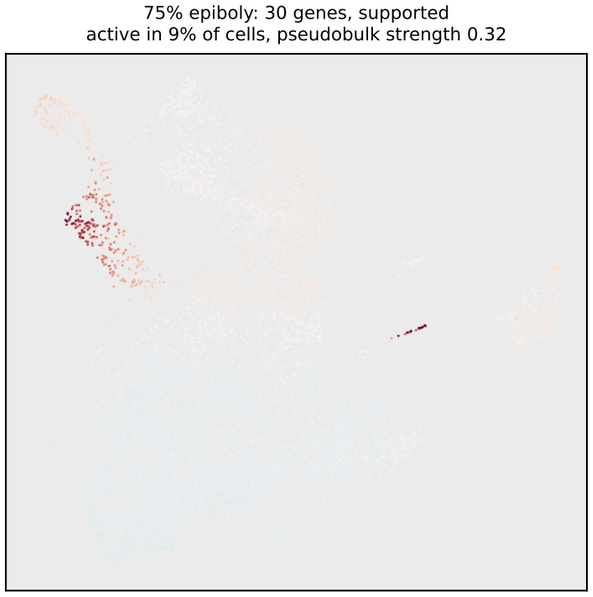
shield: 20 genes, not supported
active in 11% of cells, pseudobulk strength 0.30



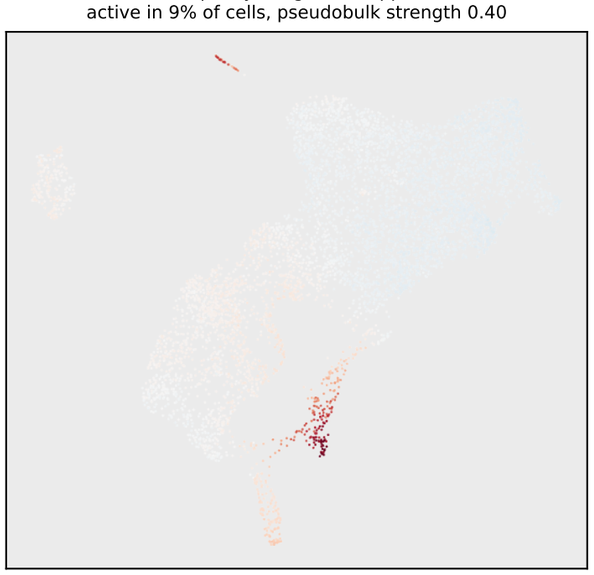
60% epiboly: 26 genes, supported
active in 13% of cells, pseudobulk strength 0.30



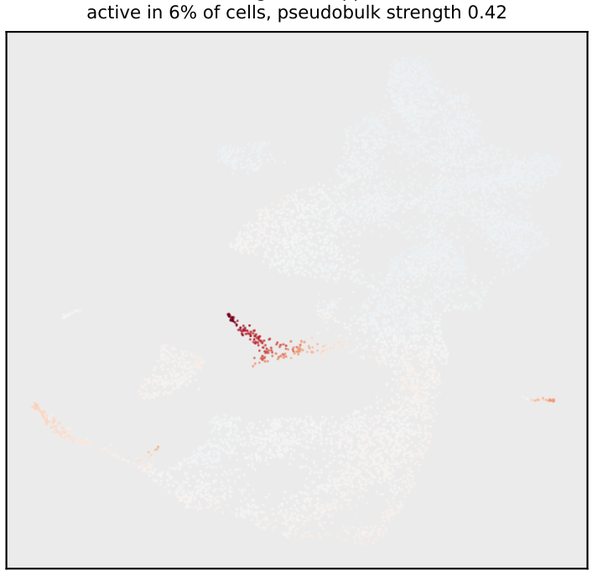
75% epiboly: 30 genes, supported
active in 9% of cells, pseudobulk strength 0.32



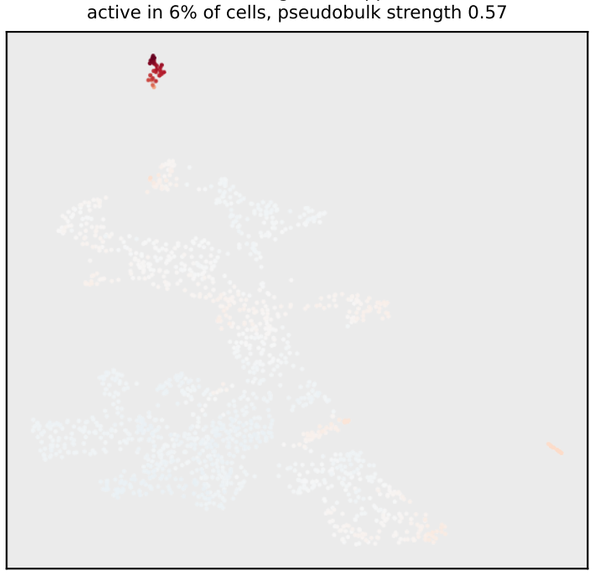
90% epiboly: 30 genes, supported
active in 9% of cells, pseudobulk strength 0.40



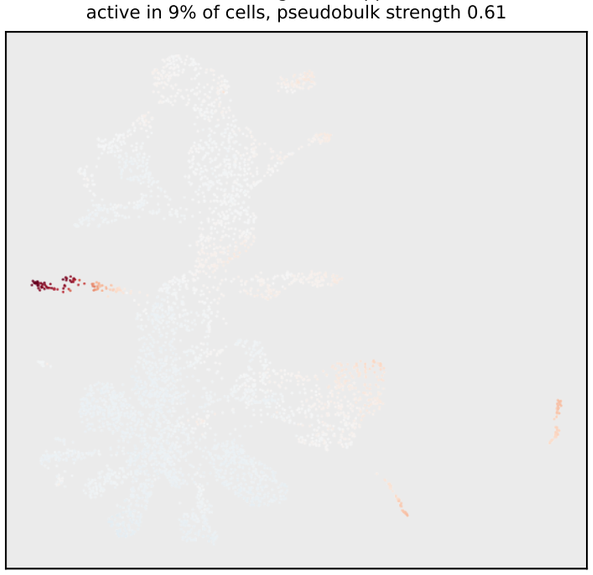
bud: 30 genes, supported
active in 6% of cells, pseudobulk strength 0.42



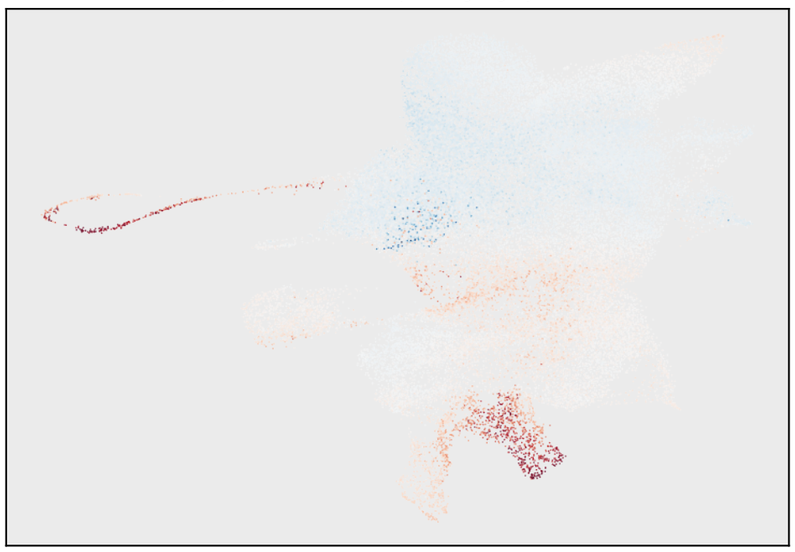
3-somite: 30 genes, supported
active in 6% of cells, pseudobulk strength 0.57



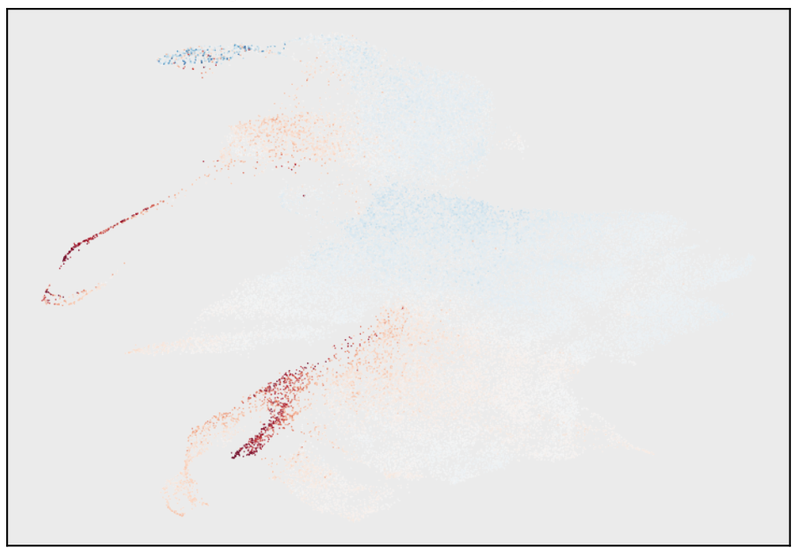
6-somite: 30 genes, supported
active in 9% of cells, pseudobulk strength 0.61



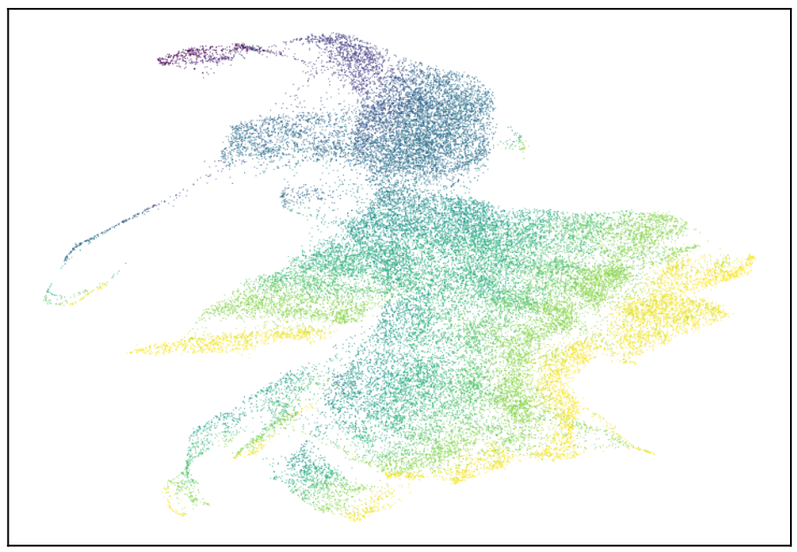
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



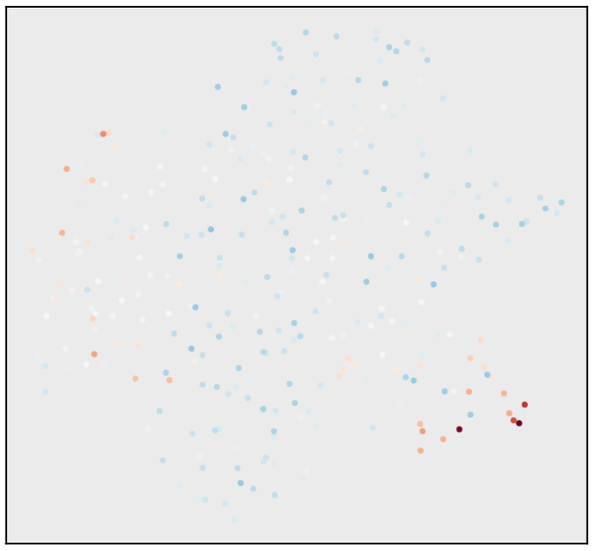
atlas, unaligned: stages



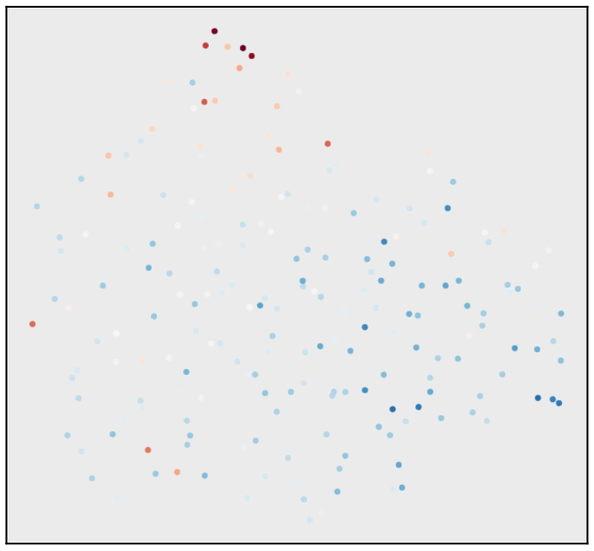
Genes (number of stages): ta (12), xbp1 (12), zgc:162964 (12), cnn2 (12), waslb (12), klf6a (12), asph (12), cast (12), rcn3 (12), pak1 (12), si:dkey-56m19.5 (12), acsbg2 (12), fkbp14 (12), cx43.4 (12), lrrc59 (12), sec61b (12), ip6k2b (11), dnajc3a (11), lhfp (11), serpinh1b (9), dsc2l (9), si:dkey-261h17.1 (8), pmp22b (8), twist2 (7), cebpd (7), aff3 (7), anxa2a (6), col8a1a (6), emilin3a (5), hfe2 (5)

presomitic_mesoderm

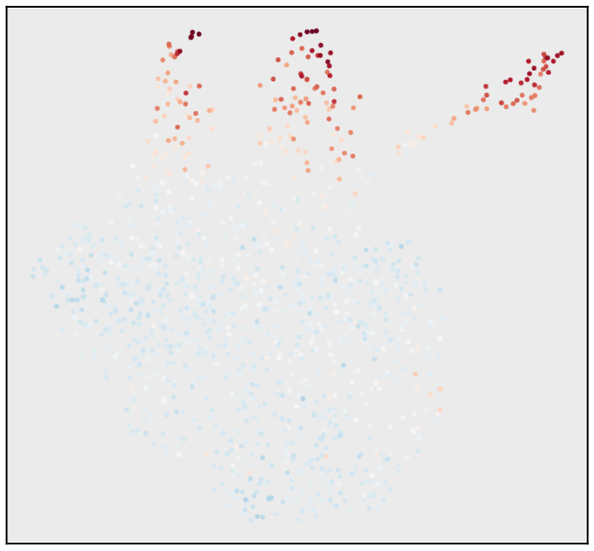
high: 9 genes, not supported
active in 1% of cells, pseudobulk strength 0.12



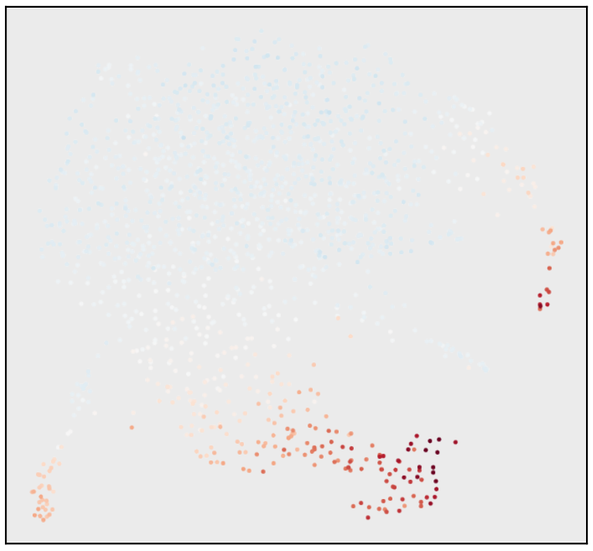
oblong: 8 genes, not supported
active in 2% of cells, pseudobulk strength 0.14



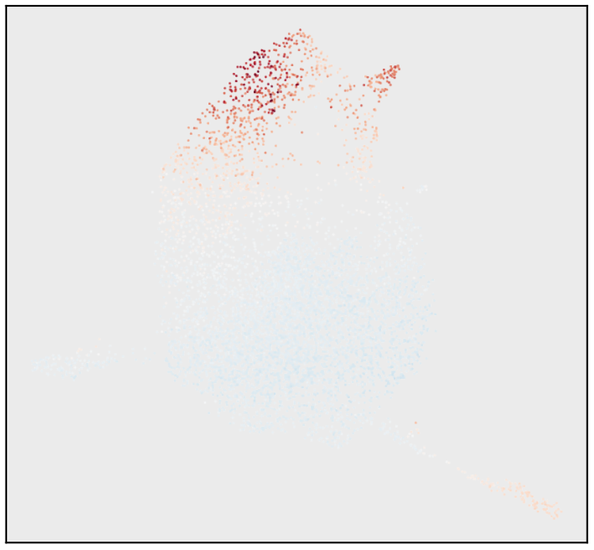
dome: 14 genes, not supported
active in 11% of cells, pseudobulk strength 0.18



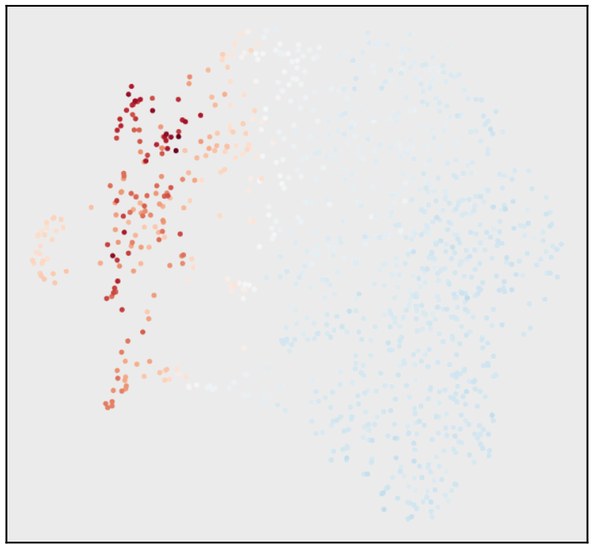
30% epiboly: 21 genes, not supported
active in 12% of cells, pseudobulk strength 0.26



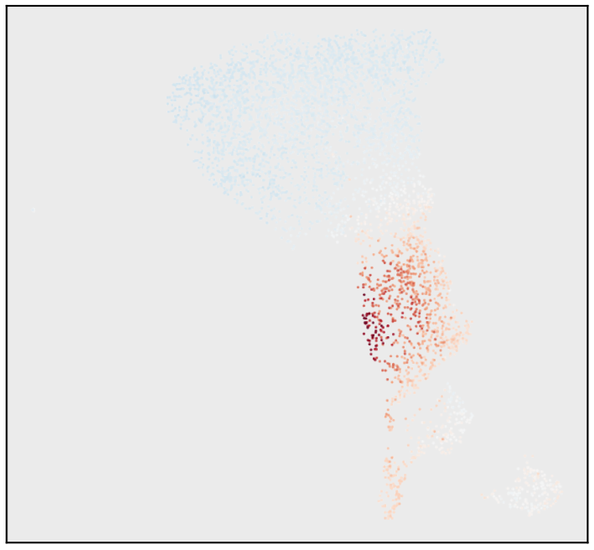
50% epiboly: 27 genes, not supported
active in 15% of cells, pseudobulk strength 0.24



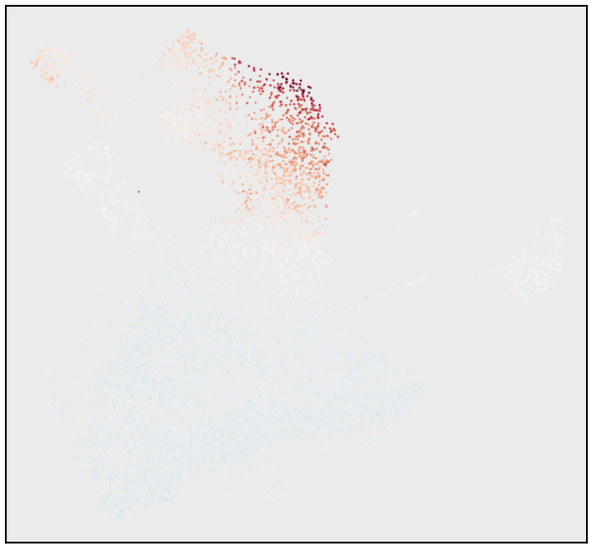
shield: 23 genes, supported
active in 20% of cells, pseudobulk strength 0.37



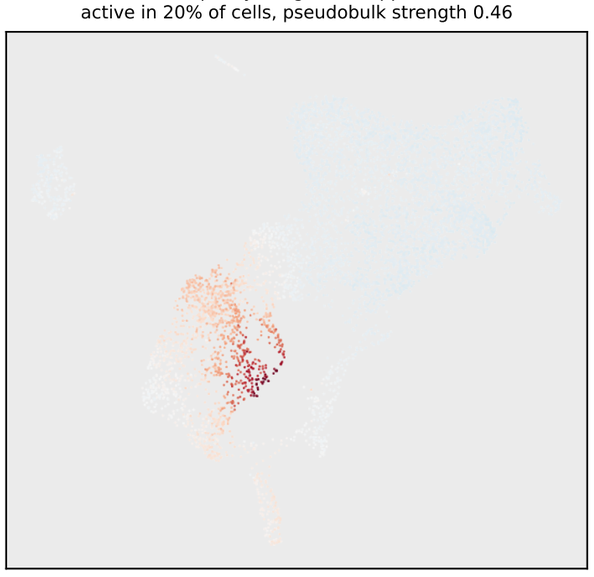
60% epiboly: 30 genes, supported
active in 22% of cells, pseudobulk strength 0.44



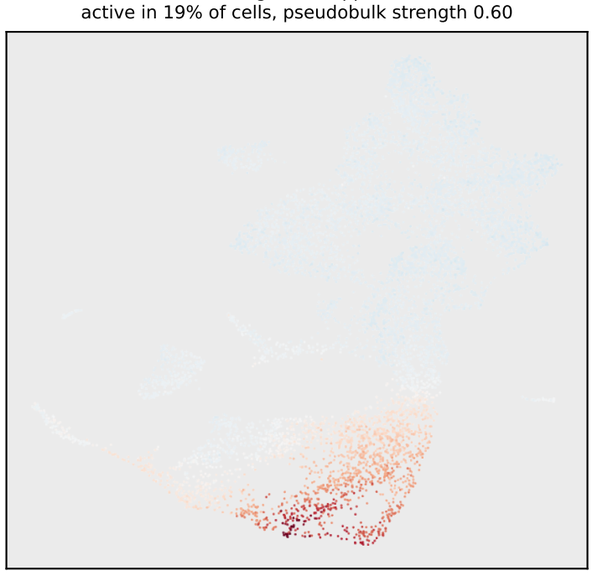
75% epiboly: 30 genes, supported
active in 22% of cells, pseudobulk strength 0.40



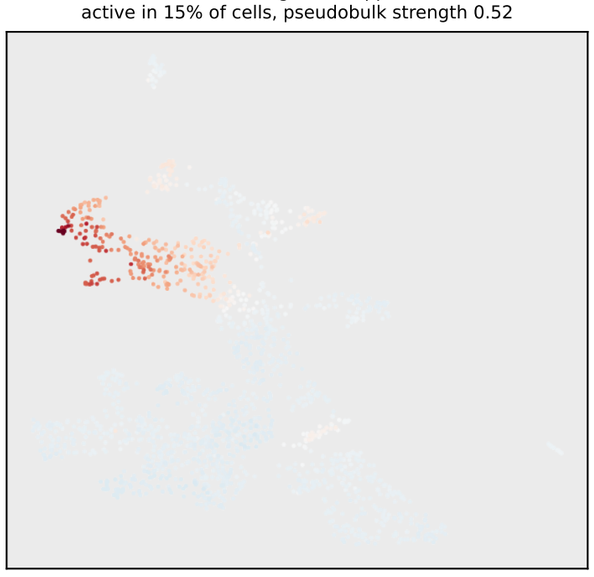
90% epiboly: 30 genes, supported
active in 20% of cells, pseudobulk strength 0.46



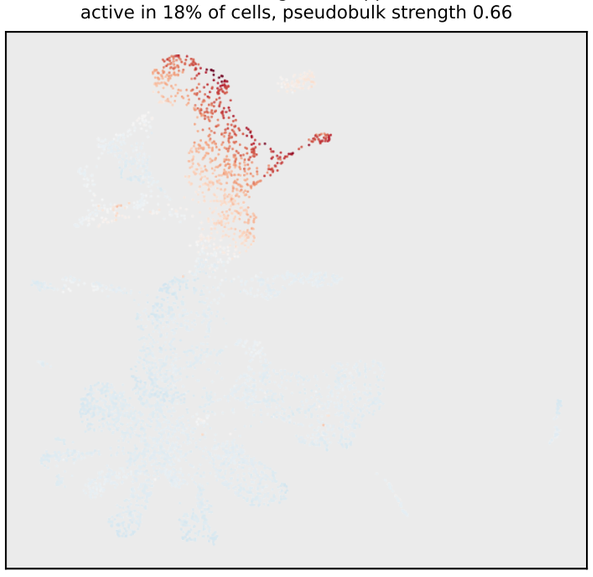
bud: 30 genes, supported
active in 19% of cells, pseudobulk strength 0.60



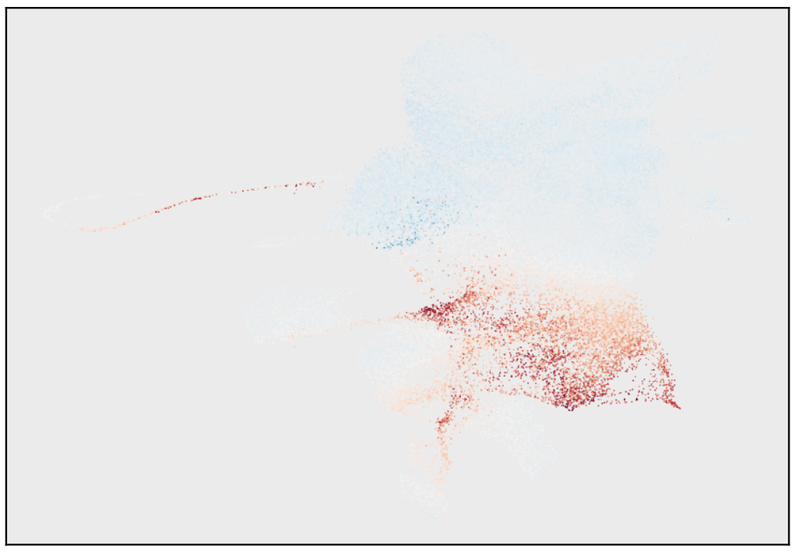
3-somite: 30 genes, supported
active in 15% of cells, pseudobulk strength 0.52



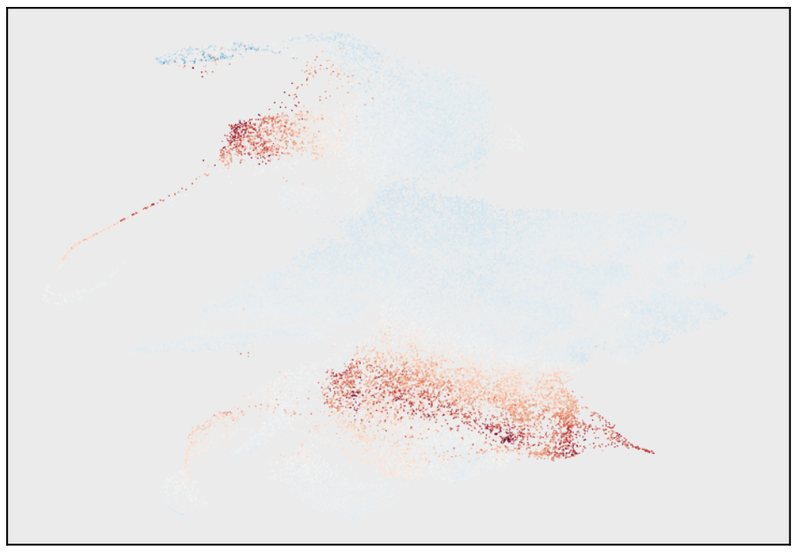
6-somite: 30 genes, supported
active in 18% of cells, pseudobulk strength 0.66



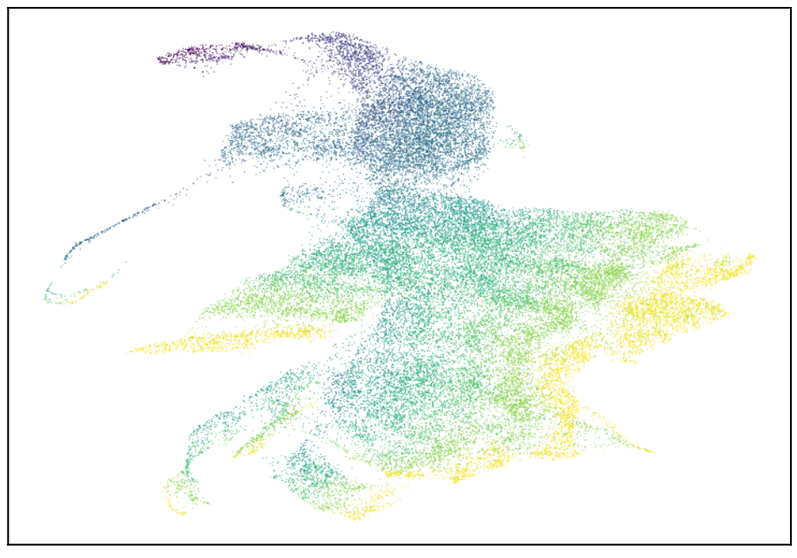
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



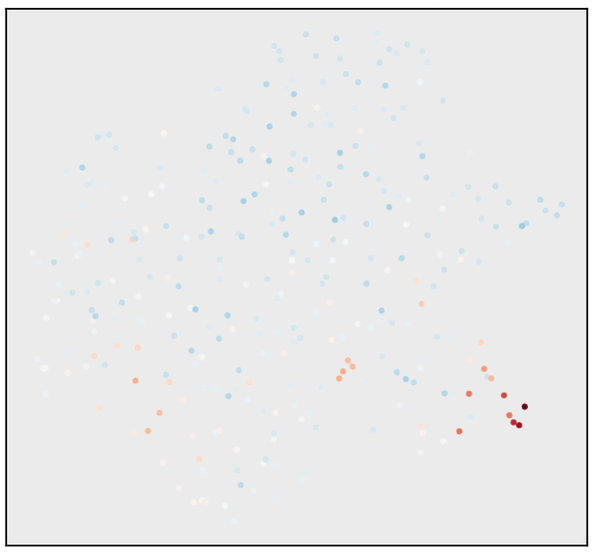
atlas, unaligned: stages



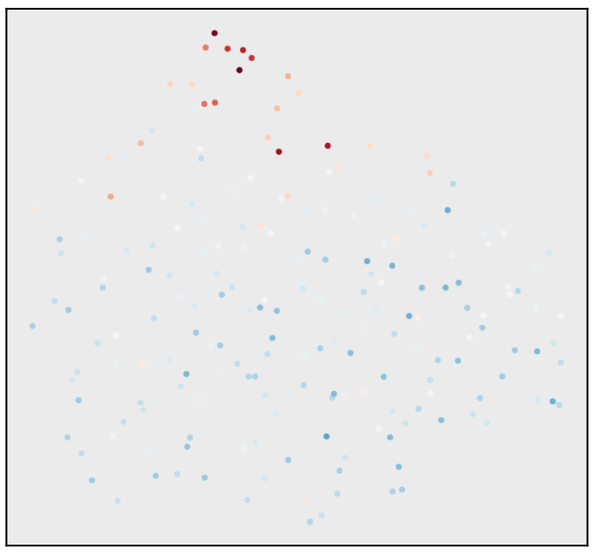
Genes (number of stages): aplnra (12), gadd45ba (12), tbx16 (12), rdh10a (12), snai1a (12), arl4aa (12), fgf17 (12), btg2 (12), mdka (11), mespab (10), ripply1 (10), her1 (10), pcdh8 (10), her7 (10), msgn1 (9), mespaa (9), mespba (9), efemp2b (9), foxc1a (9), myf5 (8), ripply2 (8), tbx6l (8), mespbb (8), ttnb (8), myod1 (7), tbx6 (7), tcf15 (7), fn1b (7), meox1 (6), lbx2 (6)

adaxial_somite

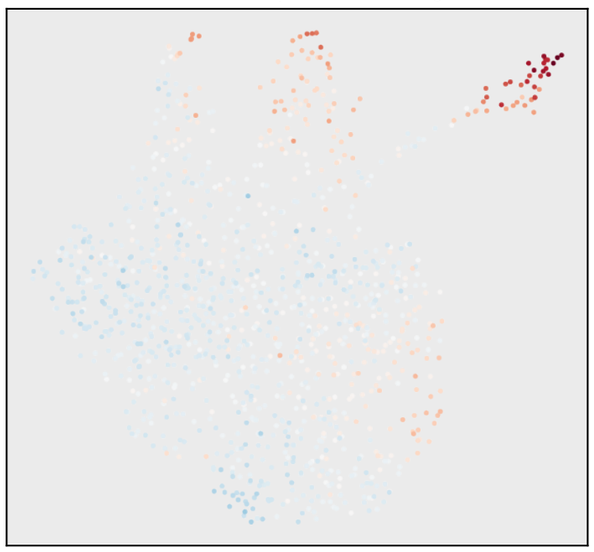
high: 11 genes, not supported
active in 2% of cells, pseudobulk strength 0.18



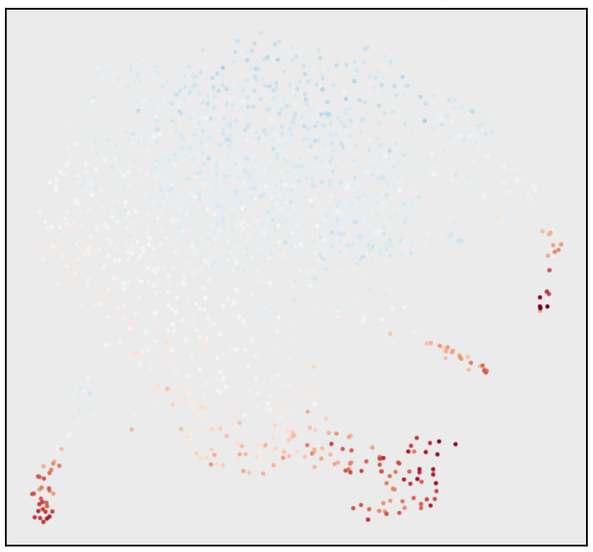
oblong: 8 genes, not supported
active in 3% of cells, pseudobulk strength 0.16



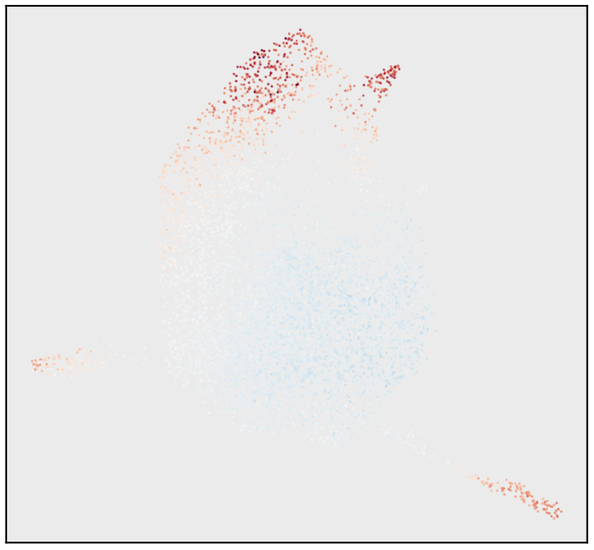
dome: 14 genes, not supported
active in 4% of cells, pseudobulk strength 0.17



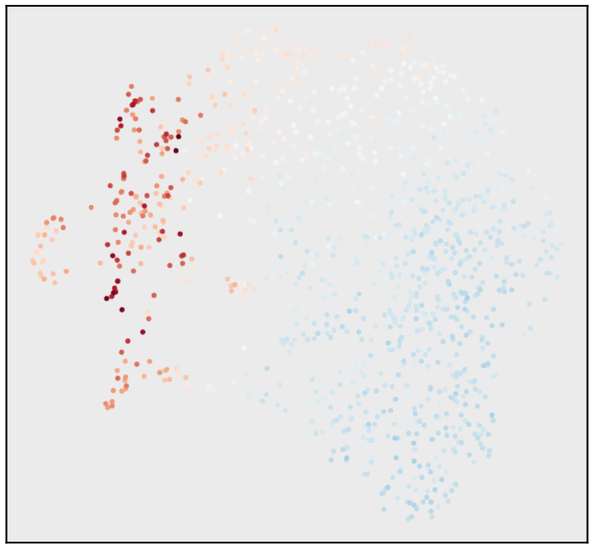
30% epiboly: 17 genes, not supported
active in 11% of cells, pseudobulk strength 0.15



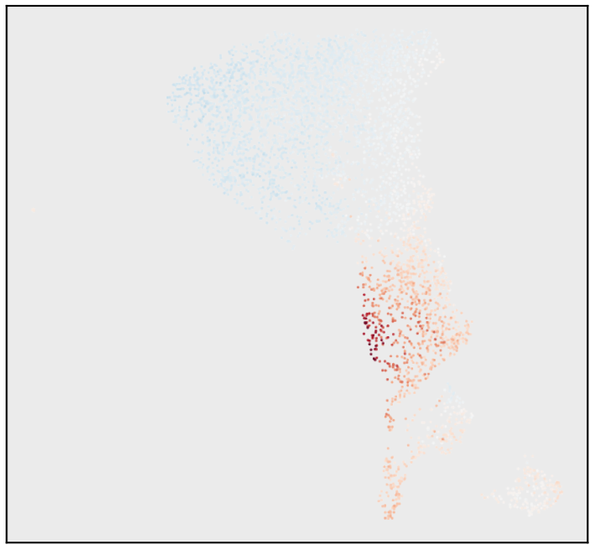
50% epiboly: 23 genes, not supported
active in 12% of cells, pseudobulk strength 0.17



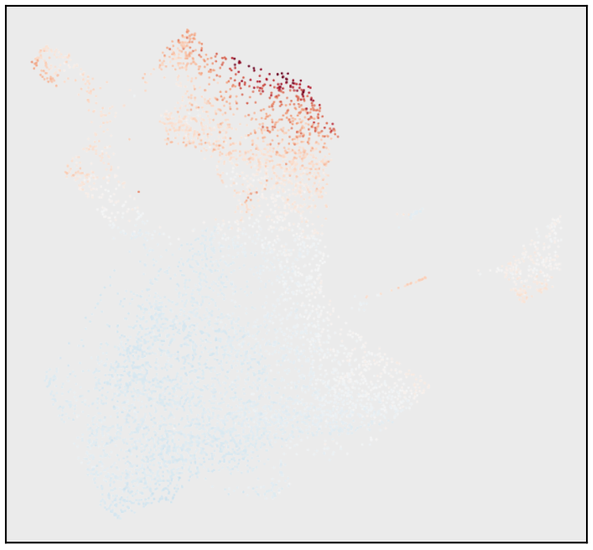
shield: 20 genes, not supported
active in 15% of cells, pseudobulk strength 0.30



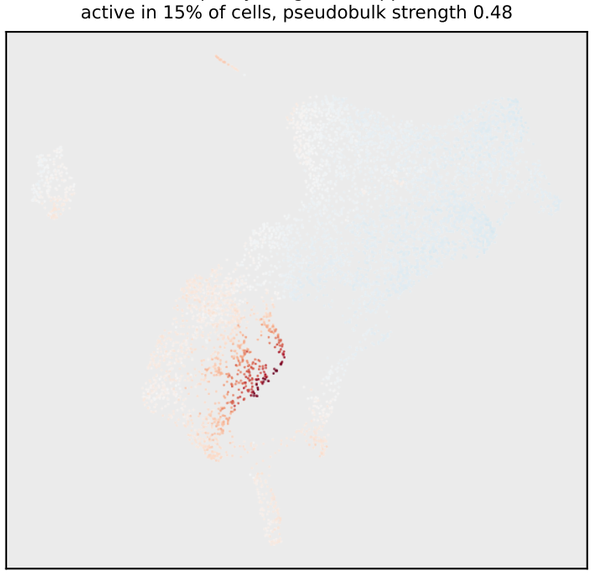
60% epiboly: 29 genes, supported
active in 20% of cells, pseudobulk strength 0.35



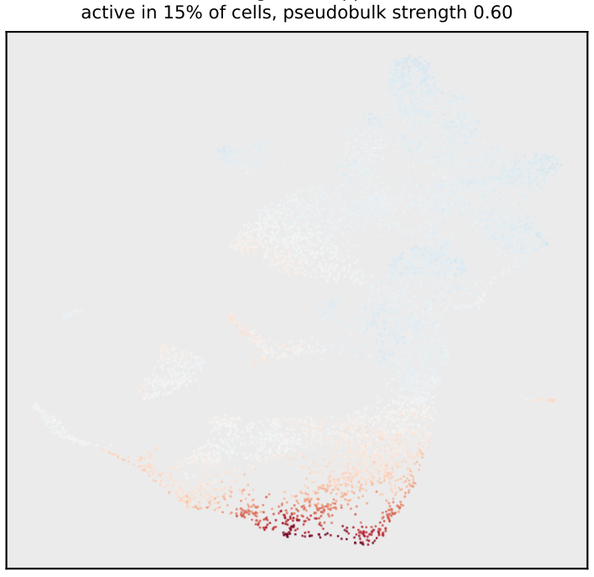
75% epiboly: 29 genes, supported
active in 19% of cells, pseudobulk strength 0.38



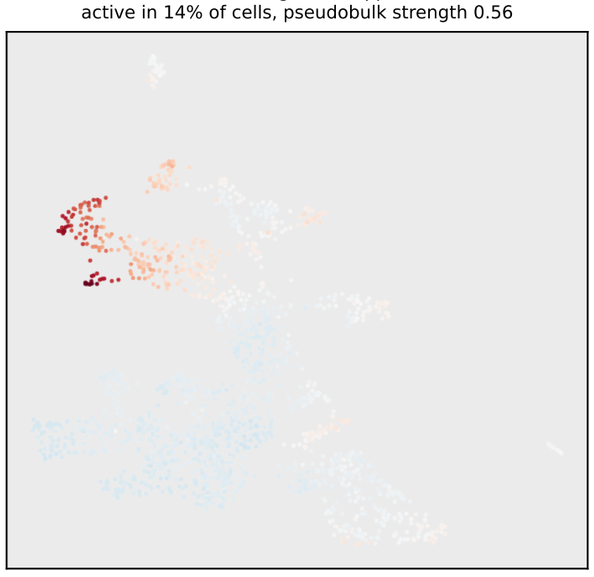
90% epiboly: 30 genes, supported
active in 15% of cells, pseudobulk strength 0.48



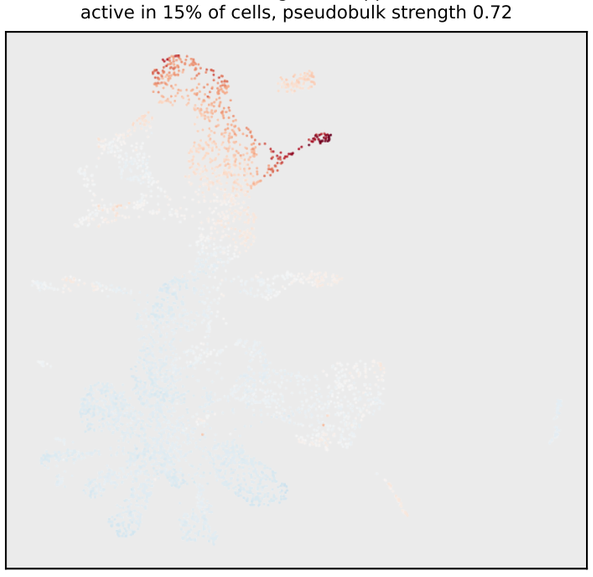
bud: 30 genes, supported
active in 15% of cells, pseudobulk strength 0.60



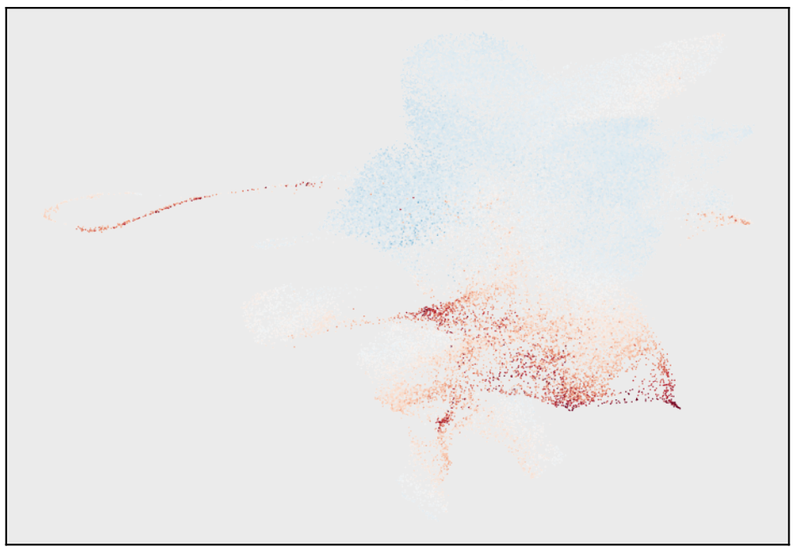
3-somite: 30 genes, supported
active in 14% of cells, pseudobulk strength 0.56



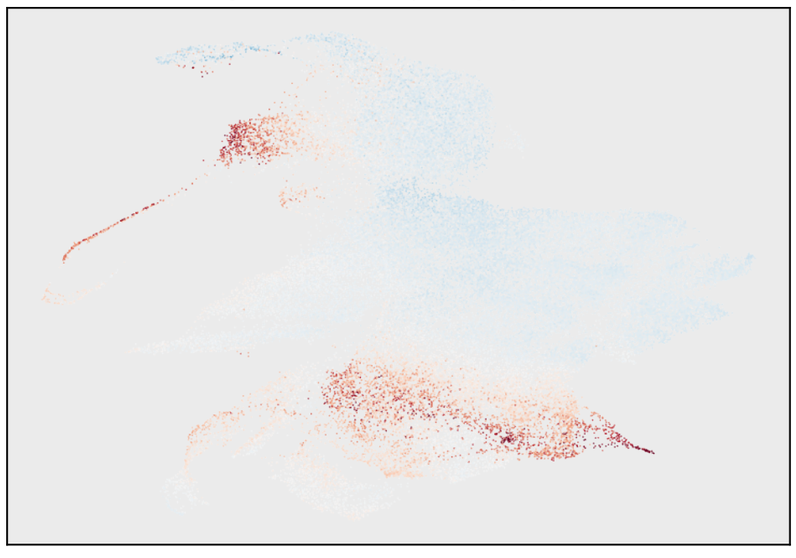
6-somite: 30 genes, supported
active in 15% of cells, pseudobulk strength 0.72



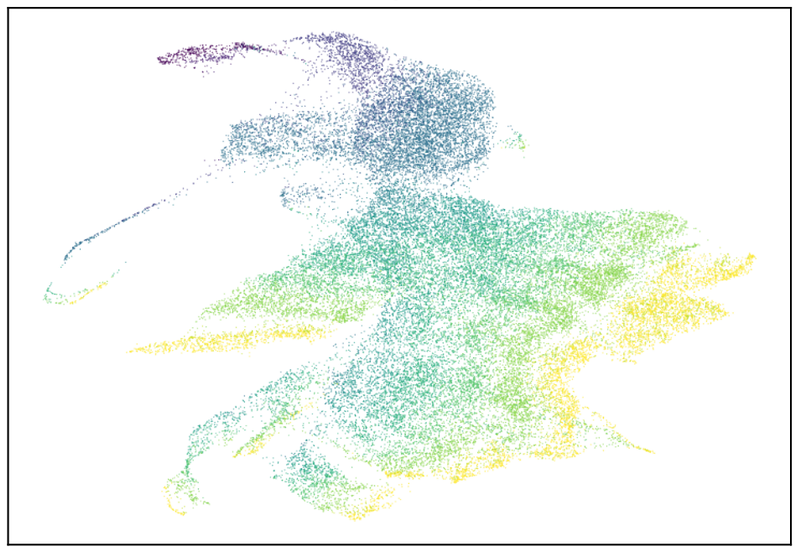
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



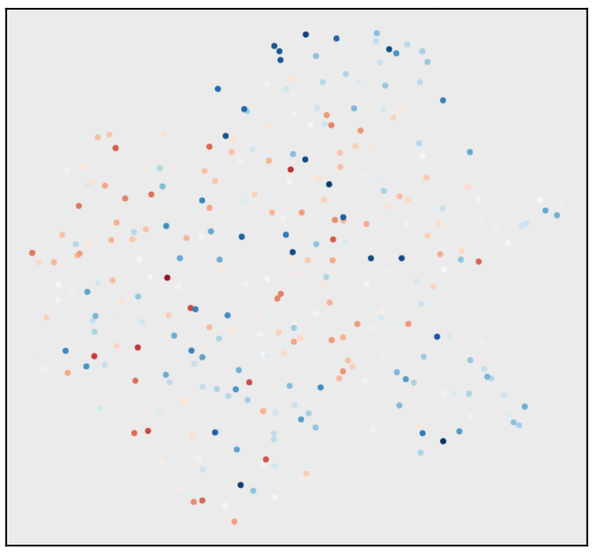
atlas, unaligned: stages



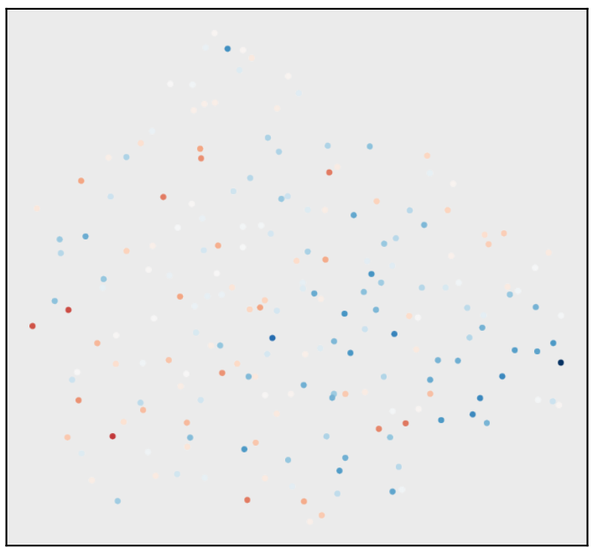
Genes (number of stages): ved (12), gadd45bb (12), cdkn1ca (12), rdh10a (12), snai1a (12), idh1 (12), arl4aa (12), igfbp1a (11), phlda2 (11), uncx4.1 (11), ripply1 (10), pcdh8 (10), zgc:110340 (10), mef2d (10), mespaa (9), cxcl12b (9), foxc1a (9), myf5 (8), ripply2 (8), cpn1 (8), ttnb (8), myod1 (7), alcama (7), fn1b (7), meox1 (6), prdm1a (6), fstl1a (6), lbx2 (6), angptl7 (6), alpi.1 (4)

tailbud

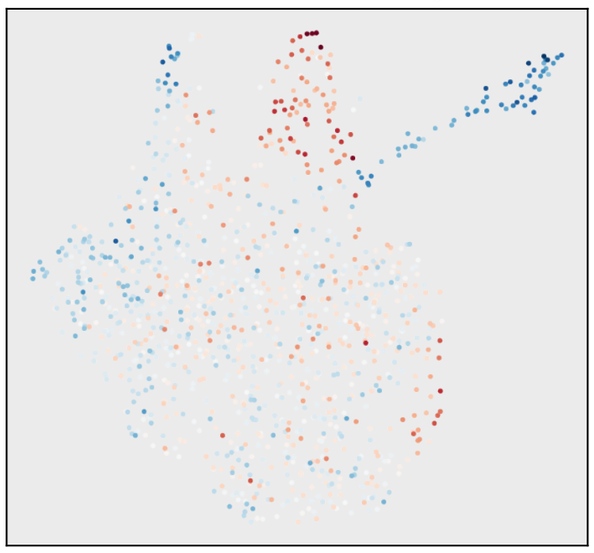
high: 21 genes, not supported
active in 0% of cells, pseudobulk strength 0.33



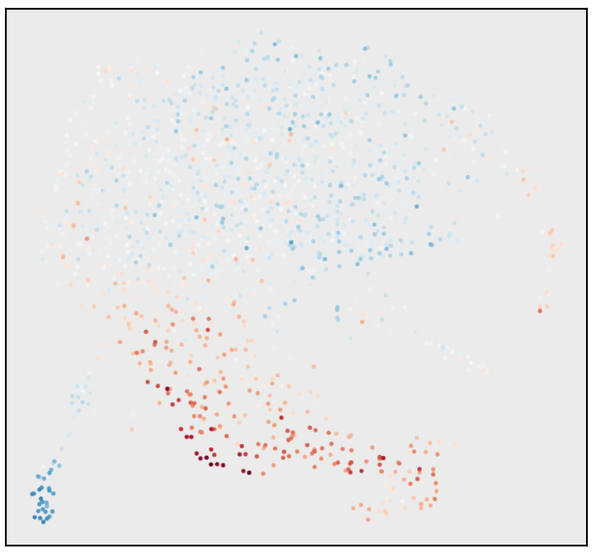
oblong: 16 genes, not supported
active in 0% of cells, pseudobulk strength 0.31



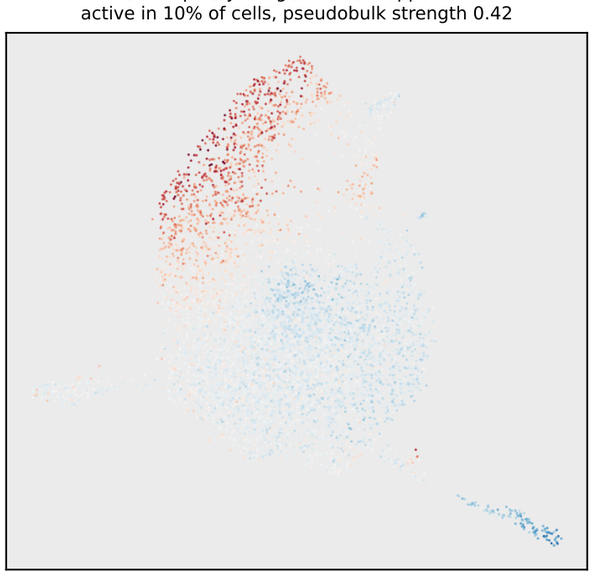
dome: 24 genes, not supported
active in 1% of cells, pseudobulk strength 0.33



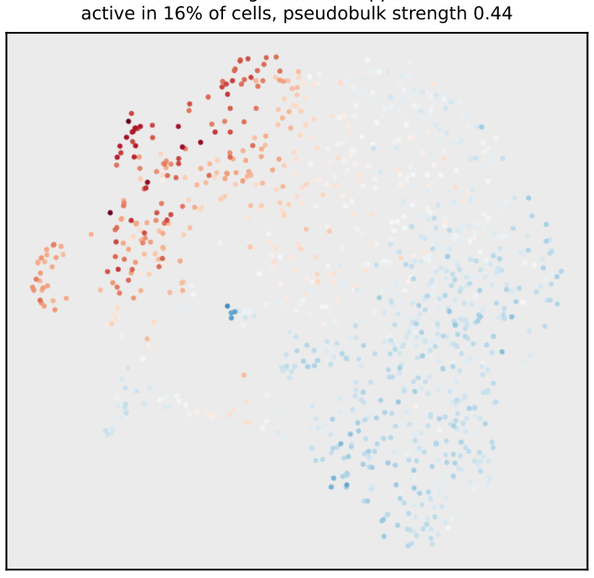
30% epiboly: 23 genes, not supported
active in 7% of cells, pseudobulk strength 0.41



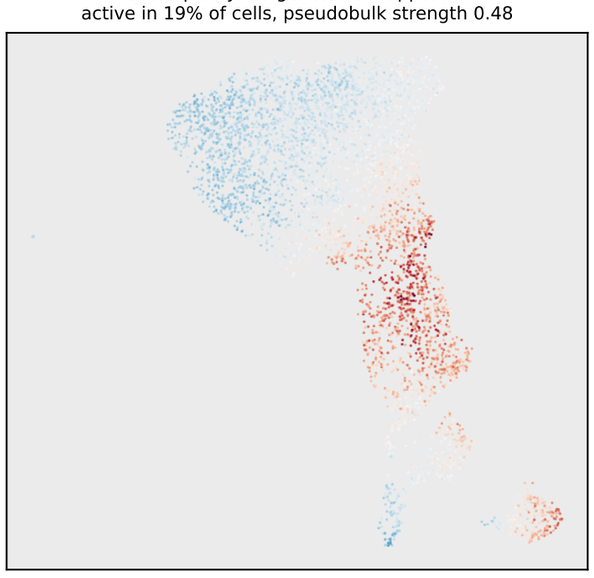
50% epiboly: 26 genes, not supported
active in 10% of cells, pseudobulk strength 0.42



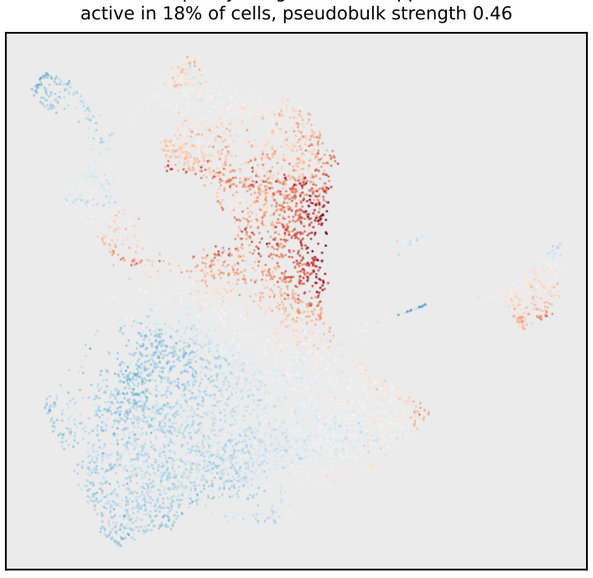
shield: 24 genes, not supported
active in 16% of cells, pseudobulk strength 0.44



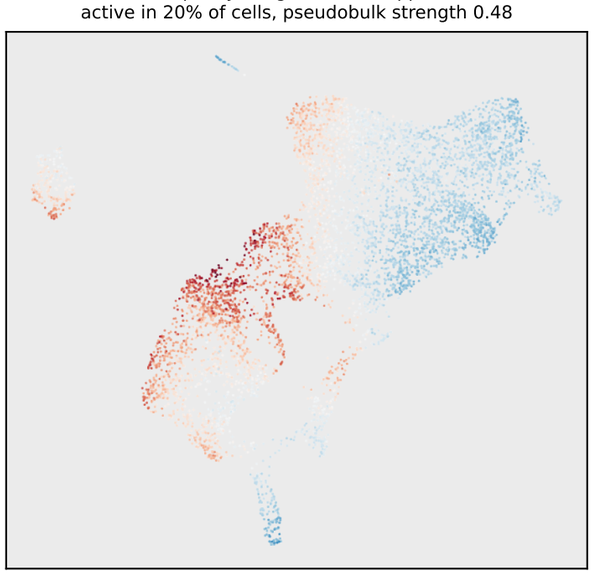
60% epiboly: 26 genes, not supported
active in 19% of cells, pseudobulk strength 0.48



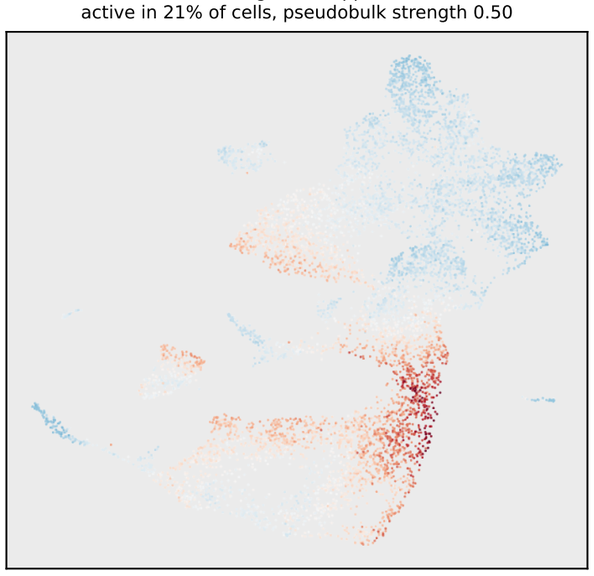
75% epiboly: 28 genes, not supported
active in 18% of cells, pseudobulk strength 0.46



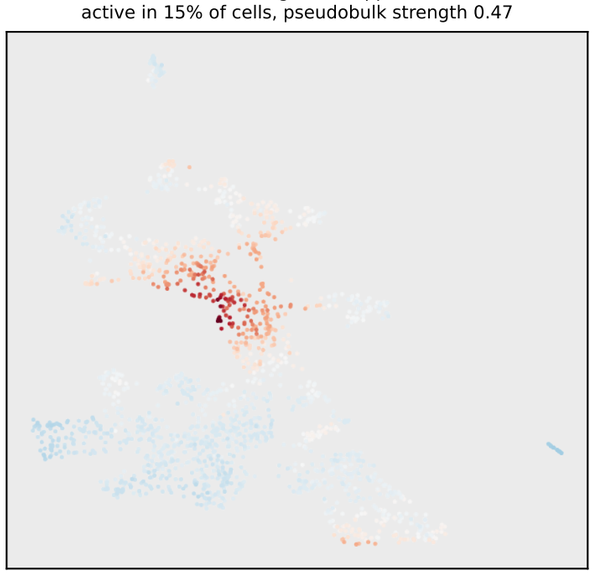
90% epiboly: 29 genes, not supported
active in 20% of cells, pseudobulk strength 0.48



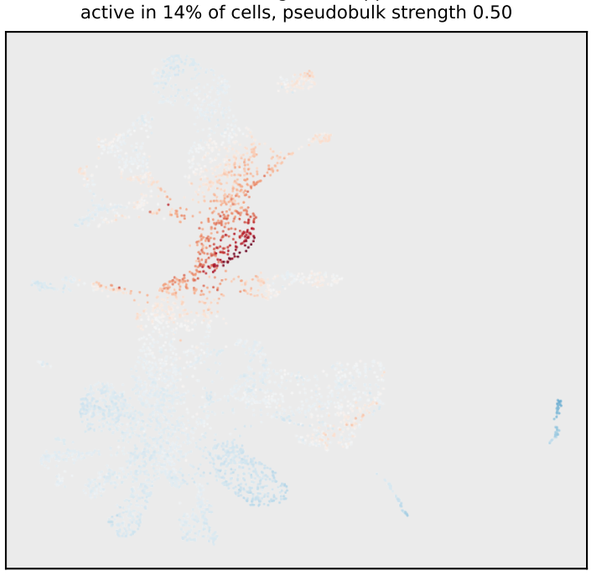
bud: 30 genes, supported
active in 21% of cells, pseudobulk strength 0.50



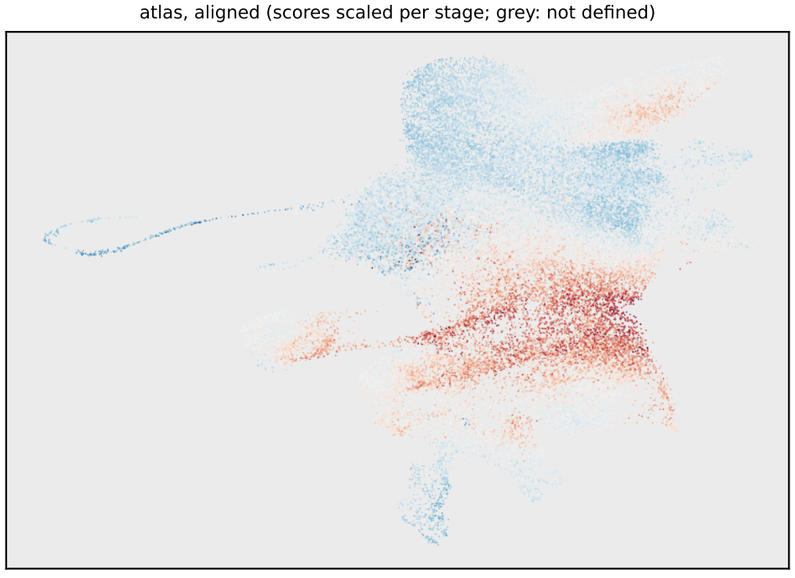
3-somite: 29 genes, supported
active in 15% of cells, pseudobulk strength 0.47



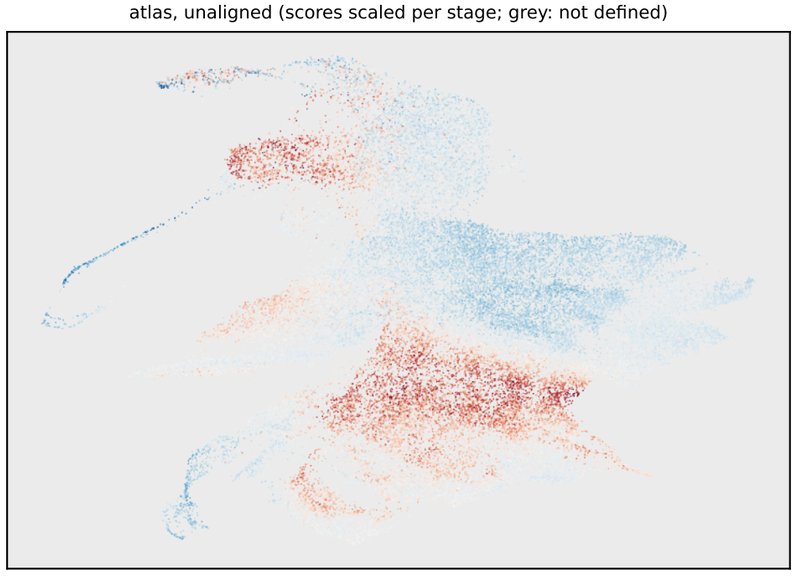
6-somite: 29 genes, supported
active in 14% of cells, pseudobulk strength 0.50



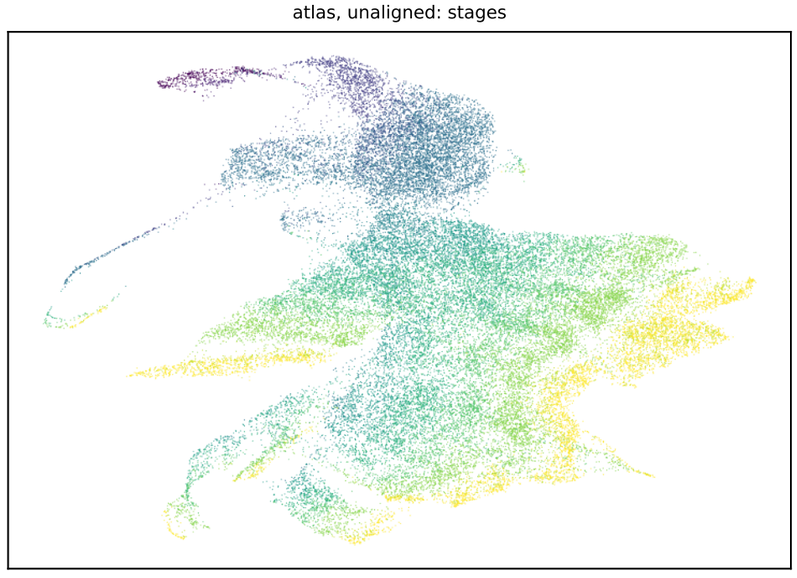
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



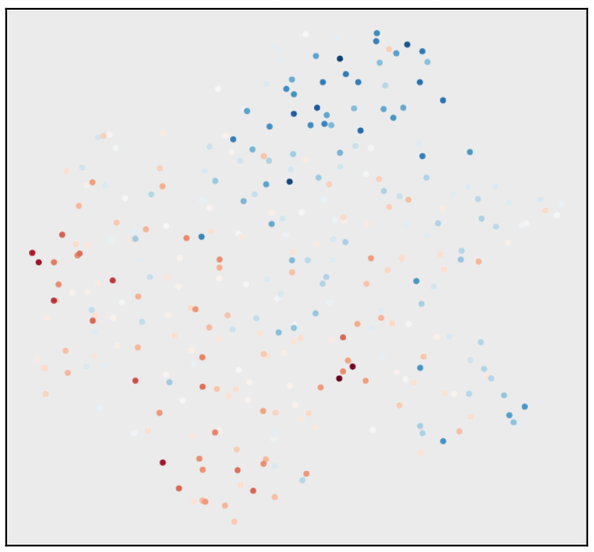
atlas, unaligned: stages



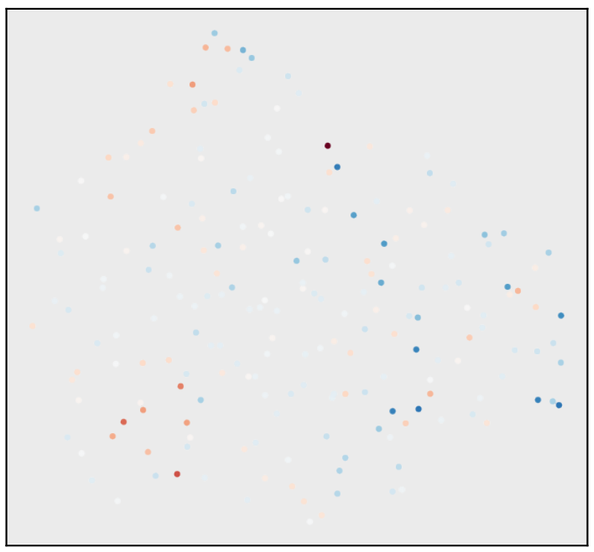
Genes (number of stages): vox (12), hes6 (12), foxd5 (12), znfl2a (12), rasl11b (12), lratb (12), rbm24a (12), stmn1a (12), si:ch211-222l21.1 (12), adam8a (12), cmtm6 (12), aif1l (12), helb (12), zswim5 (12), ccnb1 (12), lpar1 (11), vent (11), tsku (11), rarga (11), p4ha2 (10), nfkbiaa (10), fn1a (10), fzd10 (10), ube2e2 (10), smad9 (8), esrrga (8), msx2b (5), bmp2a (4), hoxd11a (3), abcc6a (3)

lateral_plate

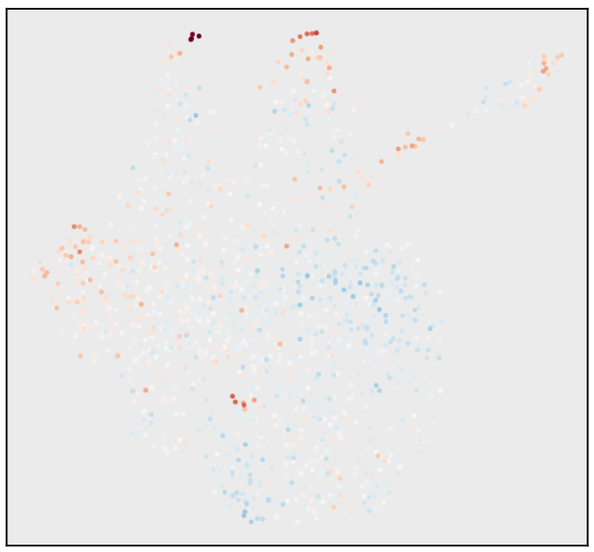
high: 8 genes, not supported
active in 1% of cells, pseudobulk strength 0.11



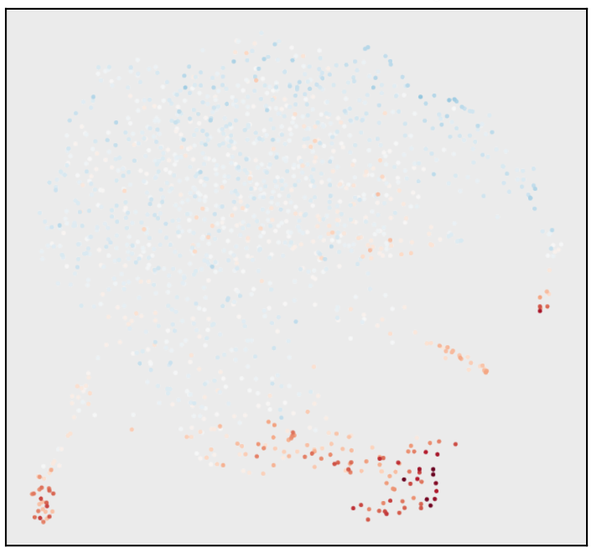
oblong: 8 genes, not supported
active in 0% of cells, pseudobulk strength 0.20



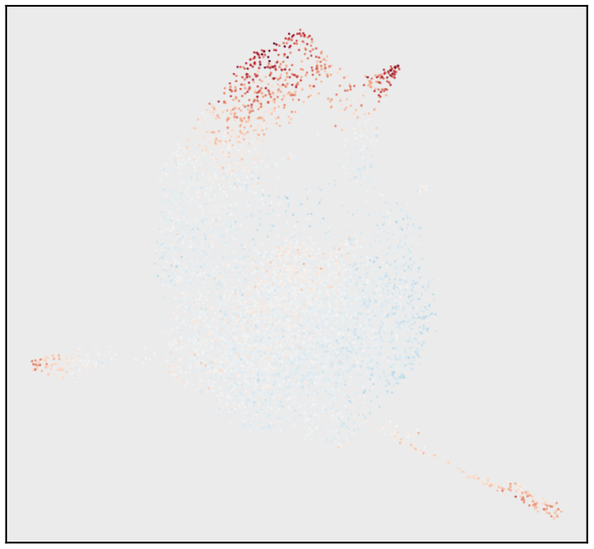
dome: 12 genes, not supported
active in 1% of cells, pseudobulk strength 0.24



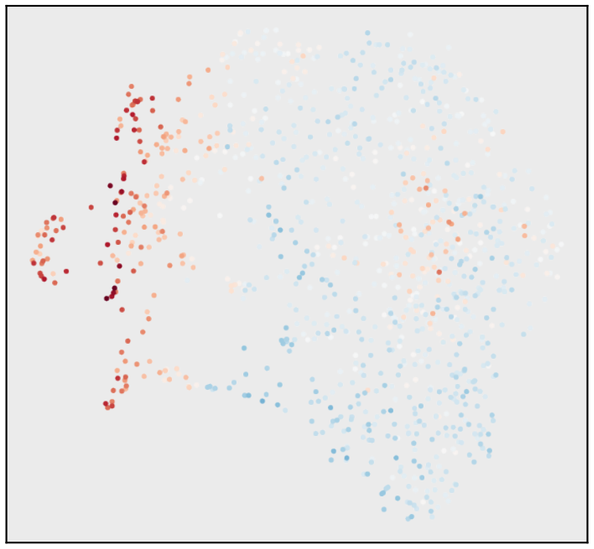
30% epiboly: 11 genes, not supported
active in 4% of cells, pseudobulk strength 0.24



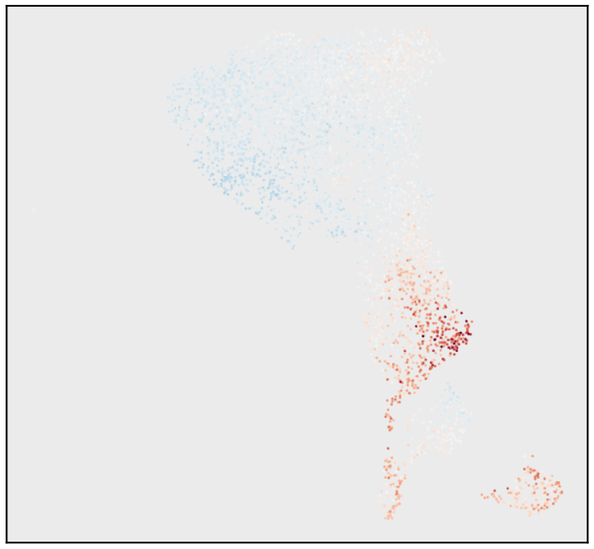
50% epiboly: 18 genes, not supported
active in 8% of cells, pseudobulk strength 0.28



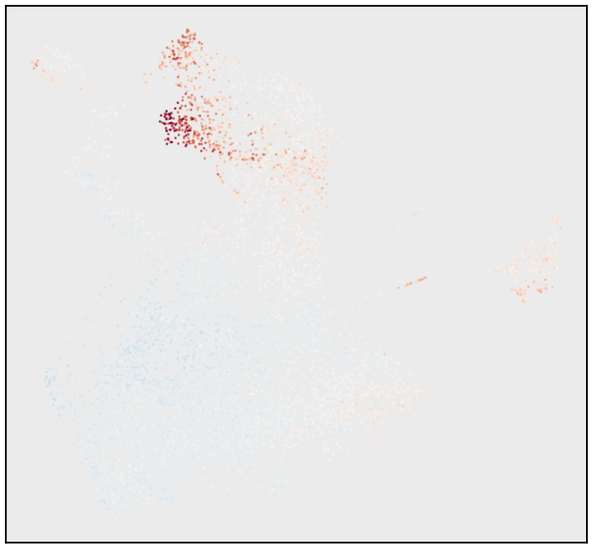
shield: 10 genes, not supported
active in 6% of cells, pseudobulk strength 0.24



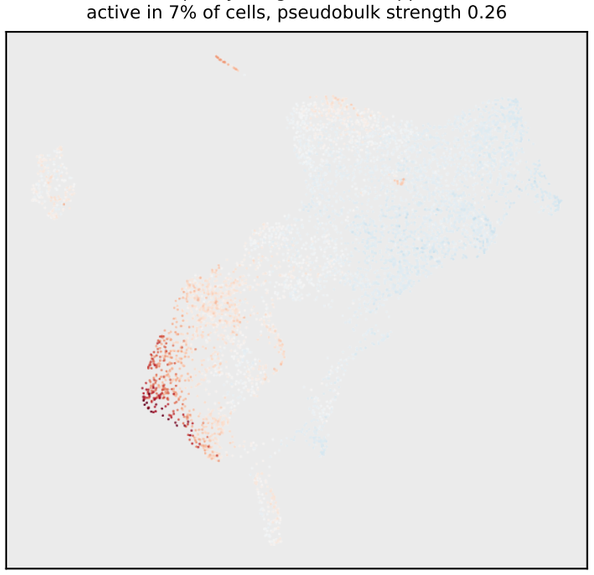
60% epiboly: 16 genes, not supported
active in 9% of cells, pseudobulk strength 0.25



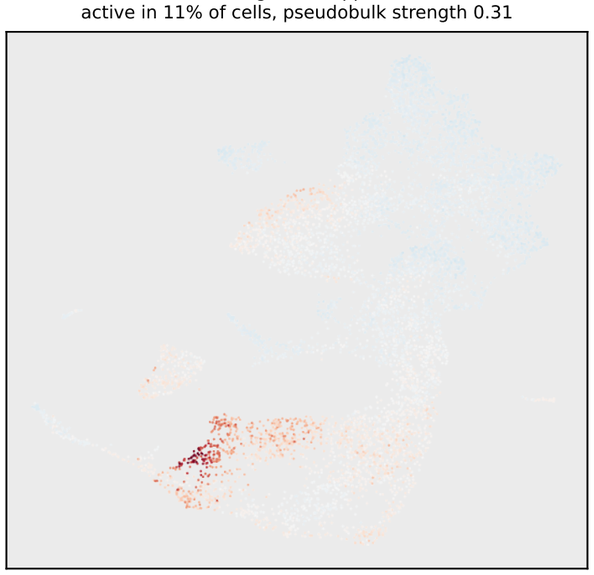
75% epiboly: 21 genes, not supported
active in 7% of cells, pseudobulk strength 0.20



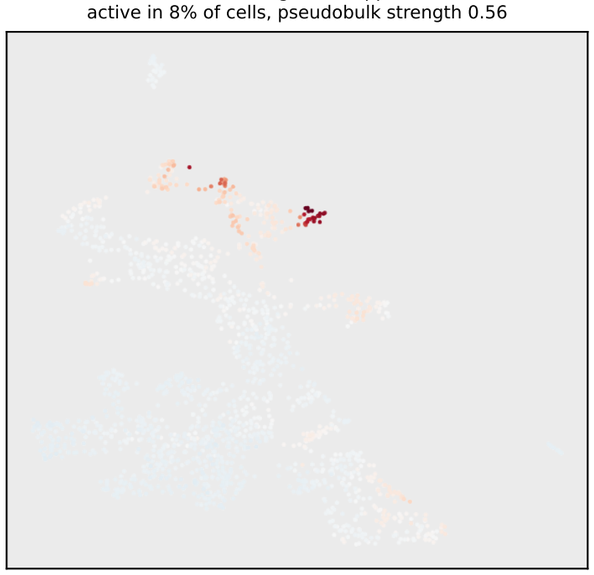
90% epiboly: 21 genes, not supported
active in 7% of cells, pseudobulk strength 0.26



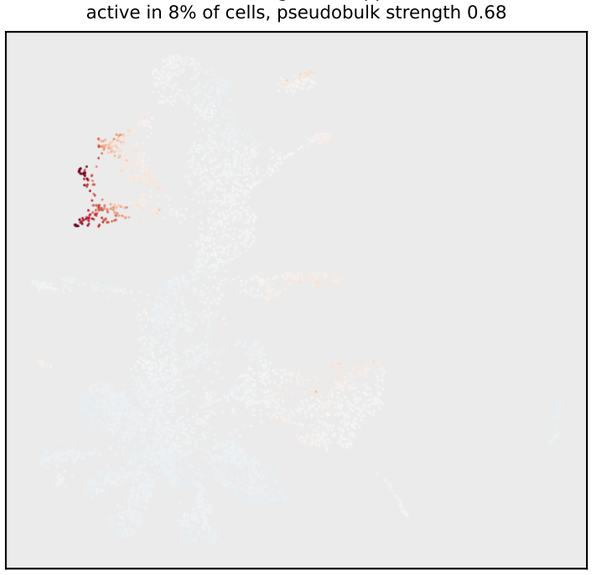
bud: 25 genes, supported
active in 11% of cells, pseudobulk strength 0.31



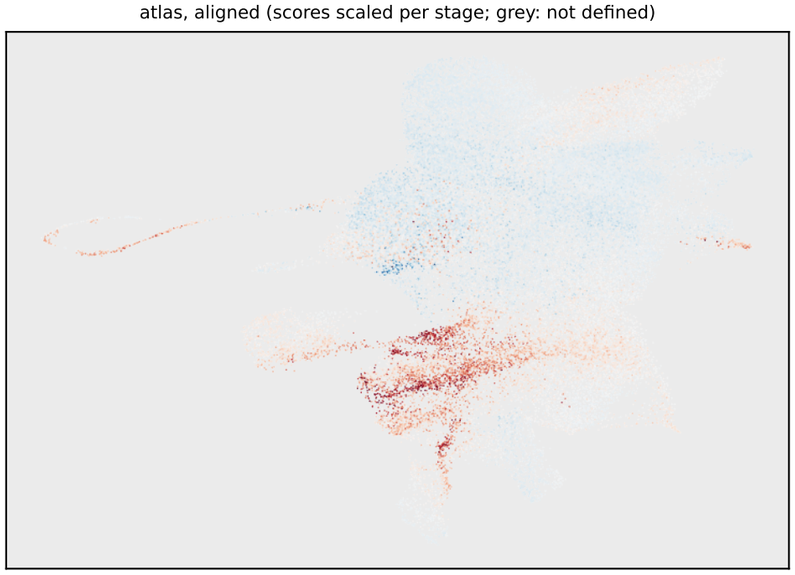
3-somite: 24 genes, supported
active in 8% of cells, pseudobulk strength 0.56



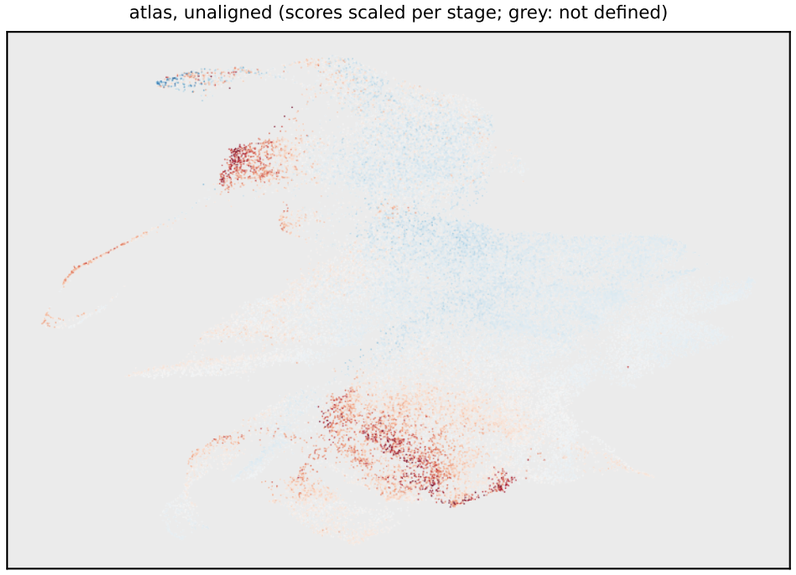
6-somite: 30 genes, supported
active in 8% of cells, pseudobulk strength 0.68



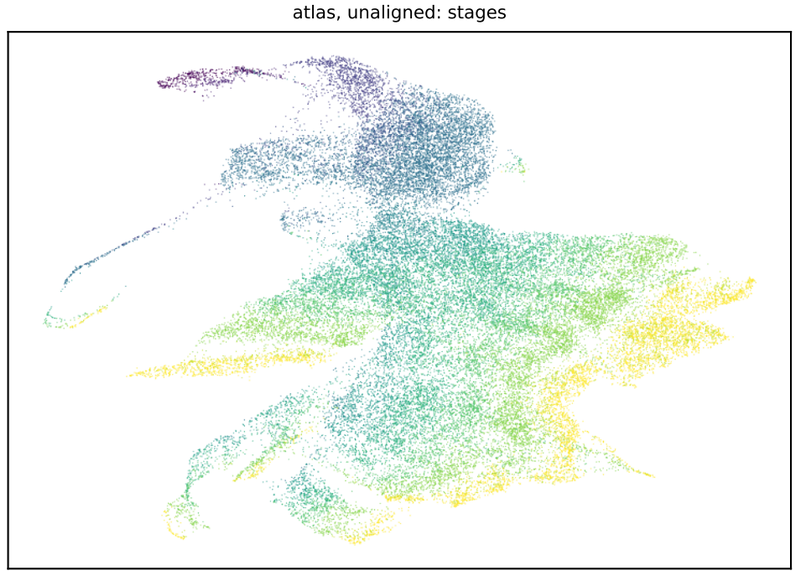
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



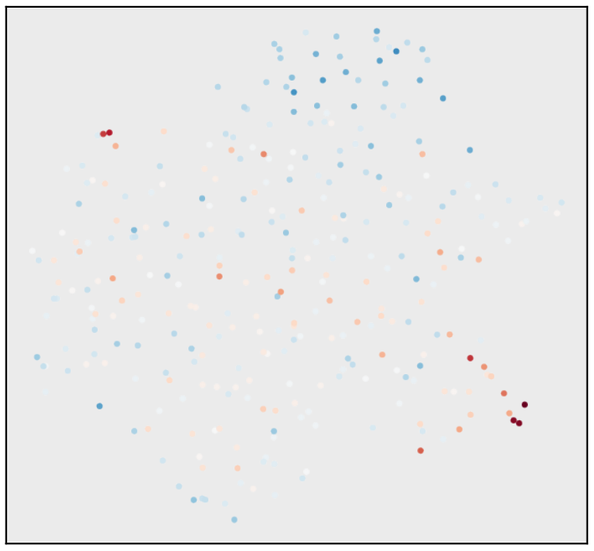
atlas, unaligned: stages



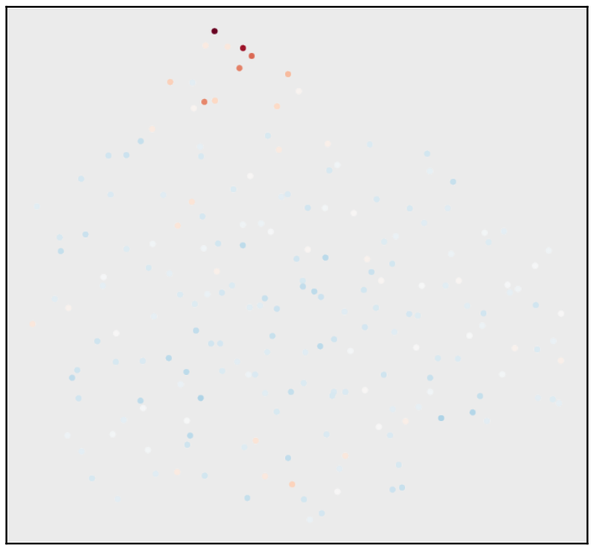
Genes (number of stages): aplnrb (12), blf (12), gata2a (12), si:ch73-299h12.3 (12), mych (12), gata6 (11), ets2 (11), lmo2 (10), ttna (10), tpm4a (10), drl (10), tbx1 (8), hand2 (7), nkx2.7 (7), tmem88a (7), sox7 (7), dusp5 (5), myb (5), hoxb5b (5), hoxb5a (5), tal1 (4), etv2 (4), tbx20 (4), hoxa13b (4), fli1a (3), gfi1aa (2), hbbe1.1 (2), gata1a (1), nkx2.5 (1), spi1b (1)

intermediate_mesoderm

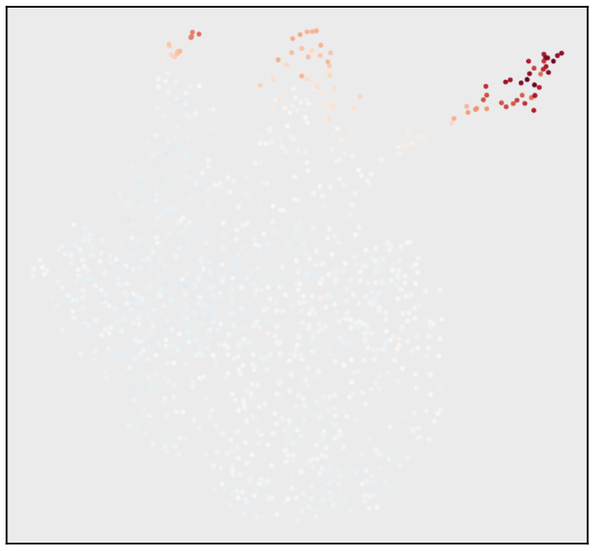
high: 7 genes, not supported
active in 0% of cells, pseudobulk strength 0.25



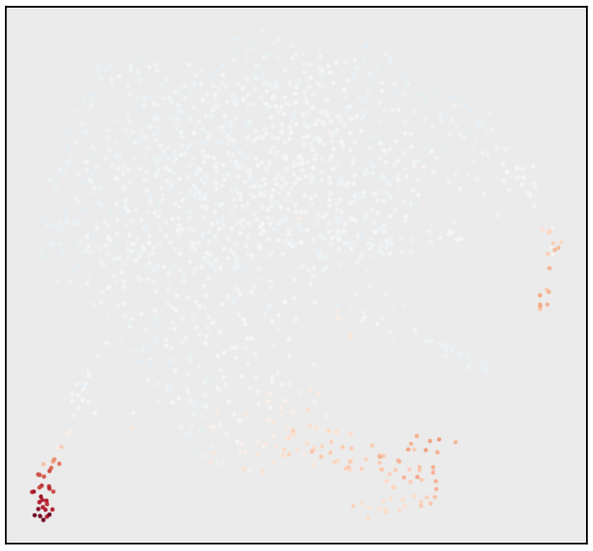
oblong: 6 genes, not supported
active in 1% of cells, pseudobulk strength 0.33



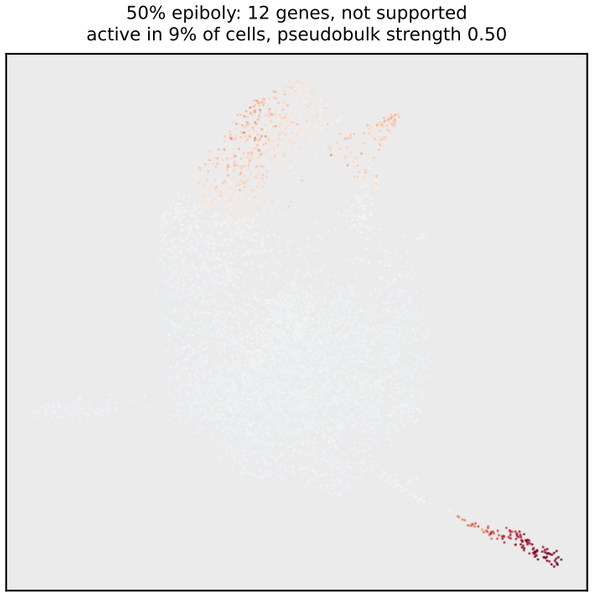
dome: 9 genes, not supported
active in 6% of cells, pseudobulk strength 0.39



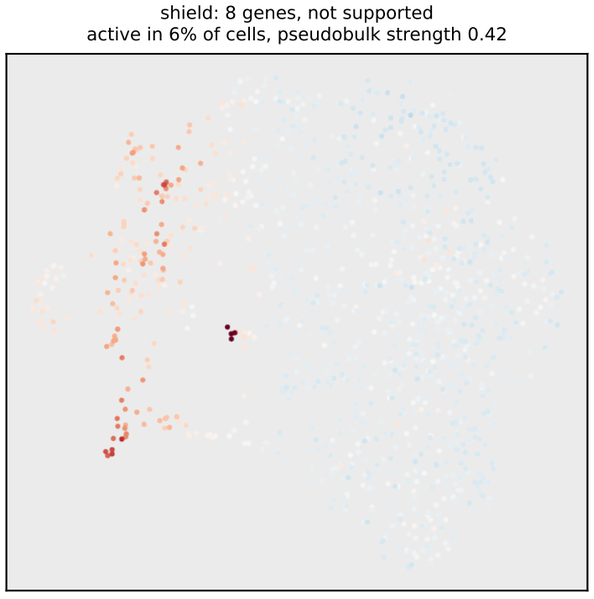
30% epiboly: 10 genes, not supported
active in 9% of cells, pseudobulk strength 0.52



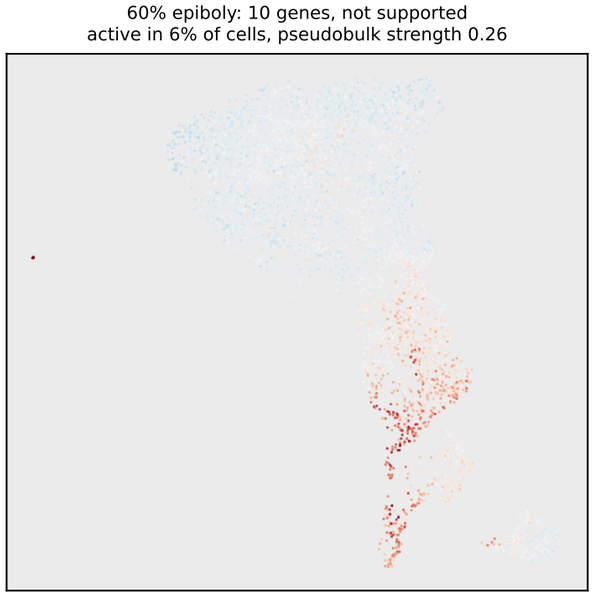
50% epiboly: 12 genes, not supported
active in 9% of cells, pseudobulk strength 0.50



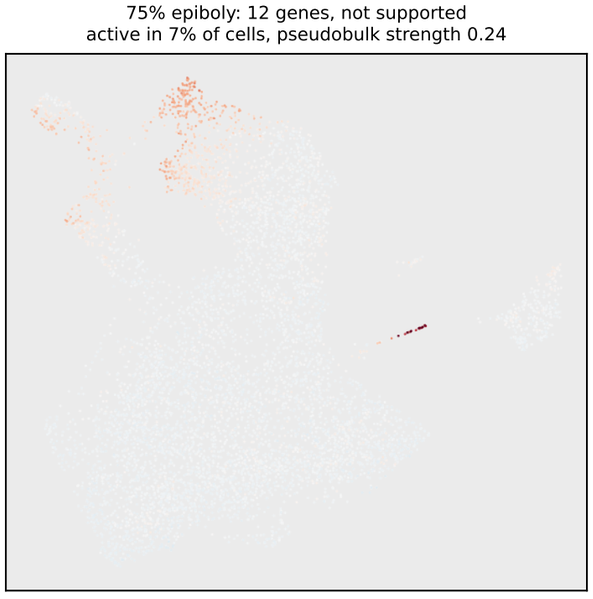
shield: 8 genes, not supported
active in 6% of cells, pseudobulk strength 0.42



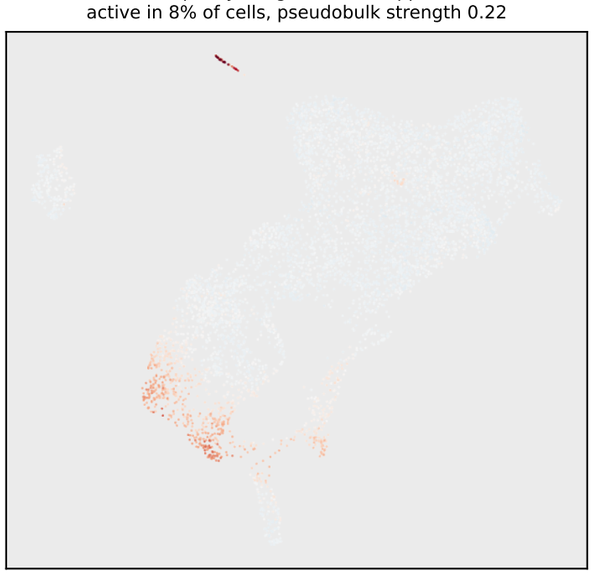
60% epiboly: 10 genes, not supported
active in 6% of cells, pseudobulk strength 0.26



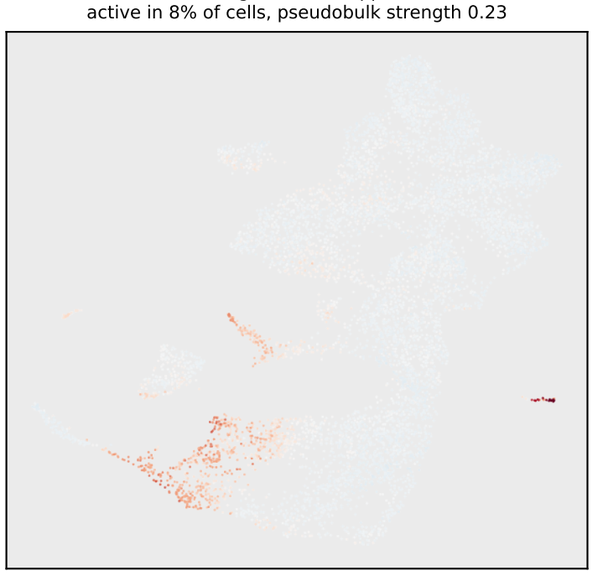
75% epiboly: 12 genes, not supported
active in 7% of cells, pseudobulk strength 0.24



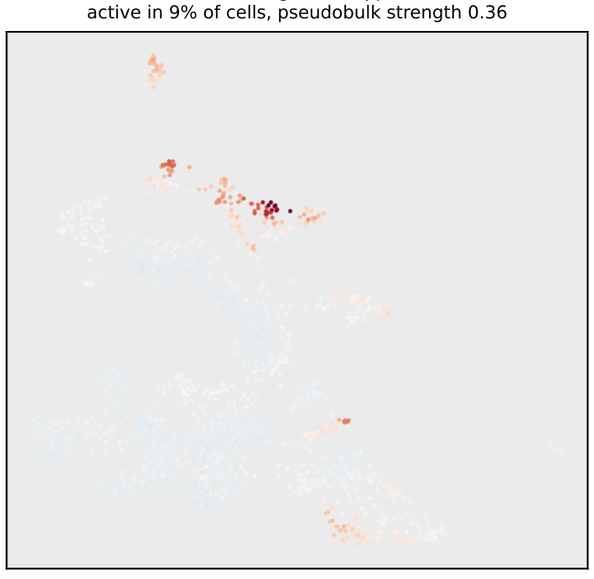
90% epiboly: 12 genes, not supported
active in 8% of cells, pseudobulk strength 0.22



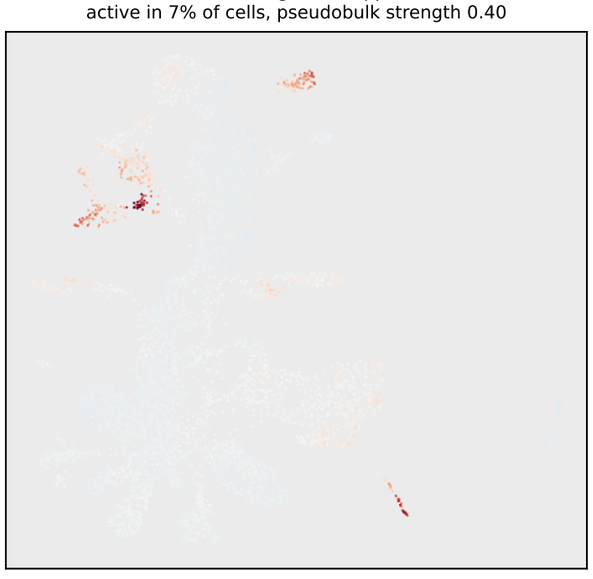
bud: 14 genes, not supported
active in 8% of cells, pseudobulk strength 0.23



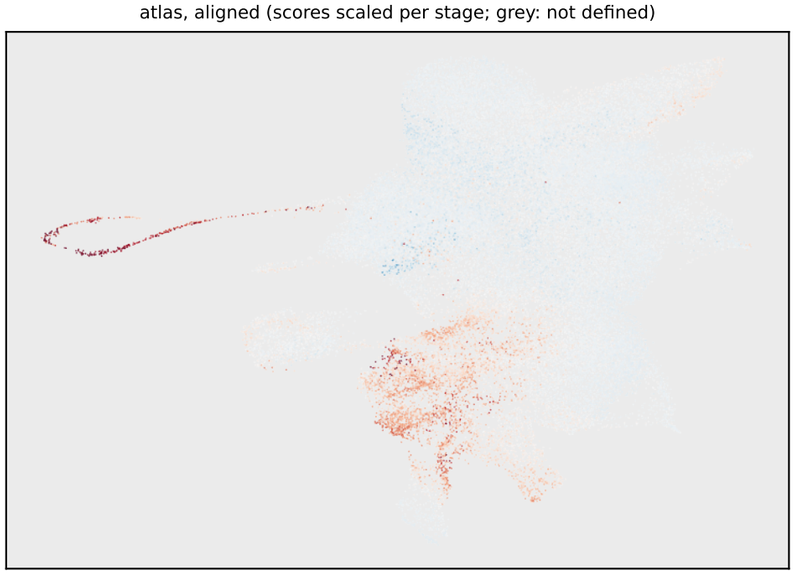
3-somite: 11 genes, supported
active in 9% of cells, pseudobulk strength 0.36



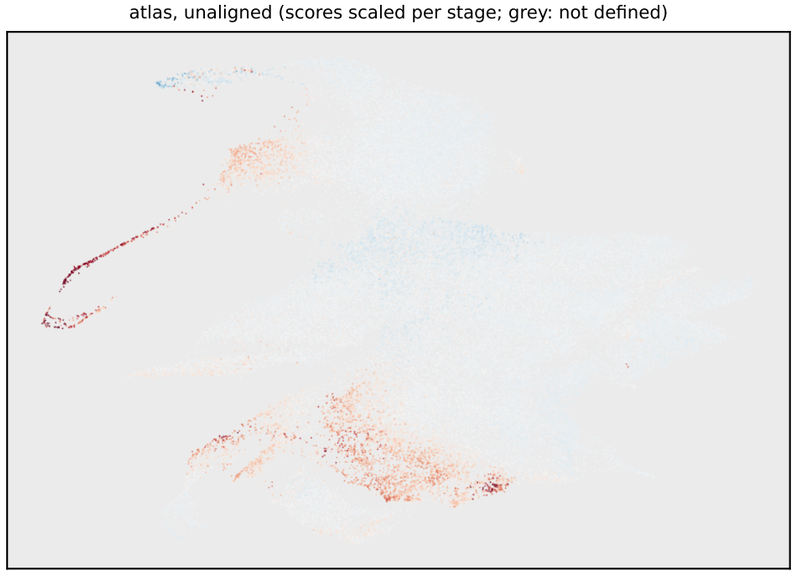
6-somite: 14 genes, supported
active in 7% of cells, pseudobulk strength 0.40



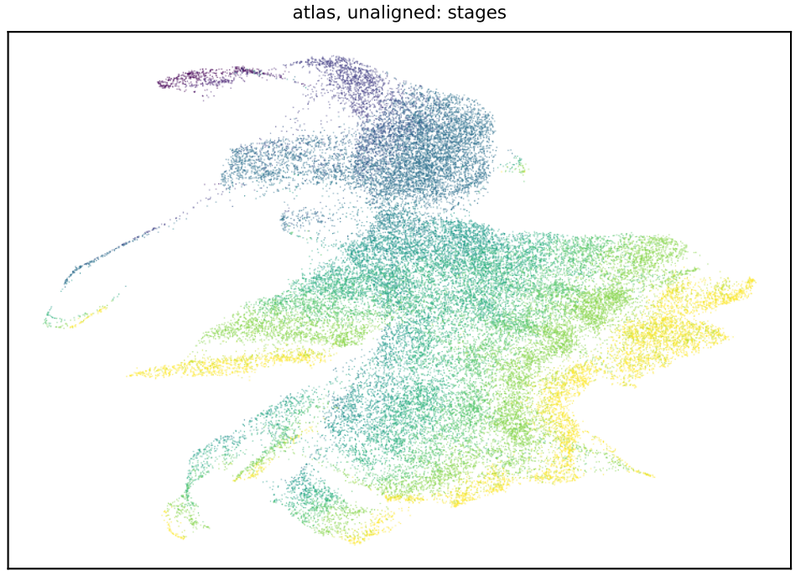
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



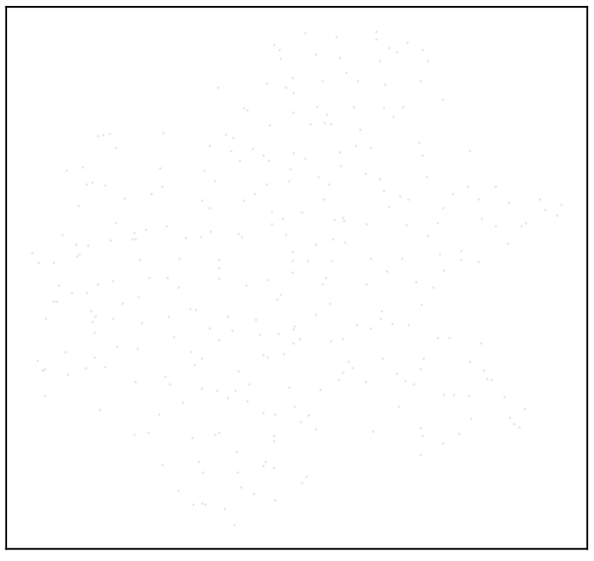
atlas, unaligned: stages



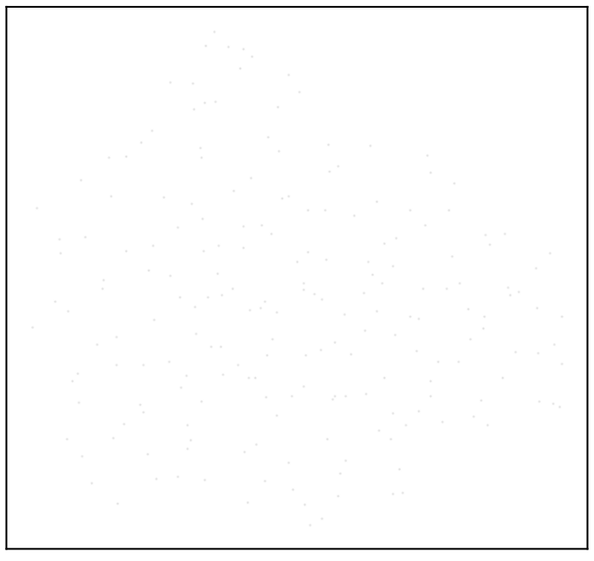
Genes (number of stages): ftr83 (12), ftr82 (12), lhx1a (12), calm1a (12), osr1 (11), ndrg1a (11), ponzr1 (11), acy3.2 (9), gadd45ab (9), twist1a (6), mafbb (6), ptgs1 (4), cldnc (4), hnf1bb (3), cebpa (3)

heart

high: not defined



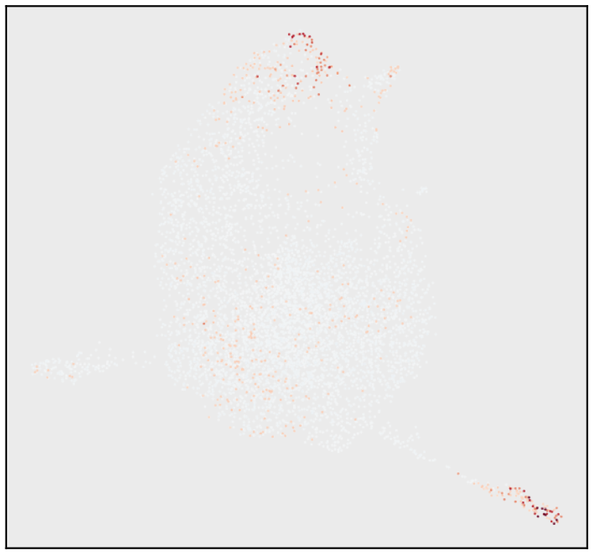
oblong: not defined



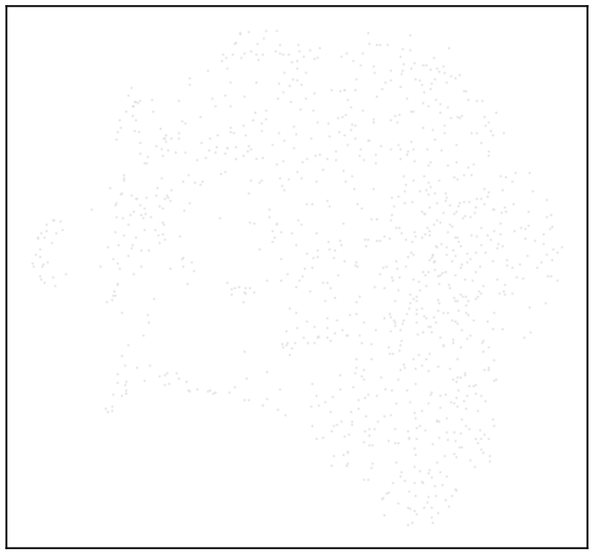
dome: not defined

30% epiboly: not defined

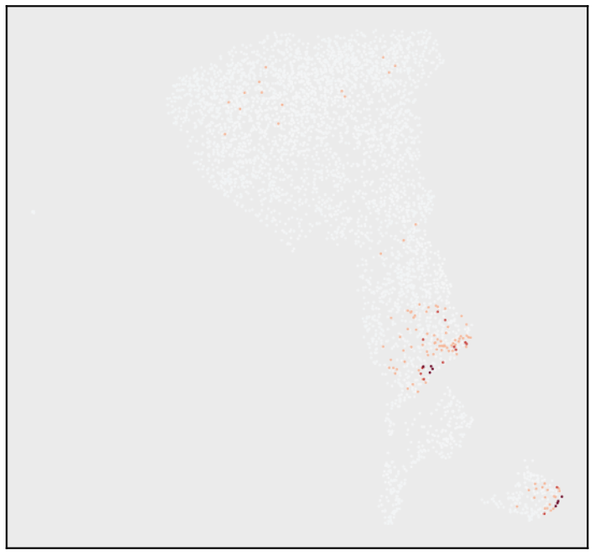
50% epiboly: 3 genes, not supported
active in 1% of cells, pseudobulk strength 0.02



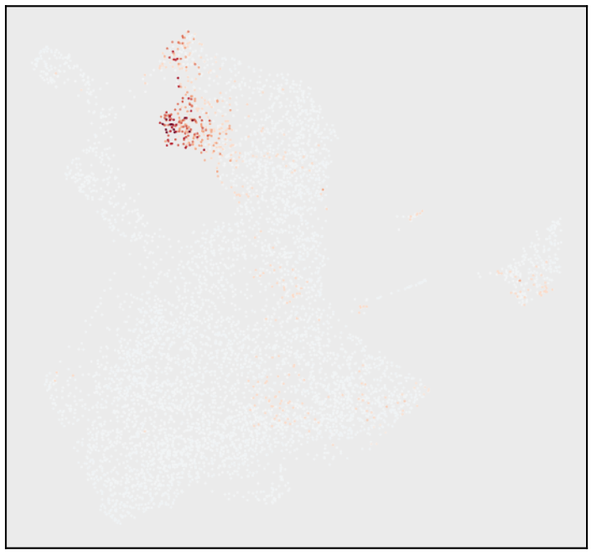
shield: not defined



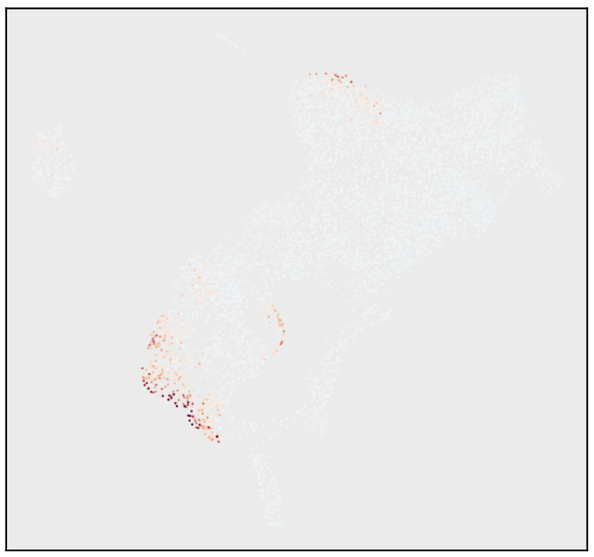
60% epiboly: 2 genes, not supported
active in 1% of cells, pseudobulk strength 0.01



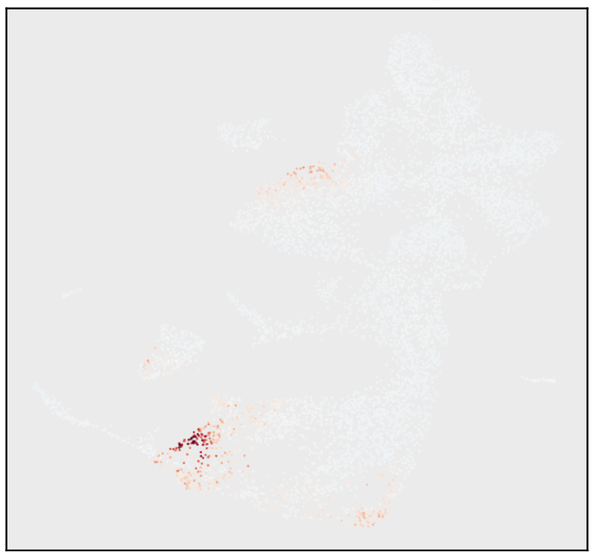
75% epiboly: 2 genes, not supported
active in 2% of cells, pseudobulk strength 0.04



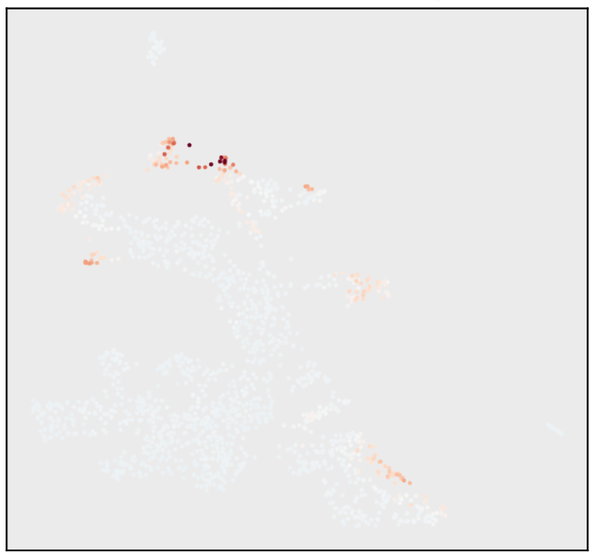
90% epiboly: 3 genes, not supported
active in 2% of cells, pseudobulk strength 0.19



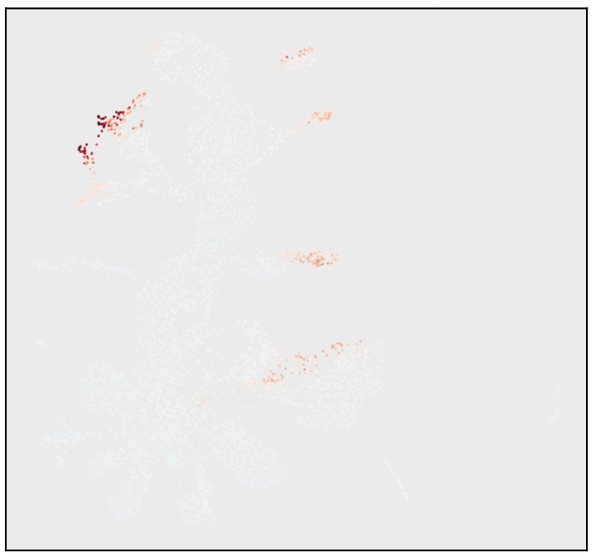
bud: 4 genes, supported
active in 4% of cells, pseudobulk strength 0.42



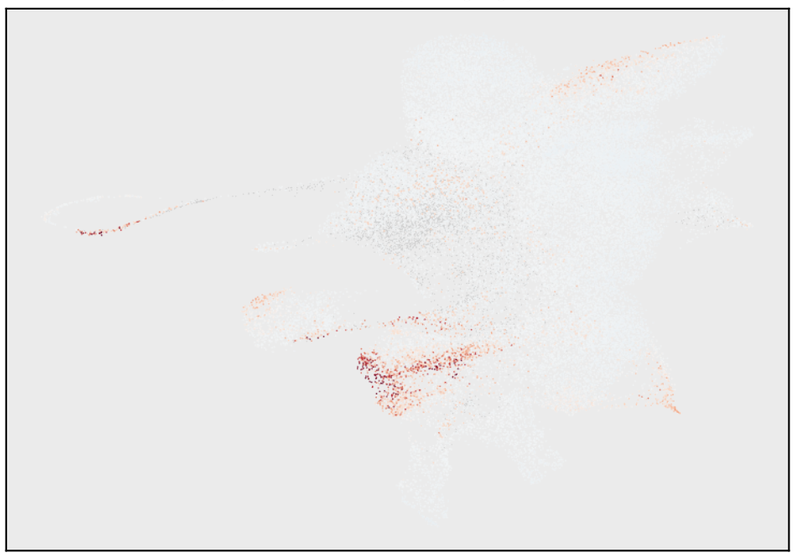
3-somite: 4 genes, supported
active in 4% of cells, pseudobulk strength 0.87



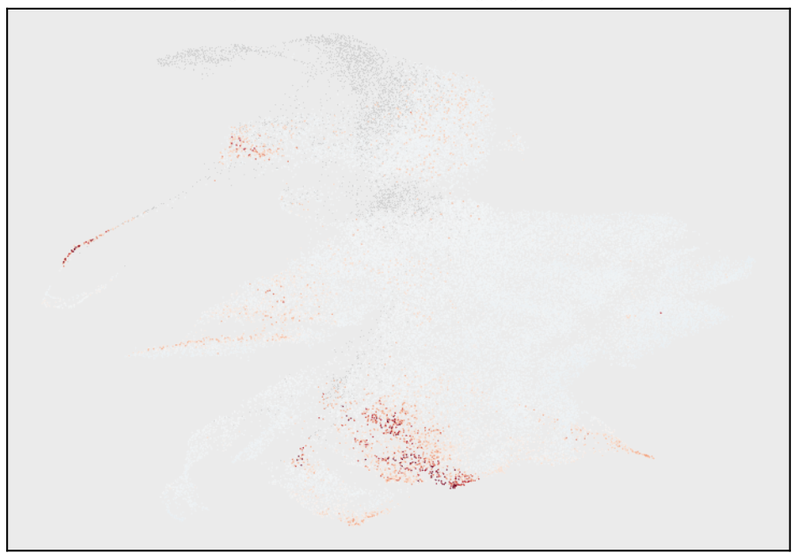
6-somite: 5 genes, supported
active in 5% of cells, pseudobulk strength 0.83



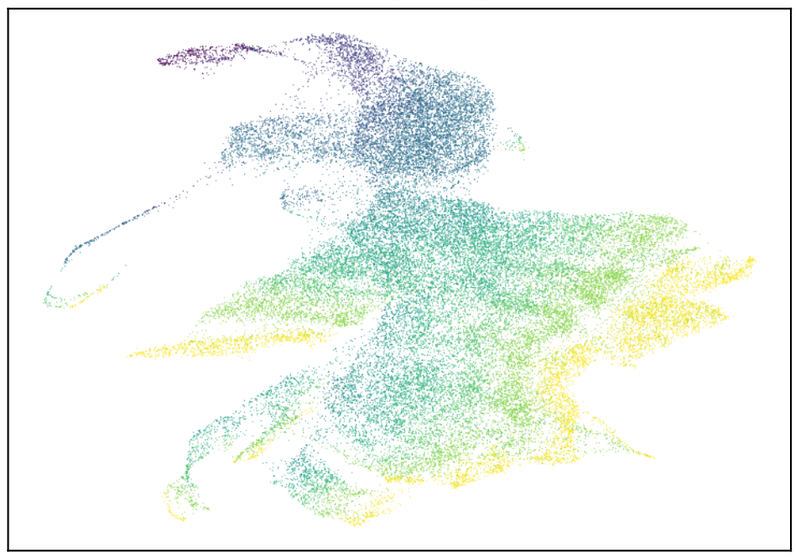
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



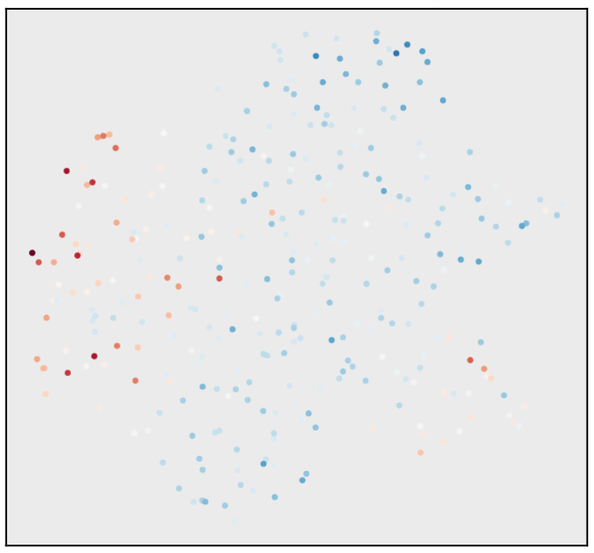
atlas, unaligned: stages



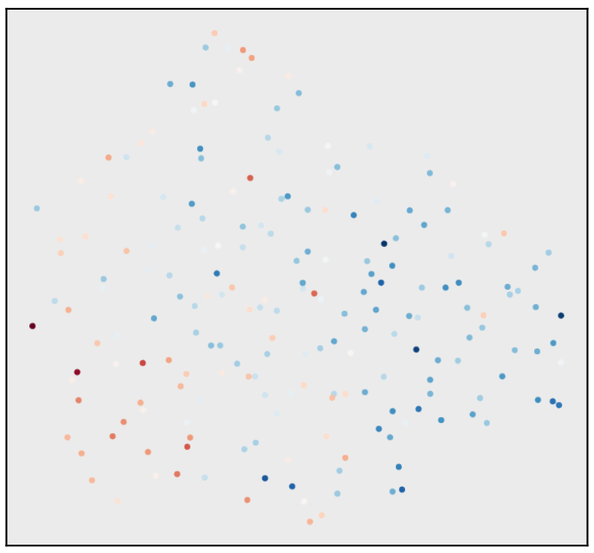
Genes (number of stages): nkx2.7 (7), tmem88a (7), tbx20 (4), rbfox1l (4), nkx2.5 (1)

forebrain_eye

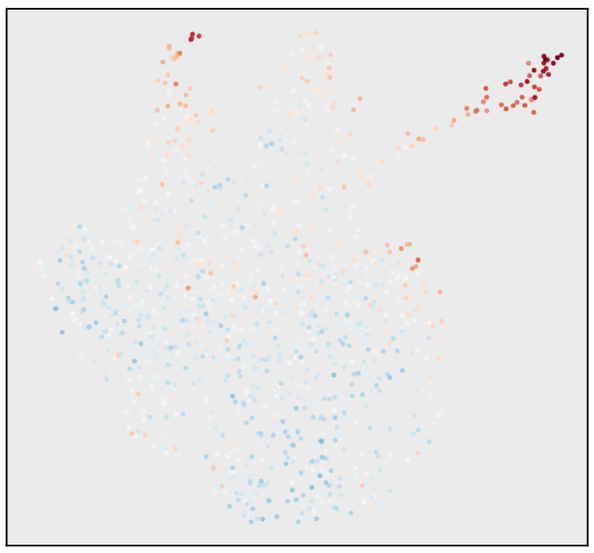
high: 10 genes, not supported
active in 1% of cells, pseudobulk strength 0.11



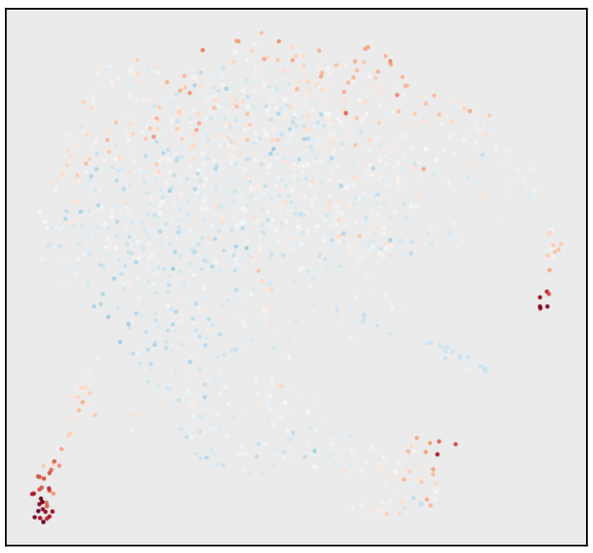
oblong: 11 genes, not supported
active in 1% of cells, pseudobulk strength 0.15



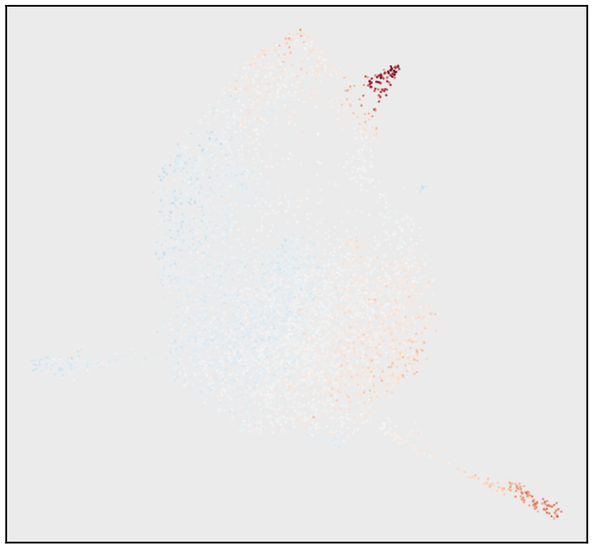
dome: 16 genes, not supported
active in 3% of cells, pseudobulk strength 0.16



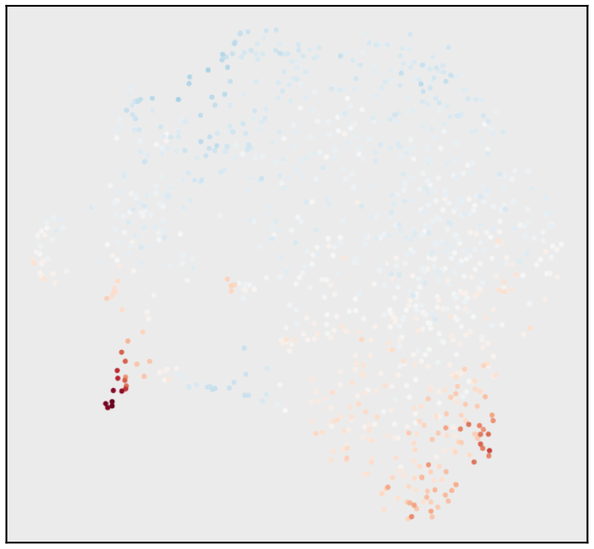
30% epiboly: 15 genes, not supported
active in 2% of cells, pseudobulk strength 0.18



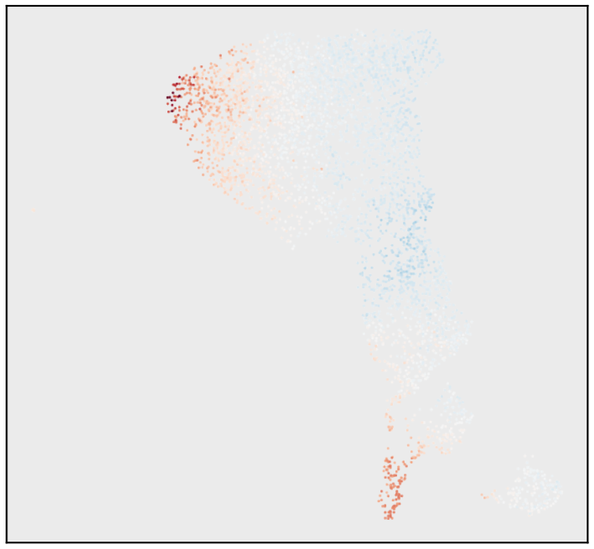
50% epiboly: 23 genes, not supported
active in 4% of cells, pseudobulk strength 0.23



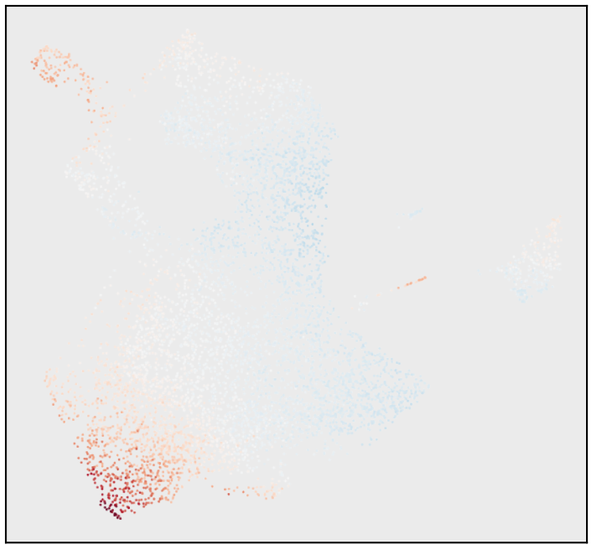
shield: 18 genes, not supported
active in 7% of cells, pseudobulk strength 0.27



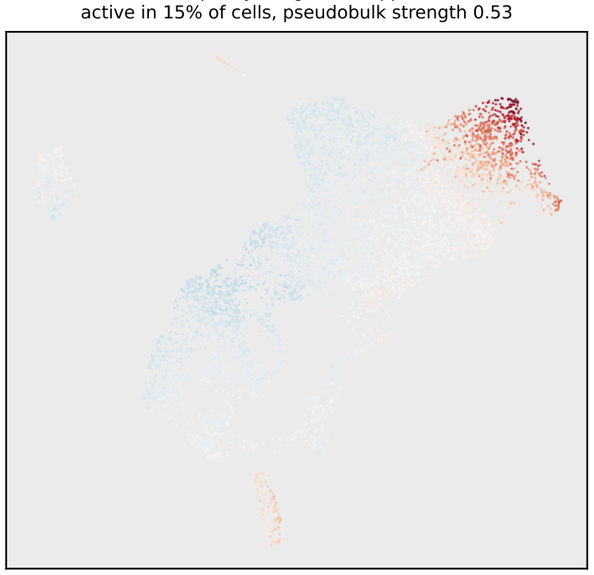
60% epiboly: 25 genes, not supported
active in 13% of cells, pseudobulk strength 0.34



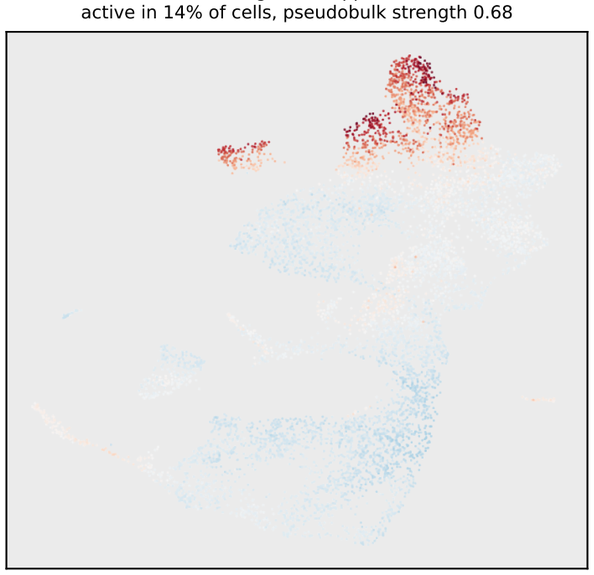
75% epiboly: 28 genes, supported
active in 16% of cells, pseudobulk strength 0.40



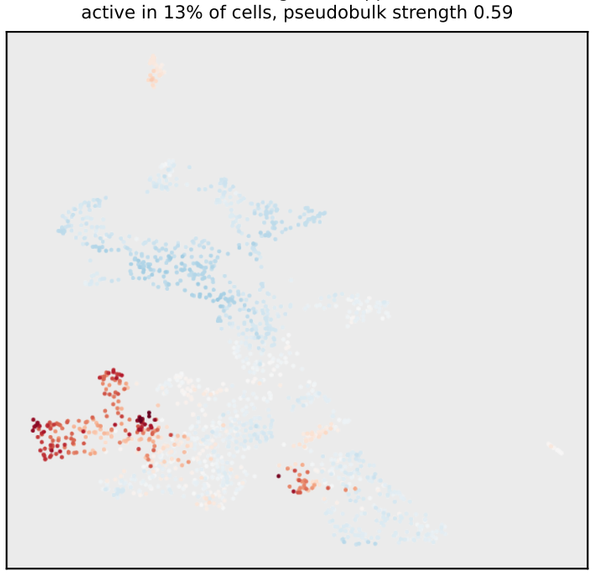
90% epiboly: 30 genes, supported
active in 15% of cells, pseudobulk strength 0.53



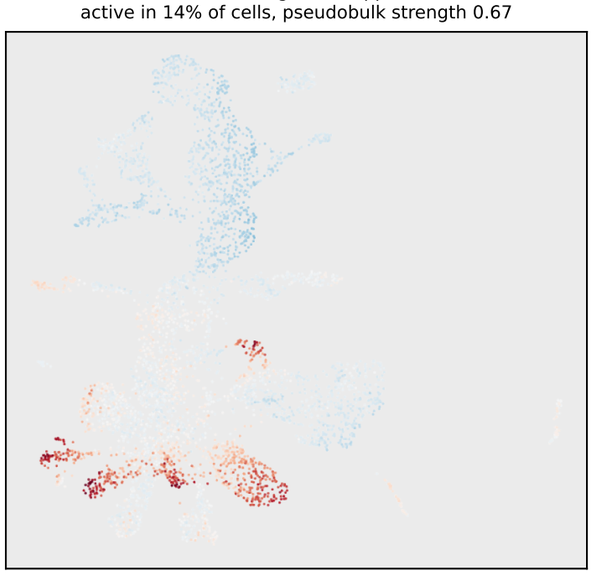
bud: 30 genes, supported
active in 14% of cells, pseudobulk strength 0.68



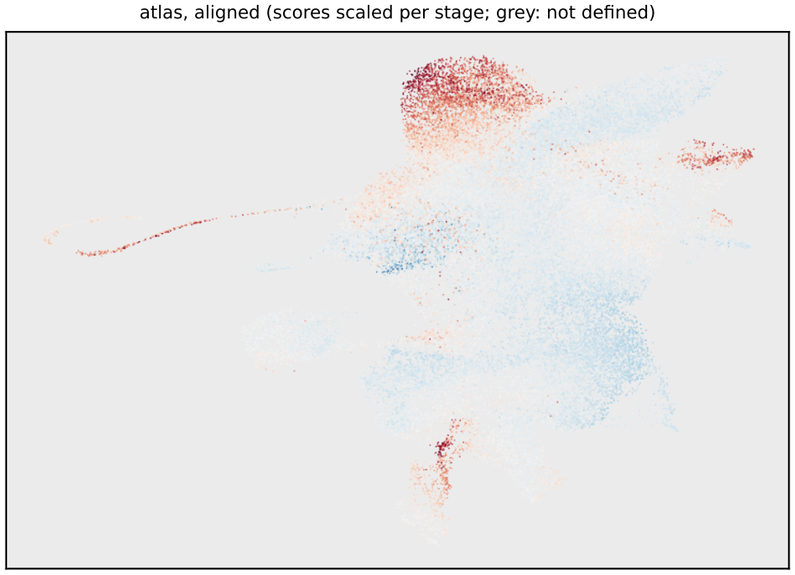
3-somite: 30 genes, supported
active in 13% of cells, pseudobulk strength 0.59



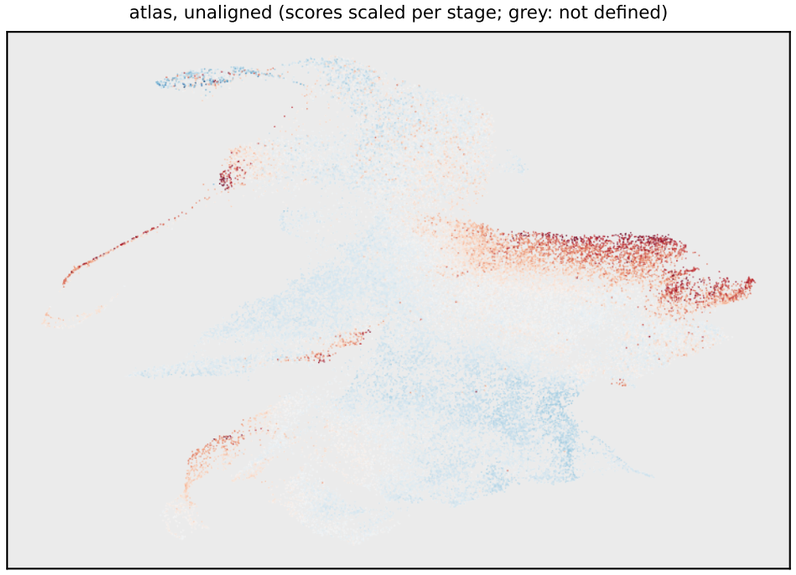
6-somite: 30 genes, supported
active in 14% of cells, pseudobulk strength 0.67



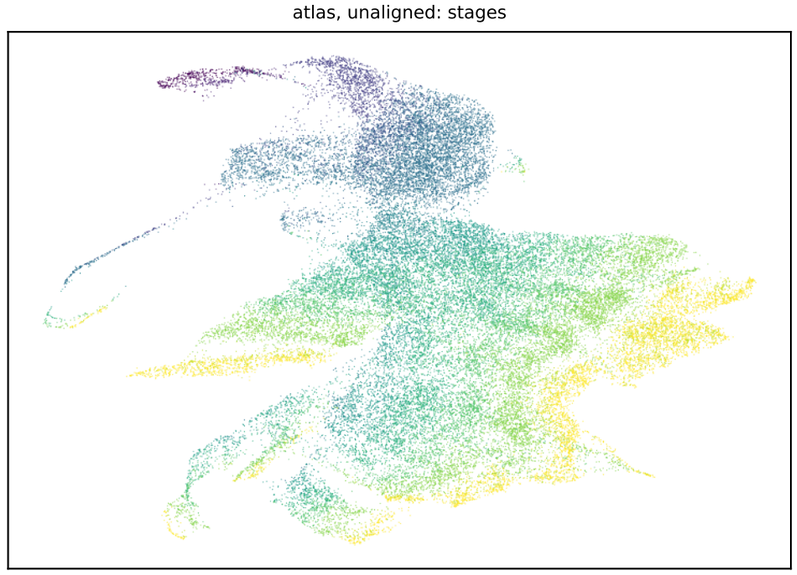
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



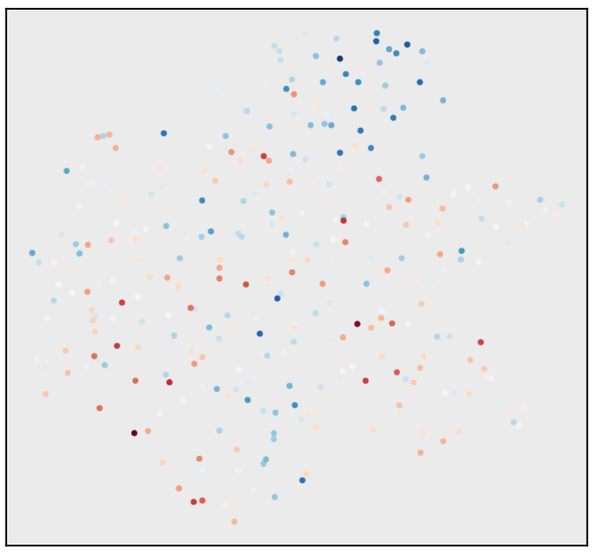
atlas, unaligned: stages



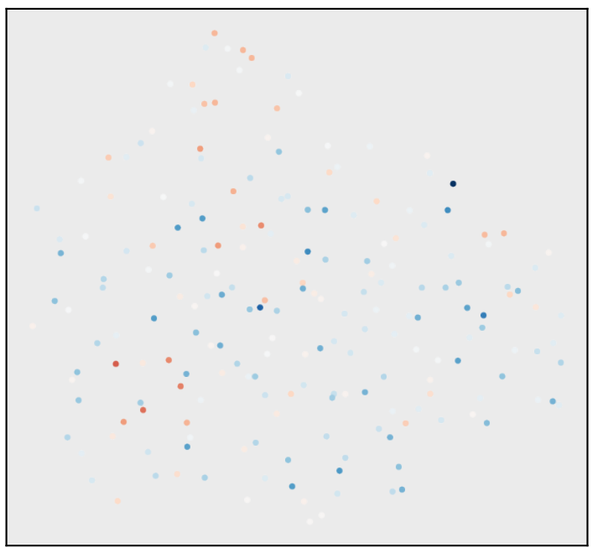
Genes (number of stages): fzd8b (12), stm (12), zic2a (12), tdgf1 (12), swap70b (12), zic2b (12), midn (12), rspo3 (11), sox4a (11), hesx1 (10), pax6b (10), alcamb (10), sox2 (10), nat8l (10), rx3 (9), emx3 (9), six3b (8), her9 (8), otx2 (8), olig2 (7), meis1b (7), pax6a (7), foxb1b (7), epha4a (7), fezf2 (6), pdgfbb (6), stc2a (6), si:ch211-153b23.5 (5), nkx2.4b (5), barhl2 (5)

midbrain_hindbrain_boundary

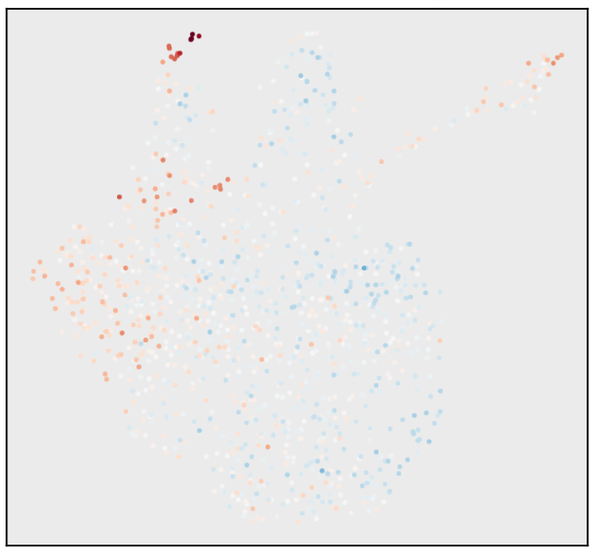
high: 15 genes, not supported
active in 0% of cells, pseudobulk strength 0.25



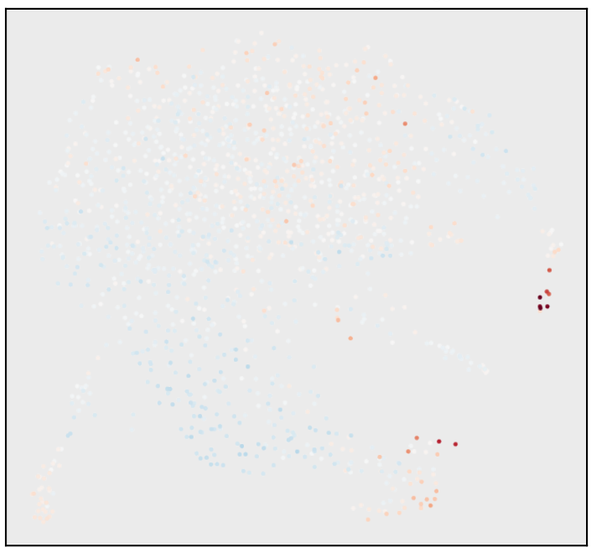
oblong: 13 genes, not supported
active in 0% of cells, pseudobulk strength 0.27



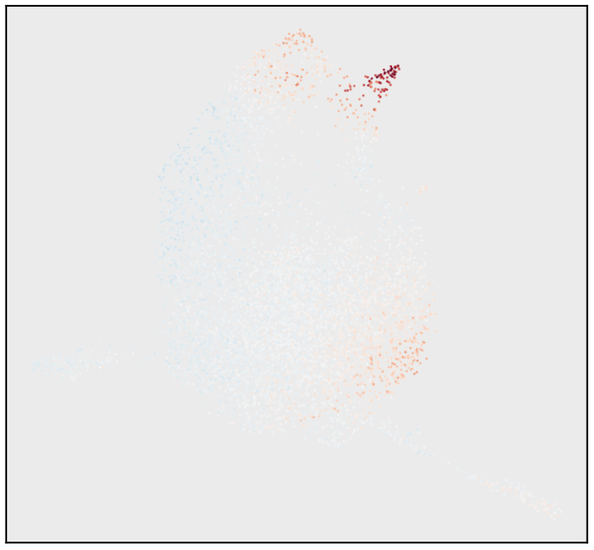
dome: 16 genes, not supported
active in 2% of cells, pseudobulk strength 0.17



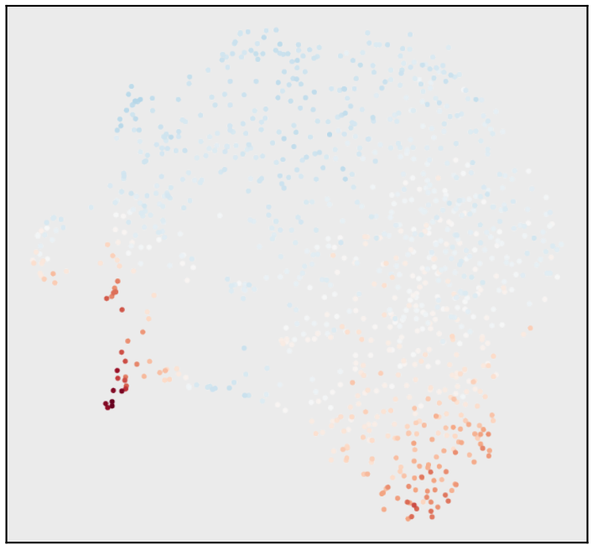
30% epiboly: 15 genes, not supported
active in 1% of cells, pseudobulk strength 0.16



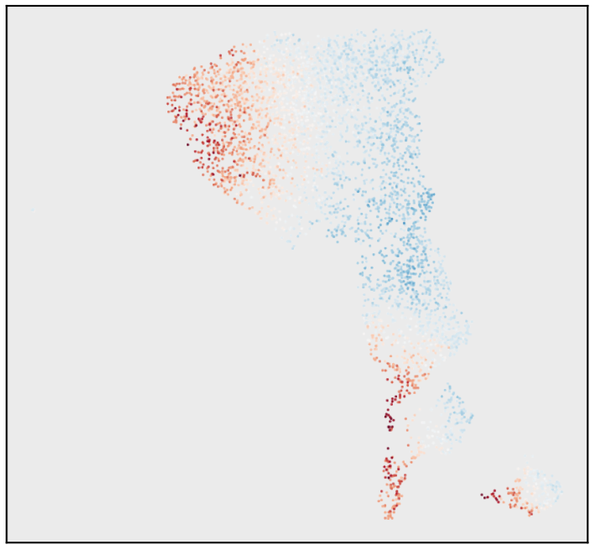
50% epiboly: 22 genes, not supported
active in 5% of cells, pseudobulk strength 0.17



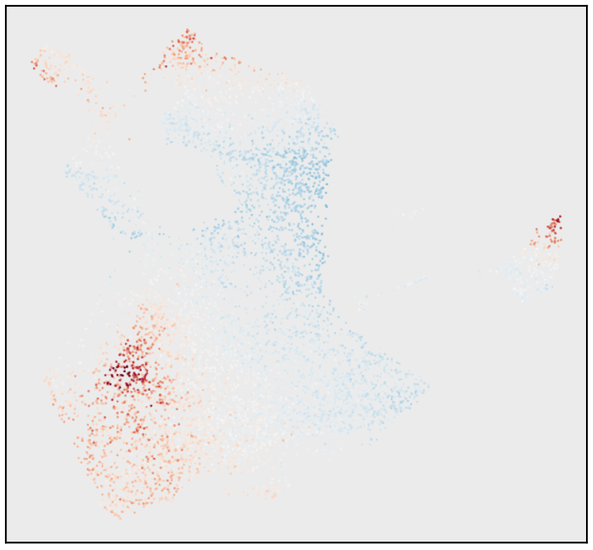
shield: 14 genes, not supported
active in 10% of cells, pseudobulk strength 0.22



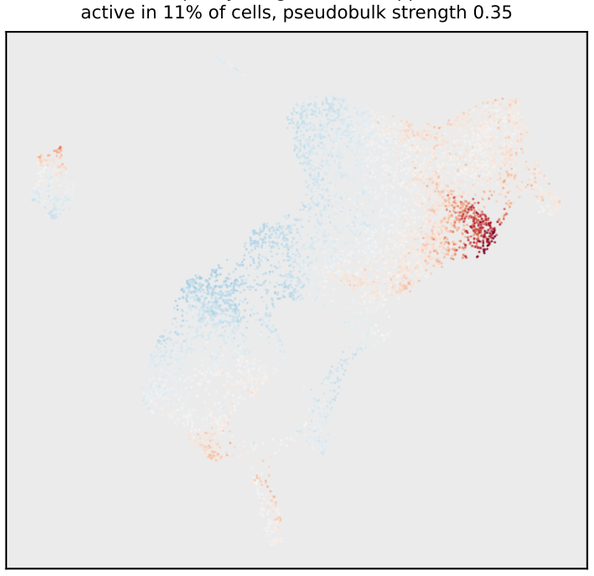
60% epiboly: 21 genes, not supported
active in 18% of cells, pseudobulk strength 0.28



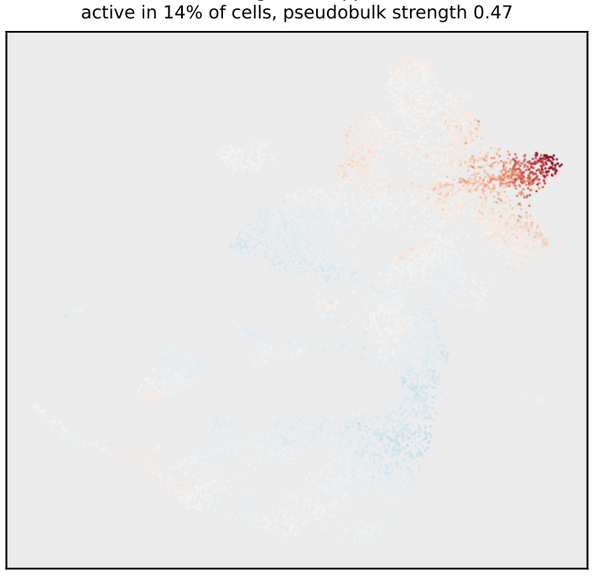
75% epiboly: 24 genes, supported
active in 14% of cells, pseudobulk strength 0.29



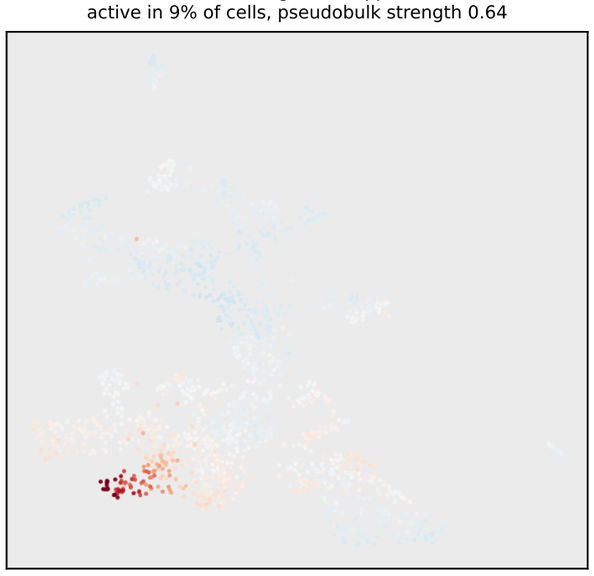
90% epiboly: 27 genes, not supported
active in 11% of cells, pseudobulk strength 0.35



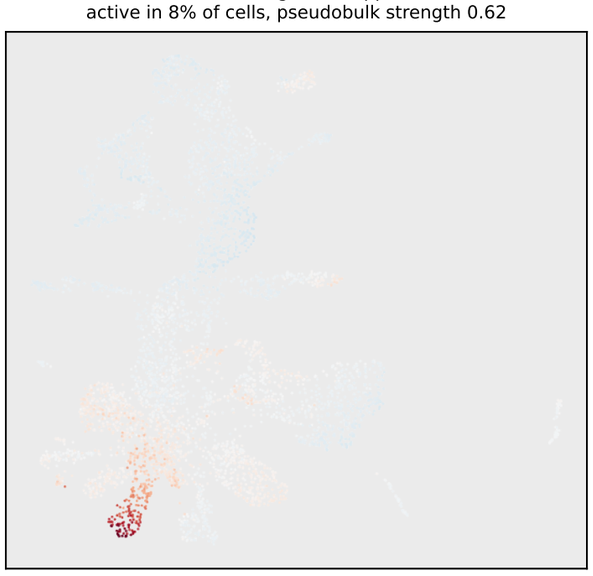
bud: 30 genes, supported
active in 14% of cells, pseudobulk strength 0.47



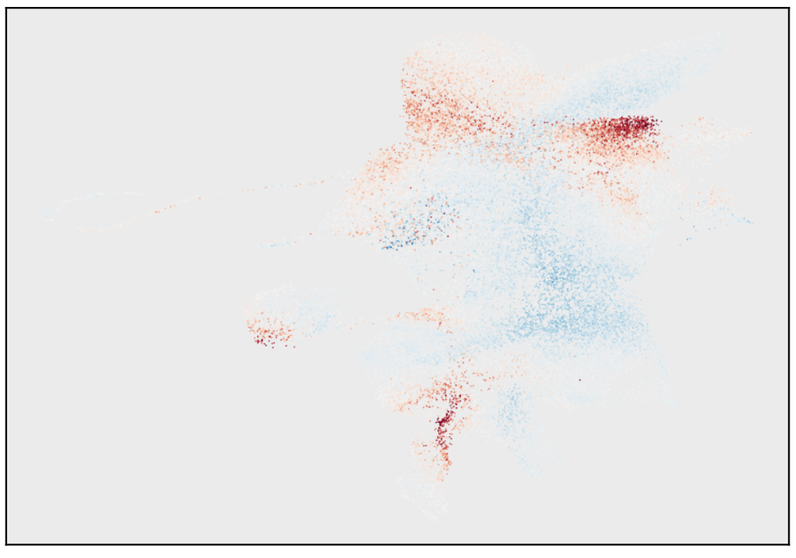
3-somite: 30 genes, supported
active in 9% of cells, pseudobulk strength 0.64



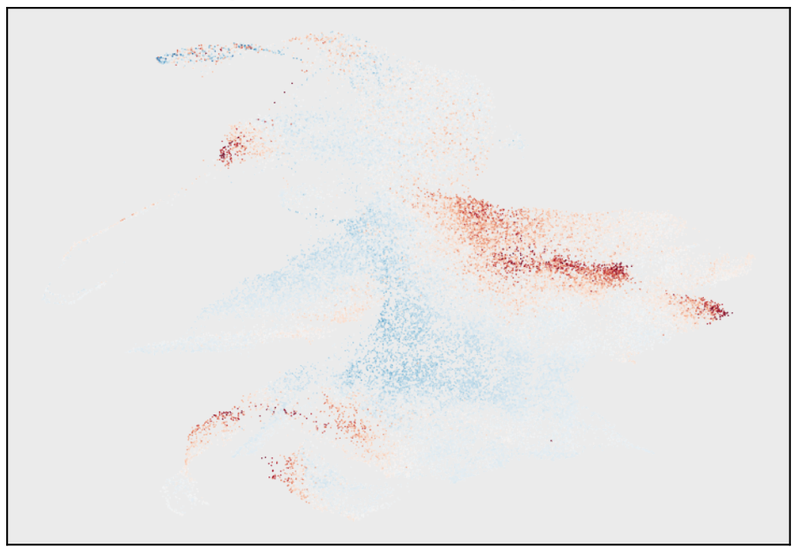
6-somite: 30 genes, supported
active in 8% of cells, pseudobulk strength 0.62



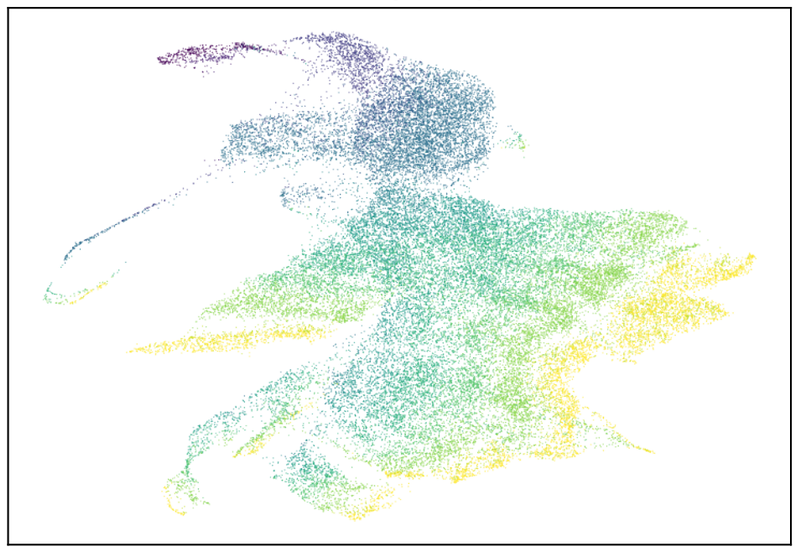
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



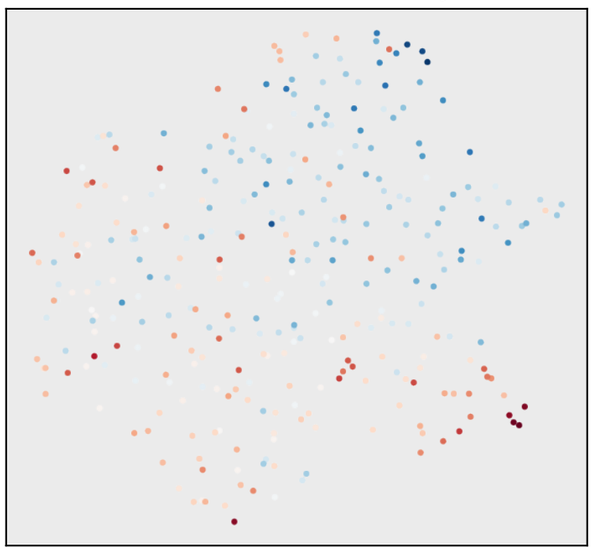
atlas, unaligned: stages



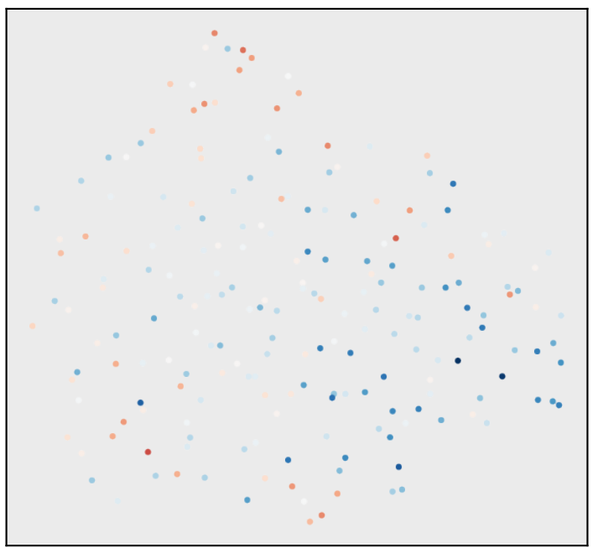
Genes (number of stages): her3 (12), her5 (12), her11 (12), shisa2 (12), otx1b (12), zic3 (12), cnpy1 (12), her8a (12), ier5l (12), tubb5 (12), uacab (12), efna5a (11), ism1 (10), otx1a (10), mdkb (8), sox21a (8), prdm12b (8), nr2f2 (8), dmbx1a (7), phox2a (7), gbx2 (7), wnt8b (6), lmx1bb (6), pax2a (5), emilin1a (5), ebf2 (5), eng2a (4), eng2b (4), mab21l2 (3), wnt7ba (3)

hindbrain

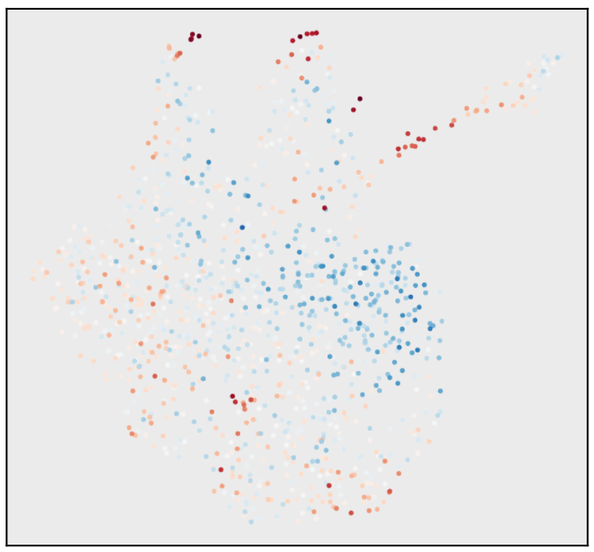
high: 11 genes, not supported
active in 12% of cells, pseudobulk strength 0.16



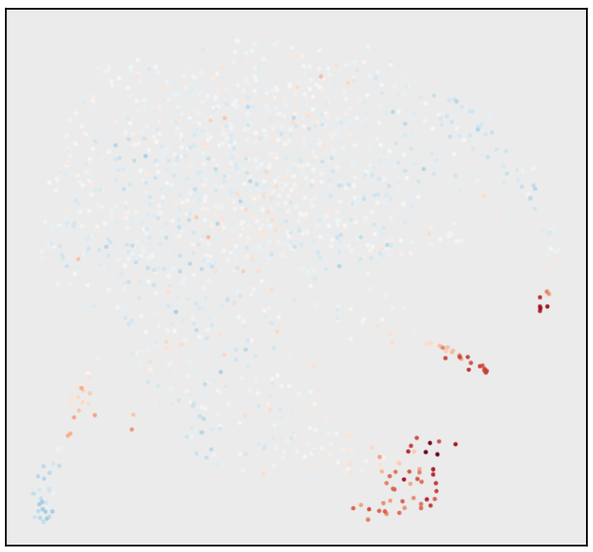
oblong: 10 genes, not supported
active in 0% of cells, pseudobulk strength 0.19



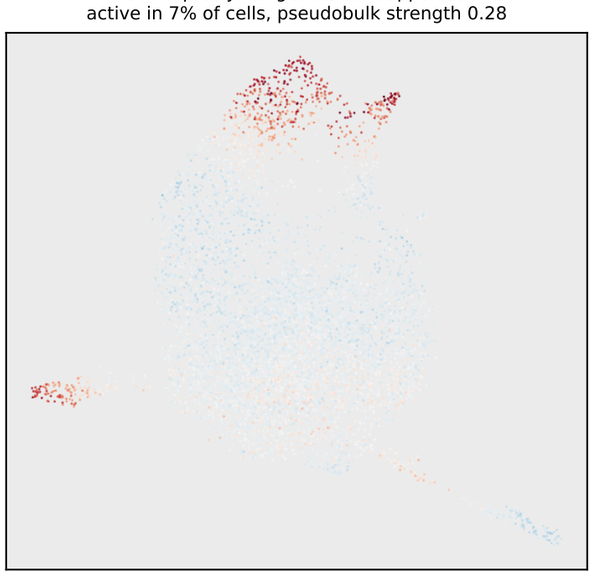
dome: 15 genes, not supported
active in 2% of cells, pseudobulk strength 0.21



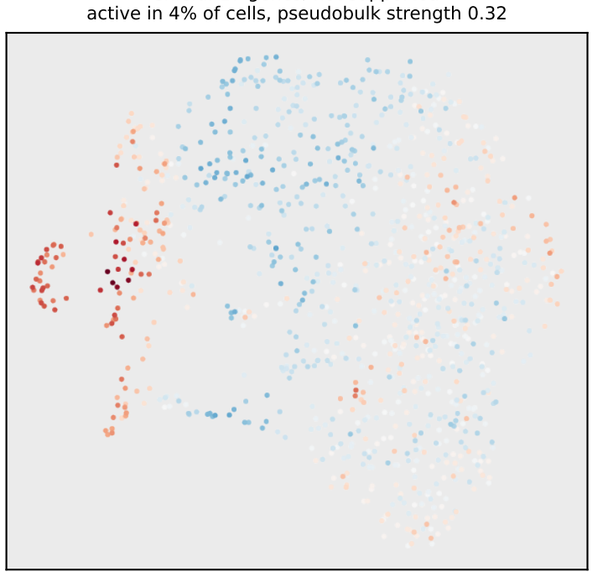
30% epiboly: 16 genes, not supported
active in 4% of cells, pseudobulk strength 0.24



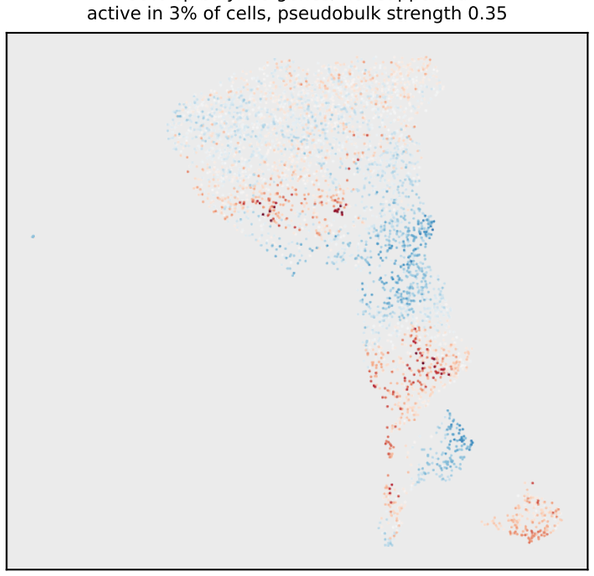
50% epiboly: 22 genes, not supported
active in 7% of cells, pseudobulk strength 0.28



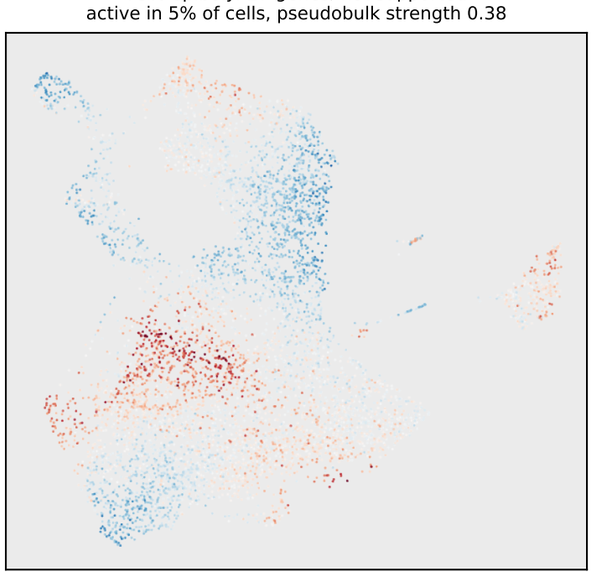
shield: 17 genes, not supported
active in 4% of cells, pseudobulk strength 0.32



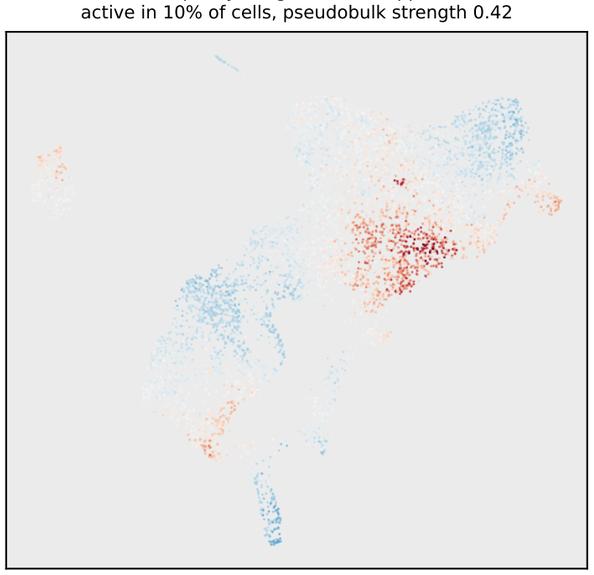
60% epiboly: 26 genes, not supported
active in 3% of cells, pseudobulk strength 0.35



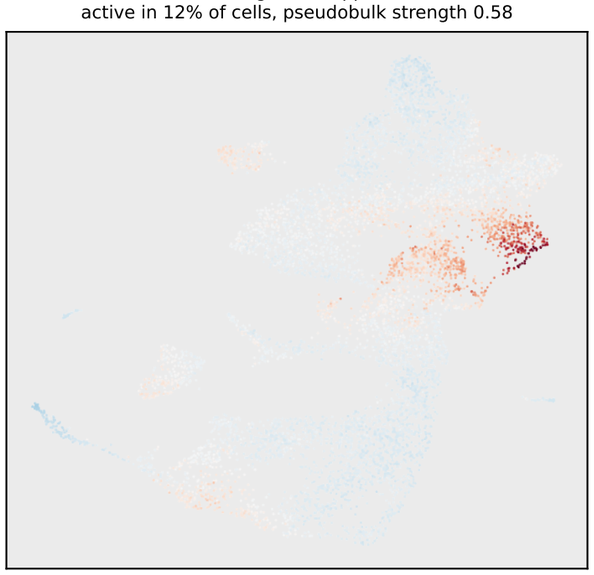
75% epiboly: 27 genes, not supported
active in 5% of cells, pseudobulk strength 0.38



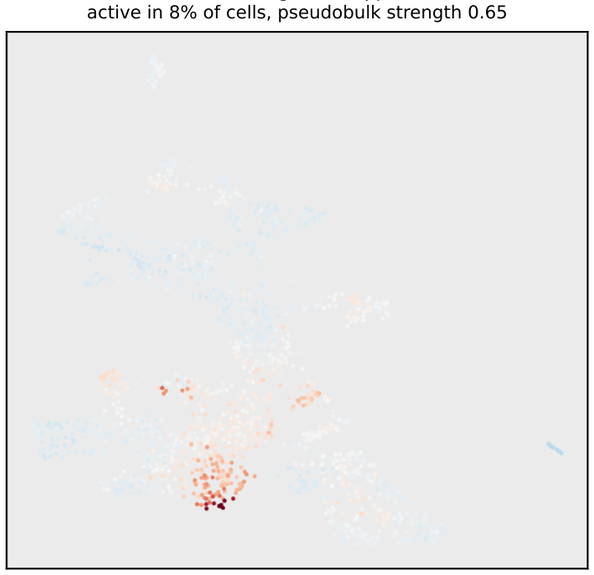
90% epiboly: 28 genes, not supported
active in 10% of cells, pseudobulk strength 0.42



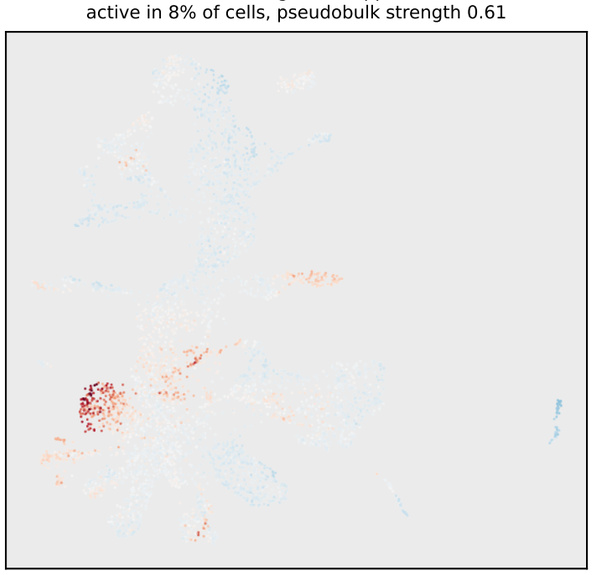
bud: 30 genes, supported
active in 12% of cells, pseudobulk strength 0.58



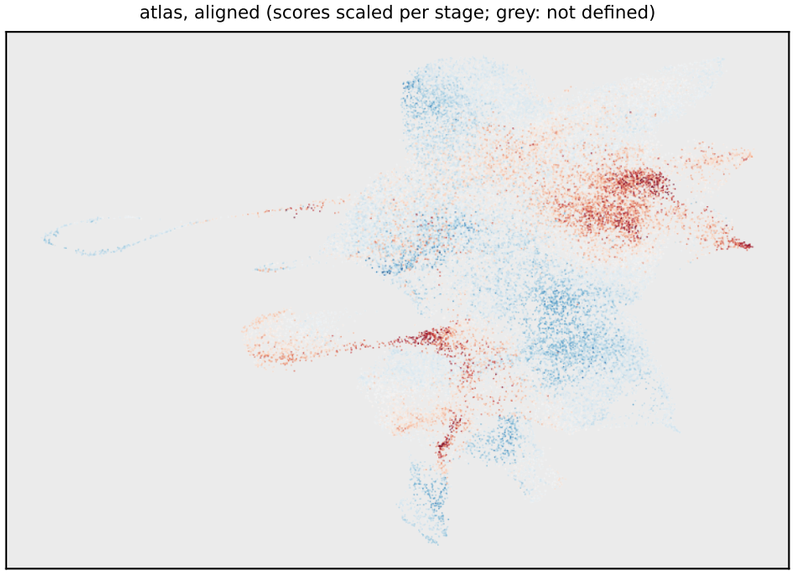
3-somite: 30 genes, supported
active in 8% of cells, pseudobulk strength 0.65



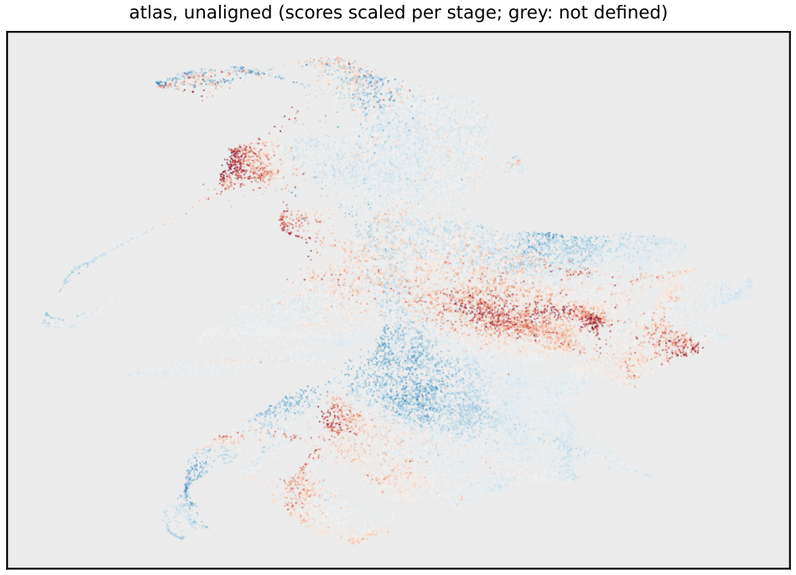
6-somite: 30 genes, supported
active in 8% of cells, pseudobulk strength 0.61



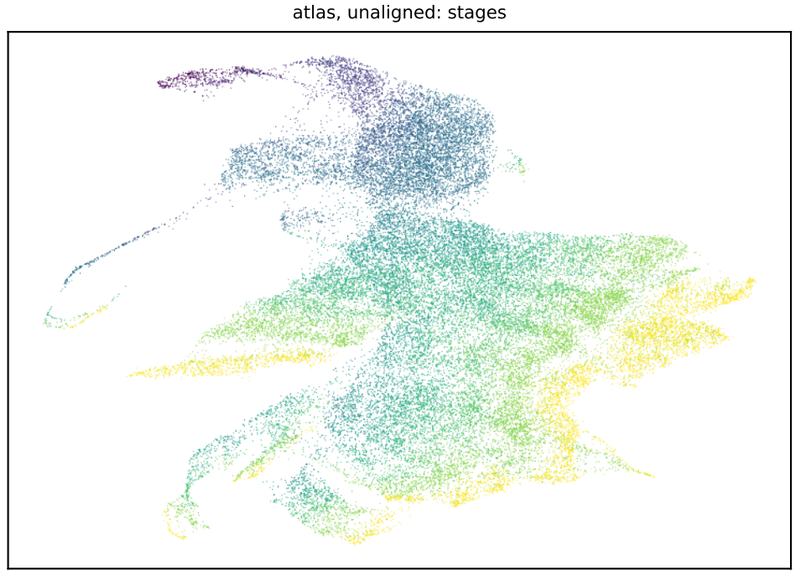
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



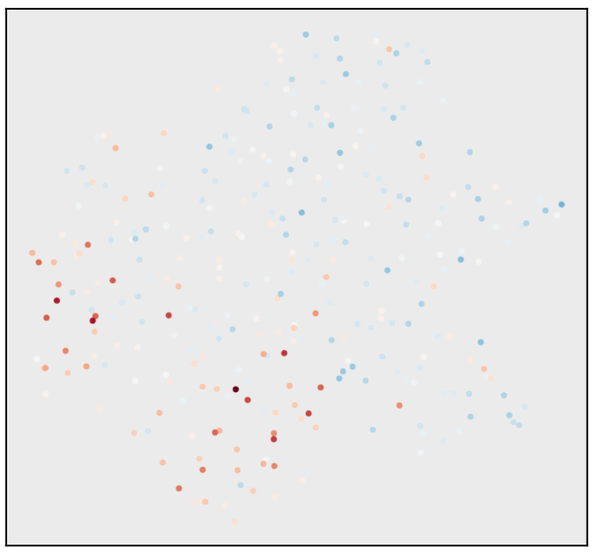
atlas, unaligned: stages



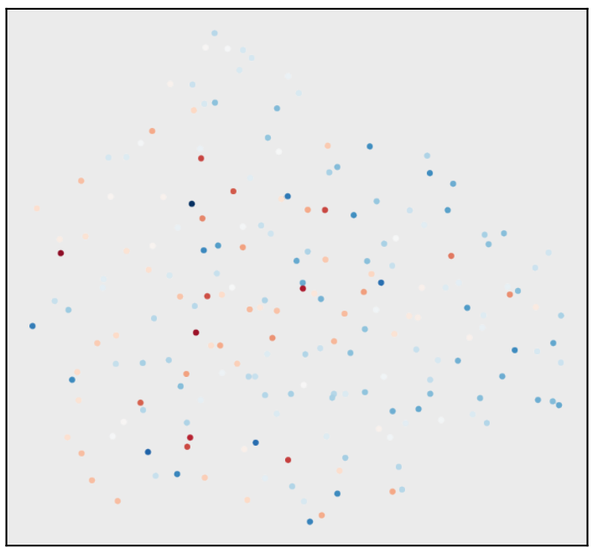
Genes (number of stages): irx7 (12), phlda1 (12), rnd1l (12), sox11a (12), fam212ab (12), gpd1b (12), lfng (12), sox11b (12), sox13 (12), btg1 (12), olig4 (11), efnb2a (10), gbx1 (10), crabp2b (10), dbx1a (9), bdnf (9), her2 (8), olig3 (8), cyp26c1 (7), mafba (7), dusp2 (7), meis3 (7), hoxb1b (6), gsx1 (6), hoxb3a (6), irx5a (6), hoxb1a (5), cyp26b1 (4), egr2b (3), otpb (3)

primary_neurons

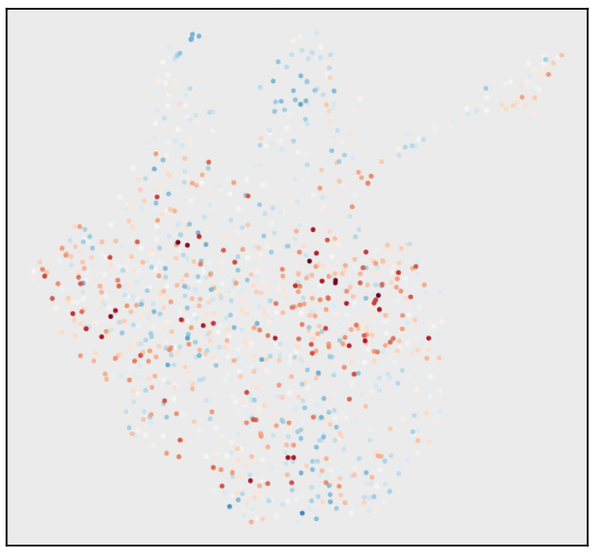
high: 6 genes, not supported
active in 1% of cells, pseudobulk strength 0.27



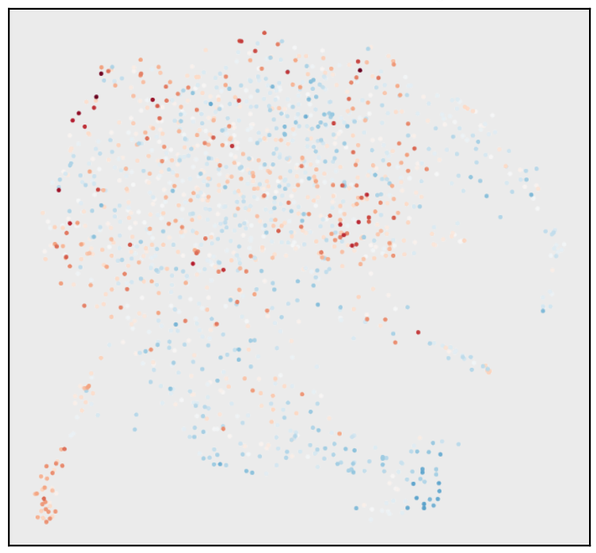
oblong: 4 genes, not supported
active in 0% of cells, pseudobulk strength 0.25



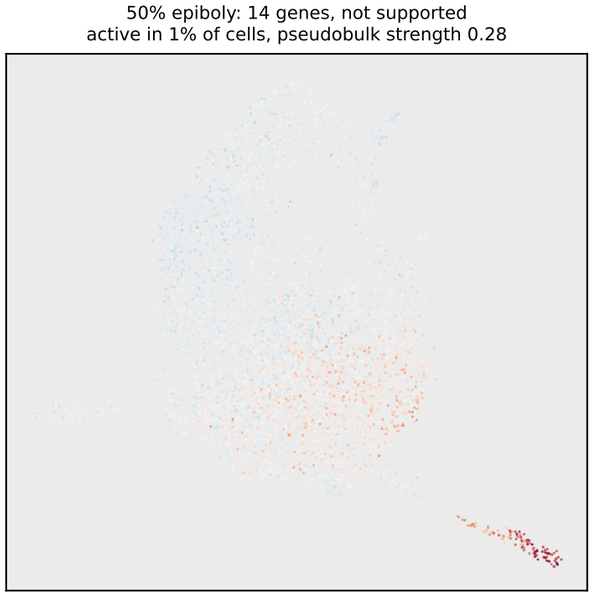
dome: 8 genes, not supported
active in 0% of cells, pseudobulk strength 0.21



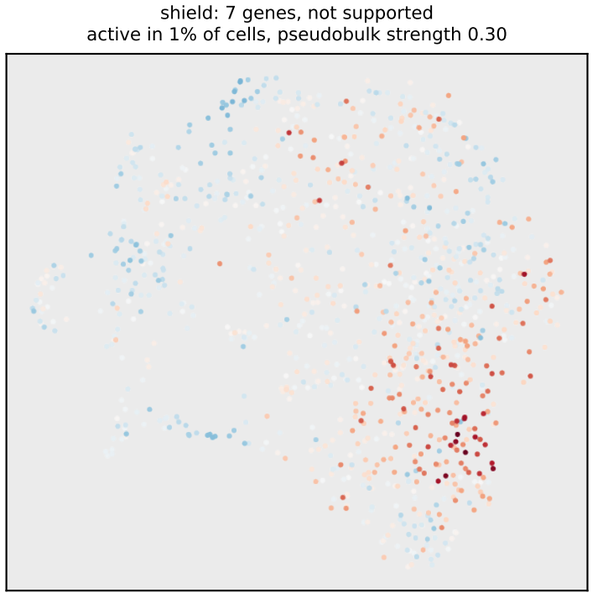
30% epiboly: 9 genes, not supported
active in 0% of cells, pseudobulk strength 0.28



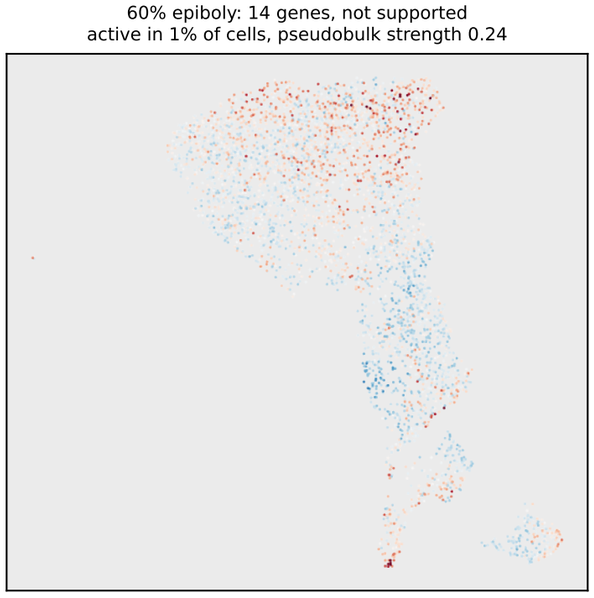
50% epiboly: 14 genes, not supported
active in 1% of cells, pseudobulk strength 0.28



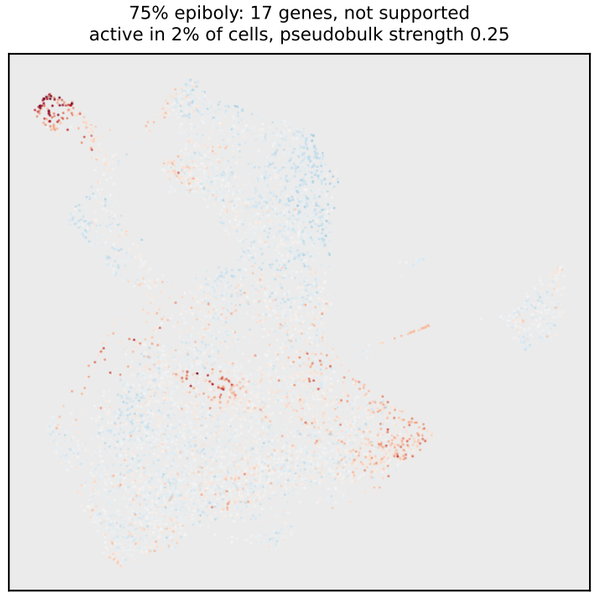
shield: 7 genes, not supported
active in 1% of cells, pseudobulk strength 0.30



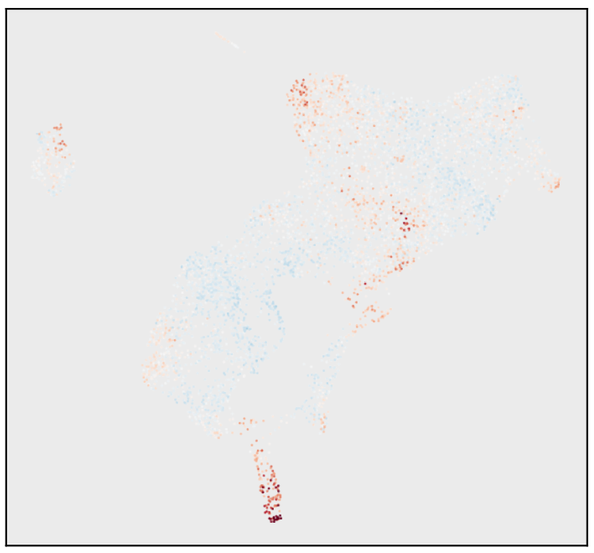
60% epiboly: 14 genes, not supported
active in 1% of cells, pseudobulk strength 0.24



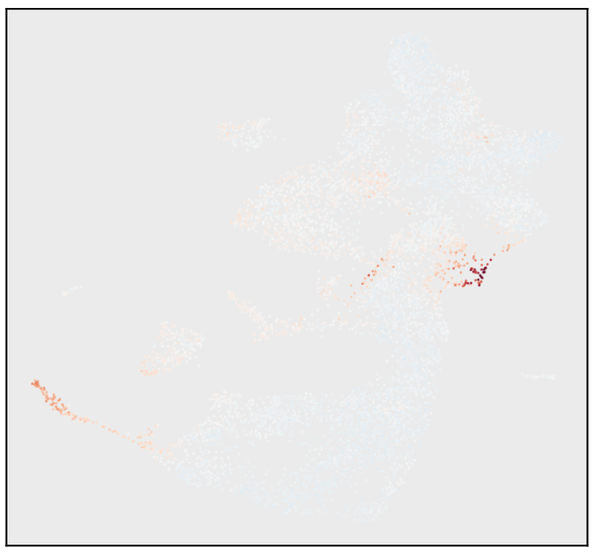
75% epiboly: 17 genes, not supported
active in 2% of cells, pseudobulk strength 0.25



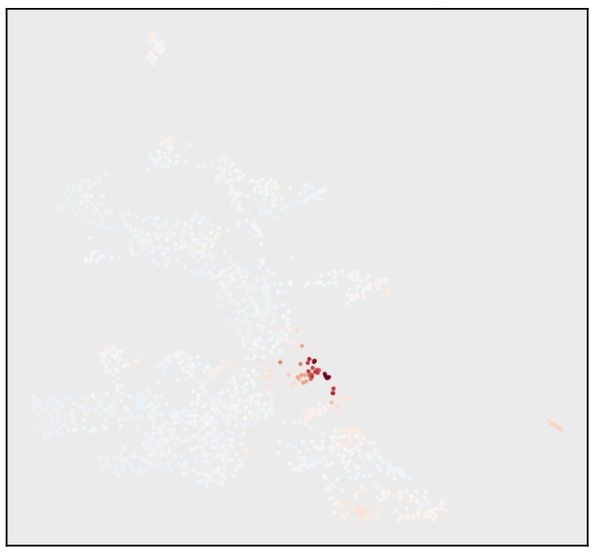
90% epiboly: 17 genes, not supported
active in 2% of cells, pseudobulk strength 0.34



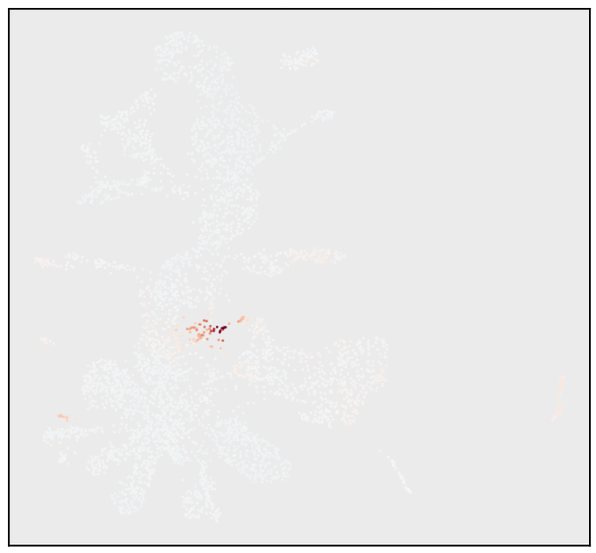
bud: 22 genes, not supported
active in 4% of cells, pseudobulk strength 0.47



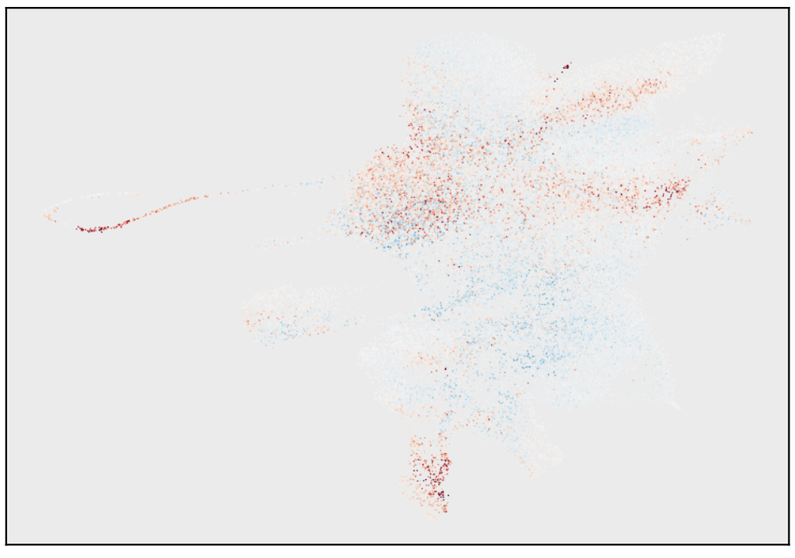
3-somite: 21 genes, supported
active in 4% of cells, pseudobulk strength 0.51



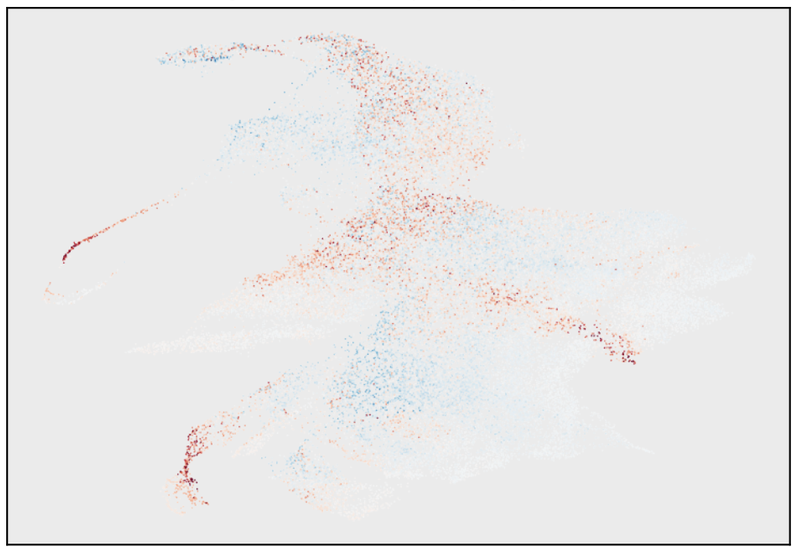
6-somite: 22 genes, supported
active in 4% of cells, pseudobulk strength 0.64



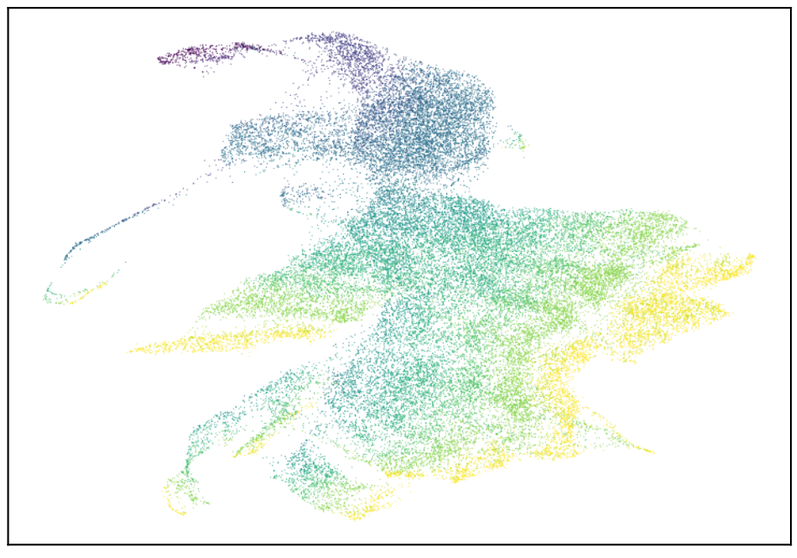
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



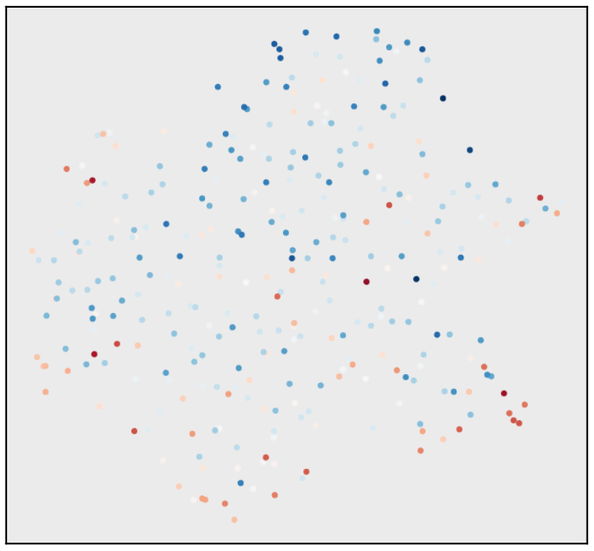
atlas, unaligned: stages



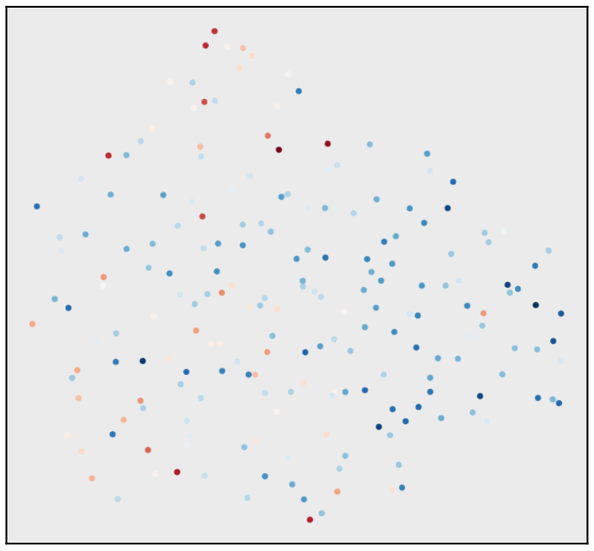
Genes (number of stages): zbtb10 (12), btbd6b (12), efna1b (12), hacd3 (12), smox (11), pnx (10), pnocb (9), gadd45gb.1 (8), pcdh18a (8), prdm14 (8), vim (7), nfasca (7), chodl (7), elavl3 (6), ascl1a (6), cbfb (6), isl1 (5), neurod4 (3), neurod1 (3), tlx2 (3), insm1a (3), tlx3b (3)

neural_crest

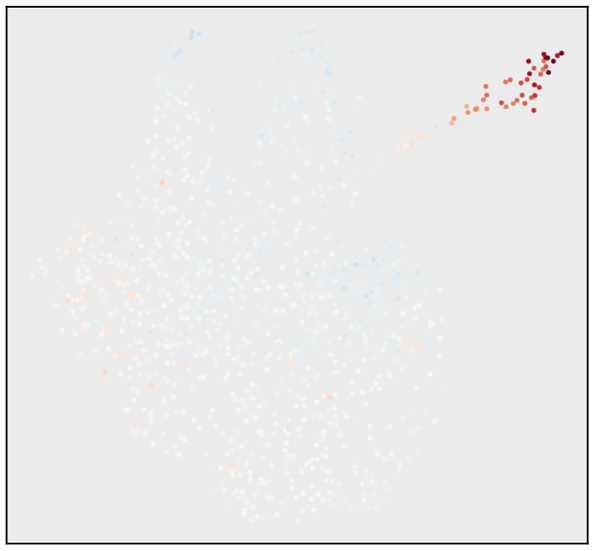
high: 9 genes, not supported
active in 0% of cells, pseudobulk strength 0.28



oblong: 9 genes, not supported
active in 0% of cells, pseudobulk strength 0.32

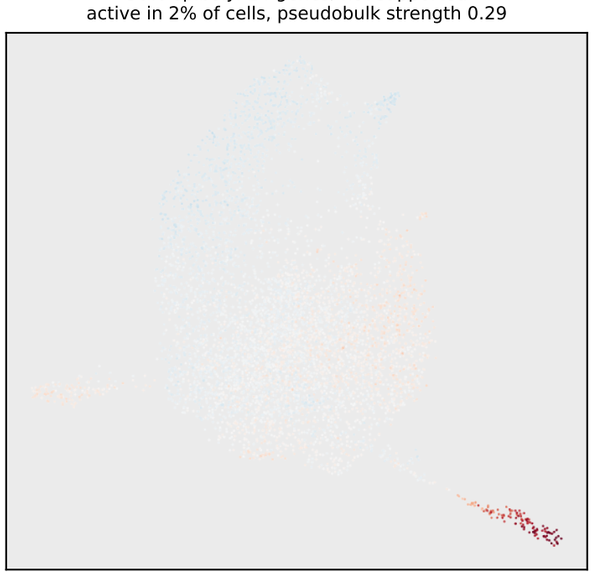


dome: 11 genes, not supported
active in 3% of cells, pseudobulk strength 0.33

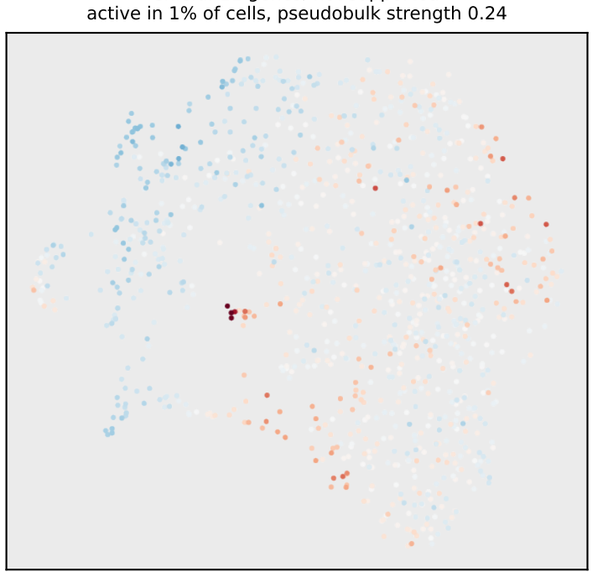


30% epiboly: 11 genes, not supported
active in 3% of cells, pseudobulk strength 0.23

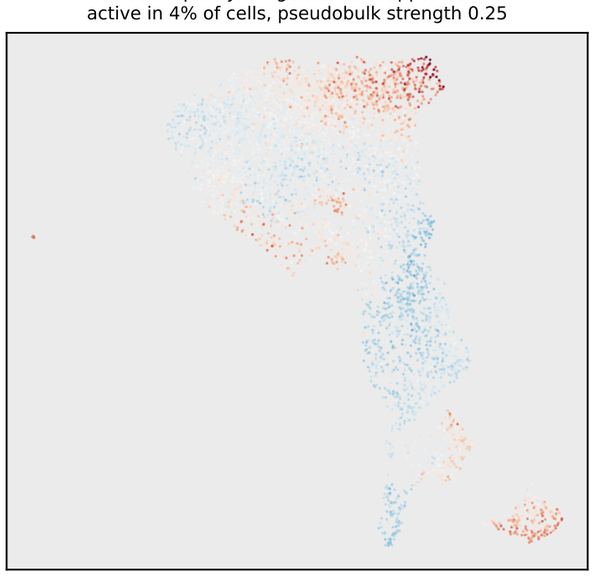
50% epiboly: 14 genes, not supported
active in 2% of cells, pseudobulk strength 0.29



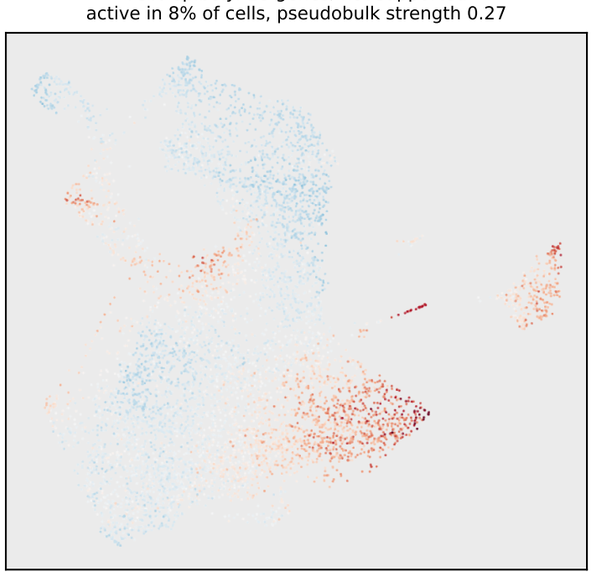
shield: 11 genes, not supported
active in 1% of cells, pseudobulk strength 0.24



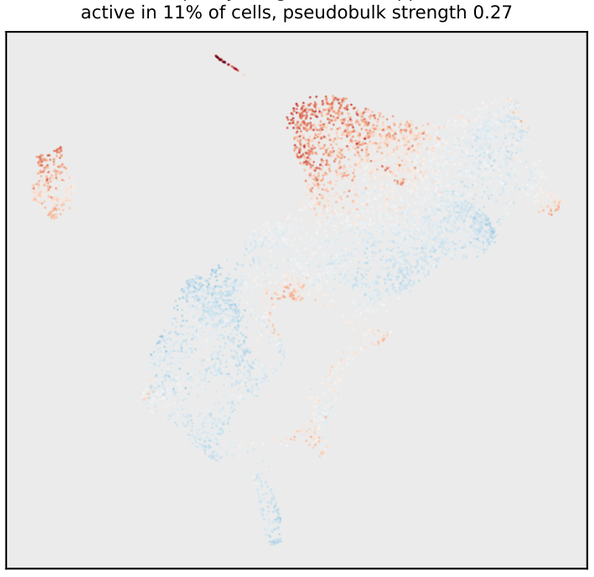
60% epiboly: 16 genes, not supported
active in 4% of cells, pseudobulk strength 0.25



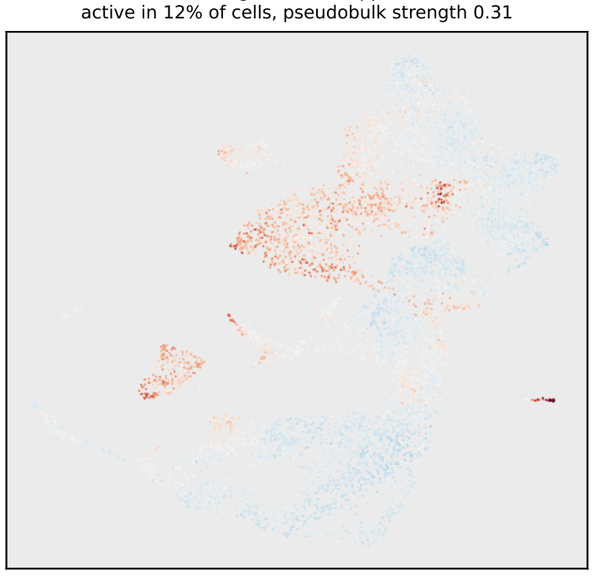
75% epiboly: 18 genes, not supported
active in 8% of cells, pseudobulk strength 0.27



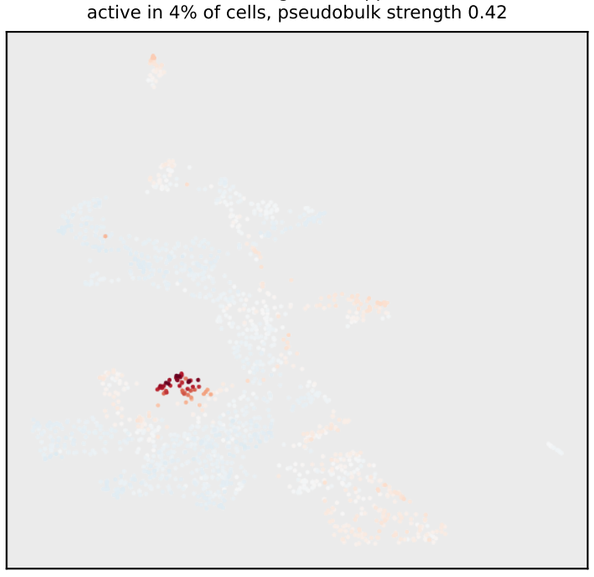
90% epiboly: 16 genes, not supported
active in 11% of cells, pseudobulk strength 0.27



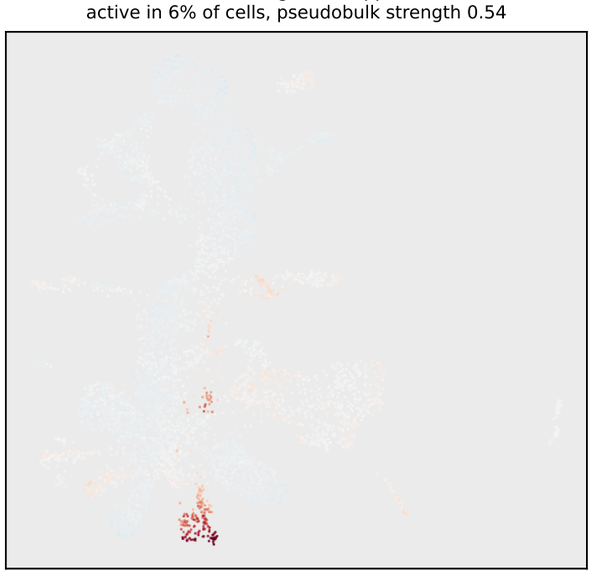
bud: 19 genes, not supported
active in 12% of cells, pseudobulk strength 0.31



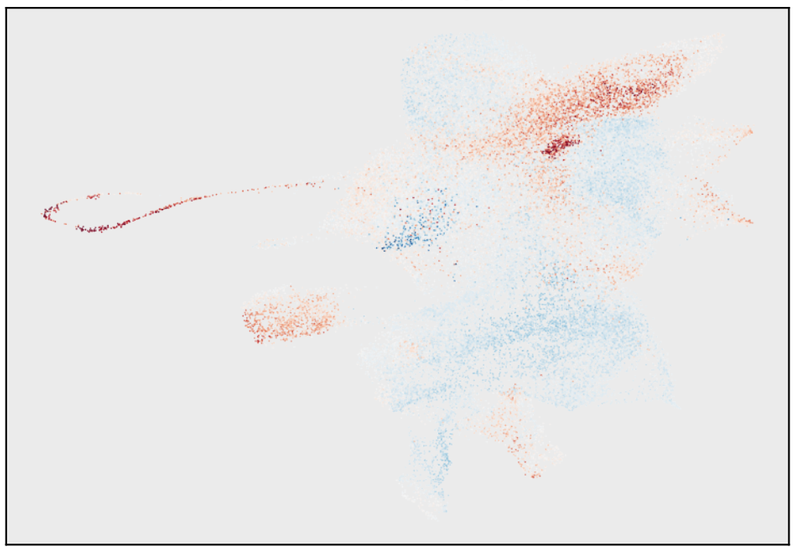
3-somite: 19 genes, supported
active in 4% of cells, pseudobulk strength 0.42



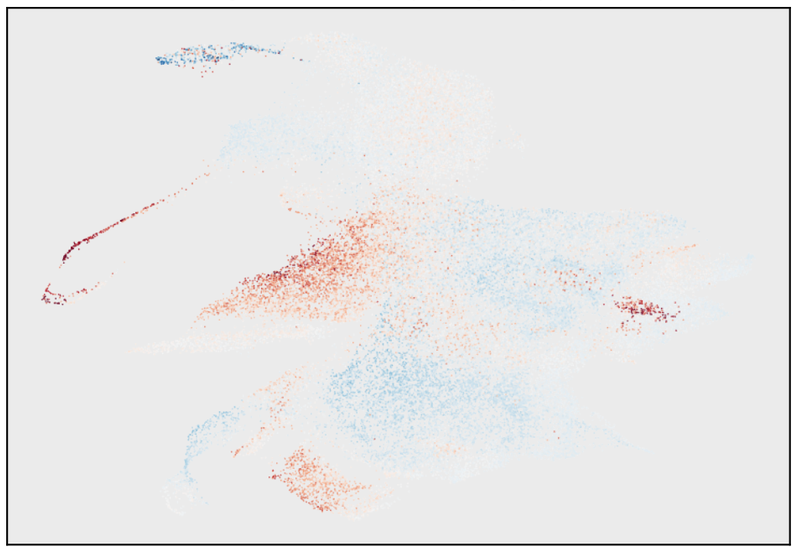
6-somite: 20 genes, supported
active in 6% of cells, pseudobulk strength 0.54



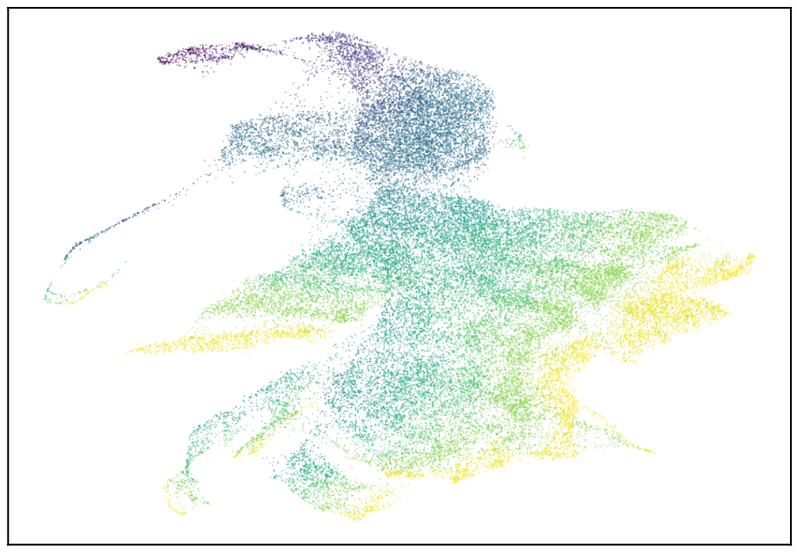
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



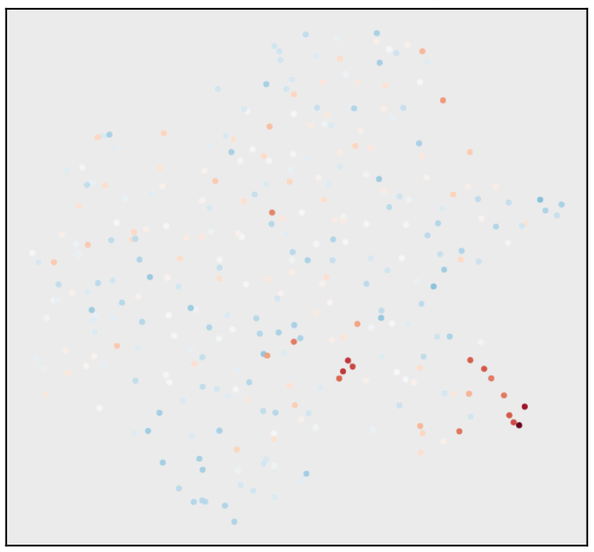
atlas, unaligned: stages



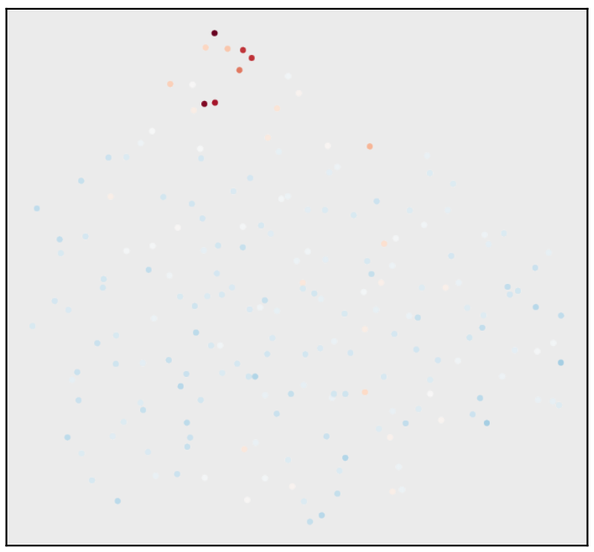
Genes (number of stages): tfap2a (12), foxd3 (12), fam212aa (12), fam46ba (12), klf2b (12), gipc2 (12), crip2 (12), ovol1 (11), msx3 (10), zfp36l1b (10), lhx5 (9), sox9b (9), cdh6 (8), nes (7), kctd15a (6), sox10 (5), twist1b (5), sox1b (5), alx1 (3), ednrab (1)

placodes

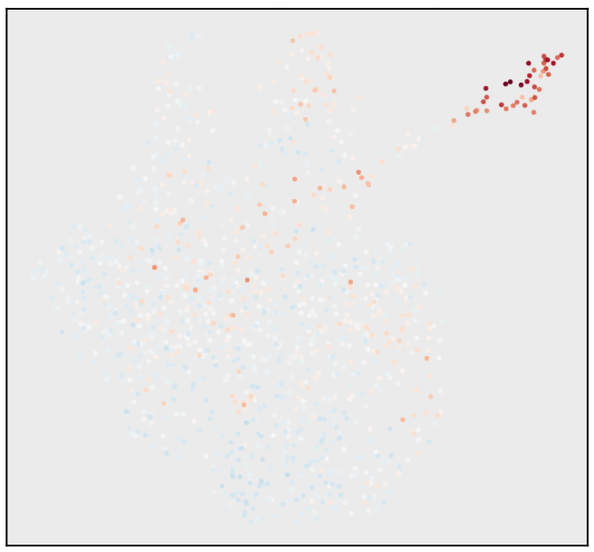
high: 5 genes, not supported
active in 0% of cells, pseudobulk strength 0.13



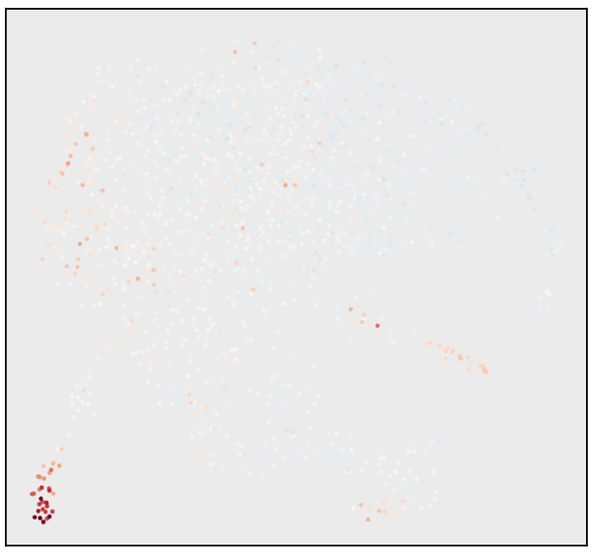
oblong: 3 genes, not supported
active in 2% of cells, pseudobulk strength 0.14



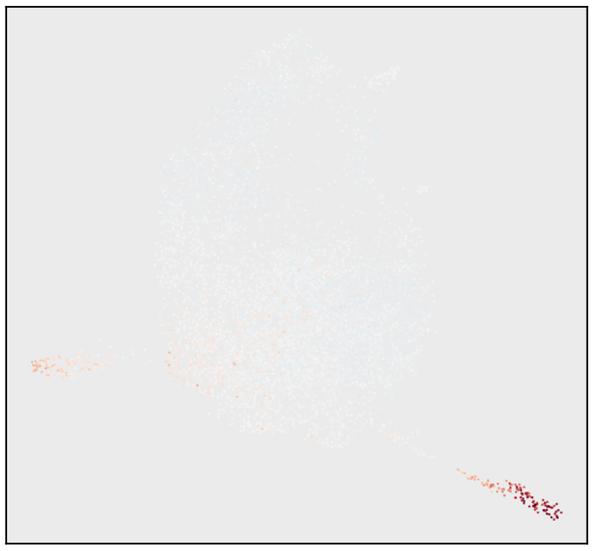
dome: 10 genes, not supported
active in 3% of cells, pseudobulk strength 0.13



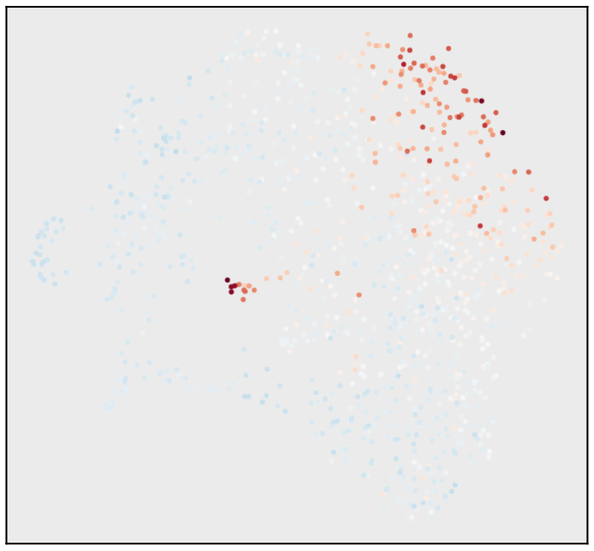
30% epiboly: 13 genes, not supported
active in 3% of cells, pseudobulk strength 0.14



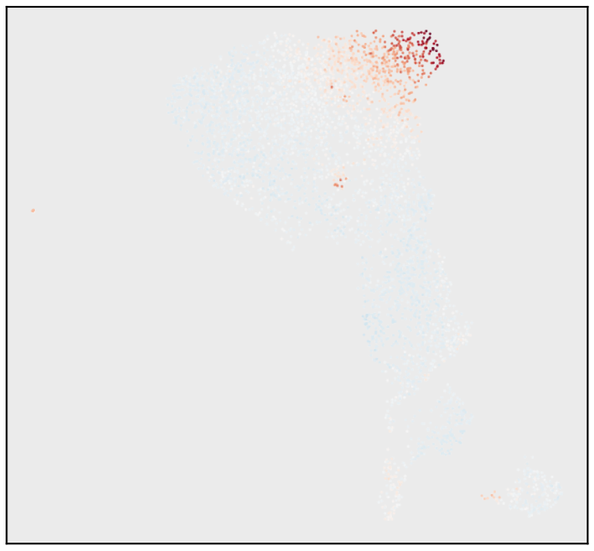
50% epiboly: 20 genes, not supported
active in 2% of cells, pseudobulk strength 0.14



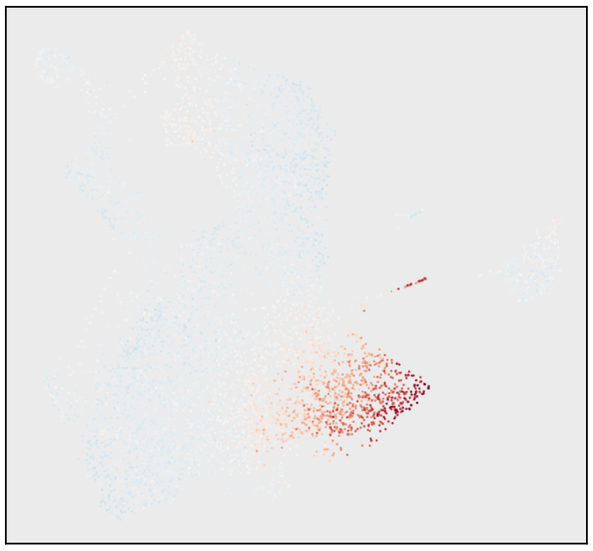
shield: 8 genes, not supported
active in 5% of cells, pseudobulk strength 0.13



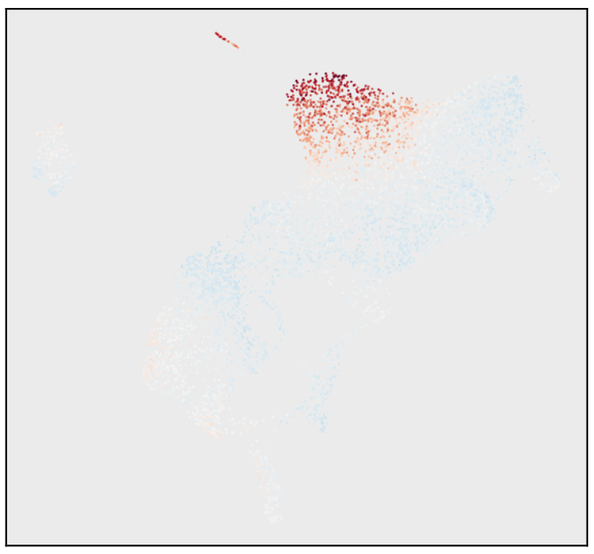
60% epiboly: 19 genes, not supported
active in 8% of cells, pseudobulk strength 0.16



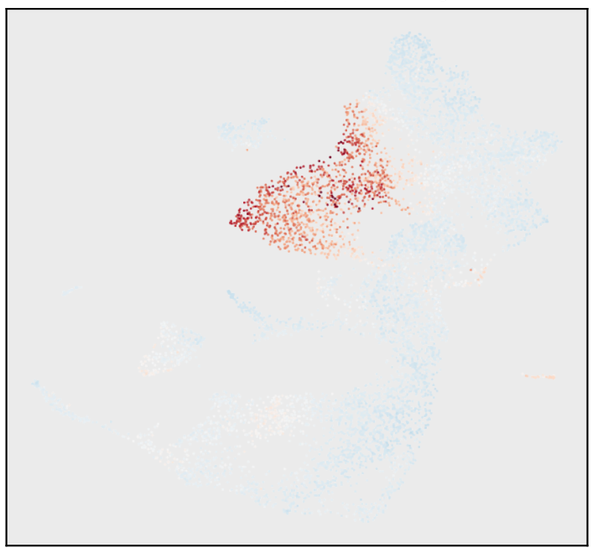
75% epiboly: 21 genes, not supported
active in 12% of cells, pseudobulk strength 0.22



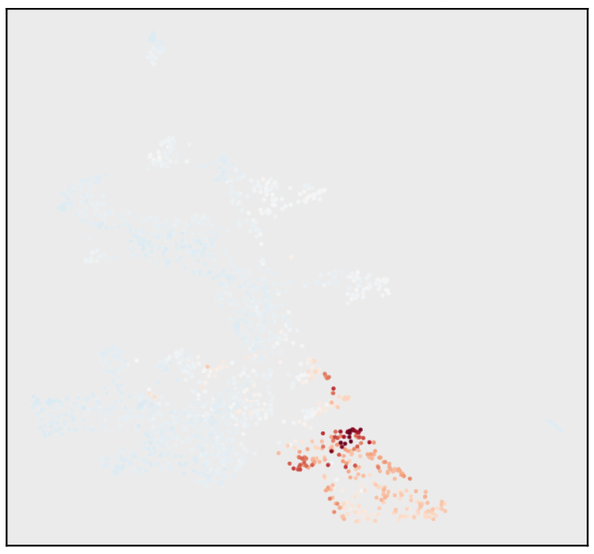
90% epiboly: 22 genes, supported
active in 15% of cells, pseudobulk strength 0.33



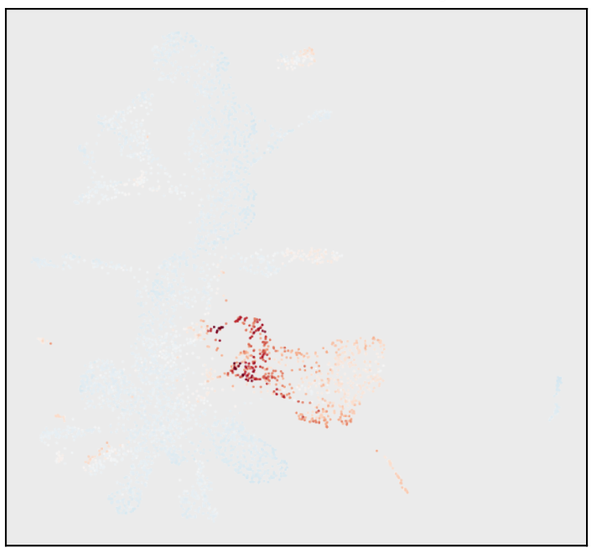
bud: 23 genes, supported
active in 17% of cells, pseudobulk strength 0.53



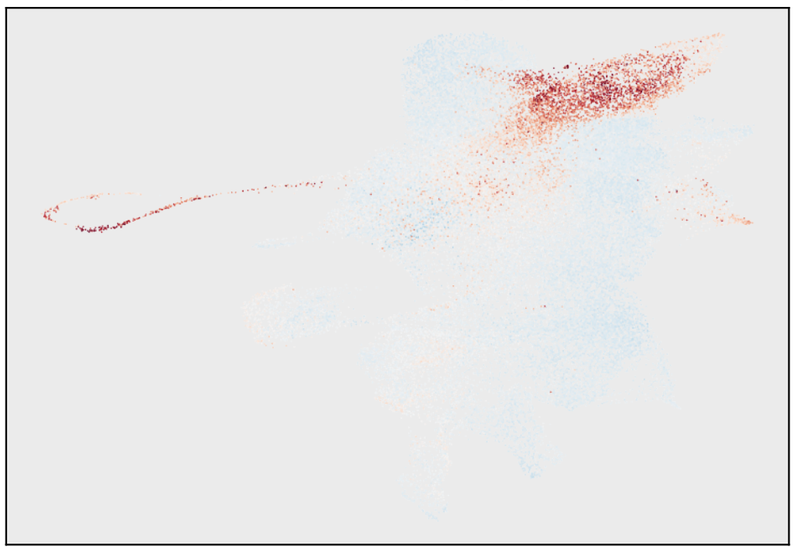
3-somite: 25 genes, supported
active in 14% of cells, pseudobulk strength 0.68



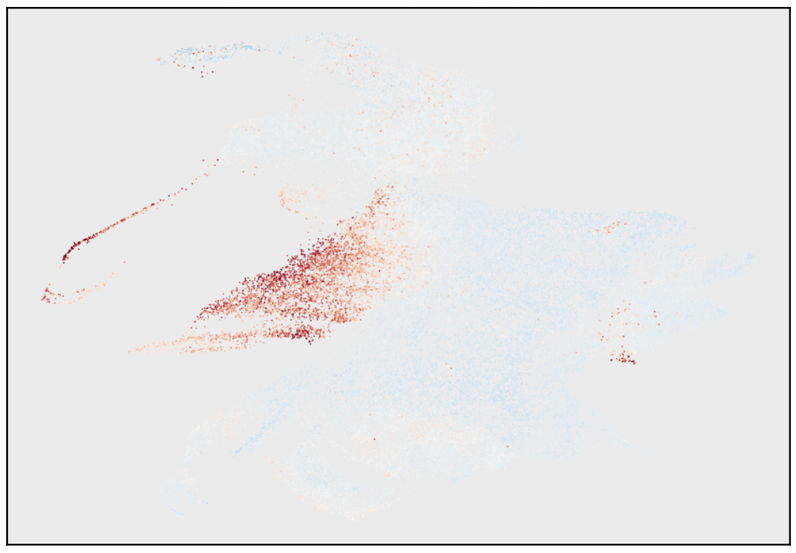
6-somite: 28 genes, supported
active in 12% of cells, pseudobulk strength 0.72



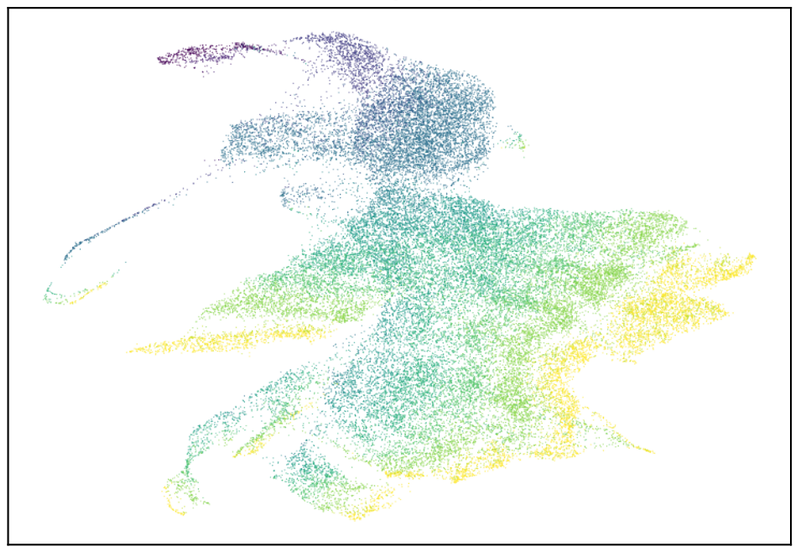
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



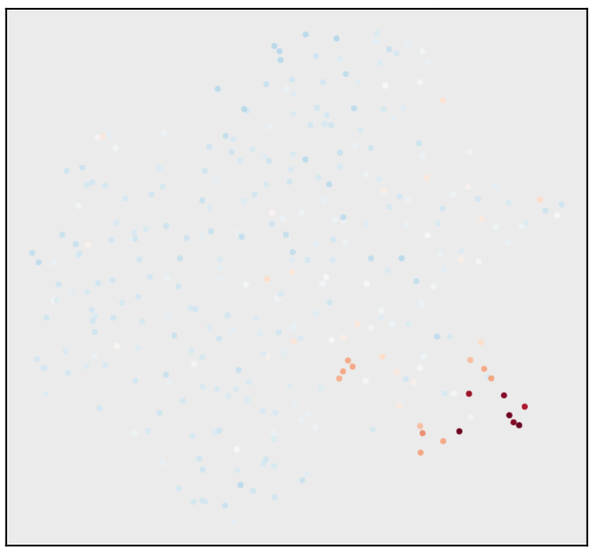
atlas, unaligned: stages



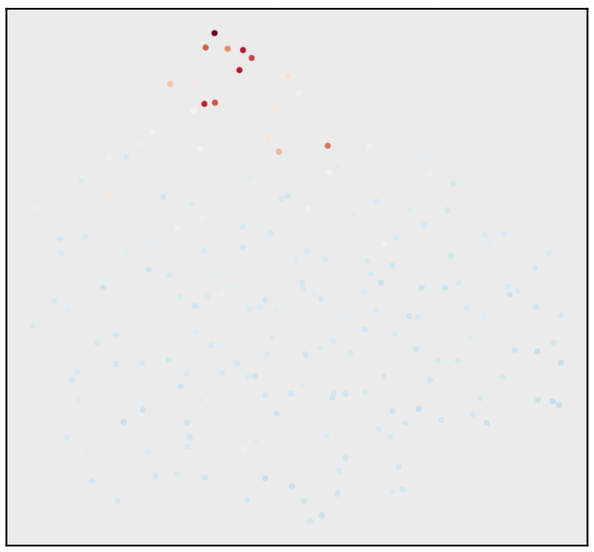
Genes (number of stages): foxi1 (12), irf6 (12), tp63 (11), pih1d3 (11), cldna (10), tfap2c (10), eya1 (10), ptgs2a (9), hk1 (9), sp7 (9), dlx3b (8), her13 (8), foxj1b (8), irx1a (8), insm1b (8), pitx3 (7), six4b (7), pax8 (6), grhl2b (6), sdpra (6), dlx4b (6), sfrp2 (4), drgx (3), scrt2 (3), atoh1b (2), hmx3a (2), atoh1a (1), isl2b (1)

non_neural_ecoderm

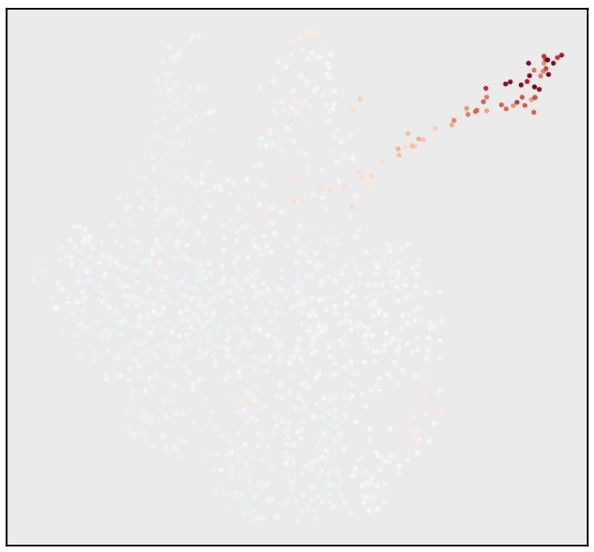
high: 6 genes, not supported
active in 3% of cells, pseudobulk strength 0.09



oblong: 3 genes, supported
active in 4% of cells, pseudobulk strength 0.12

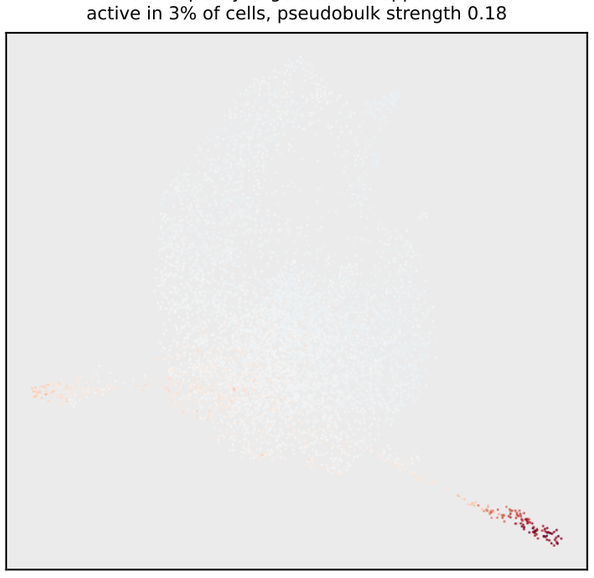


dome: 7 genes, not supported
active in 4% of cells, pseudobulk strength 0.09

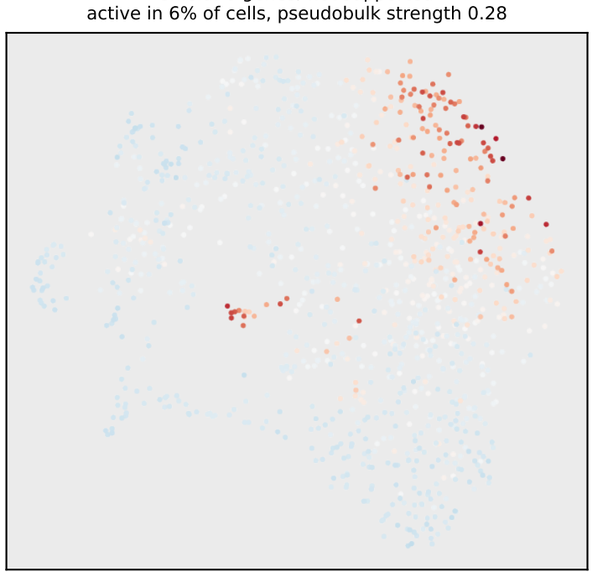


30% epiboly: 7 genes, not supported
active in 2% of cells, pseudobulk strength 0.16

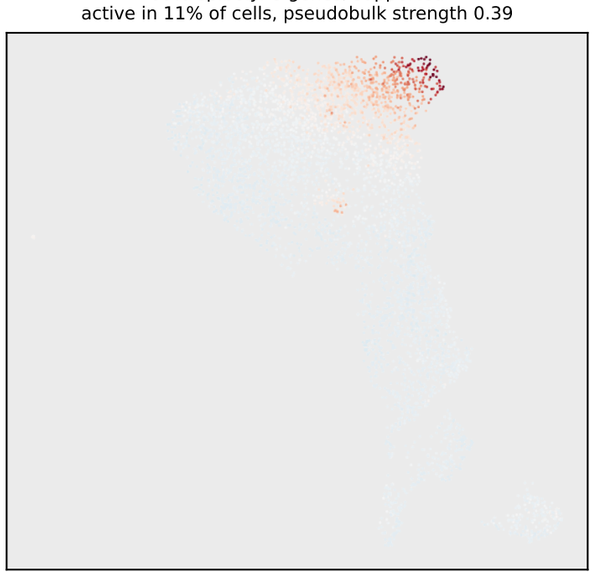
50% epiboly: 9 genes, not supported
active in 3% of cells, pseudobulk strength 0.18



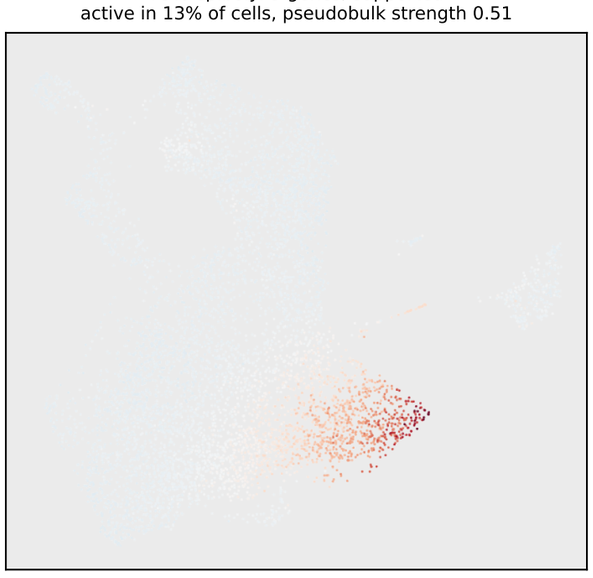
shield: 6 genes, not supported
active in 6% of cells, pseudobulk strength 0.28



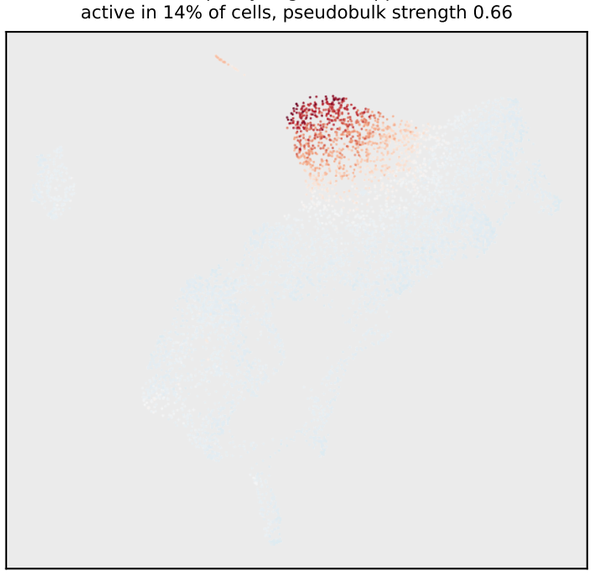
60% epiboly: 8 genes, supported
active in 11% of cells, pseudobulk strength 0.39



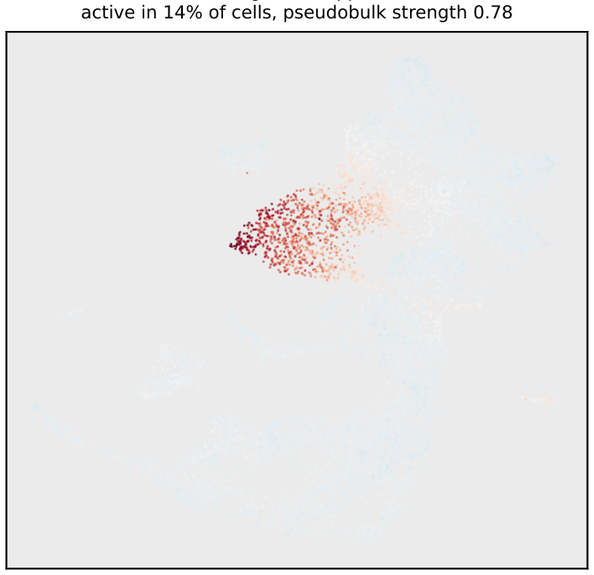
75% epiboly: 9 genes, supported
active in 13% of cells, pseudobulk strength 0.51



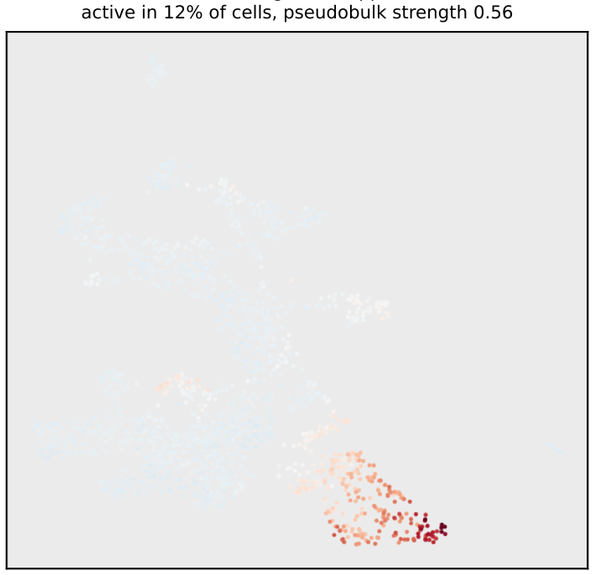
90% epiboly: 9 genes, supported
active in 14% of cells, pseudobulk strength 0.66



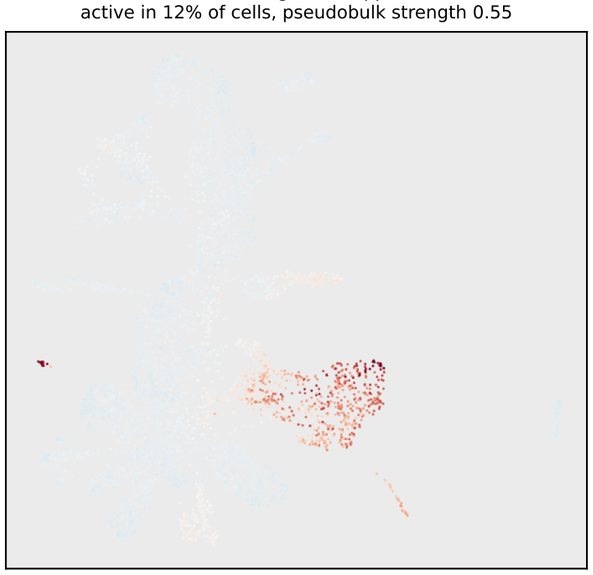
bud: 8 genes, supported
active in 14% of cells, pseudobulk strength 0.78



3-somite: 9 genes, supported
active in 12% of cells, pseudobulk strength 0.56

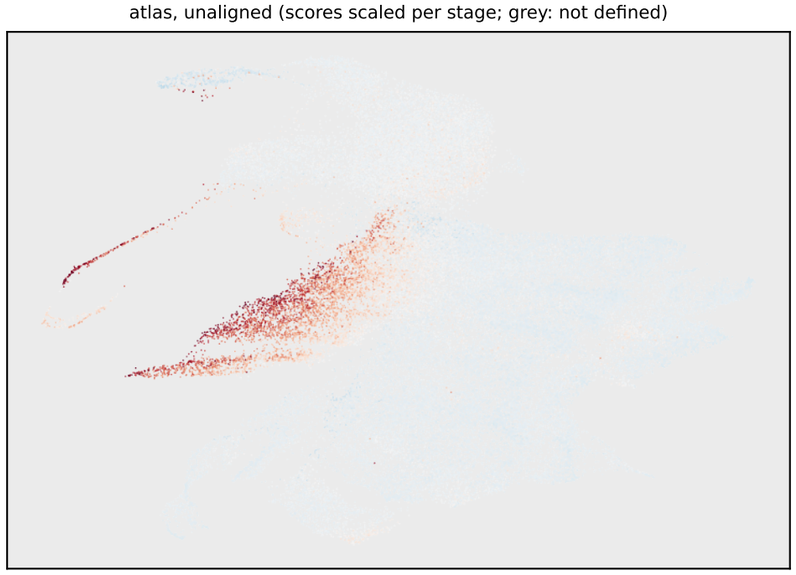


6-somite: 10 genes, supported
active in 12% of cells, pseudobulk strength 0.55

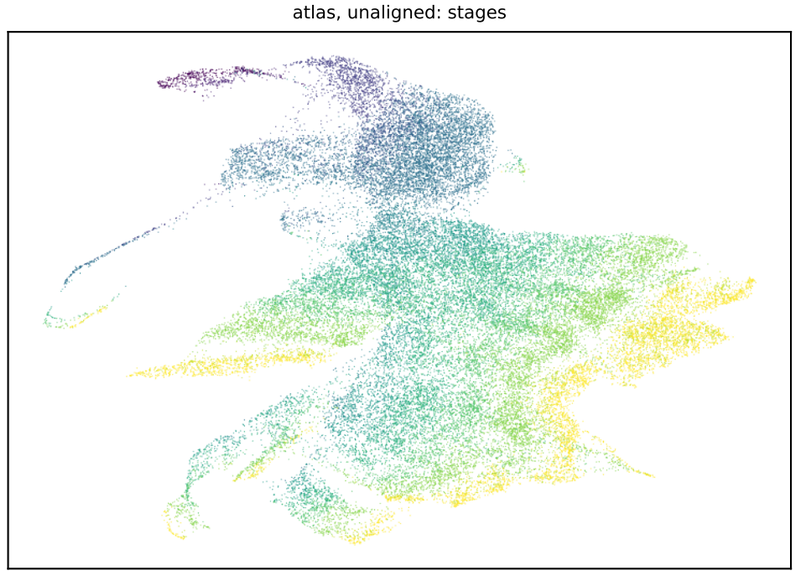


atlas, aligned (scores scaled per stage; grey: not defined)

atlas, unaligned (scores scaled per stage; grey: not defined)



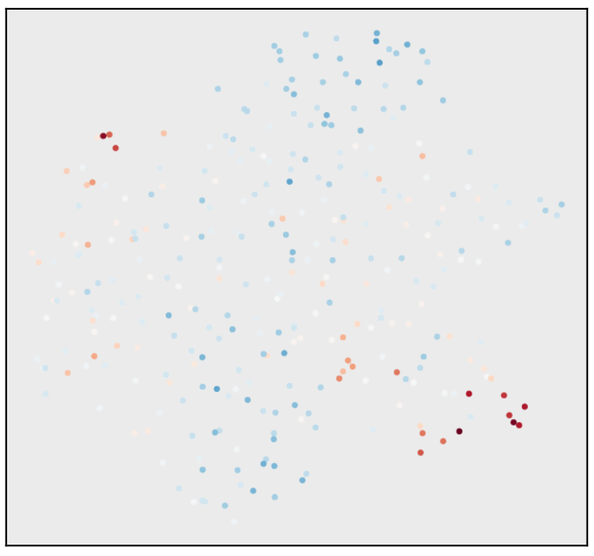
atlas, unaligned: stages



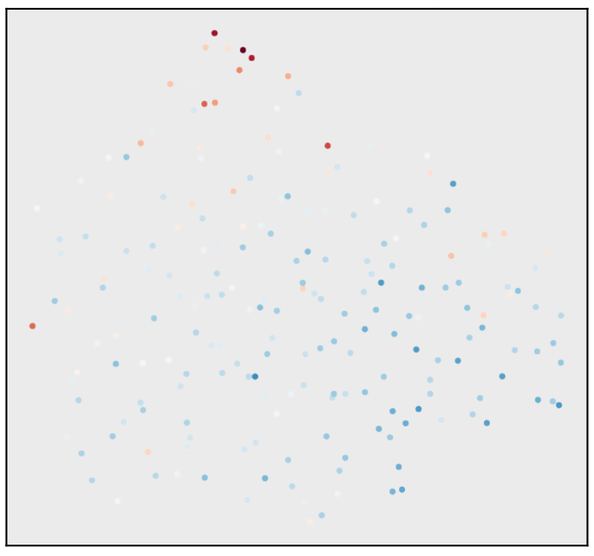
Genes (number of stages): gata3 (12), foxi1 (12), tfap2a (12), tp63 (11), fermt1 (11), tfap2c (10), bambib (8), foxi3a (7), grhl1 (7), foxi3b (1)

axial_mesoderm

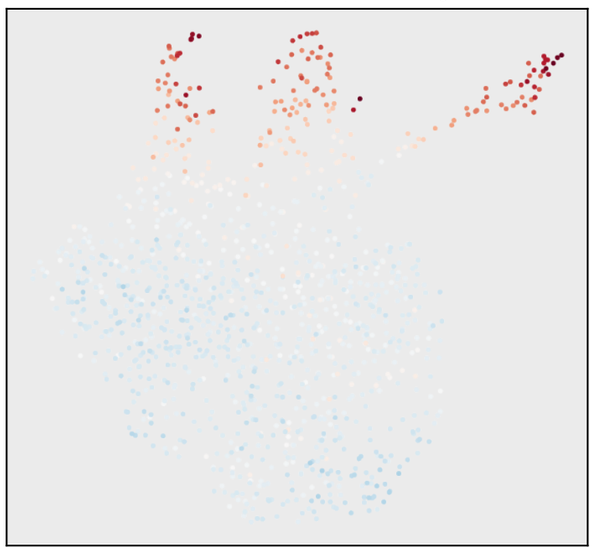
high: 19 genes, not supported
active in 3% of cells, pseudobulk strength 0.40



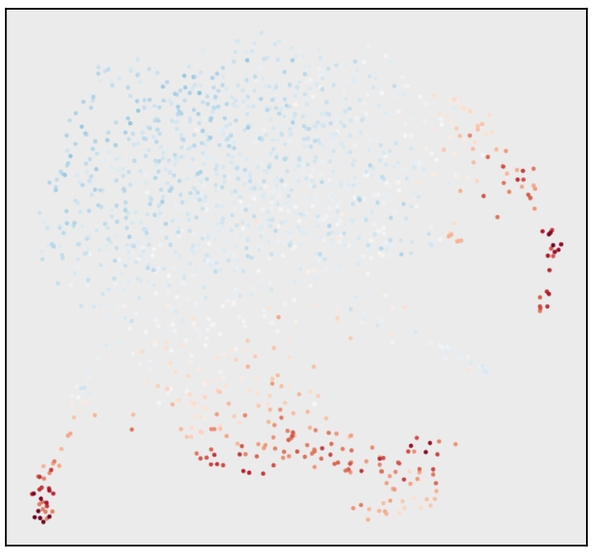
oblong: 18 genes, not supported
active in 2% of cells, pseudobulk strength 0.43



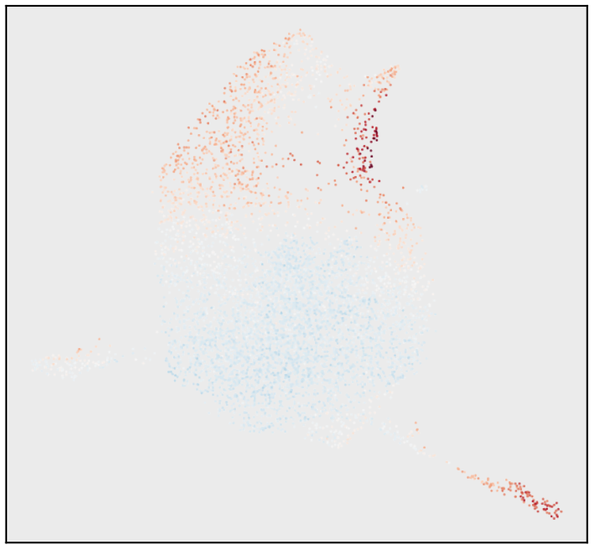
dome: 23 genes, not supported
active in 11% of cells, pseudobulk strength 0.55



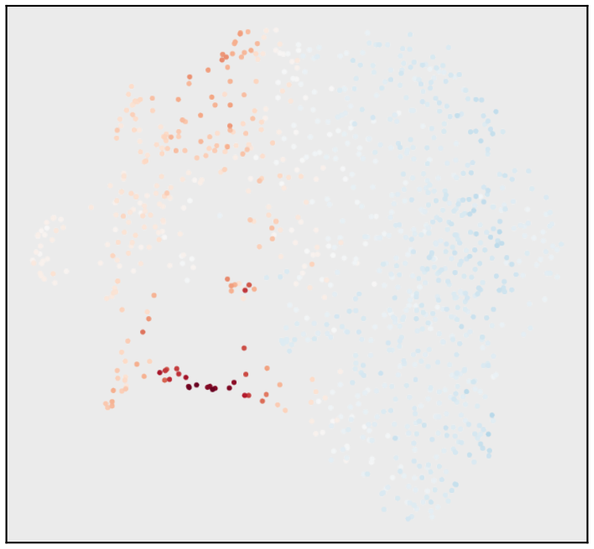
30% epiboly: 25 genes, not supported
active in 13% of cells, pseudobulk strength 0.48



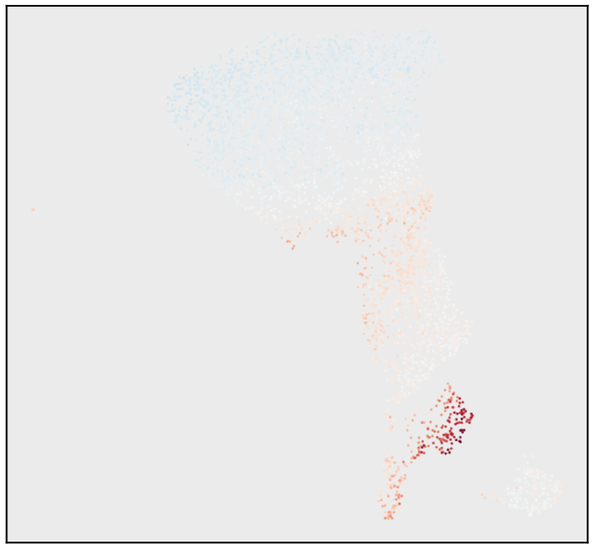
50% epiboly: 28 genes, not supported
active in 14% of cells, pseudobulk strength 0.49



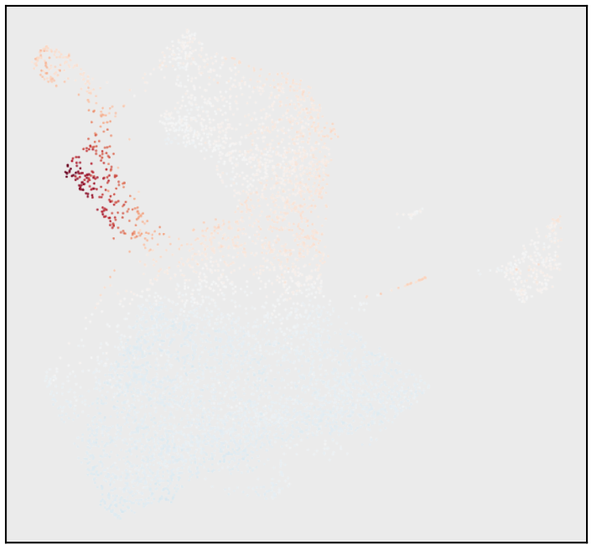
shield: 26 genes, not supported
active in 10% of cells, pseudobulk strength 0.45



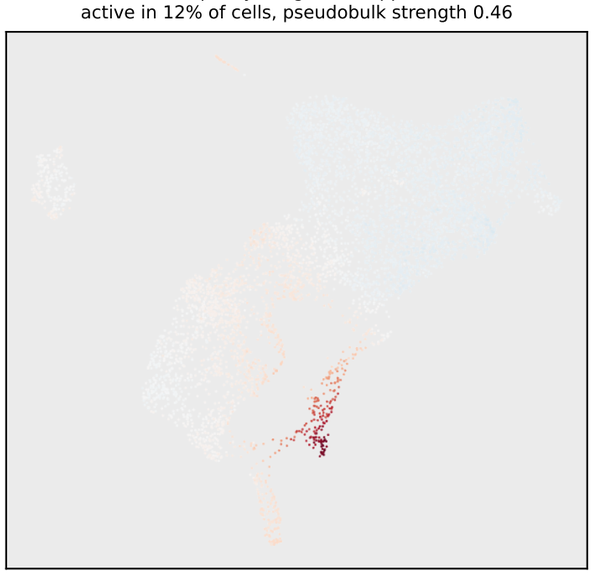
60% epiboly: 28 genes, supported
active in 16% of cells, pseudobulk strength 0.48



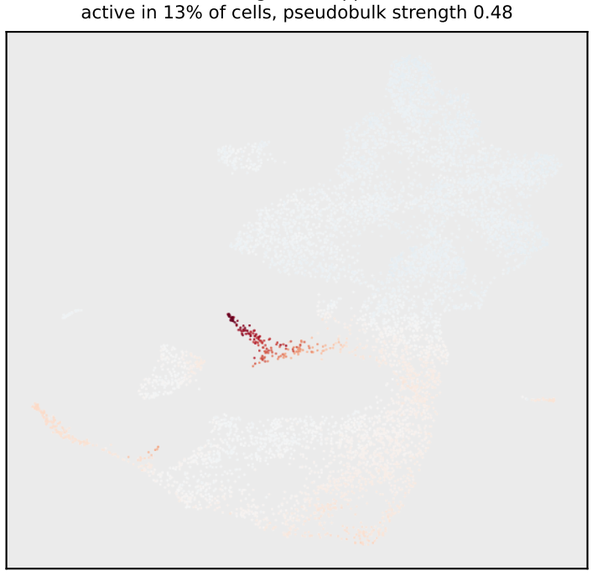
75% epiboly: 29 genes, supported
active in 15% of cells, pseudobulk strength 0.45



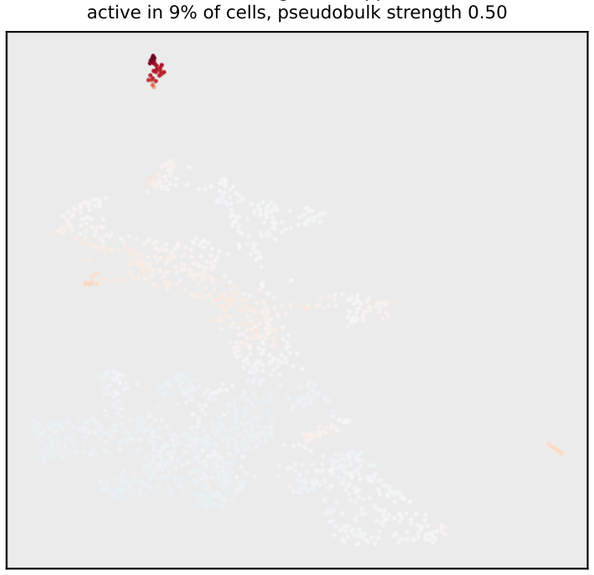
90% epiboly: 29 genes, supported
active in 12% of cells, pseudobulk strength 0.46



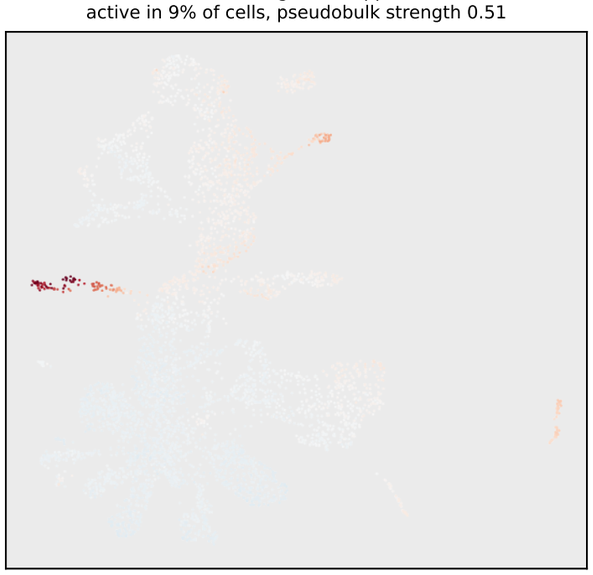
bud: 29 genes, supported
active in 13% of cells, pseudobulk strength 0.48



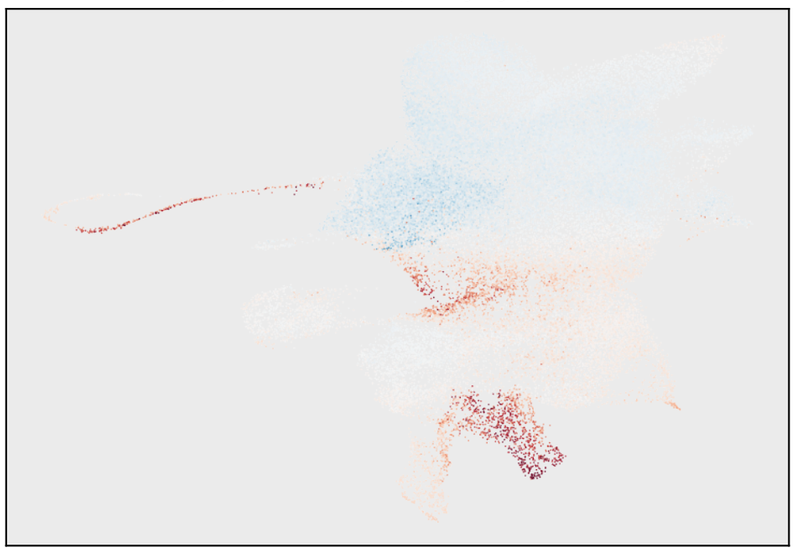
3-somite: 29 genes, supported
active in 9% of cells, pseudobulk strength 0.50



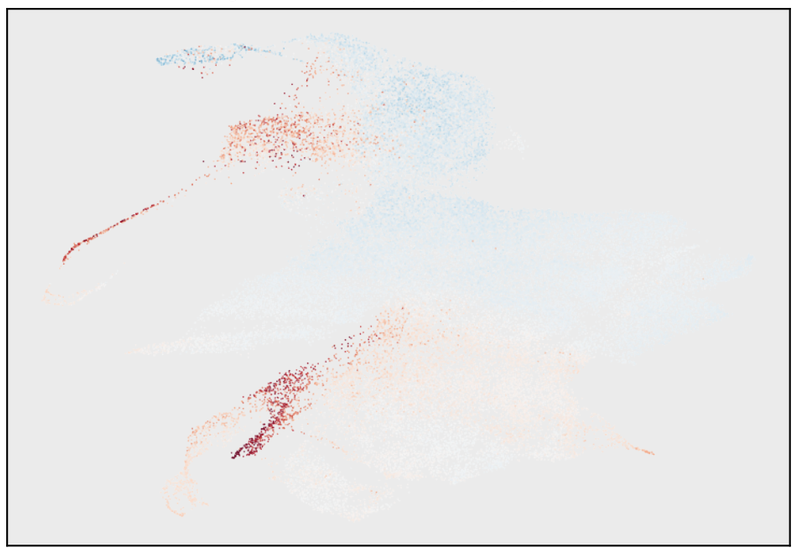
6-somite: 29 genes, supported
active in 9% of cells, pseudobulk strength 0.51



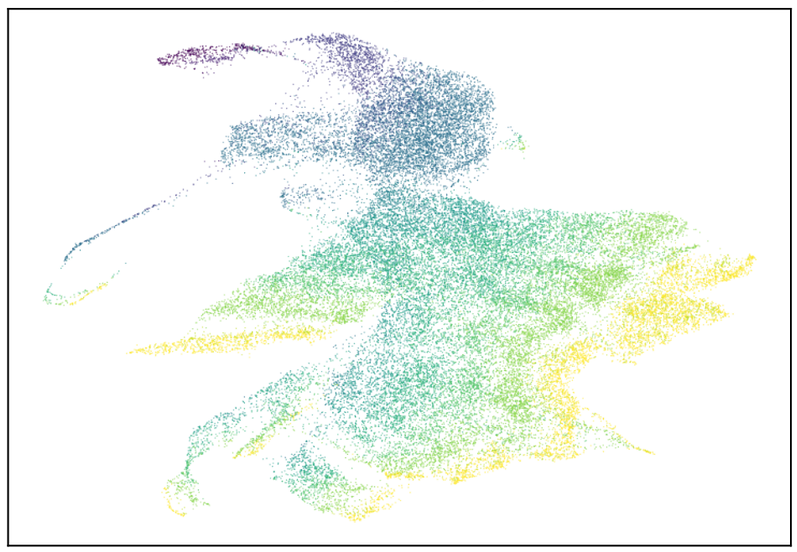
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



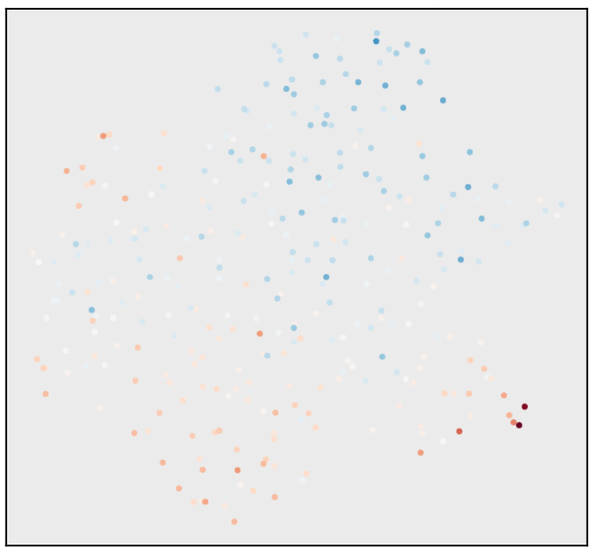
atlas, unaligned: stages



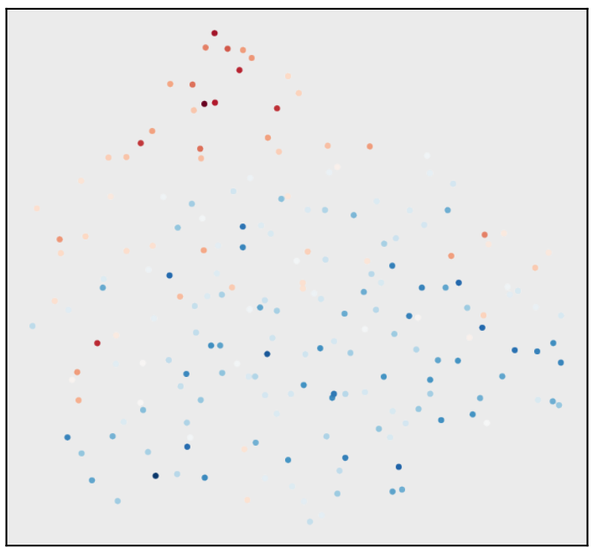
Genes (number of stages): noto (12), ta (12), tob1a (12), cnn2 (12), tram1 (12), rcn3 (12), net1 (12), pak1 (12), csrp1a (12), sec61a1l (12), cx43.4 (12), prickle1b (12), zfand5a (12), stox1 (12), trib3 (12), fstl1b (12), cnn3a (12), eif4ebp3l (12), ip6k2b (11), perp (10), ckap4 (10), kirrel3l (10), serpinh1b (9), egln3 (9), ntd5 (8), tph1b (8), twist2 (7), fbln2 (7), hfe2 (5), apobec2a (2)

hatching_gland

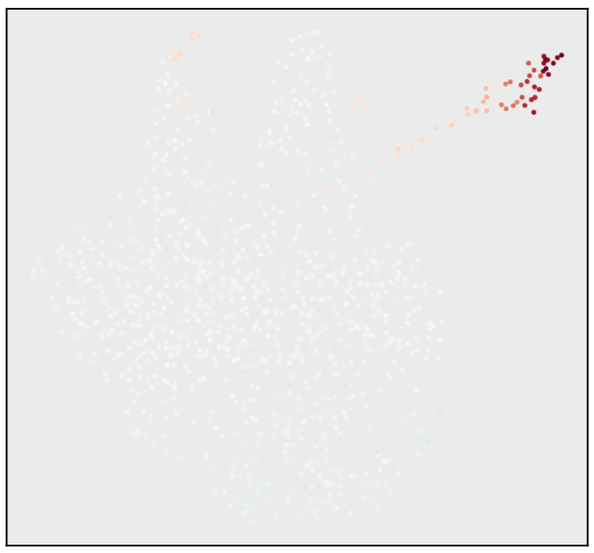
high: 22 genes, not supported
active in 0% of cells, pseudobulk strength 0.47



oblong: 22 genes, not supported
active in 0% of cells, pseudobulk strength 0.53

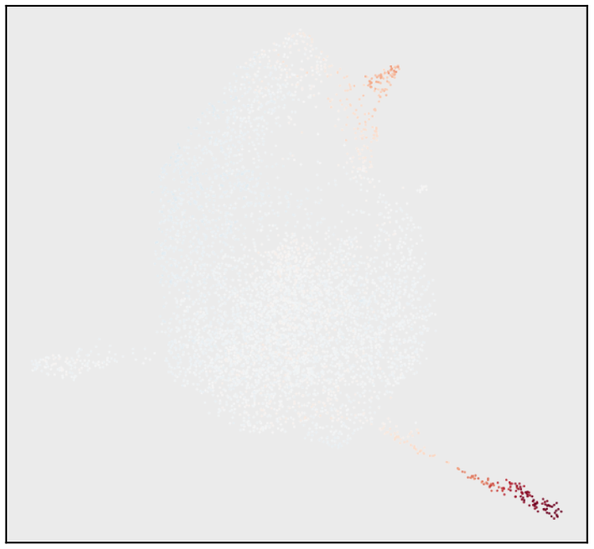


dome: 24 genes, not supported
active in 4% of cells, pseudobulk strength 0.50

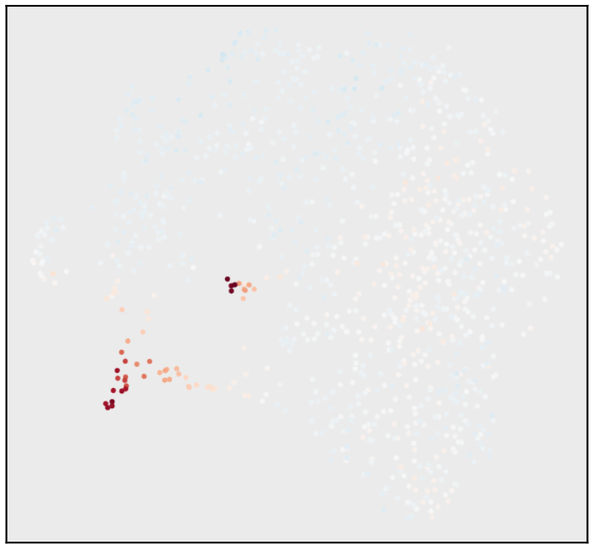


30% epiboly: 24 genes, not supported
active in 4% of cells, pseudobulk strength 0.48

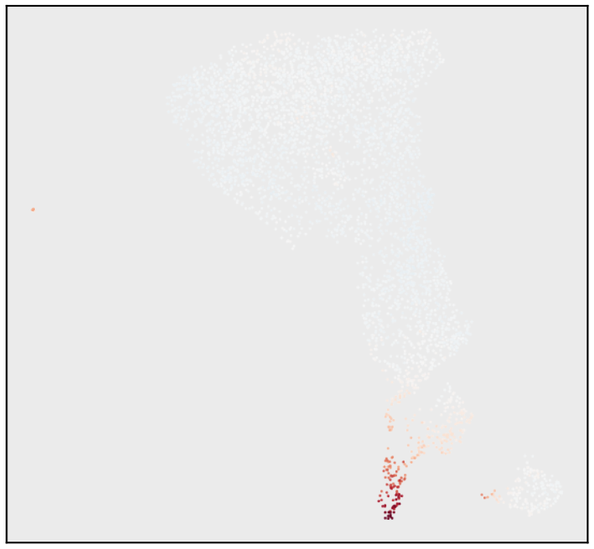
50% epiboly: 26 genes, not supported
active in 5% of cells, pseudobulk strength 0.42



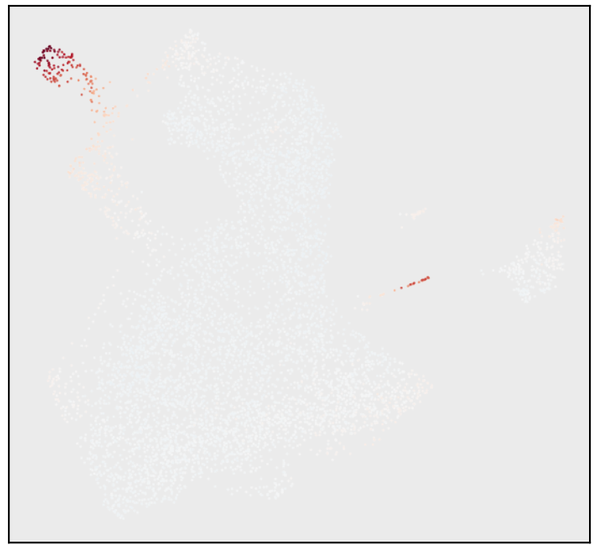
shield: 24 genes, not supported
active in 4% of cells, pseudobulk strength 0.32



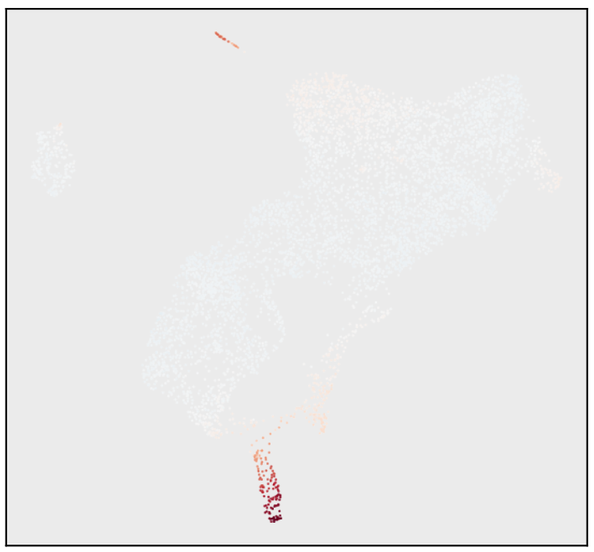
60% epiboly: 28 genes, supported
active in 6% of cells, pseudobulk strength 0.27



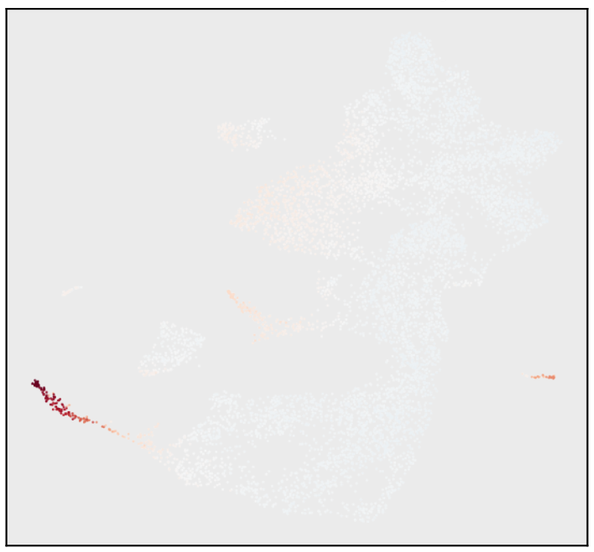
75% epiboly: 30 genes, supported
active in 6% of cells, pseudobulk strength 0.26



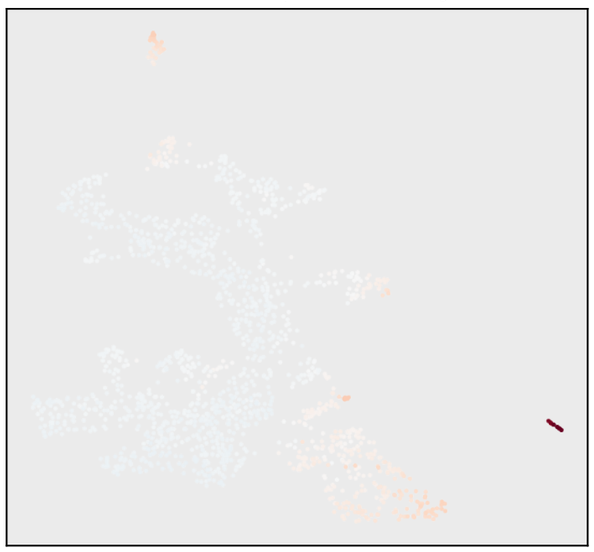
90% epiboly: 30 genes, supported
active in 6% of cells, pseudobulk strength 0.34



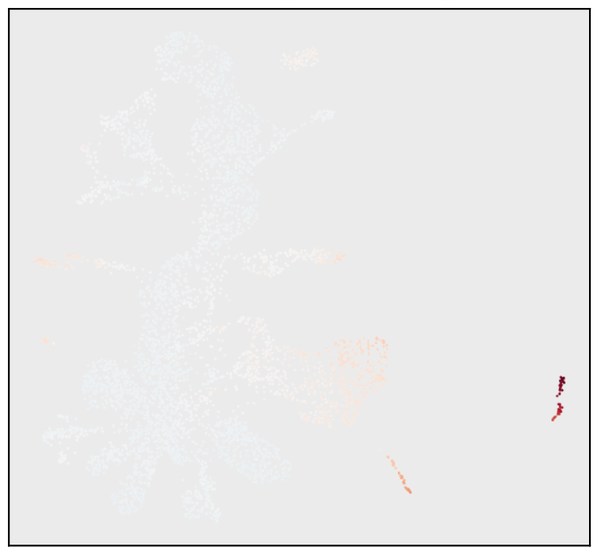
bud: 30 genes, supported
active in 10% of cells, pseudobulk strength 0.37



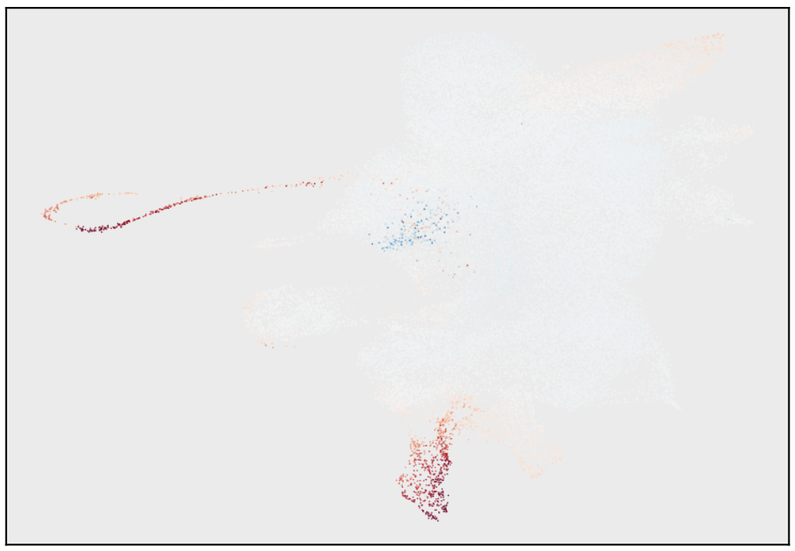
3-somite: 29 genes, supported
active in 13% of cells, pseudobulk strength 0.38



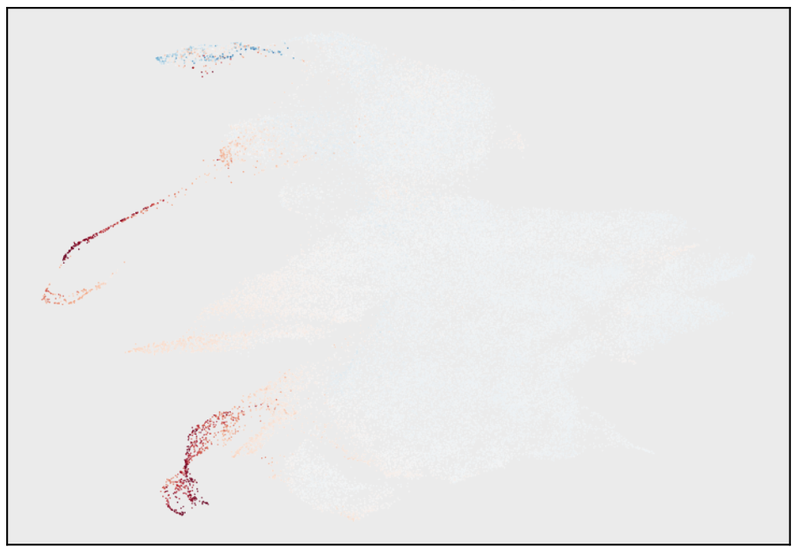
6-somite: 30 genes, supported
active in 13% of cells, pseudobulk strength 0.49



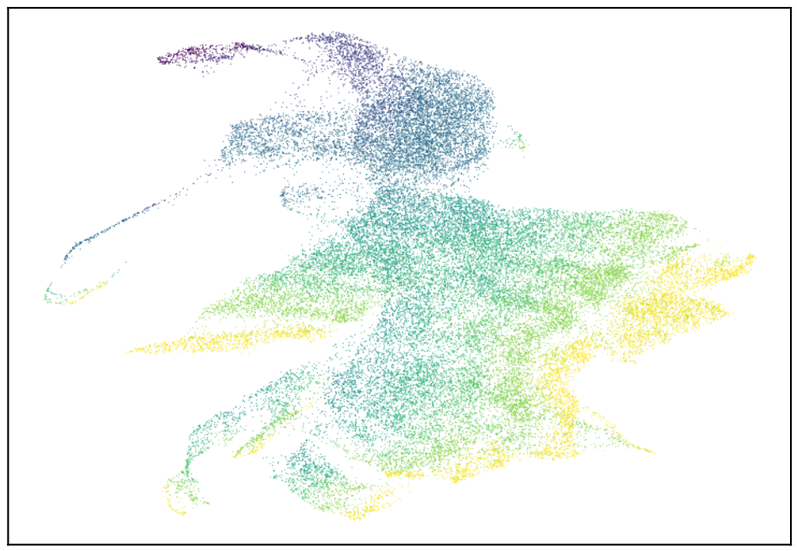
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



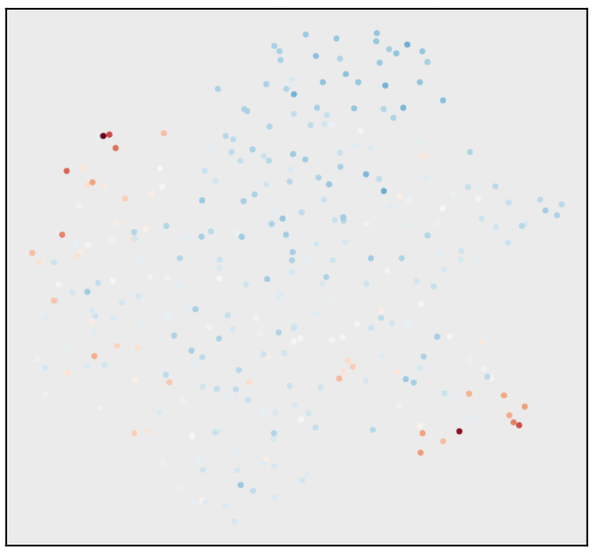
atlas, unaligned: stages



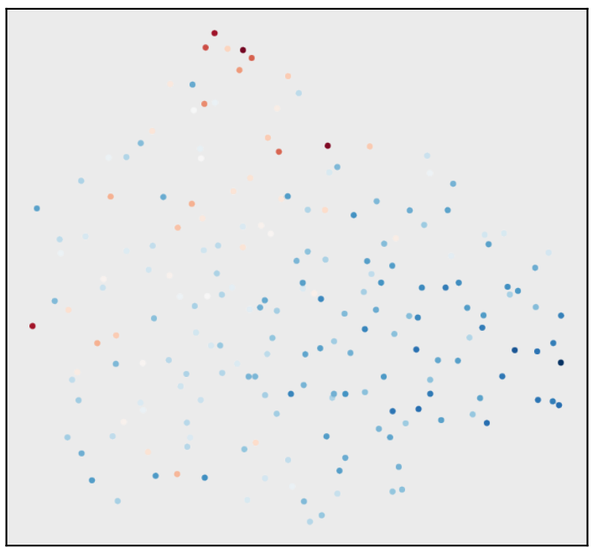
Genes (number of stages): gsc (12), klf17 (12), pfn1 (12), cldn7b (12), ell2 (12), calr3b (12), gadd45ga (12), rrbp1b (12), stk10 (12), hspa5 (12), svopl (12), cited4b (12), fbxo2 (12), sec61g (12), cd63 (12), asb11 (12), hsp90b1 (12), mbpb (12), lgals3l (12), capns1a (12), spint2 (12), tagln2 (11), slc3a2b (11), pitx2 (10), rdh8a (7), ctslb (6), bhlha15 (6), si:ch211-163l21.8 (6), inhbaa (5), he1a (5)

margin

high: 16 genes, supported
active in 3% of cells, pseudobulk strength 0.18

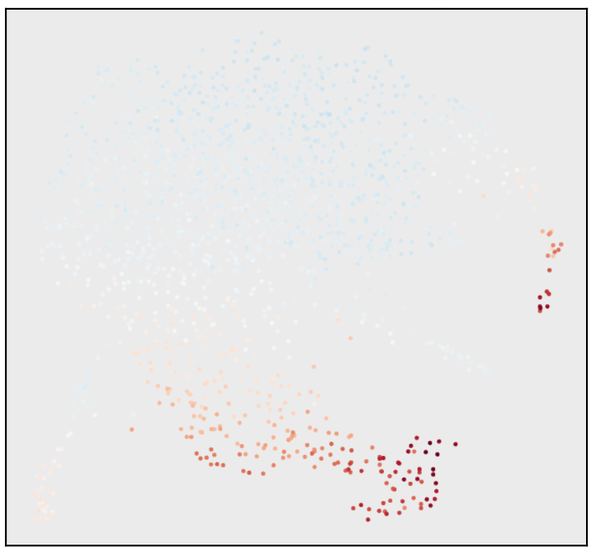


oblong: 14 genes, not supported
active in 1% of cells, pseudobulk strength 0.24

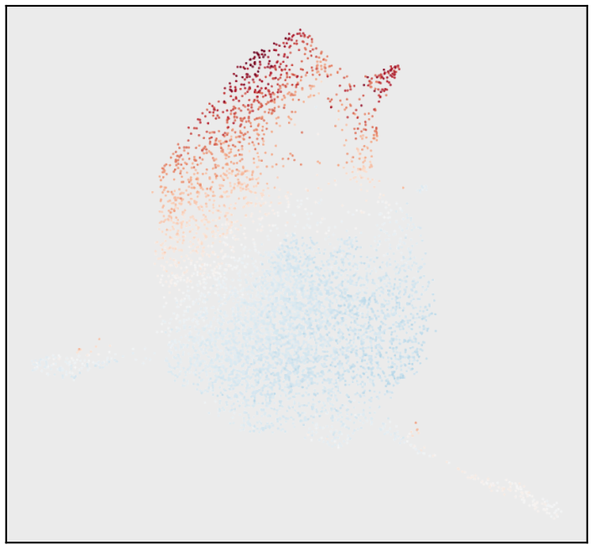


dome: 27 genes, not supported
active in 12% of cells, pseudobulk strength 0.33

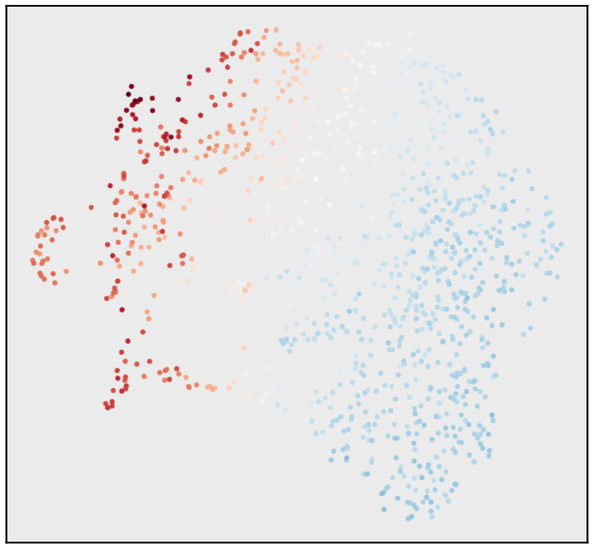
30% epiboly: 30 genes, supported
active in 15% of cells, pseudobulk strength 0.50



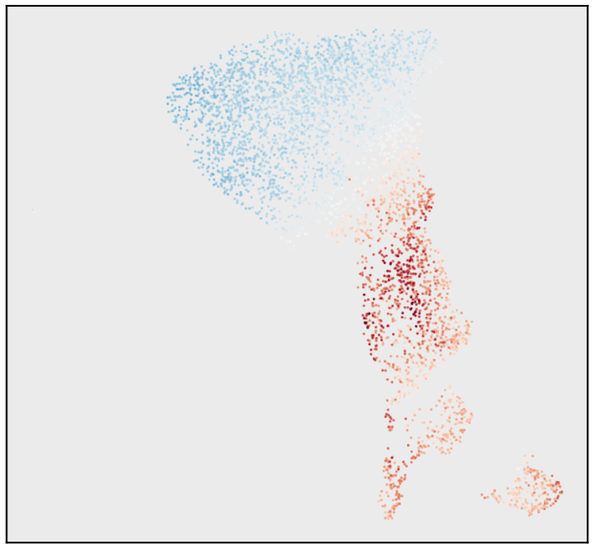
50% epiboly: 30 genes, supported
active in 20% of cells, pseudobulk strength 0.52



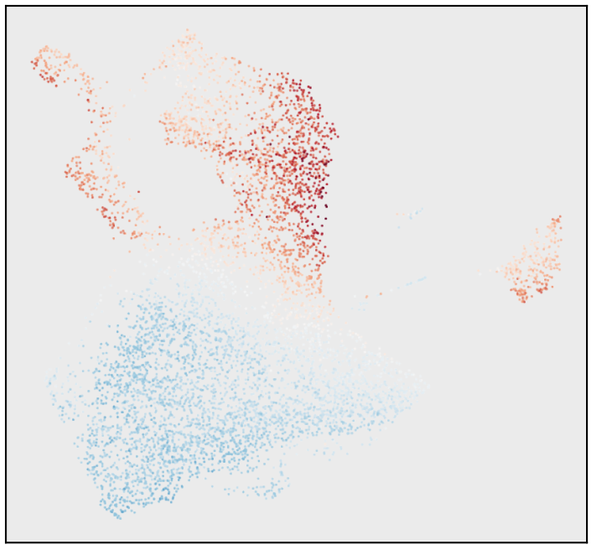
shield: 30 genes, supported
active in 27% of cells, pseudobulk strength 0.68



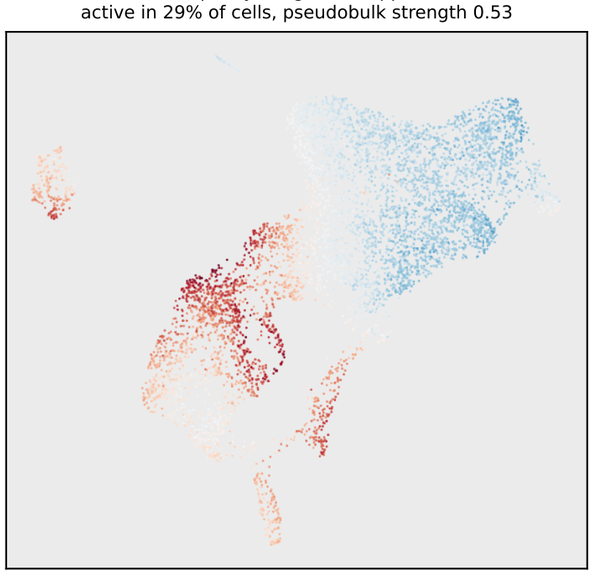
60% epiboly: 30 genes, supported
active in 34% of cells, pseudobulk strength 0.68



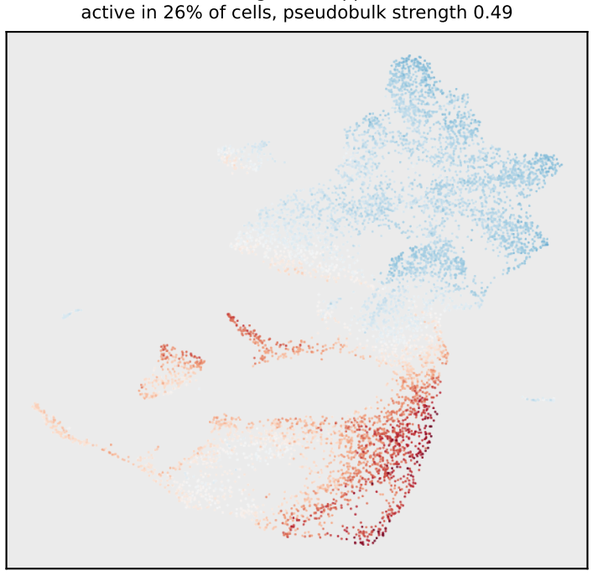
75% epiboly: 30 genes, supported
active in 33% of cells, pseudobulk strength 0.56



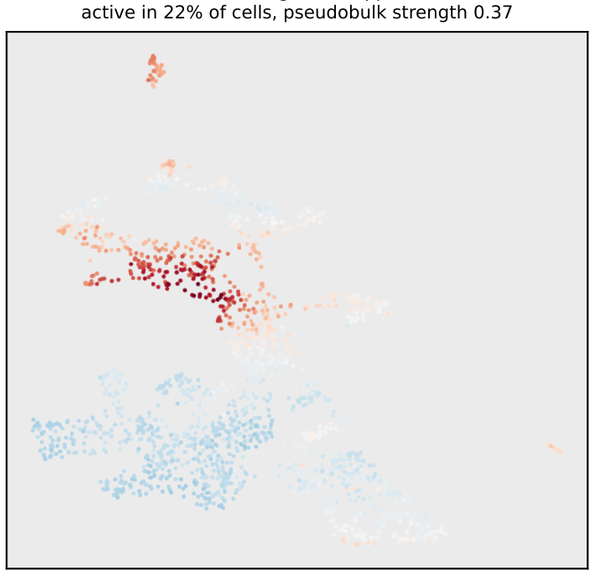
90% epiboly: 30 genes, supported
active in 29% of cells, pseudobulk strength 0.53



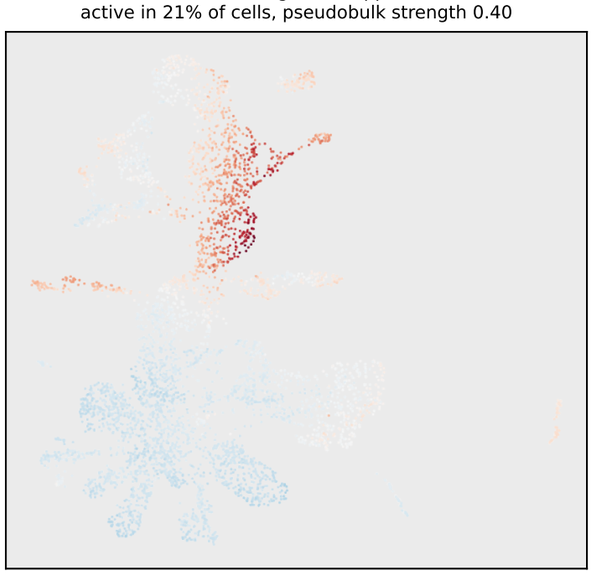
bud: 30 genes, supported
active in 26% of cells, pseudobulk strength 0.49



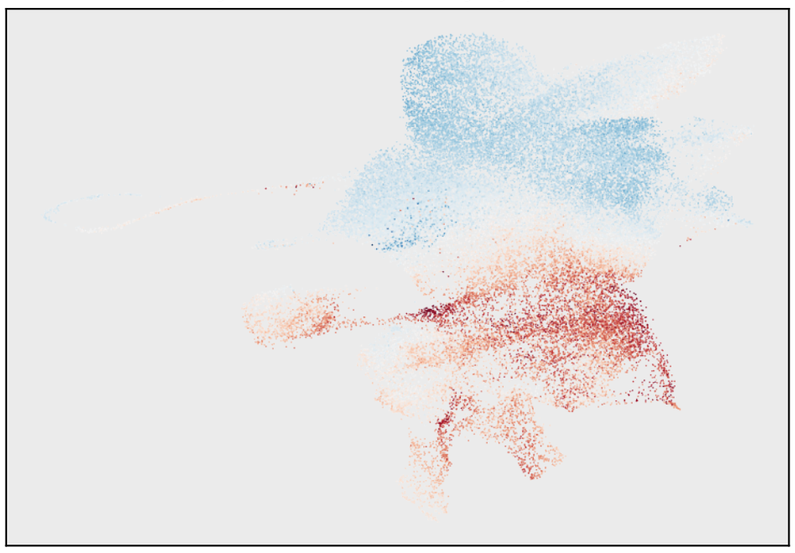
3-somite: 29 genes, supported
active in 22% of cells, pseudobulk strength 0.37



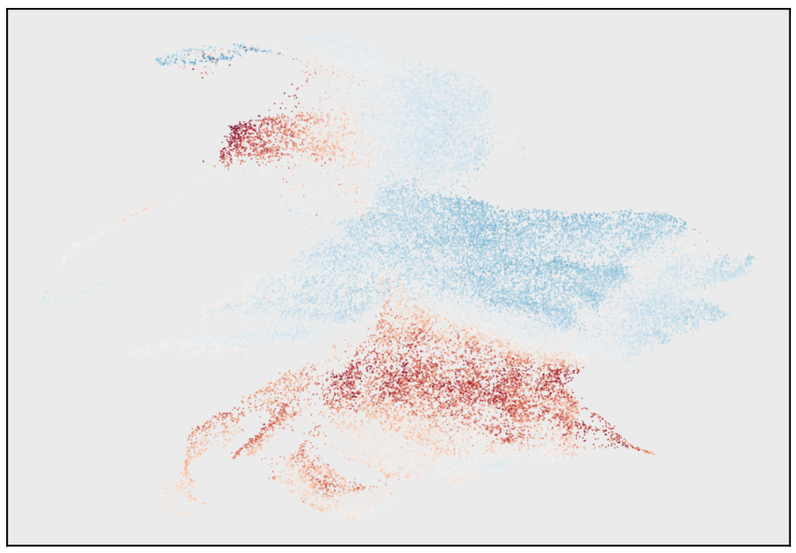
6-somite: 29 genes, supported
active in 21% of cells, pseudobulk strength 0.40



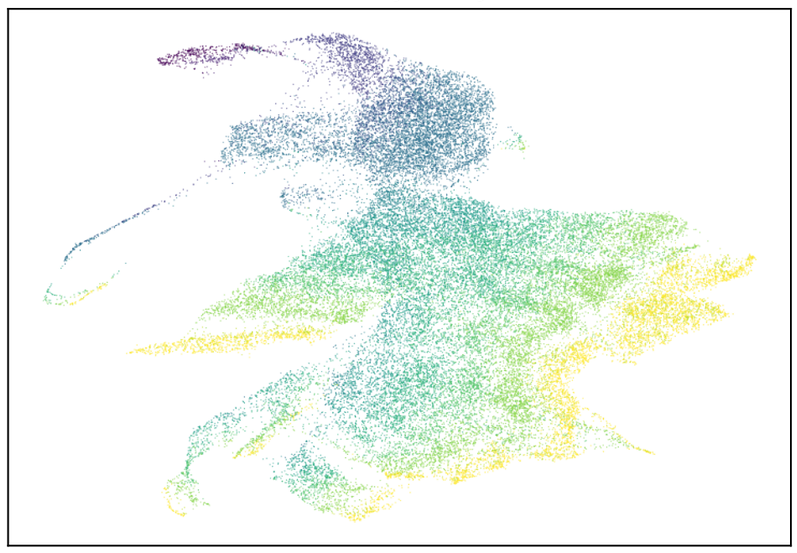
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



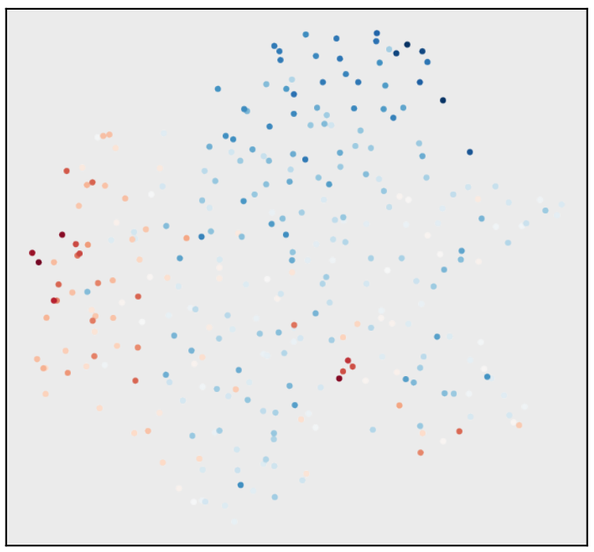
atlas, unaligned: stages



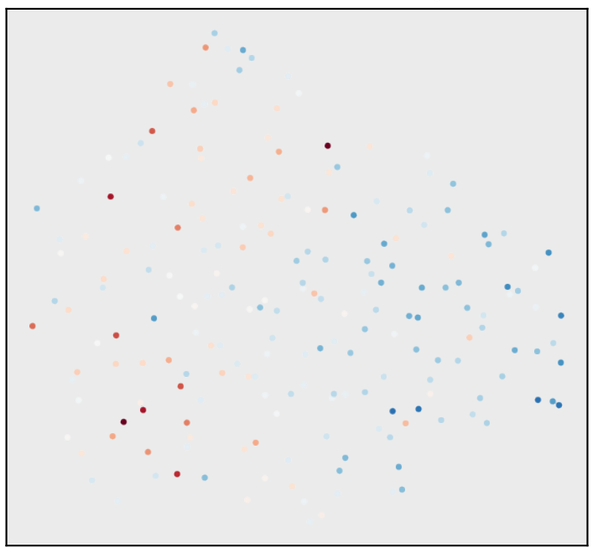
Genes (number of stages): gsc (12), ved (12), ta (12), aplnra (12), tbx16 (12), foxa (12), xbp1 (12), vox (12), rdh10a (12), gadd45bb (12), hes6 (12), arl4ab (12), tob1a (12), snai1a (12), osr1 (11), gata6 (11), mespab (10), eve1 (10), cdx4 (10), gata5 (10), pitx2 (10), ripply1 (10), foxa2 (10), her1 (10), pcdh8 (10), her7 (10), msgn1 (9), mespaa (9), serpinh1b (9), sox32 (8)

mesoderm

high: 11 genes, not supported
active in 2% of cells, pseudobulk strength 0.13

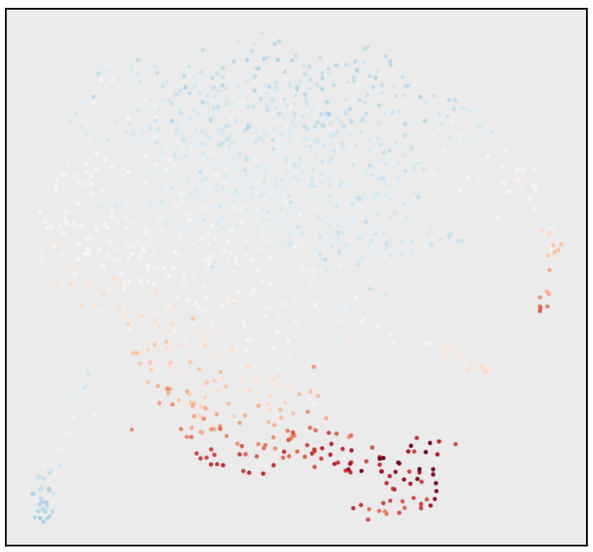


oblong: 11 genes, not supported
active in 2% of cells, pseudobulk strength 0.17

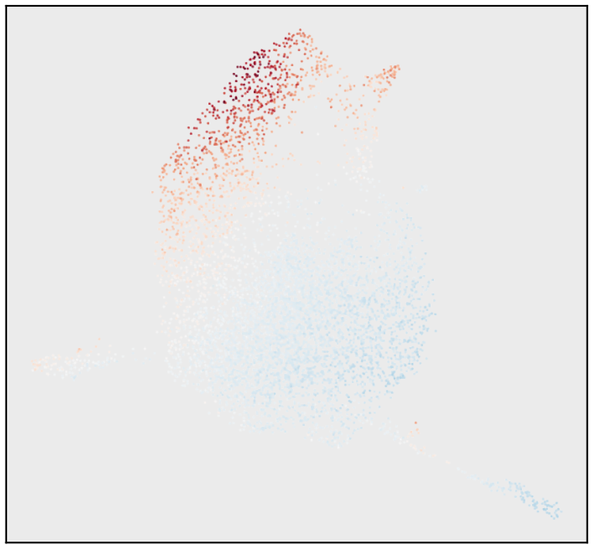


dome: 16 genes, not supported
active in 6% of cells, pseudobulk strength 0.23

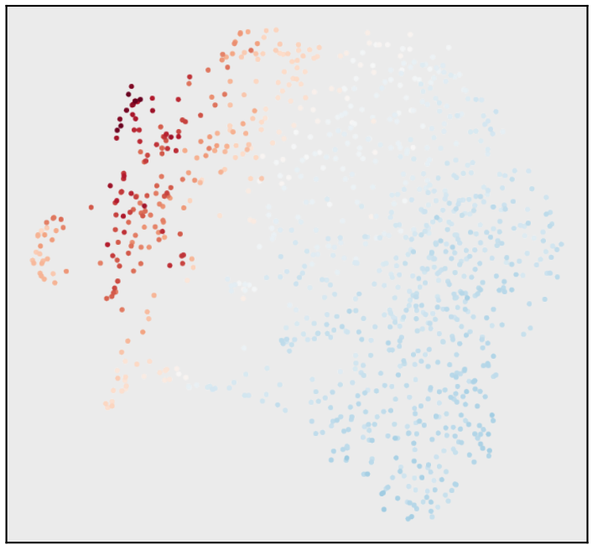
30% epiboly: 21 genes, not supported
active in 13% of cells, pseudobulk strength 0.33



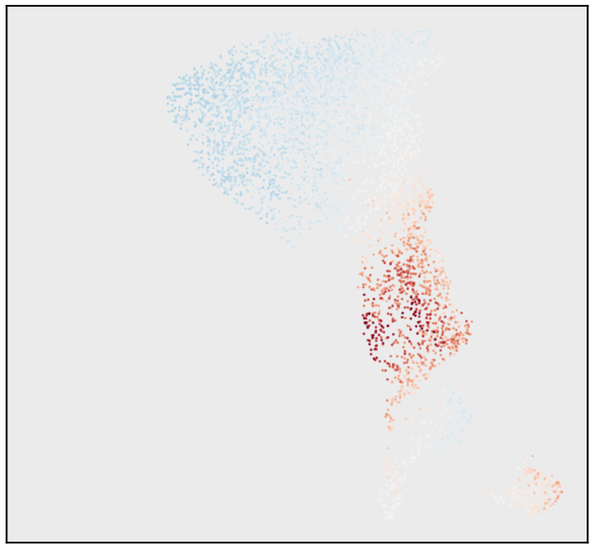
50% epiboly: 25 genes, not supported
active in 17% of cells, pseudobulk strength 0.31



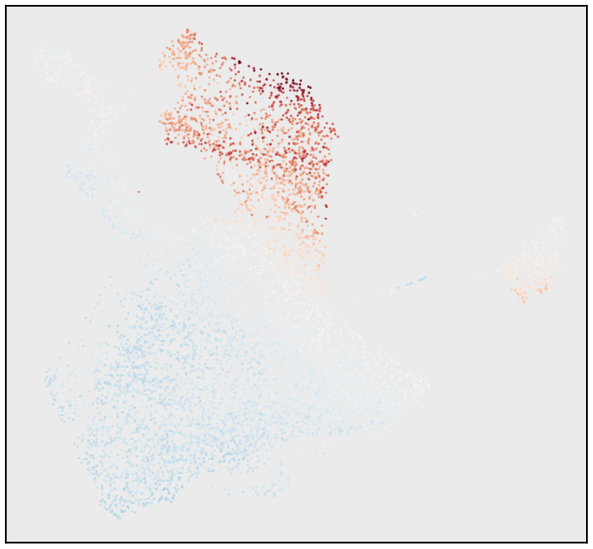
shield: 23 genes, supported
active in 24% of cells, pseudobulk strength 0.46



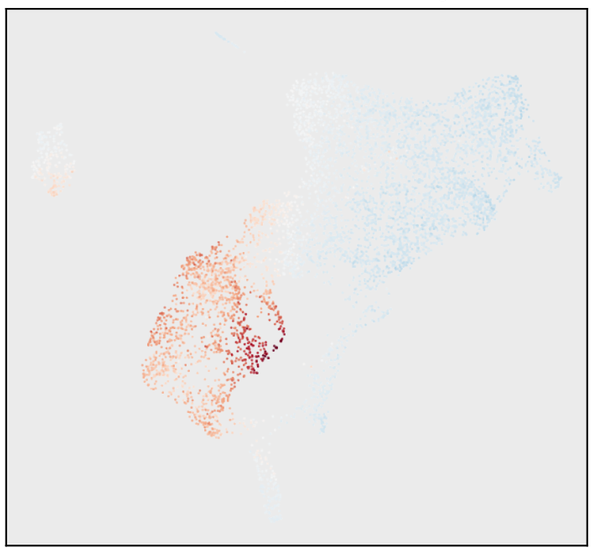
60% epiboly: 29 genes, supported
active in 25% of cells, pseudobulk strength 0.52



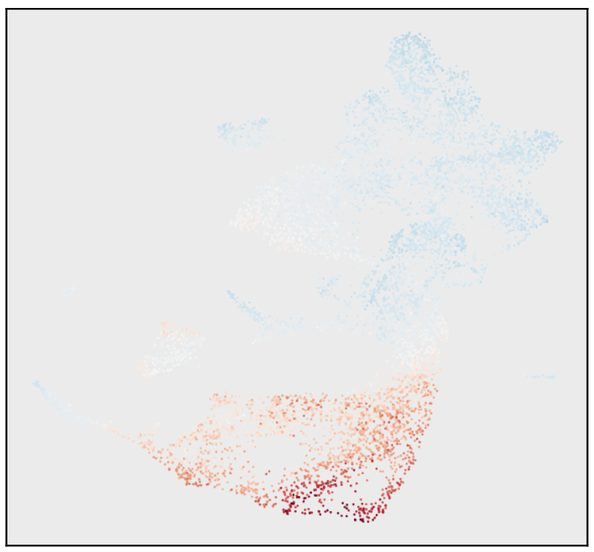
75% epiboly: 30 genes, supported
active in 24% of cells, pseudobulk strength 0.48



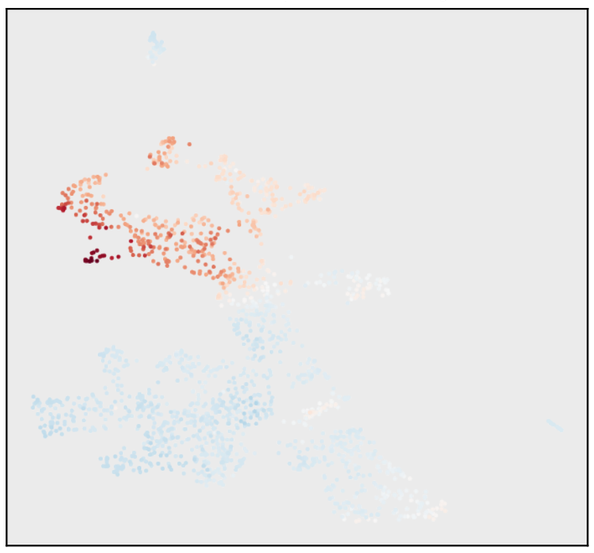
90% epiboly: 30 genes, supported
active in 24% of cells, pseudobulk strength 0.52



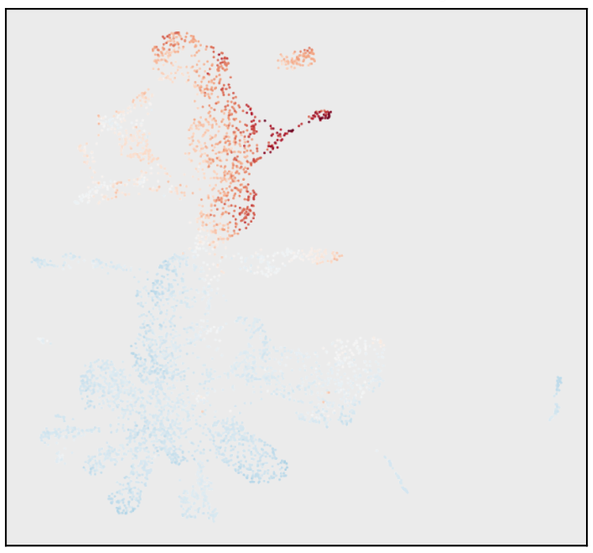
bud: 30 genes, supported
active in 25% of cells, pseudobulk strength 0.55



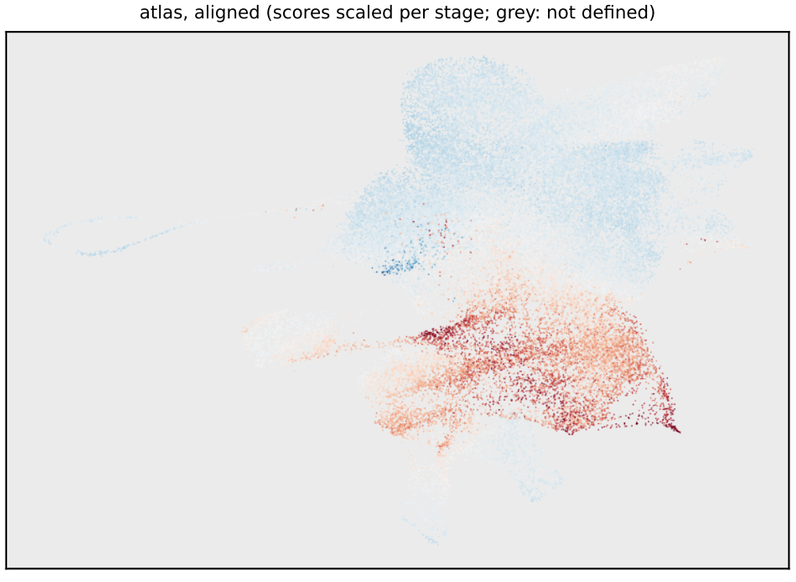
3-somite: 30 genes, supported
active in 21% of cells, pseudobulk strength 0.45



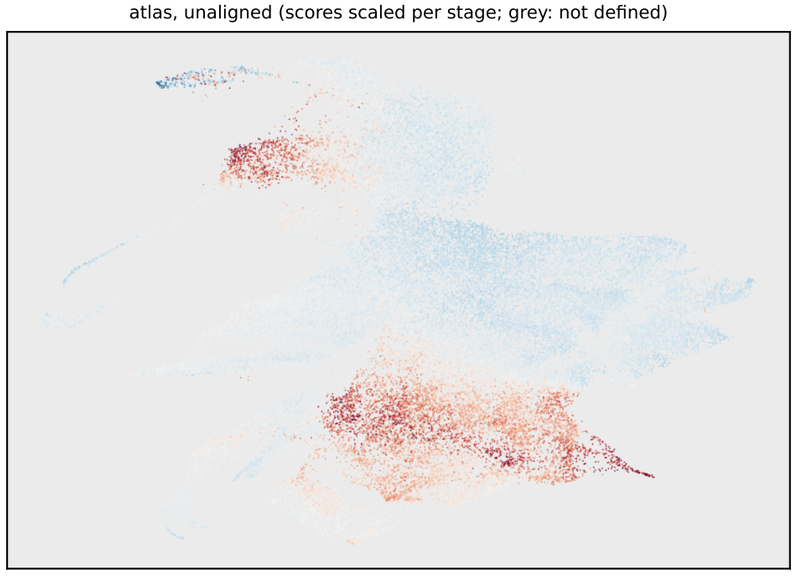
6-somite: 30 genes, supported
active in 22% of cells, pseudobulk strength 0.51



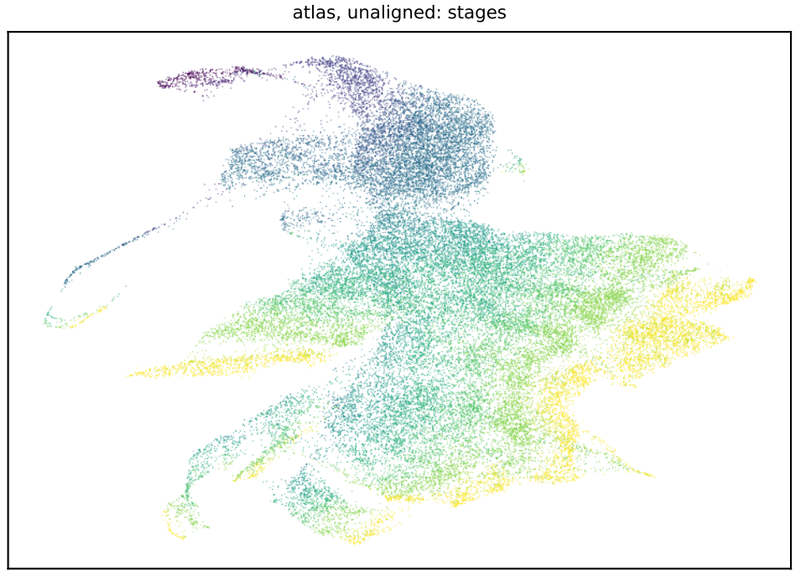
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



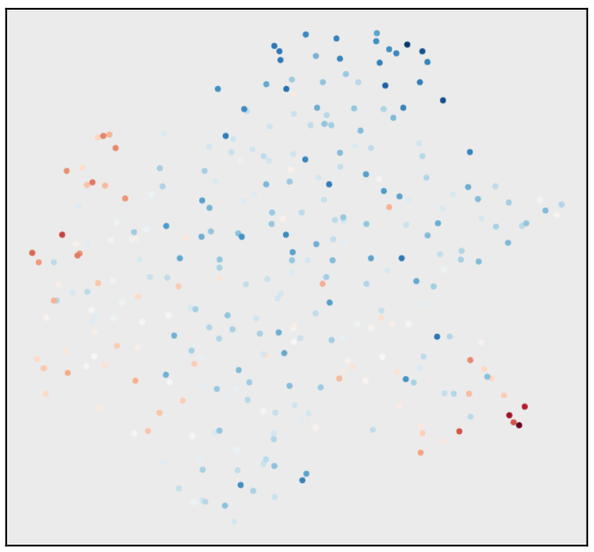
atlas, unaligned: stages



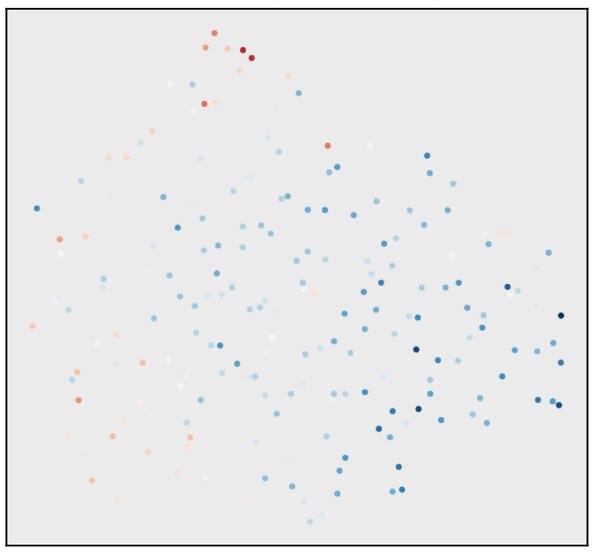
Genes (number of stages): ved (12), tbx16 (12), rdh10a (12), snai1a (12), aplnrb (12), fscn1a (12), si:dkey-261j4.5 (12), cxcl12a (12), blf (12), gata2a (12), btg2 (12), mespab (10), eve1 (10), her1 (10), pcdh8 (10), her7 (10), mespaa (9), cxcl12b (9), efemp2b (9), foxc1a (9), tbx6l (8), tbx1 (8), ttnb (8), myod1 (7), tbx6 (7), fn1b (7), twist1a (6), lbx2 (6), angptl7 (6), hbae3 (5)

neural_plate

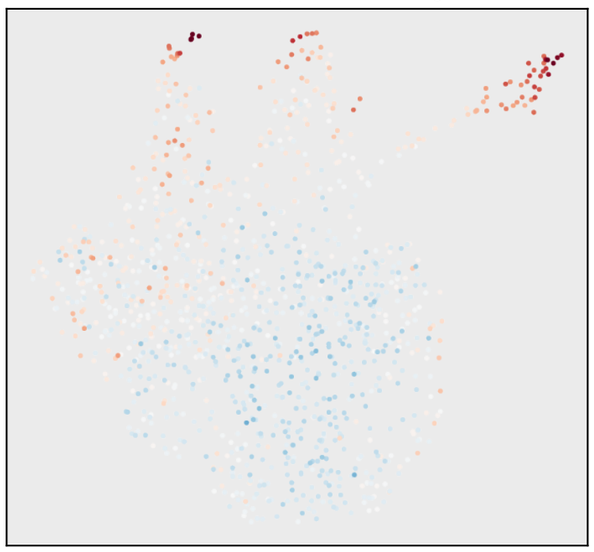
high: 18 genes, not supported
active in 1% of cells, pseudobulk strength 0.20



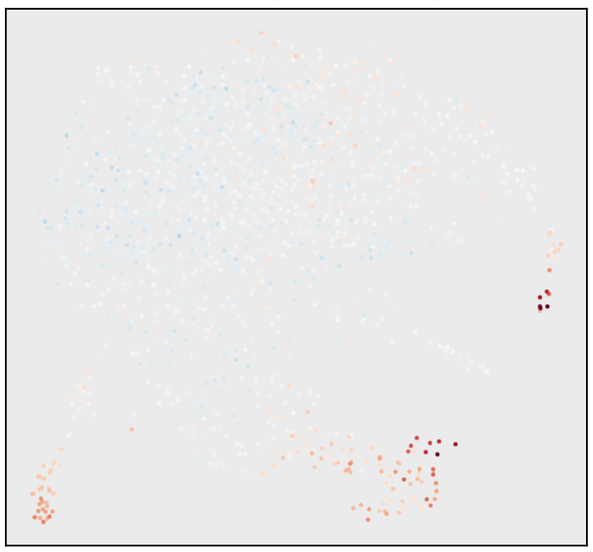
oblong: 17 genes, not supported
active in 0% of cells, pseudobulk strength 0.22



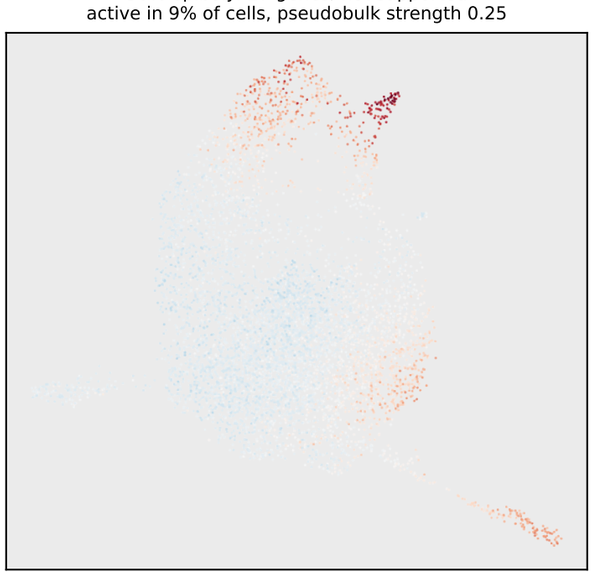
dome: 21 genes, not supported
active in 3% of cells, pseudobulk strength 0.20



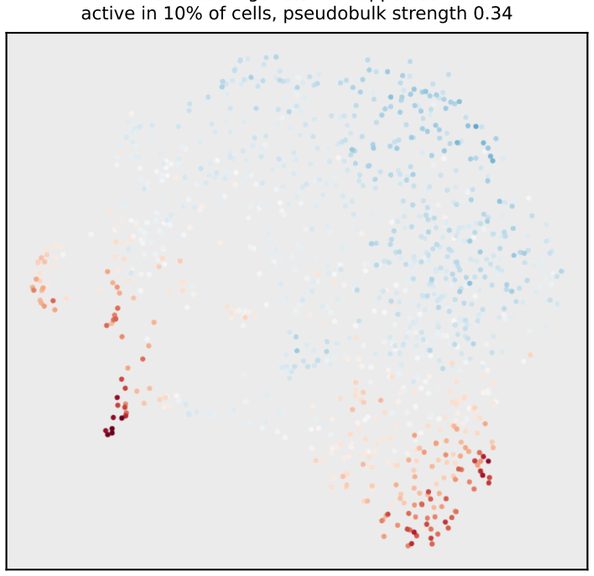
30% epiboly: 18 genes, not supported
active in 5% of cells, pseudobulk strength 0.22



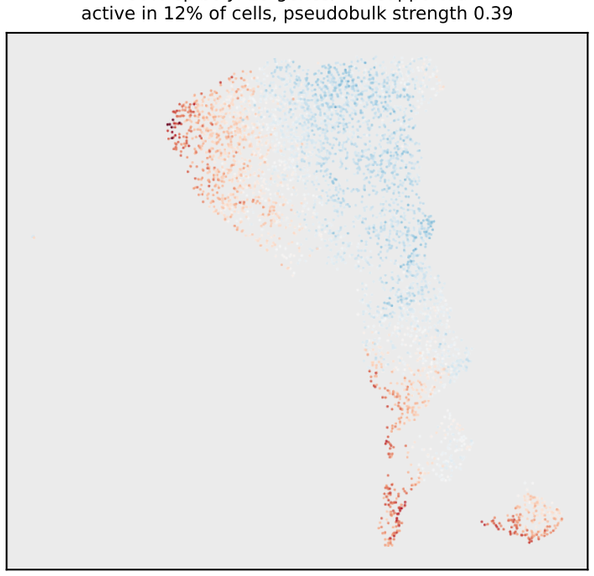
50% epiboly: 27 genes, not supported
active in 9% of cells, pseudobulk strength 0.25



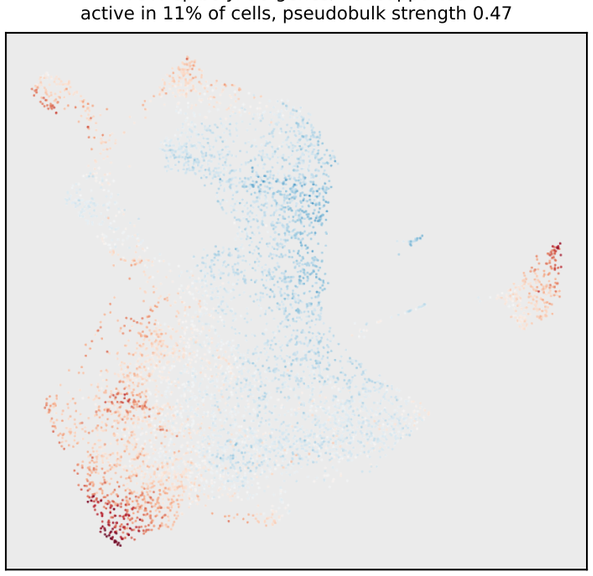
shield: 24 genes, not supported
active in 10% of cells, pseudobulk strength 0.34



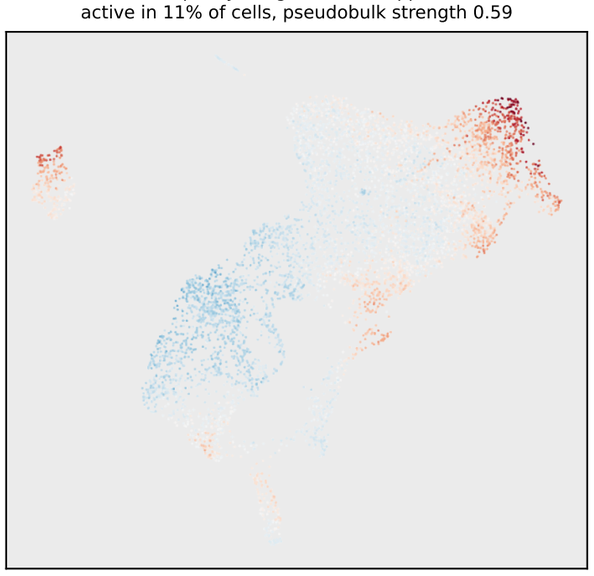
60% epiboly: 29 genes, not supported
active in 12% of cells, pseudobulk strength 0.39



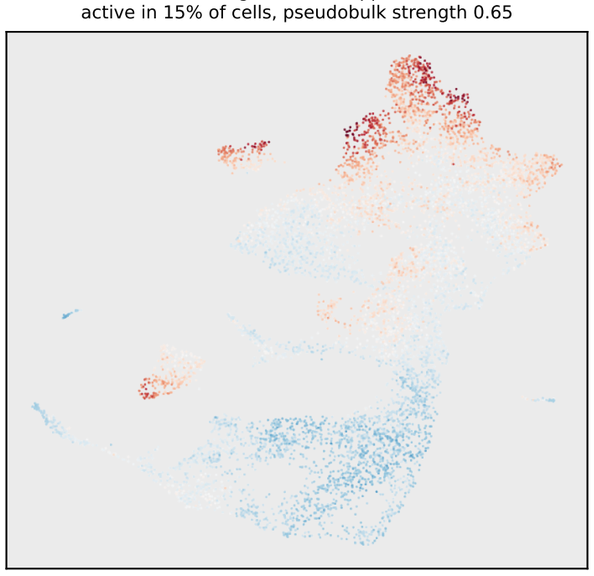
75% epiboly: 30 genes, not supported
active in 11% of cells, pseudobulk strength 0.47



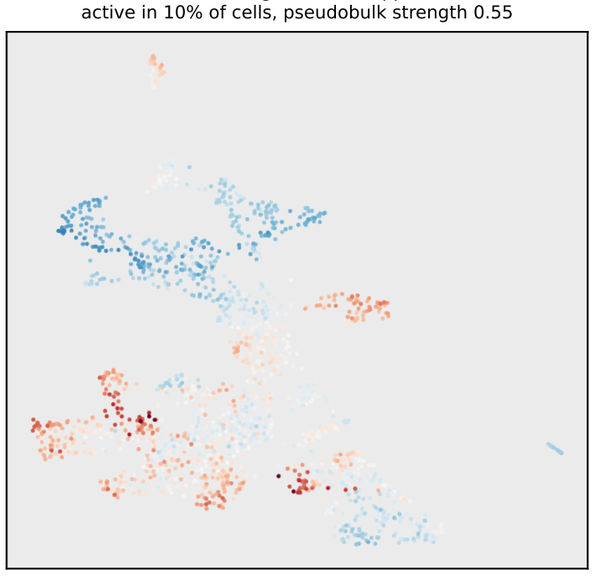
90% epiboly: 30 genes, not supported
active in 11% of cells, pseudobulk strength 0.59



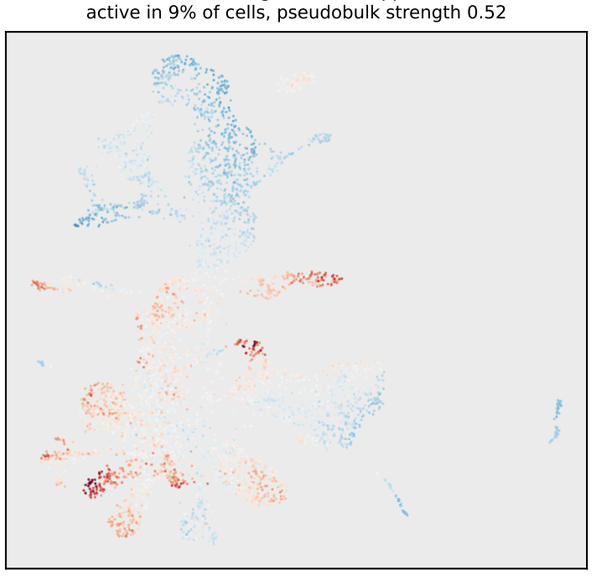
bud: 30 genes, not supported
active in 15% of cells, pseudobulk strength 0.65



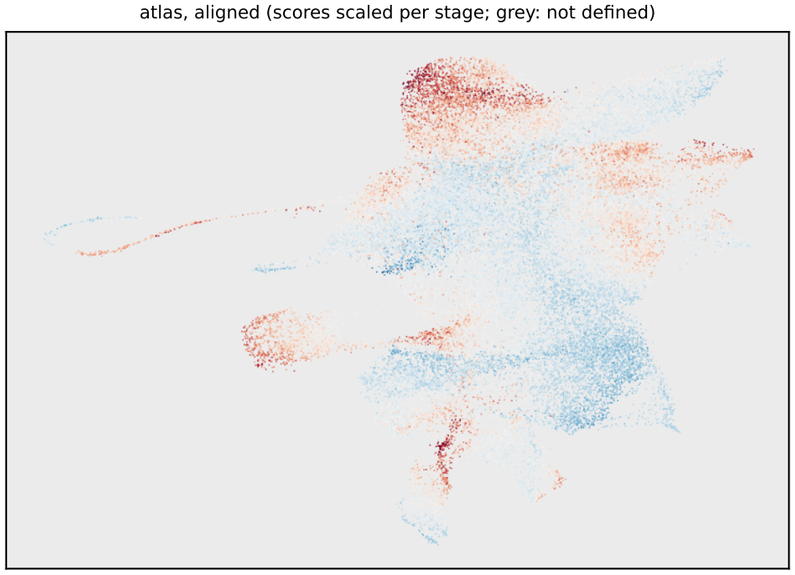
3-somite: 29 genes, not supported
active in 10% of cells, pseudobulk strength 0.55



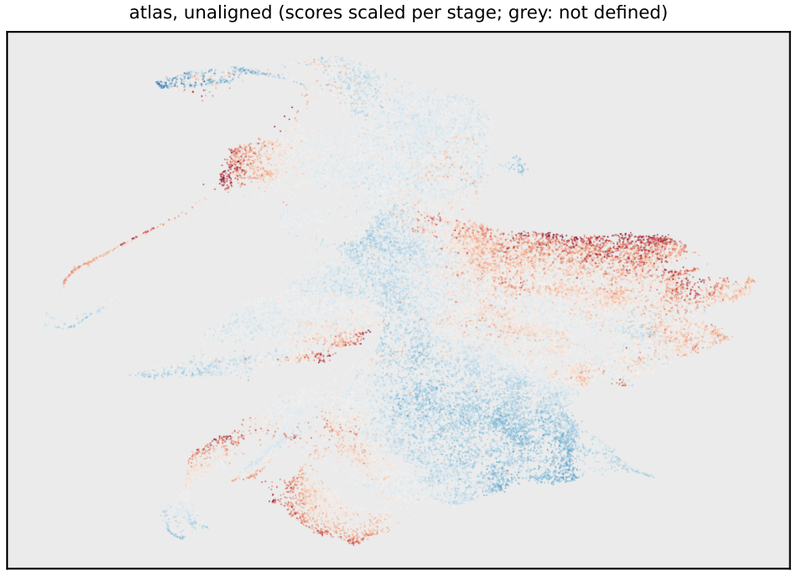
6-somite: 30 genes, not supported
active in 9% of cells, pseudobulk strength 0.52



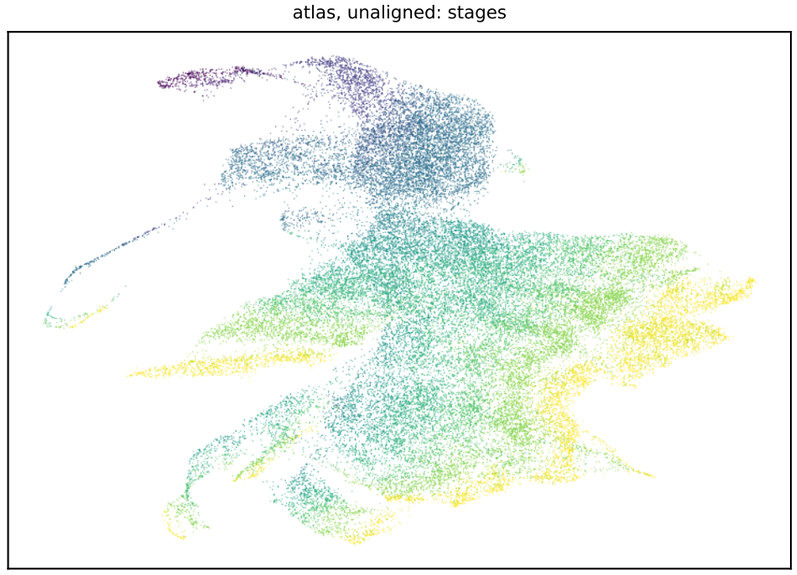
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



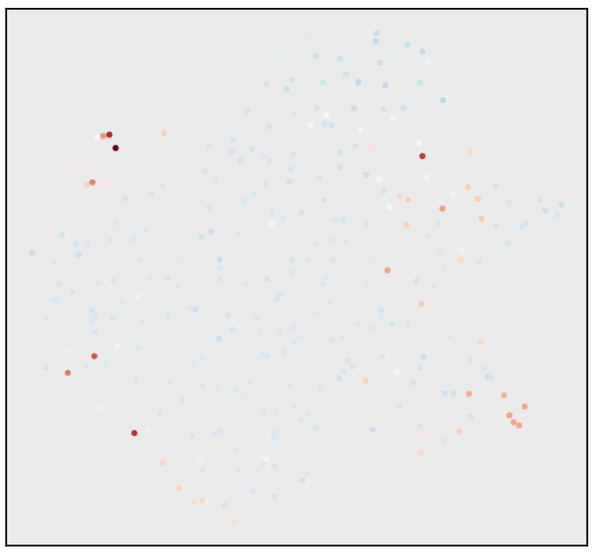
atlas, unaligned: stages



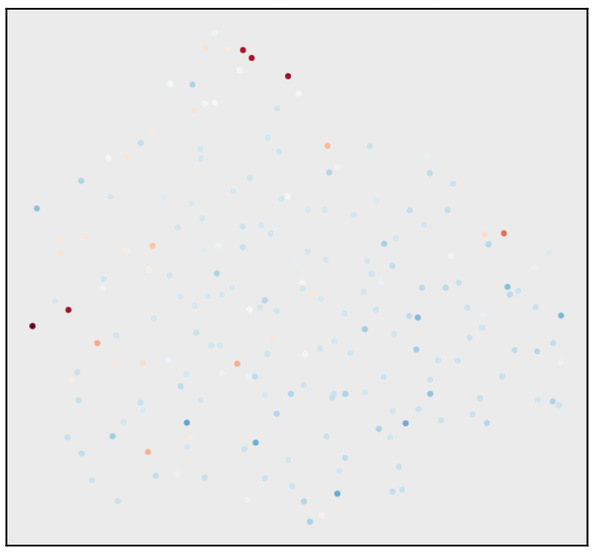
Genes (number of stages): fzd8b (12), her3 (12), her6 (12), hspb1 (12), irx7 (12), akap12b (12), zic2a (12), tdgf1 (12), rgcc (12), her5 (12), id2a (12), shisa2 (12), otx1b (12), irx3a (11), irx1b (11), slc2a12 (11), hesx1 (10), rxraa (10), nkx1.2la (10), pax6b (10), rx3 (9), her2 (8), dlx3b (8), cdh6 (8), mdkb (8), her9 (8), olig2 (7), fezf2 (6), hoxb1b (6), nkx2.9 (6)

blocking:BMP

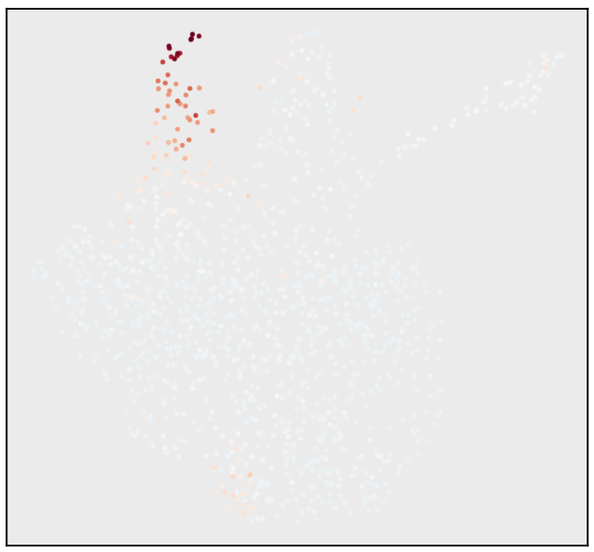
high: 2 genes, not supported
active in 0% of cells, pseudobulk strength 0.05



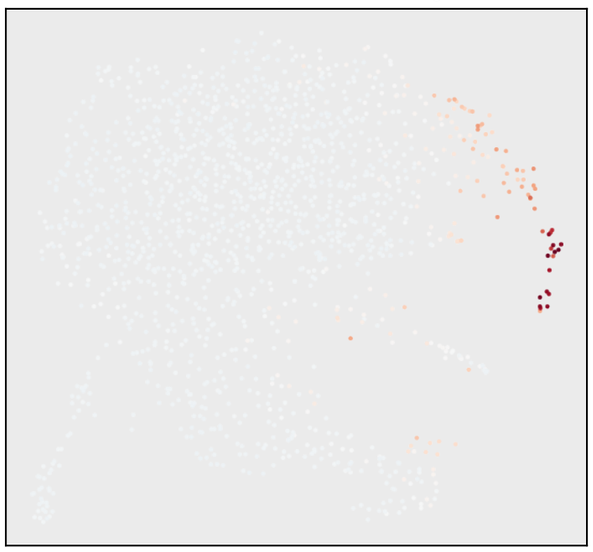
oblong: 2 genes, not supported
active in 1% of cells, pseudobulk strength 0.06



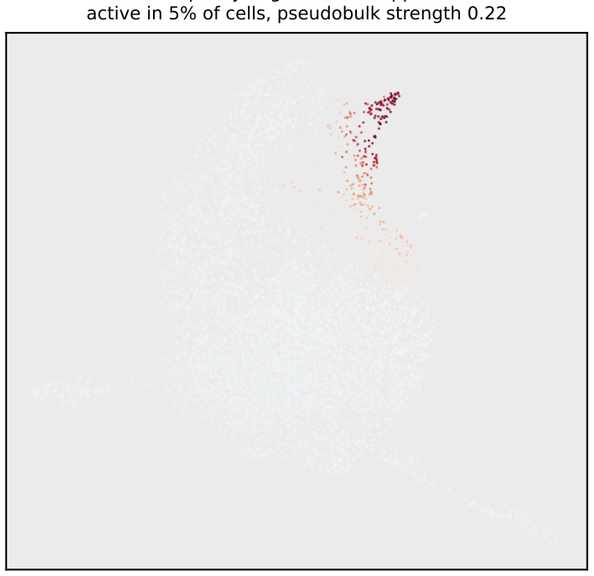
dome: 2 genes, not supported
active in 4% of cells, pseudobulk strength 0.12



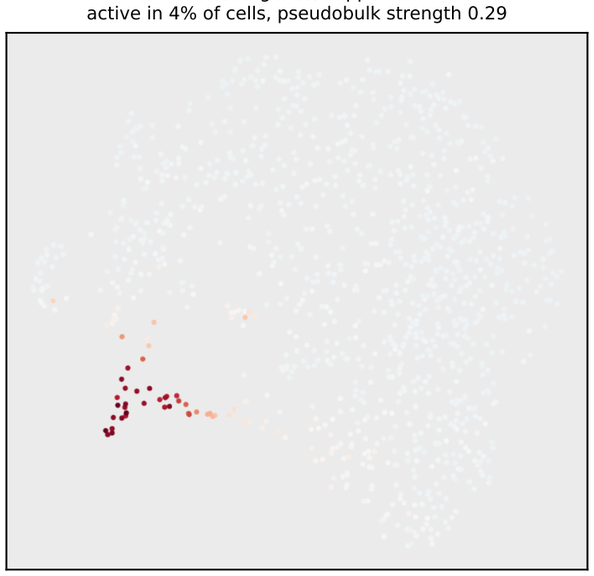
30% epiboly: 2 genes, not supported
active in 4% of cells, pseudobulk strength 0.11



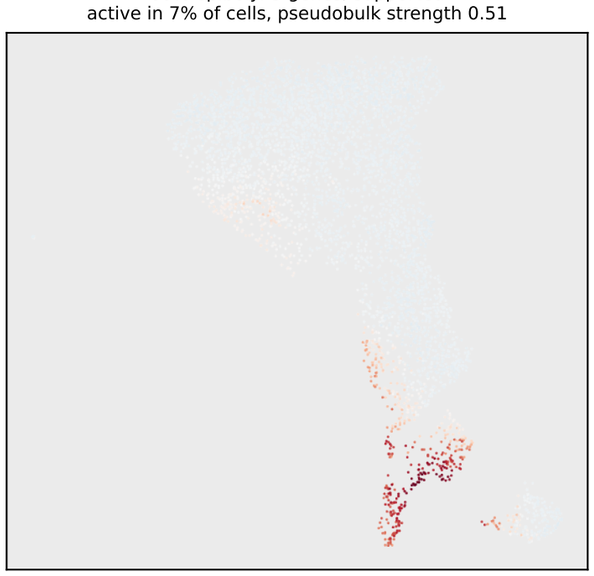
50% epiboly: 3 genes, not supported
active in 5% of cells, pseudobulk strength 0.22



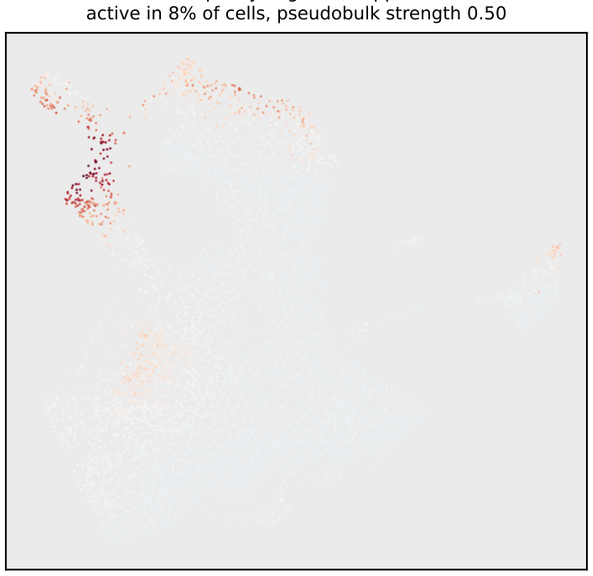
shield: 2 genes, supported
active in 4% of cells, pseudobulk strength 0.29



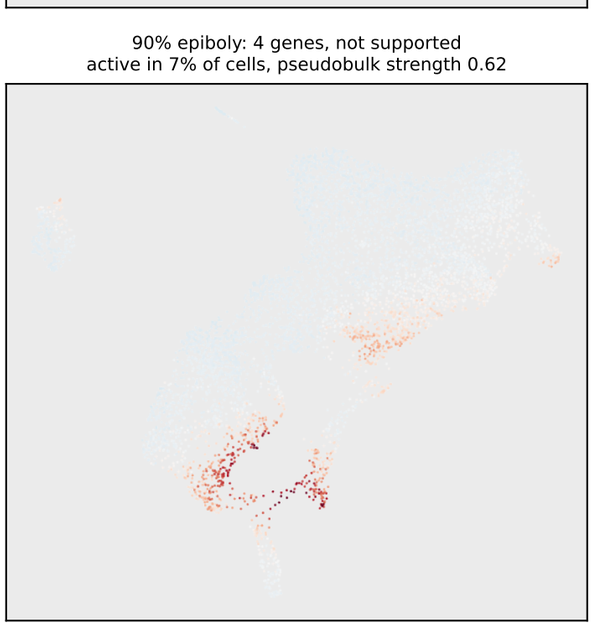
60% epiboly: 3 genes, supported
active in 7% of cells, pseudobulk strength 0.51



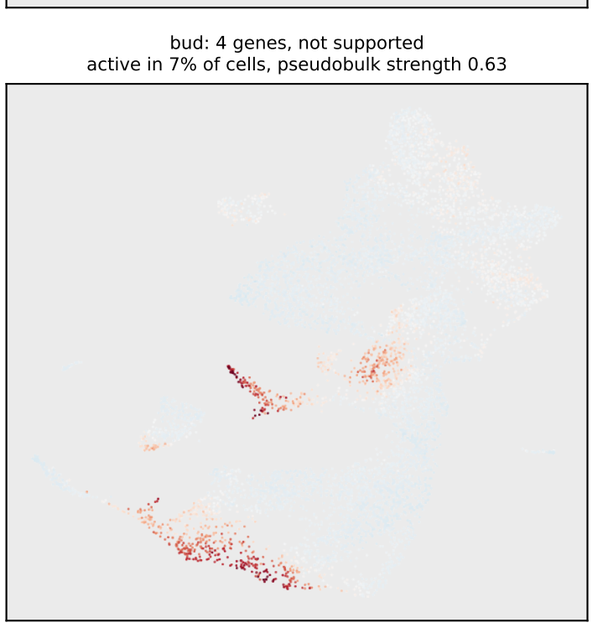
75% epiboly: 3 genes, supported
active in 8% of cells, pseudobulk strength 0.50



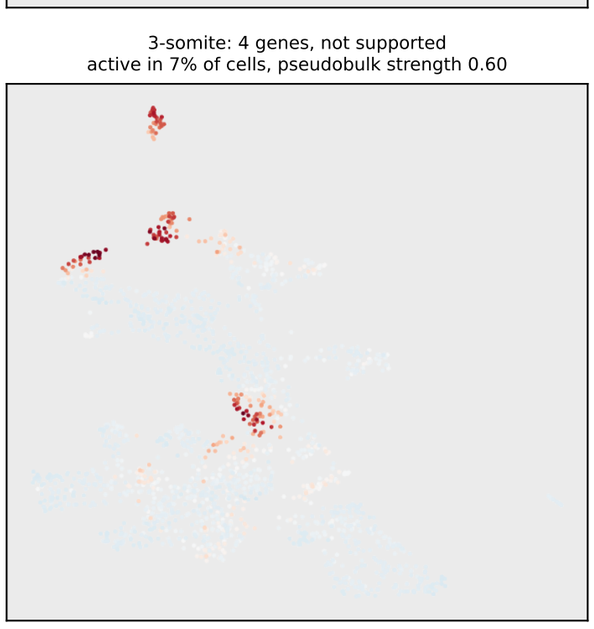
90% epiboly: 4 genes, not supported
active in 7% of cells, pseudobulk strength 0.62



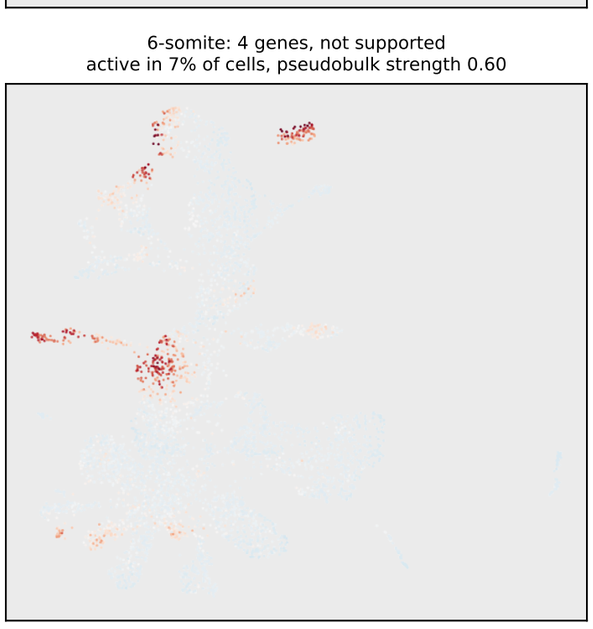
bud: 4 genes, not supported
active in 7% of cells, pseudobulk strength 0.63



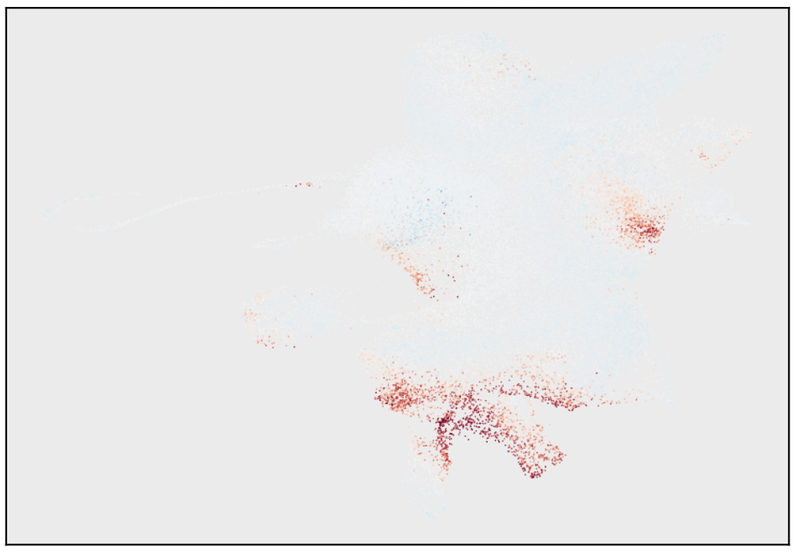
3-somite: 4 genes, not supported
active in 7% of cells, pseudobulk strength 0.60



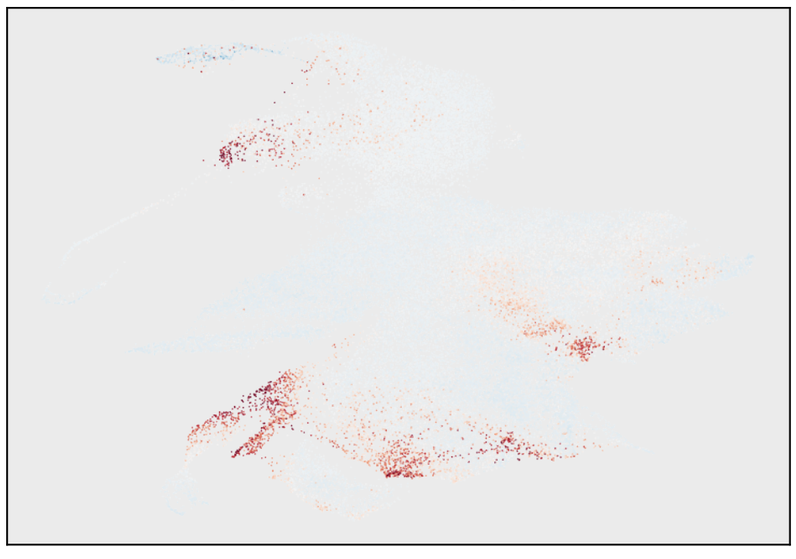
6-somite: 4 genes, not supported
active in 7% of cells, pseudobulk strength 0.60



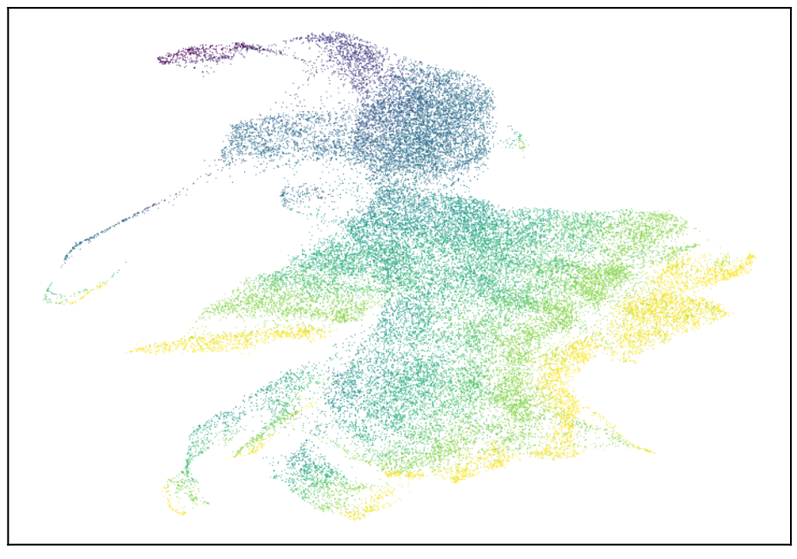
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



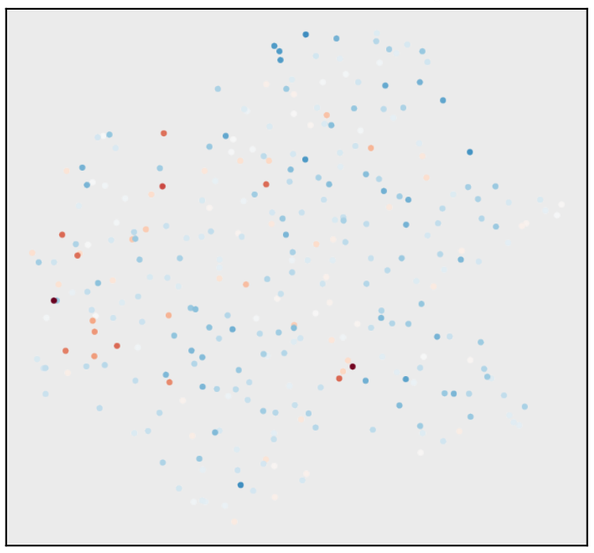
atlas, unaligned: stages



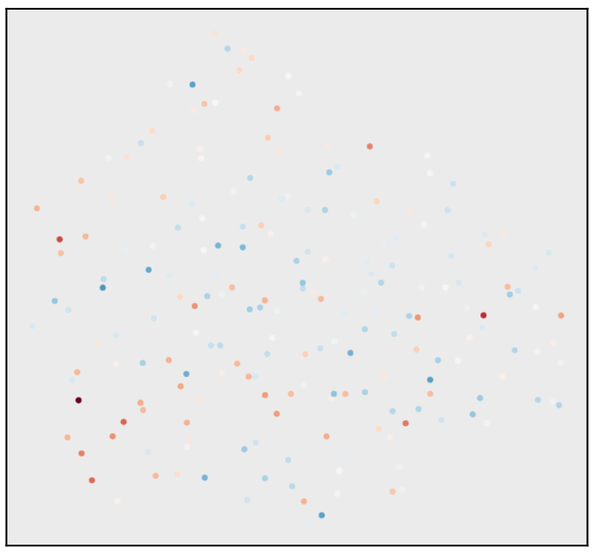
Genes (number of stages): chd (12), nog1 (12), fsta (7), nog2 (4)

blocking:RA

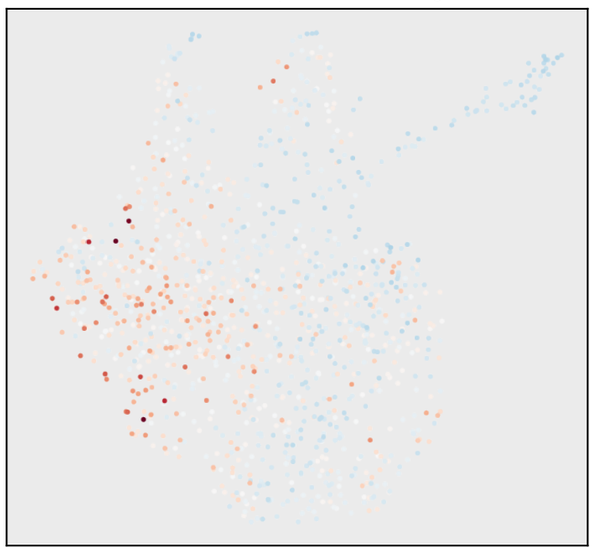
high: 1 genes, not supported
active in 0% of cells, pseudobulk strength 0.25



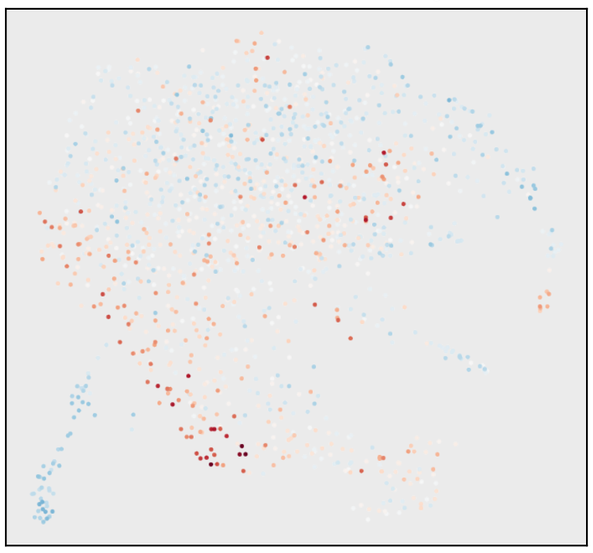
oblong: 1 genes, not supported
active in 0% of cells, pseudobulk strength 0.43



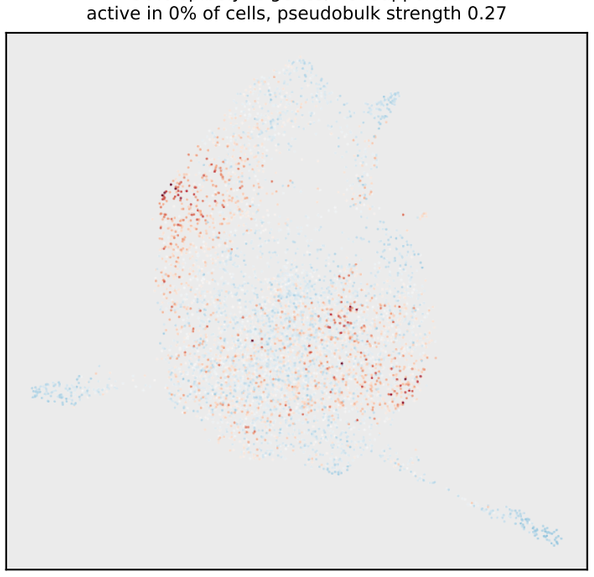
dome: 1 genes, not supported
active in 0% of cells, pseudobulk strength 0.26



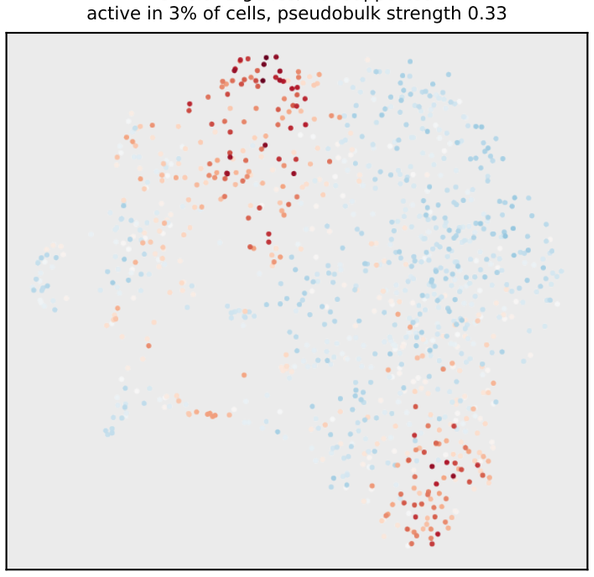
30% epiboly: 1 genes, not supported
active in 1% of cells, pseudobulk strength 0.41



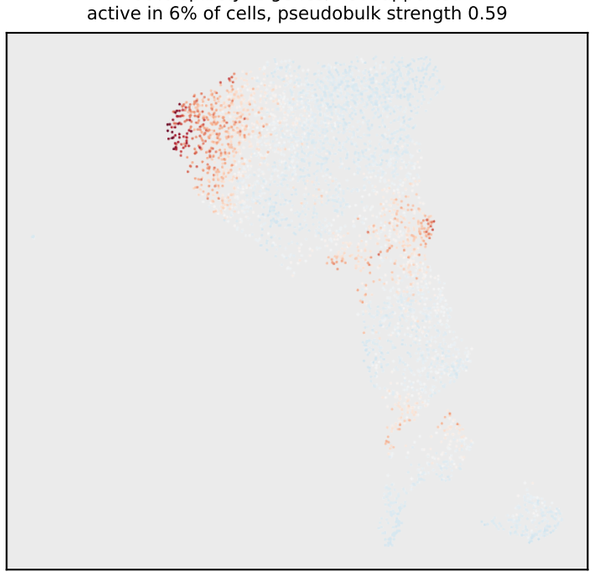
50% epiboly: 1 genes, not supported
active in 0% of cells, pseudobulk strength 0.27



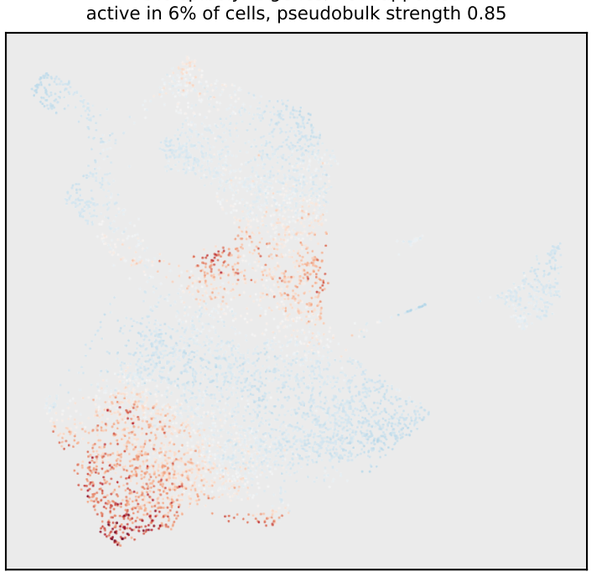
shield: 1 genes, not supported
active in 3% of cells, pseudobulk strength 0.33



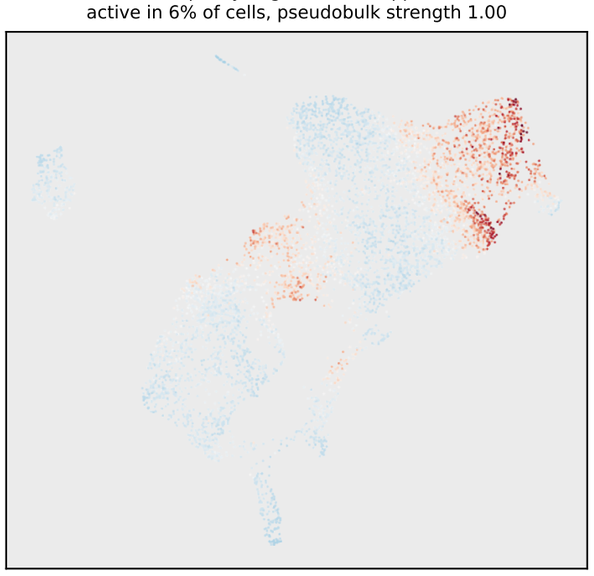
60% epiboly: 1 genes, not supported
active in 6% of cells, pseudobulk strength 0.59



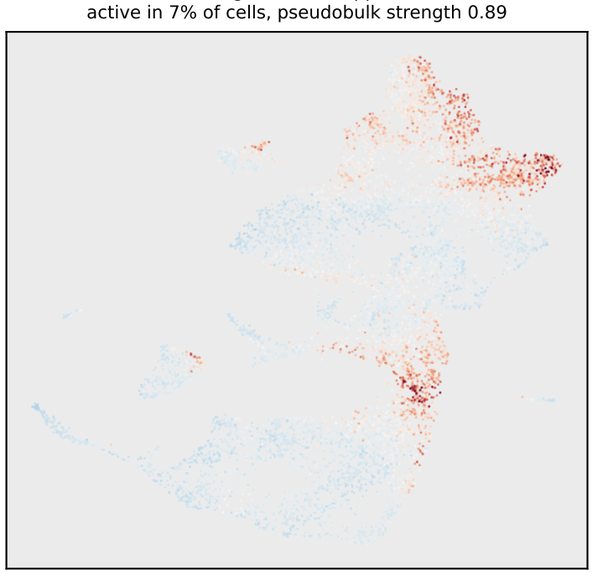
75% epiboly: 1 genes, not supported
active in 6% of cells, pseudobulk strength 0.85



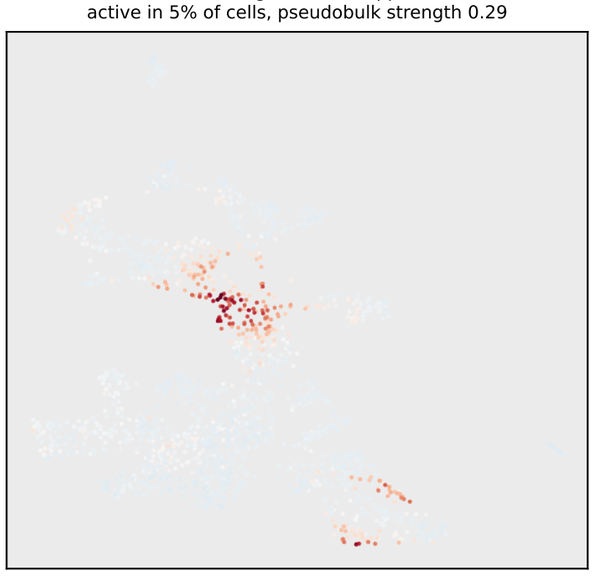
90% epiboly: 1 genes, not supported
active in 6% of cells, pseudobulk strength 1.00



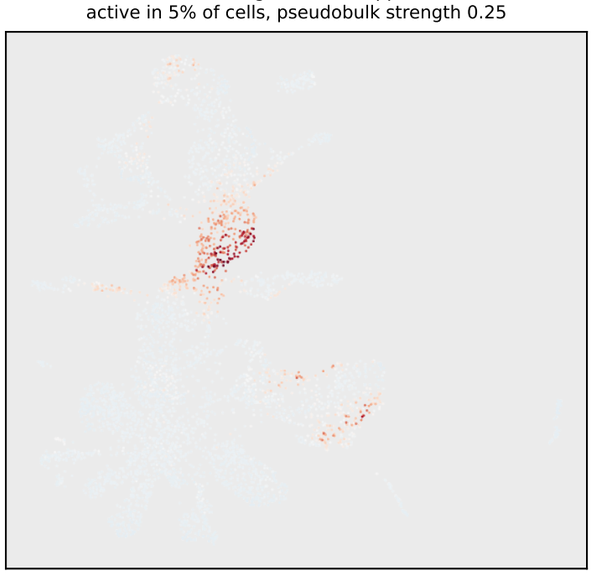
bud: 1 genes, not supported
active in 7% of cells, pseudobulk strength 0.89



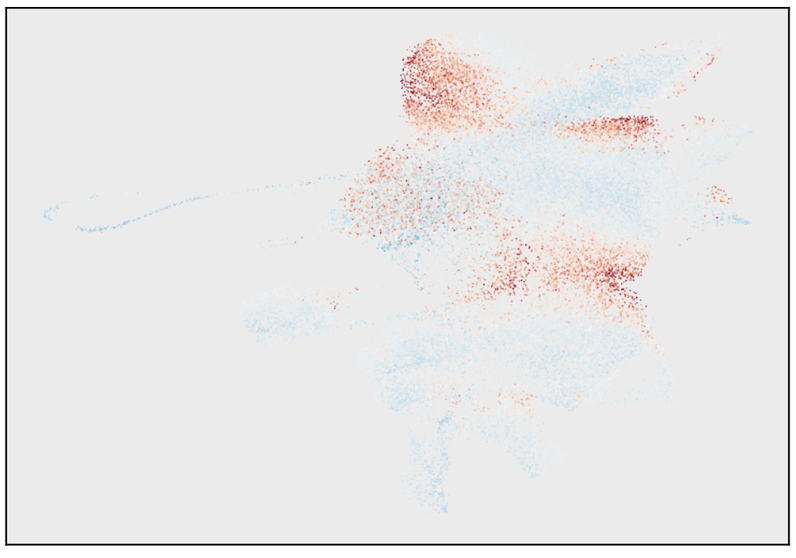
3-somite: 1 genes, not supported
active in 5% of cells, pseudobulk strength 0.29



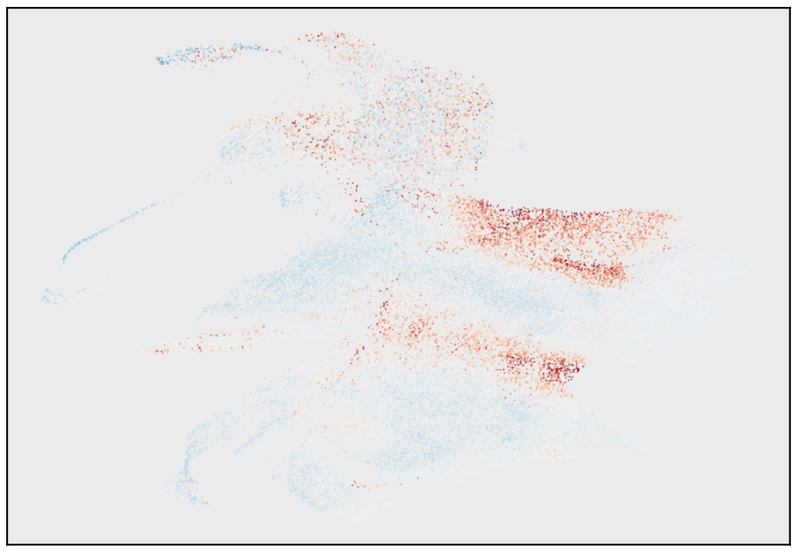
6-somite: 1 genes, not supported
active in 5% of cells, pseudobulk strength 0.25



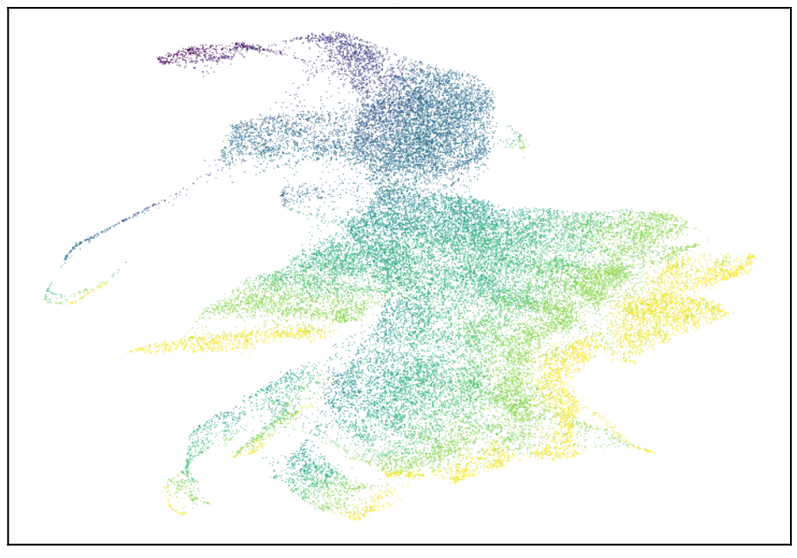
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



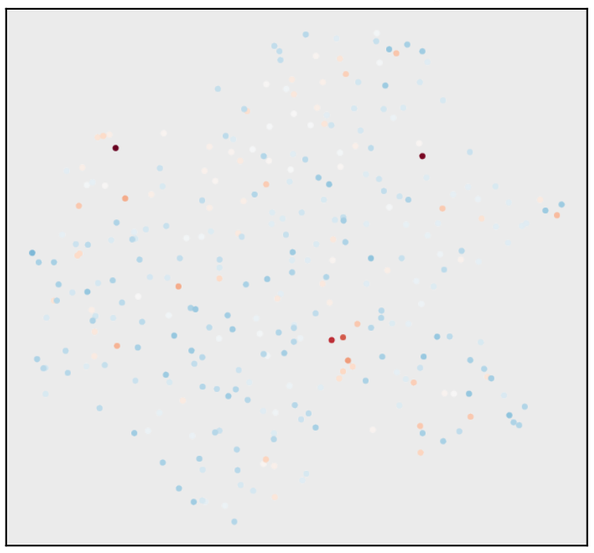
atlas, unaligned: stages



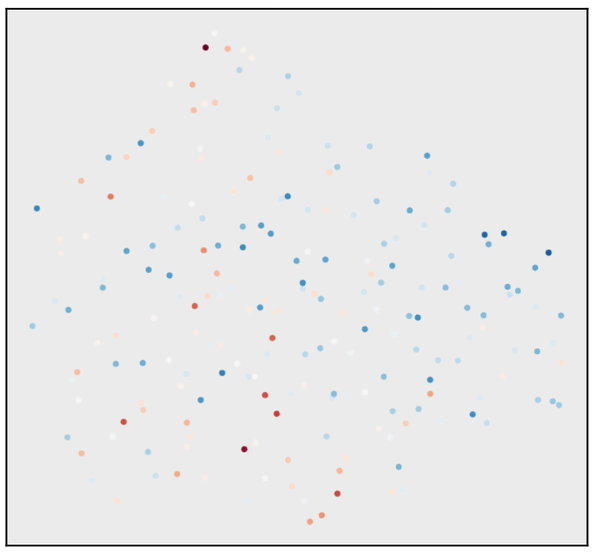
Genes (number of stages): cyp26a1 (12)

blocking:Wnt

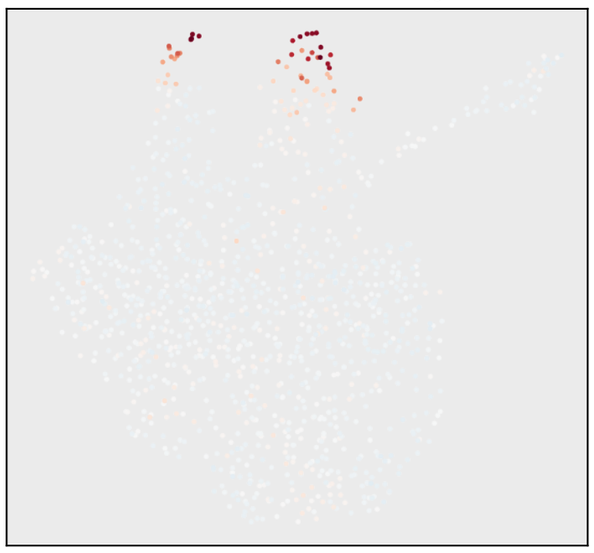
high: 2 genes, not supported
active in 0% of cells, pseudobulk strength 0.02



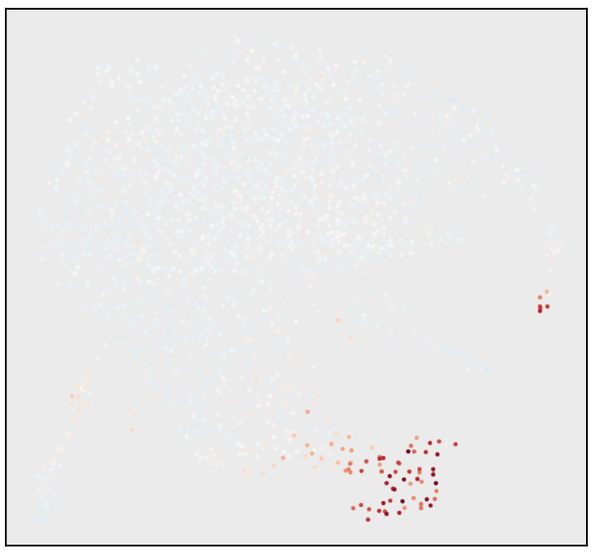
oblong: 1 genes, not supported
active in 0% of cells, pseudobulk strength 0.06



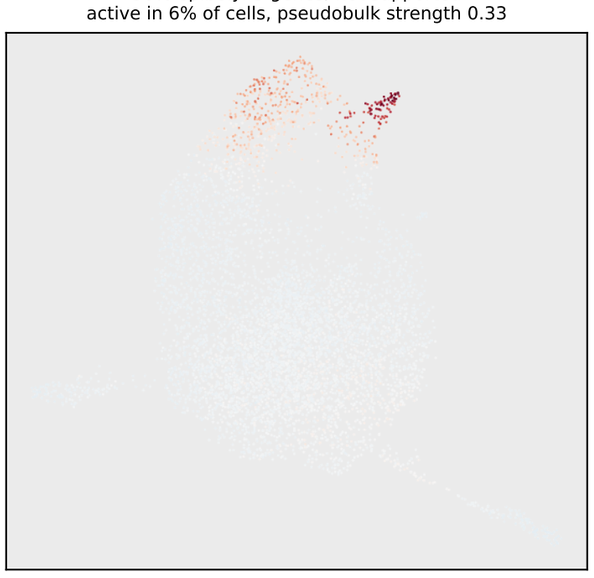
dome: 3 genes, not supported
active in 3% of cells, pseudobulk strength 0.13



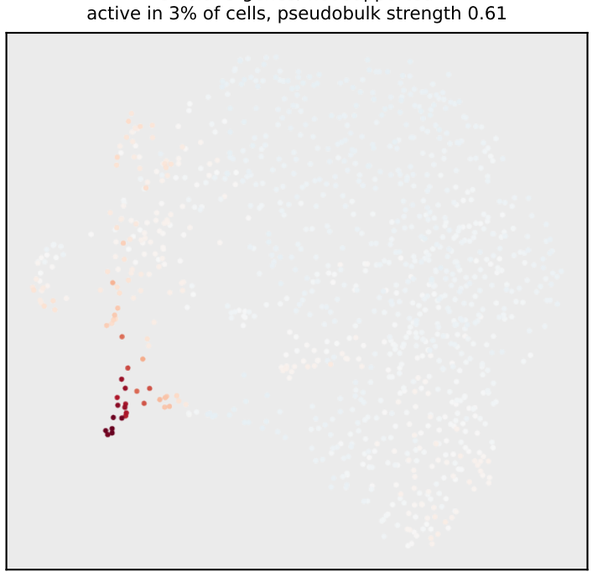
30% epiboly: 3 genes, not supported
active in 5% of cells, pseudobulk strength 0.31



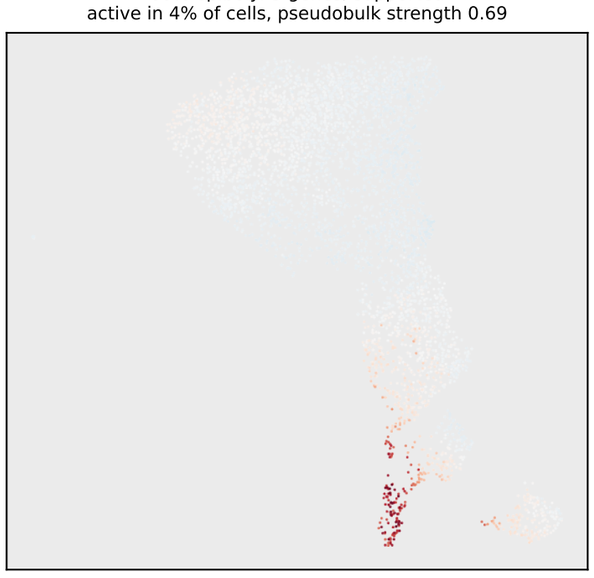
50% epiboly: 3 genes, not supported
active in 6% of cells, pseudobulk strength 0.33



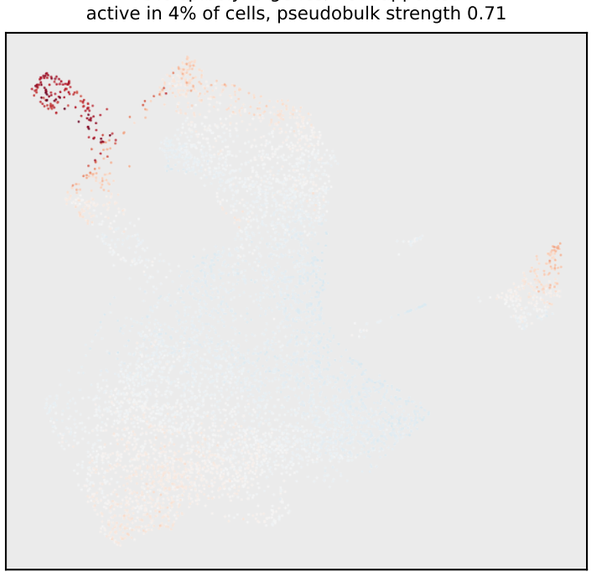
shield: 3 genes, not supported
active in 3% of cells, pseudobulk strength 0.61



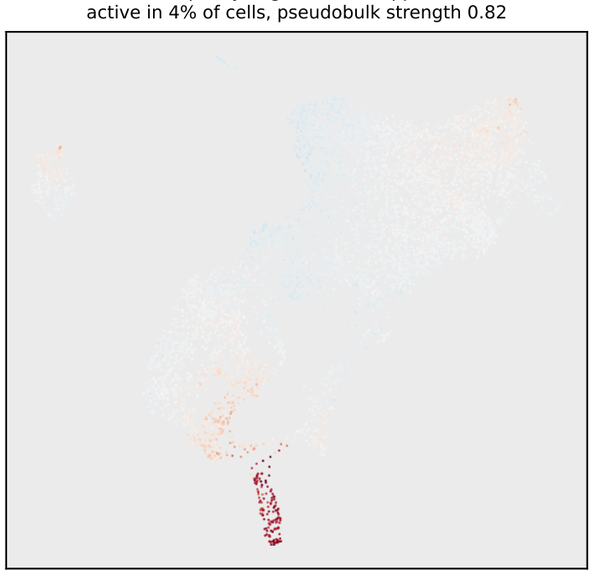
60% epiboly: 3 genes, supported
active in 4% of cells, pseudobulk strength 0.69



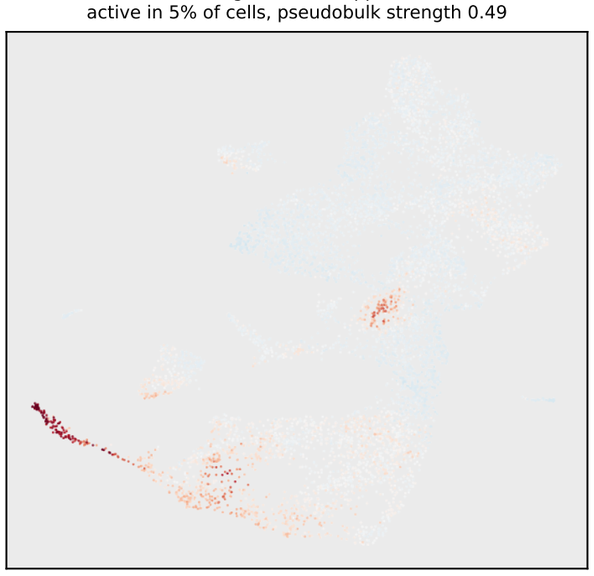
75% epiboly: 3 genes, not supported
active in 4% of cells, pseudobulk strength 0.71



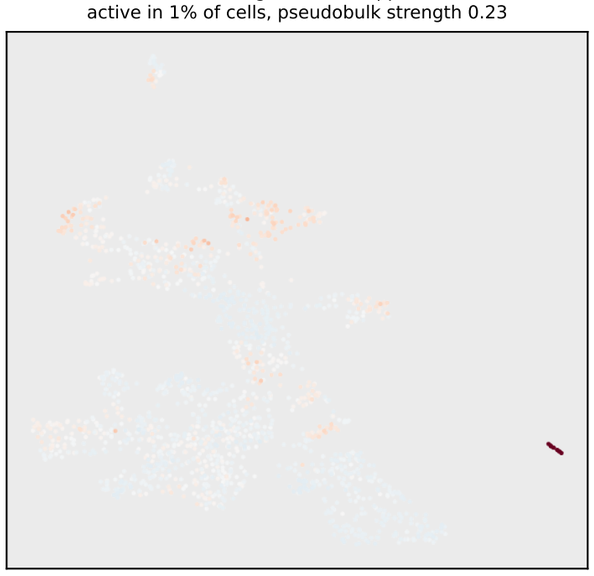
90% epiboly: 3 genes, not supported
active in 4% of cells, pseudobulk strength 0.82



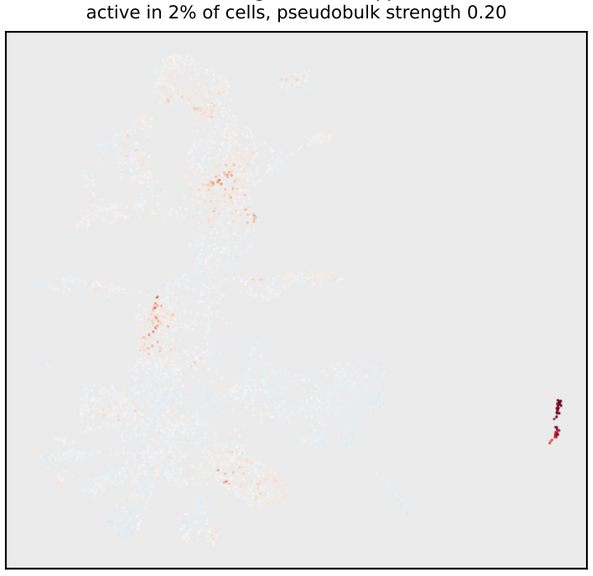
bud: 3 genes, not supported
active in 5% of cells, pseudobulk strength 0.49



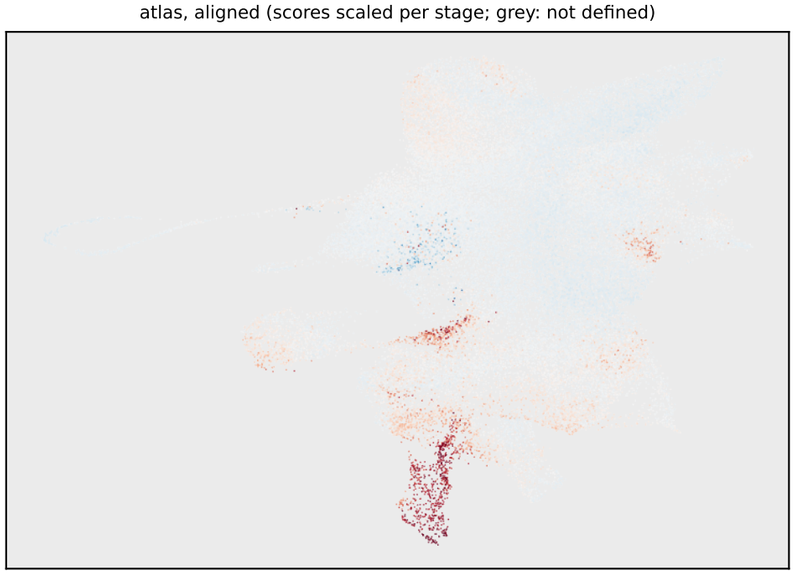
3-somite: 3 genes, not supported
active in 1% of cells, pseudobulk strength 0.23



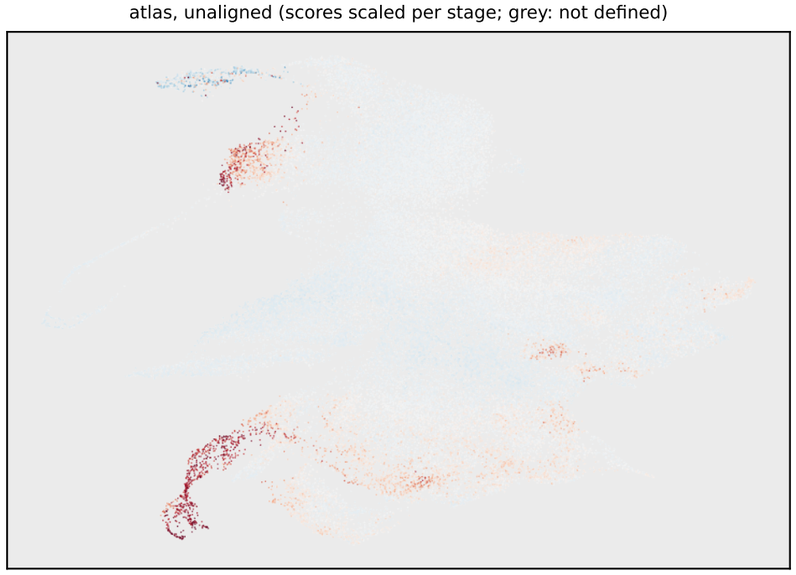
6-somite: 3 genes, not supported
active in 2% of cells, pseudobulk strength 0.20



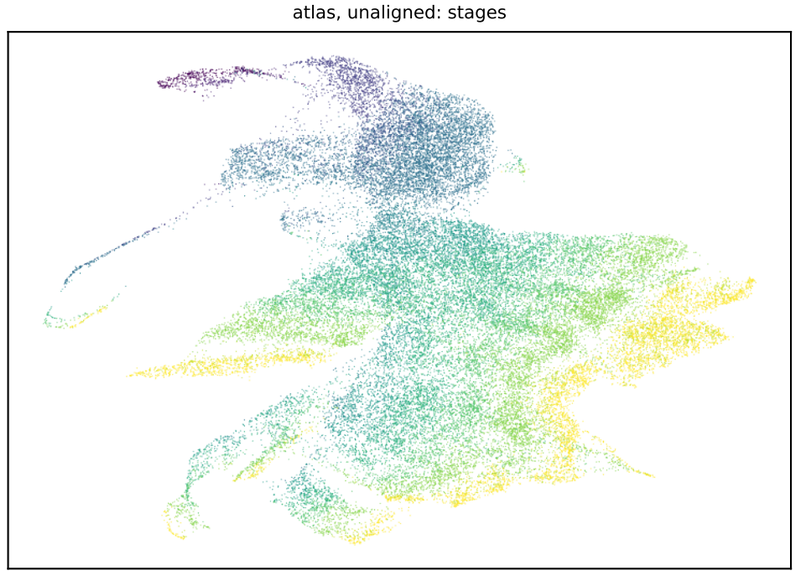
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



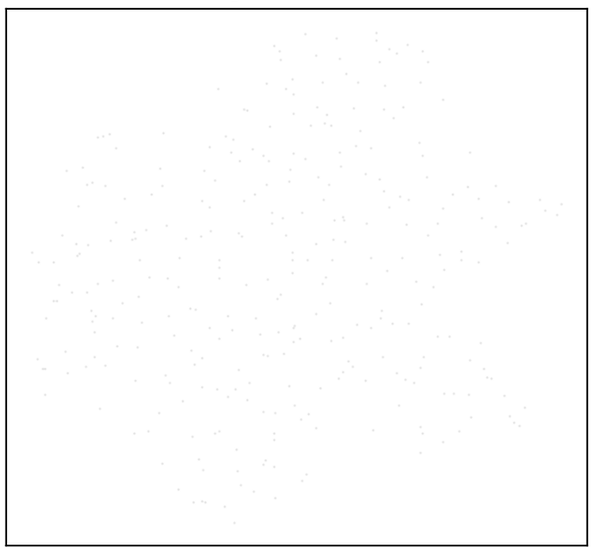
atlas, unaligned: stages



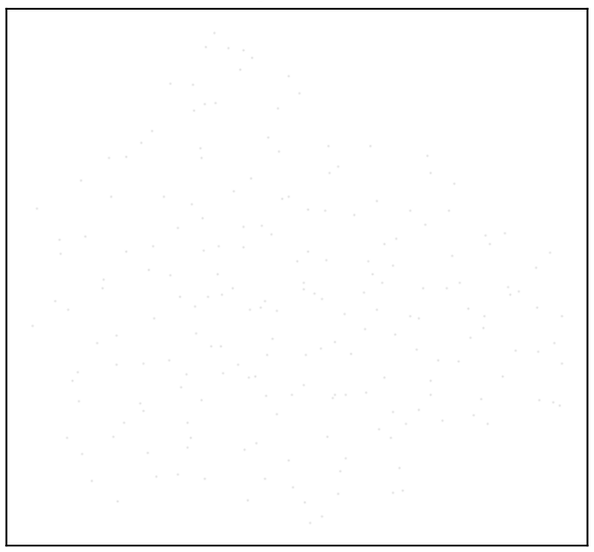
Genes (number of stages): sfrp1a (12), dkk1b (11), frzb (10)

receiving:Notch

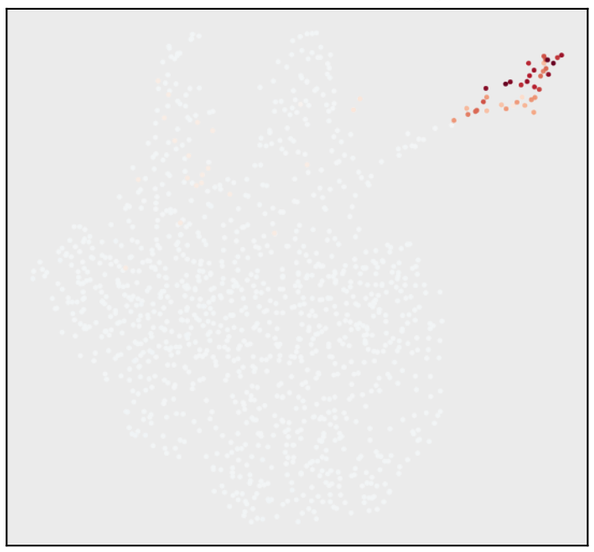
high: not defined



oblong: not defined

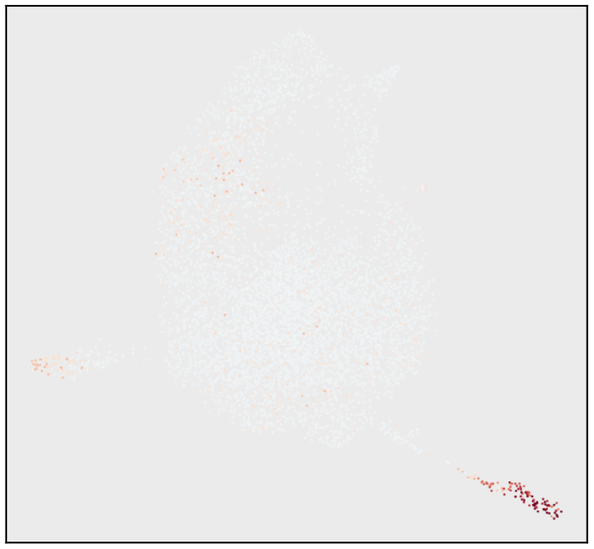


dome: 1 genes, not supported
active in 3% of cells, pseudobulk strength 0.09

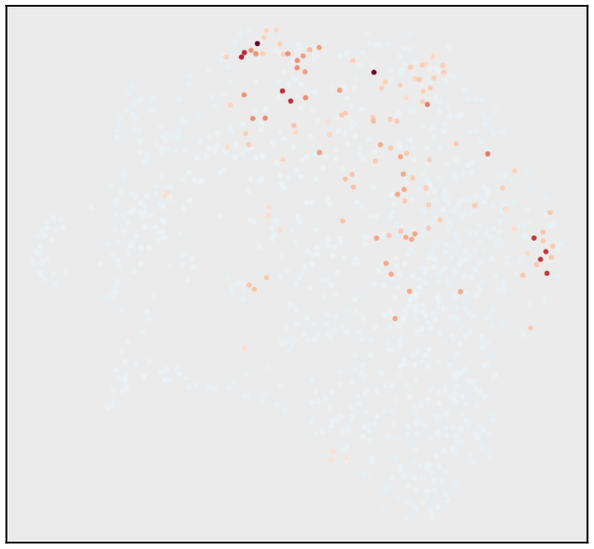


30% epiboly: 1 genes, not supported
active in 2% of cells, pseudobulk strength 0.07

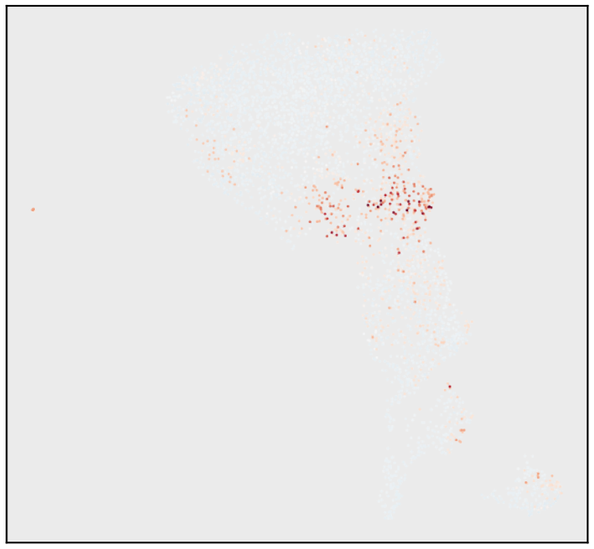
50% epiboly: 2 genes, not supported
active in 1% of cells, pseudobulk strength 0.03



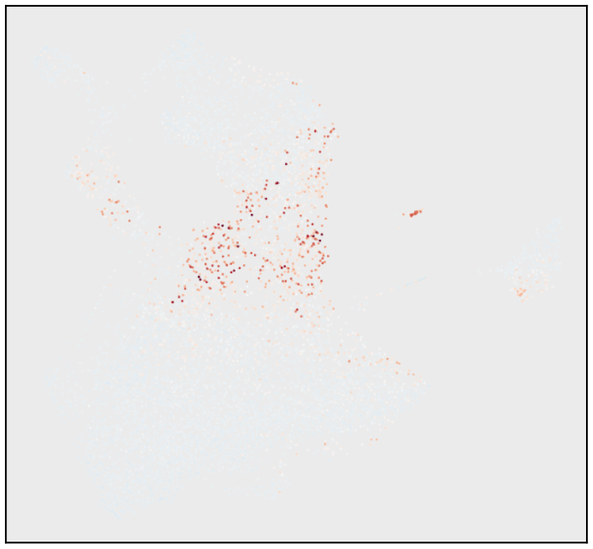
shield: 1 genes, not supported
active in 0% of cells, pseudobulk strength 0.05



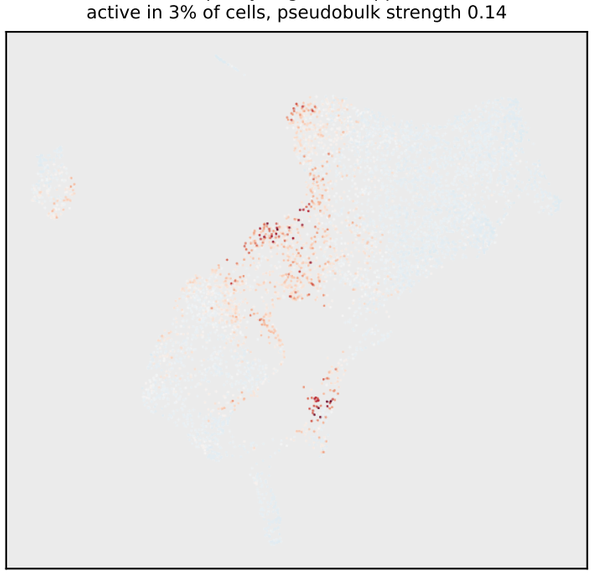
60% epiboly: 4 genes, supported
active in 1% of cells, pseudobulk strength 0.04



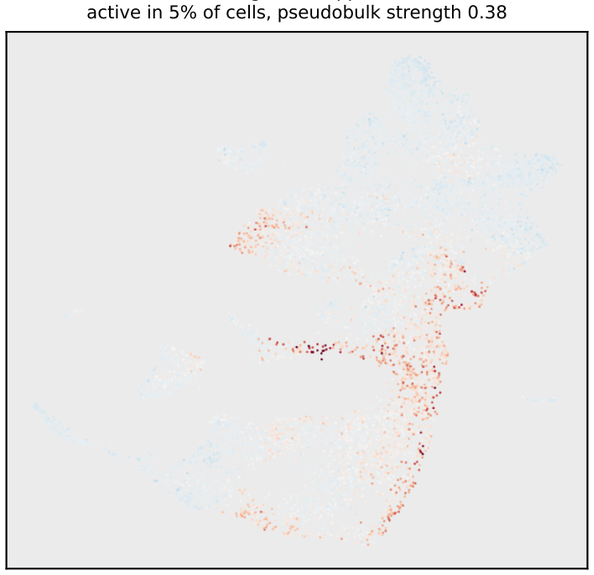
75% epiboly: 7 genes, supported
active in 3% of cells, pseudobulk strength 0.08



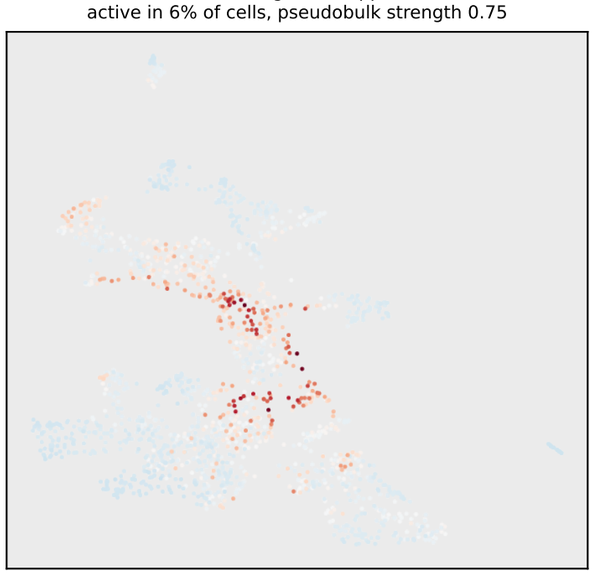
90% epiboly: 7 genes, supported
active in 3% of cells, pseudobulk strength 0.14



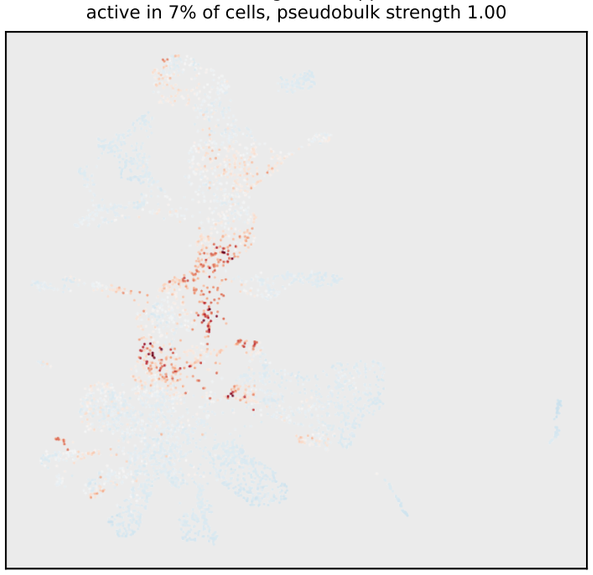
bud: 7 genes, supported
active in 5% of cells, pseudobulk strength 0.38



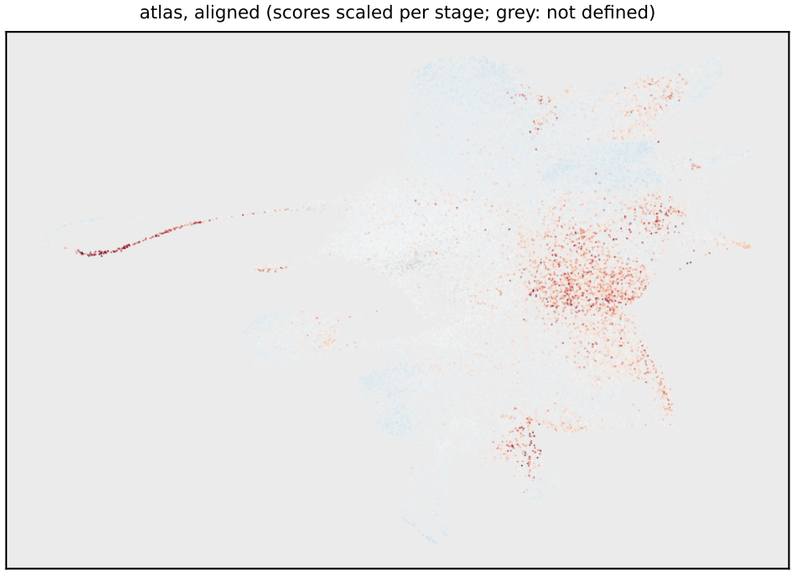
3-somite: 7 genes, supported
active in 6% of cells, pseudobulk strength 0.75



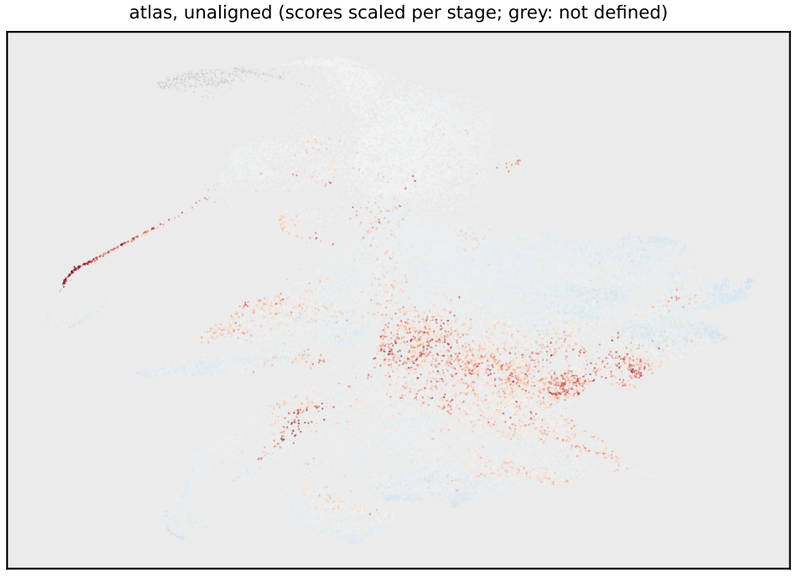
6-somite: 7 genes, supported
active in 7% of cells, pseudobulk strength 1.00



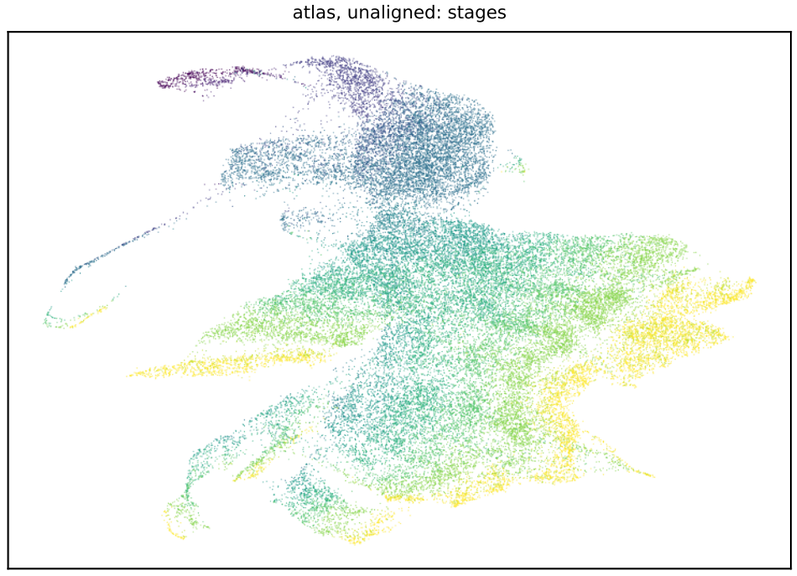
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



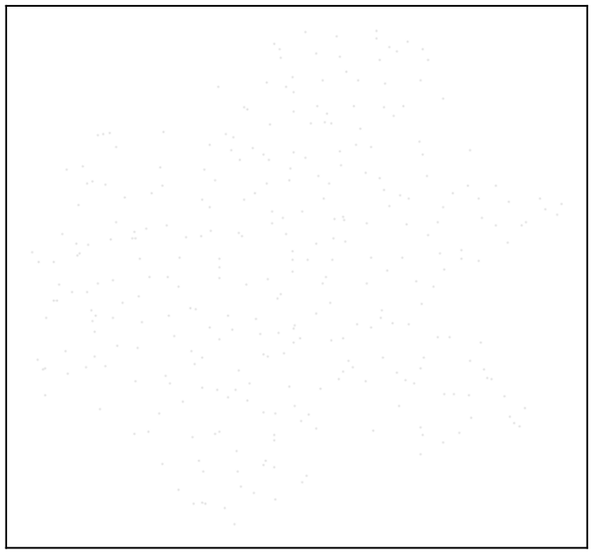
atlas, unaligned: stages



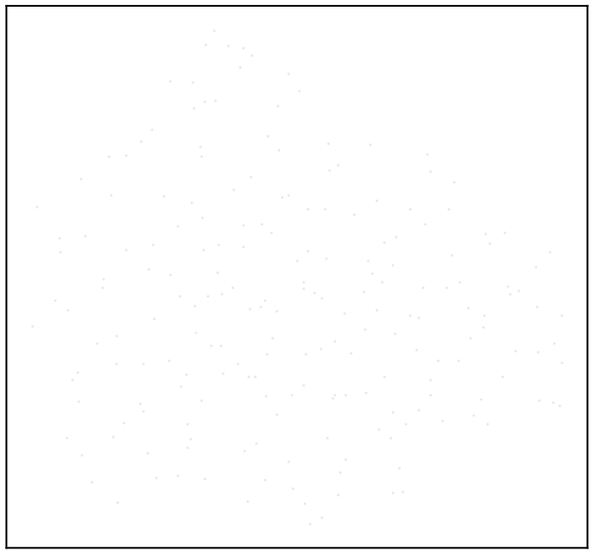
Genes (number of stages): hey1 (9), her12 (8), her15.1 (6), her15.2 (6), her4.1 (5), her4.2 (5), her4.3 (5)

receiving:RA

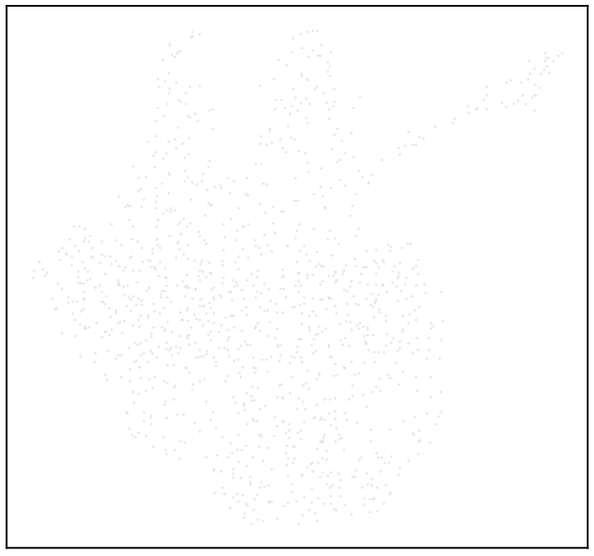
high: not defined



oblong: not defined

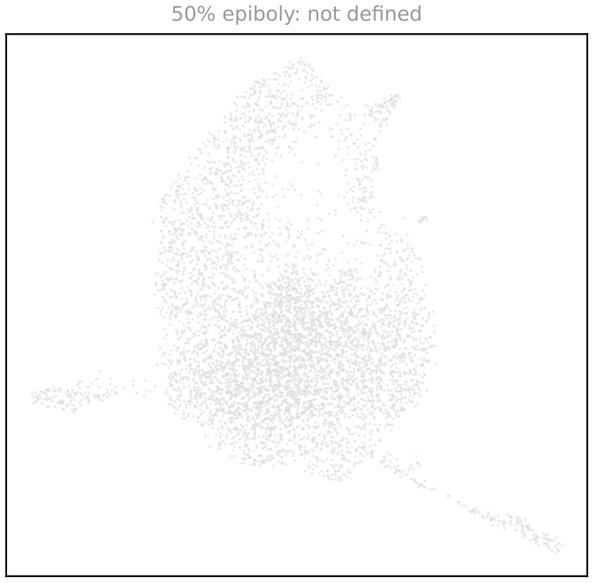


dome: not defined

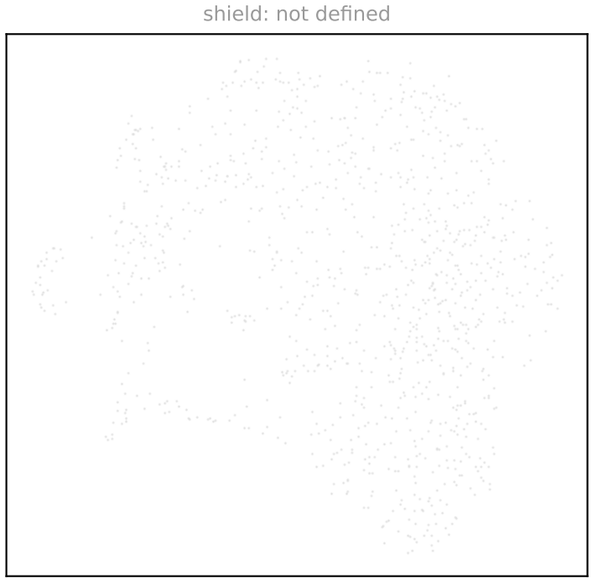


30% epiboly: not defined

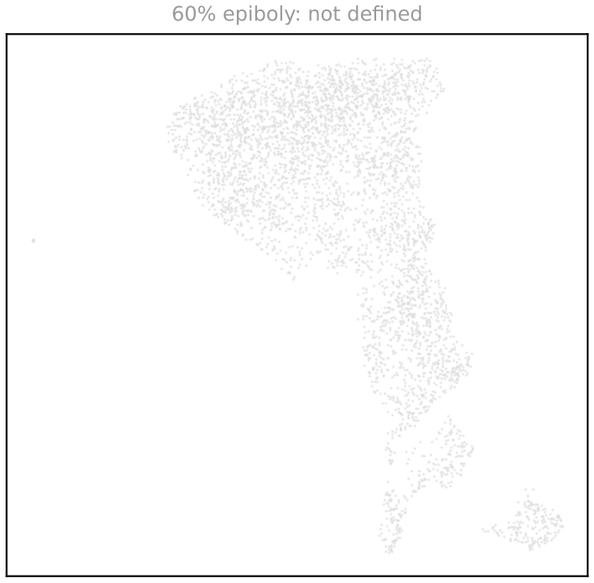
50% epiboly: not defined



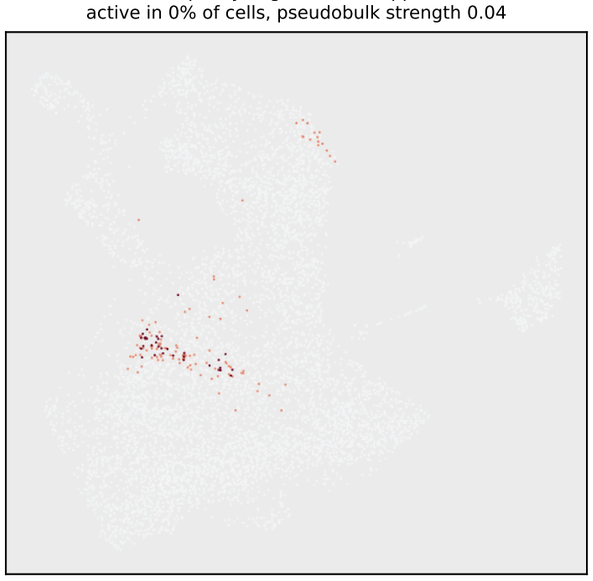
shield: not defined



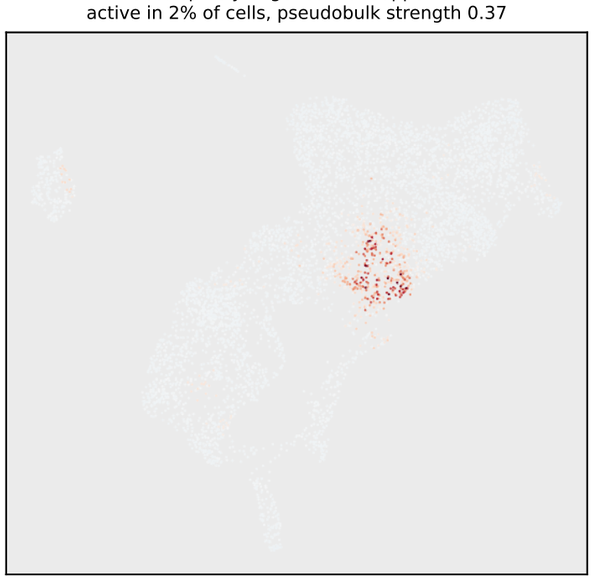
60% epiboly: not defined



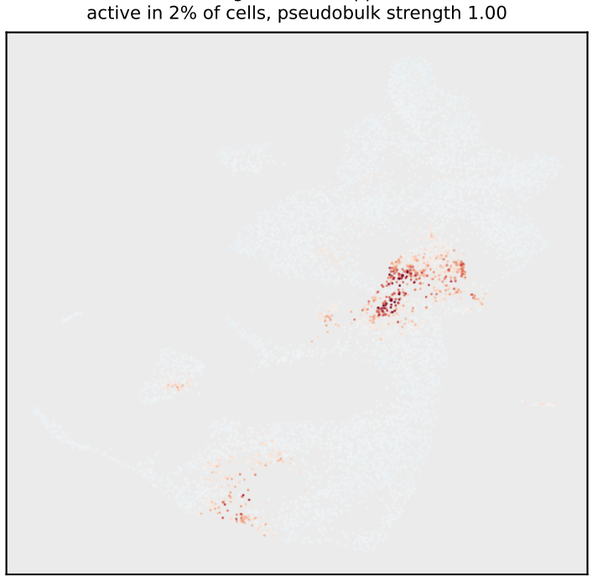
75% epiboly: 1 genes, not supported
active in 0% of cells, pseudobulk strength 0.04



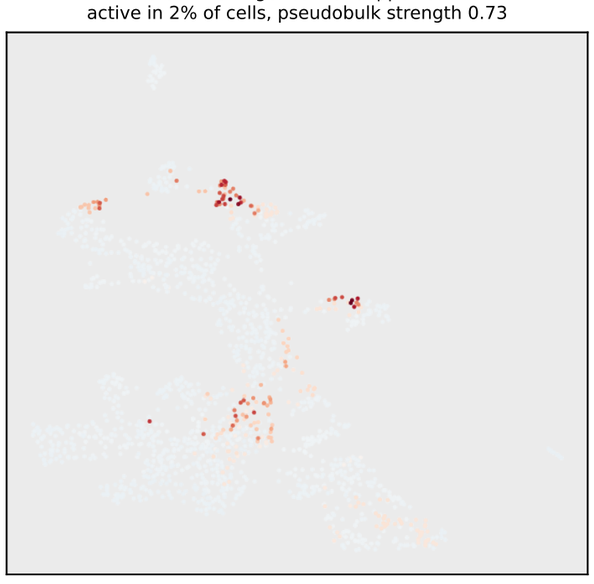
90% epiboly: 1 genes, not supported
active in 2% of cells, pseudobulk strength 0.37



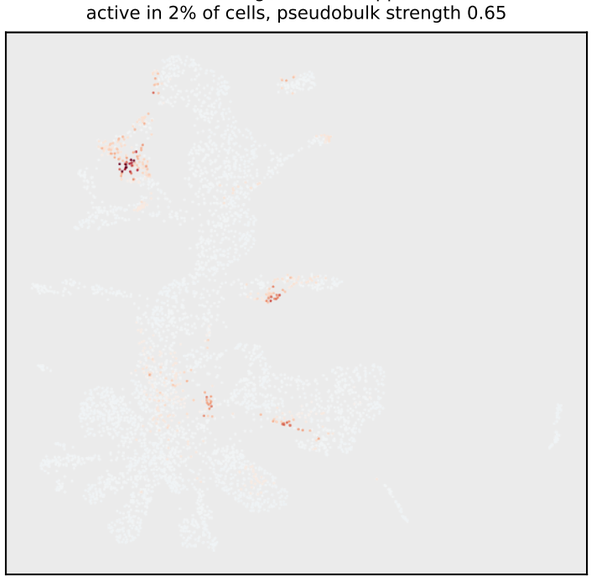
bud: 1 genes, not supported
active in 2% of cells, pseudobulk strength 1.00



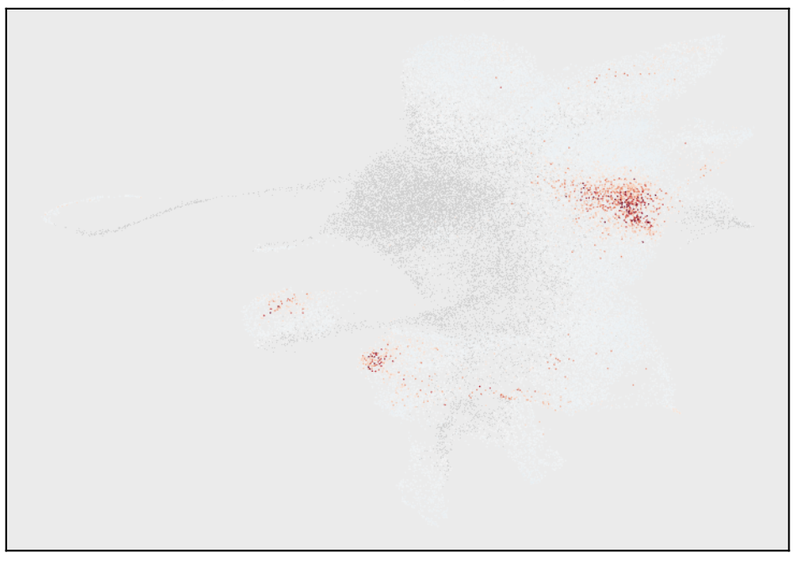
3-somite: 1 genes, not supported
active in 2% of cells, pseudobulk strength 0.73



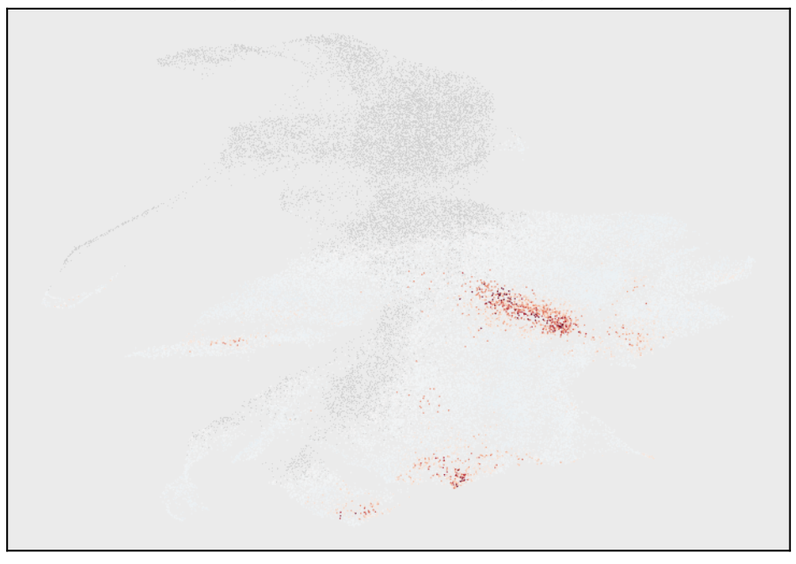
6-somite: 1 genes, not supported
active in 2% of cells, pseudobulk strength 0.65



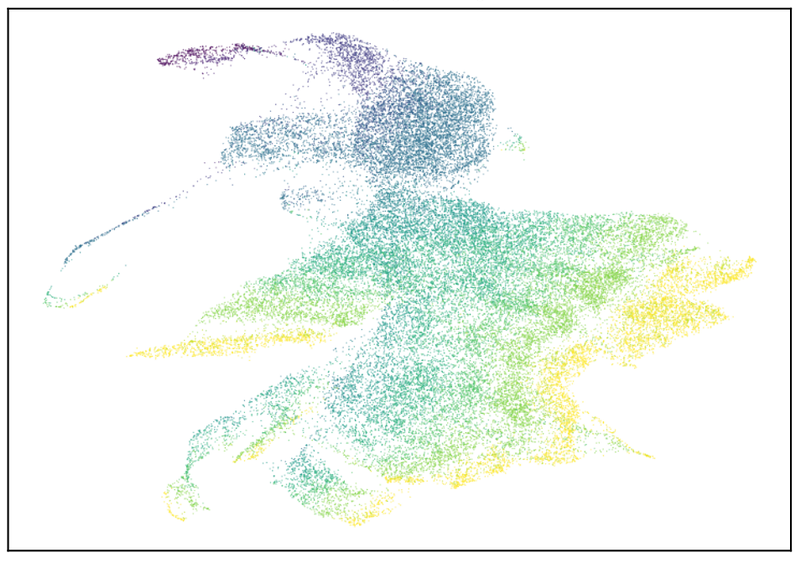
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



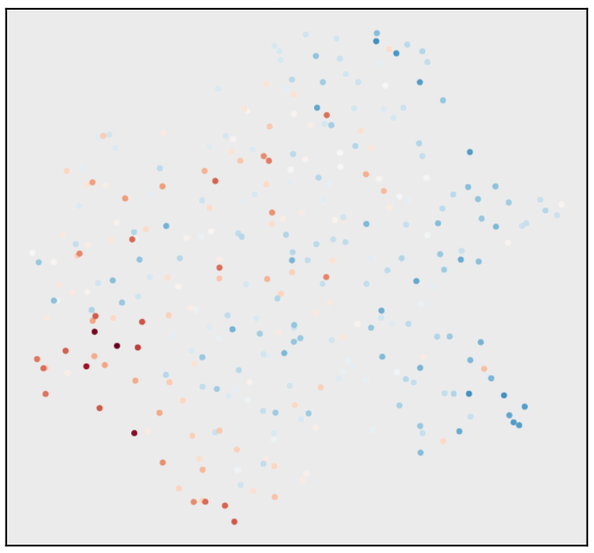
atlas, unaligned: stages



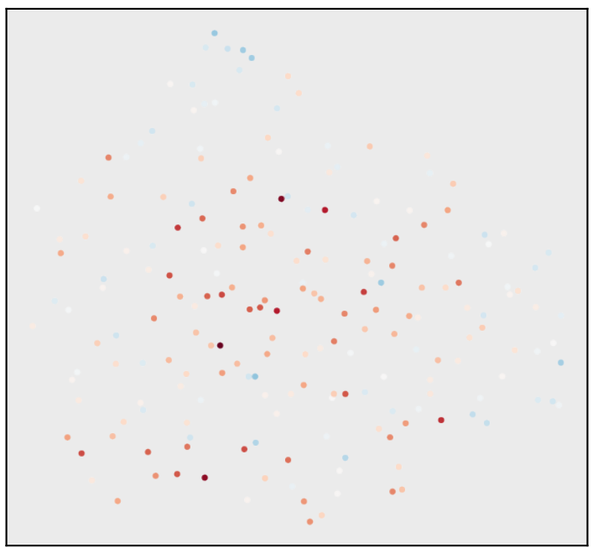
Genes (number of stages): dhrs3a (5)

sending:BMP_organizer

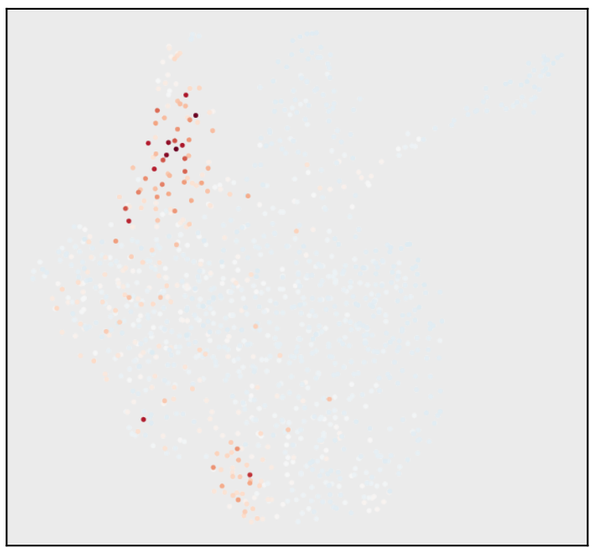
high: 1 genes, not supported
active in 0% of cells, pseudobulk strength 0.22



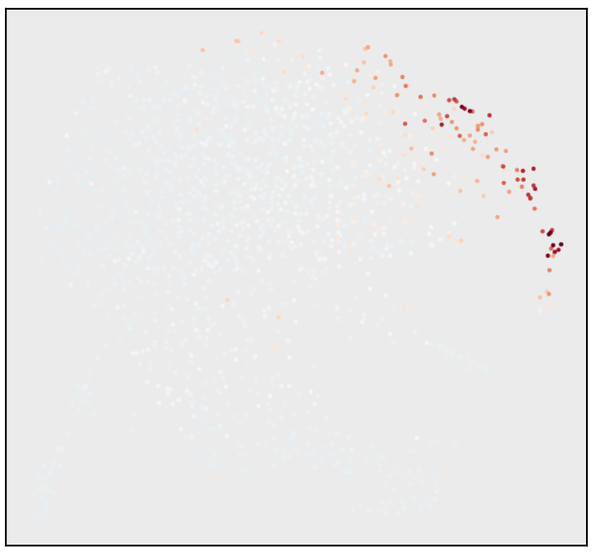
oblong: 1 genes, not supported
active in 2% of cells, pseudobulk strength 0.45



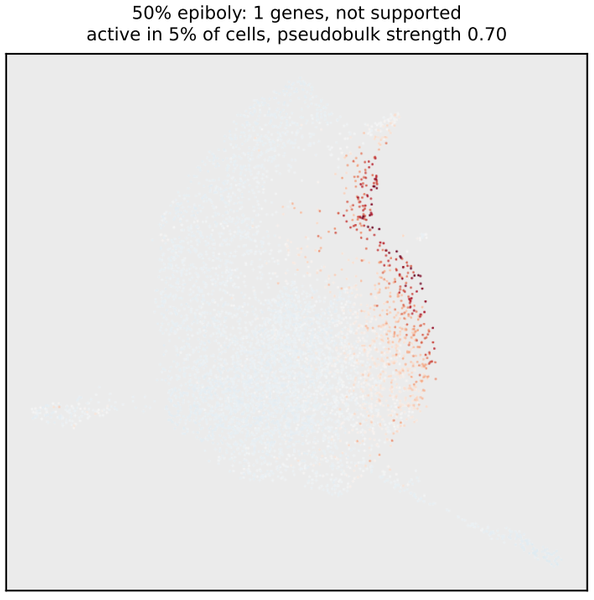
dome: 1 genes, not supported
active in 2% of cells, pseudobulk strength 0.38



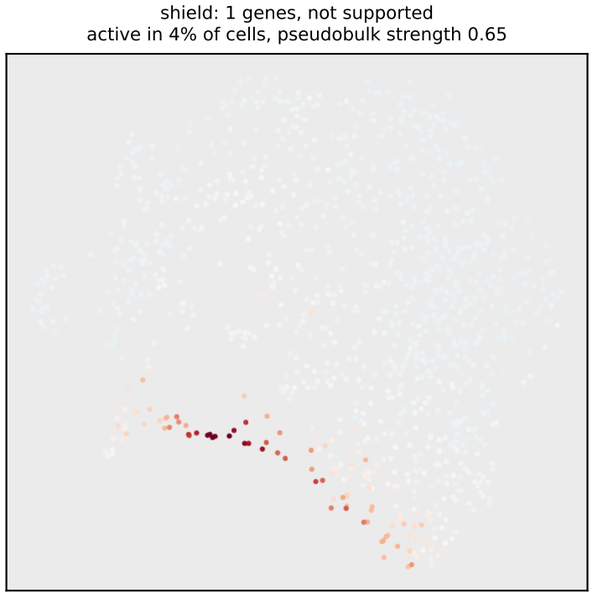
30% epiboly: 1 genes, not supported
active in 4% of cells, pseudobulk strength 0.33



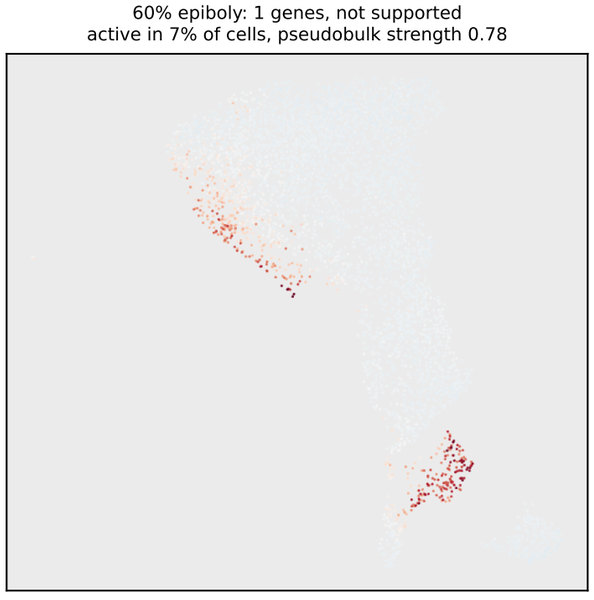
50% epiboly: 1 genes, not supported
active in 5% of cells, pseudobulk strength 0.70



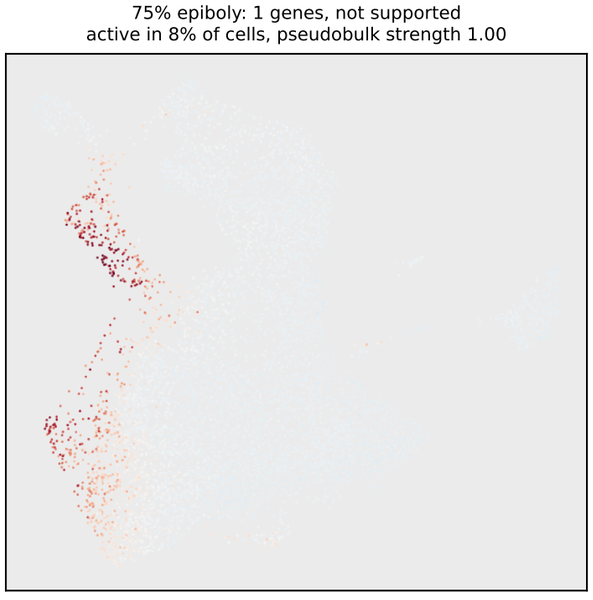
shield: 1 genes, not supported
active in 4% of cells, pseudobulk strength 0.65



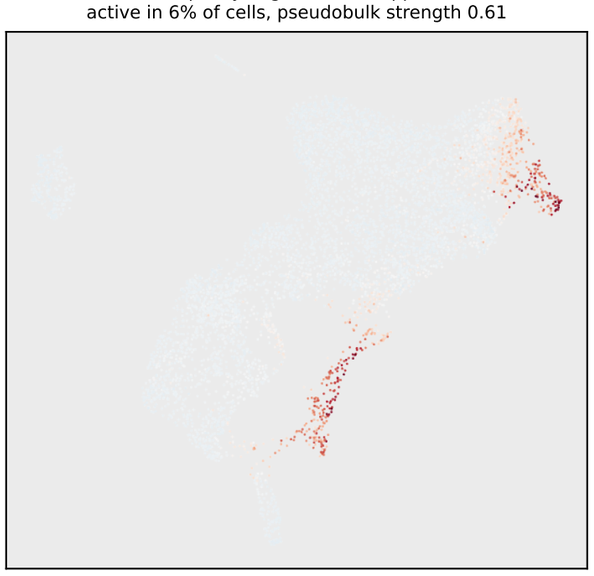
60% epiboly: 1 genes, not supported
active in 7% of cells, pseudobulk strength 0.78



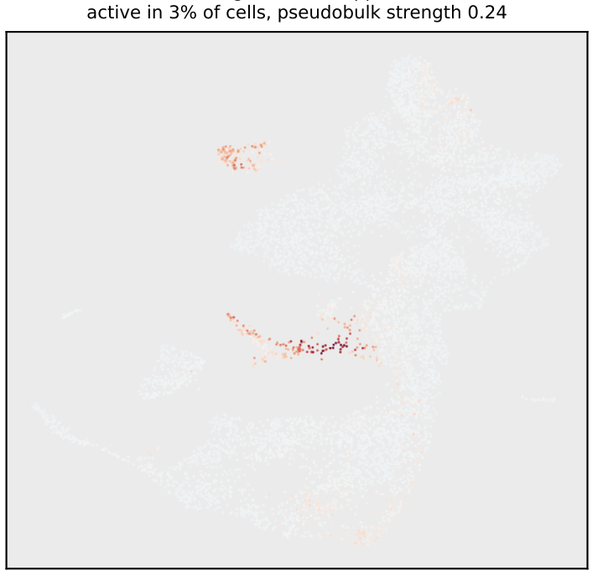
75% epiboly: 1 genes, not supported
active in 8% of cells, pseudobulk strength 1.00



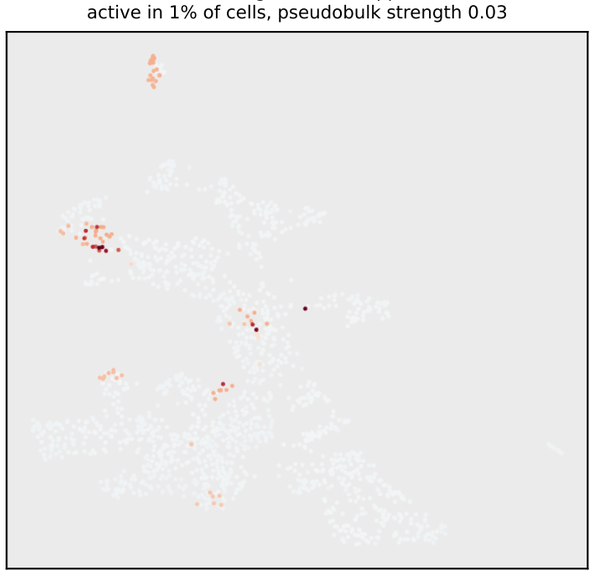
90% epiboly: 1 genes, not supported
active in 6% of cells, pseudobulk strength 0.61



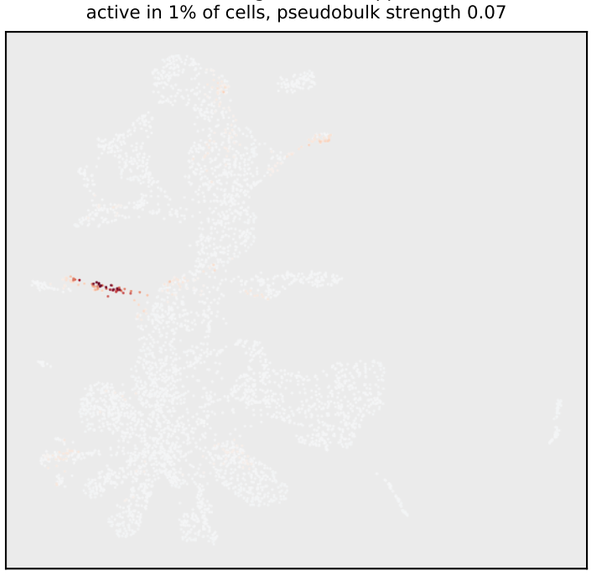
bud: 1 genes, not supported
active in 3% of cells, pseudobulk strength 0.24



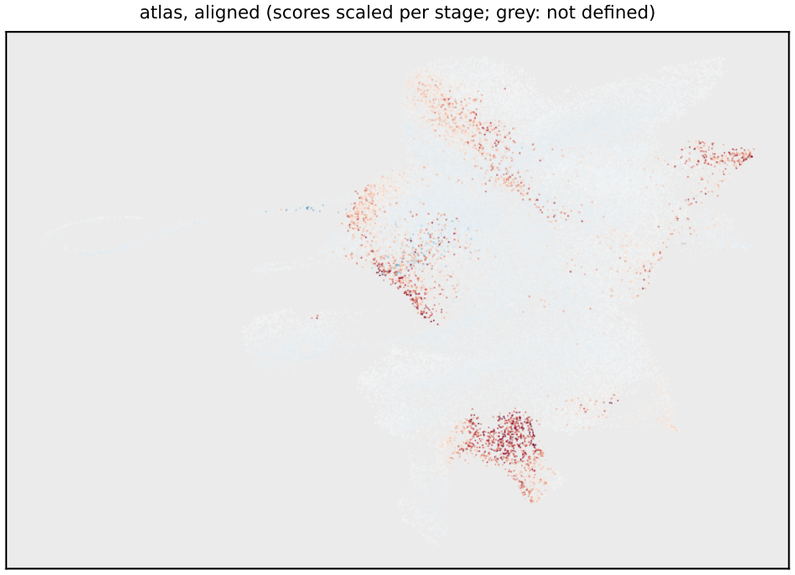
3-somite: 1 genes, not supported
active in 1% of cells, pseudobulk strength 0.03



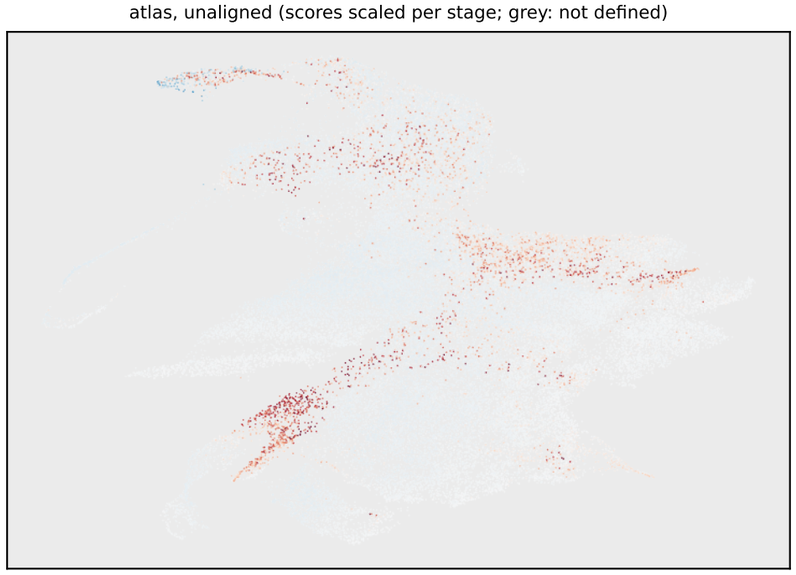
6-somite: 1 genes, not supported
active in 1% of cells, pseudobulk strength 0.07



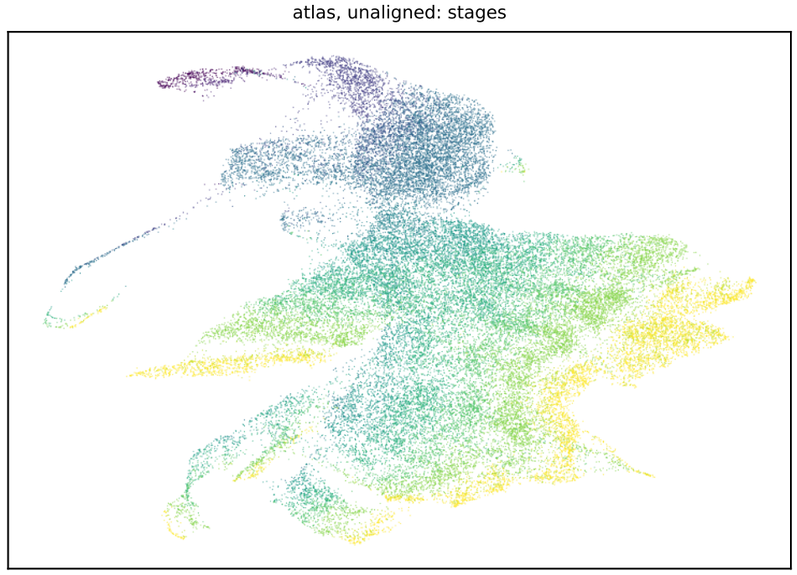
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



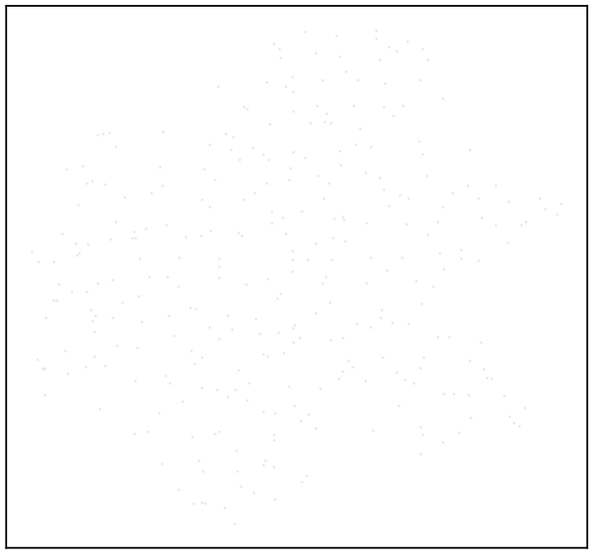
atlas, unaligned: stages



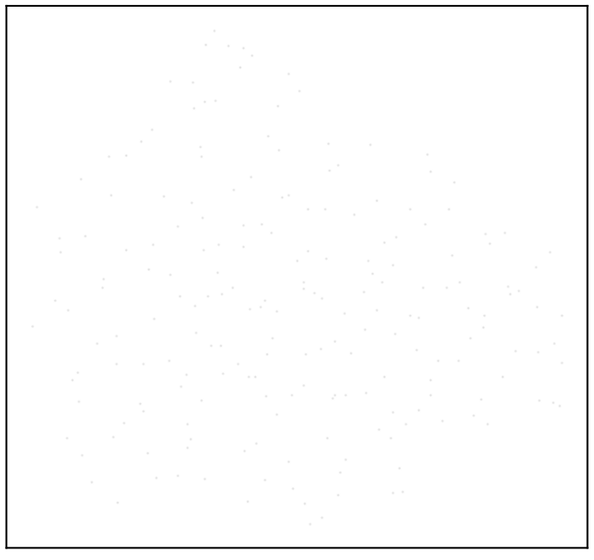
Genes (number of stages): admp (12)

sending:Hedgehog

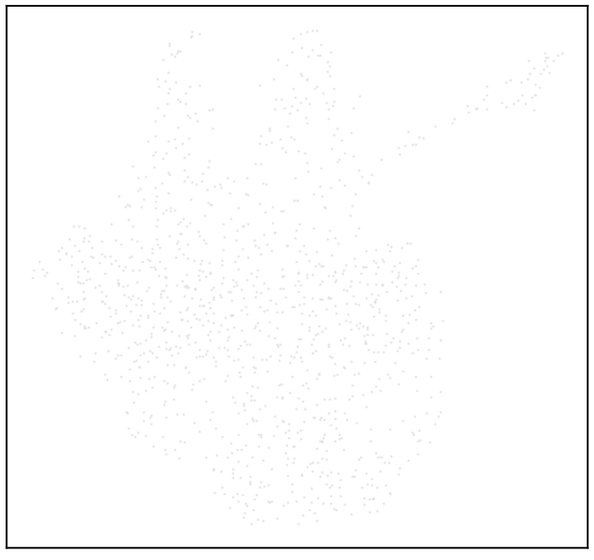
high: not defined



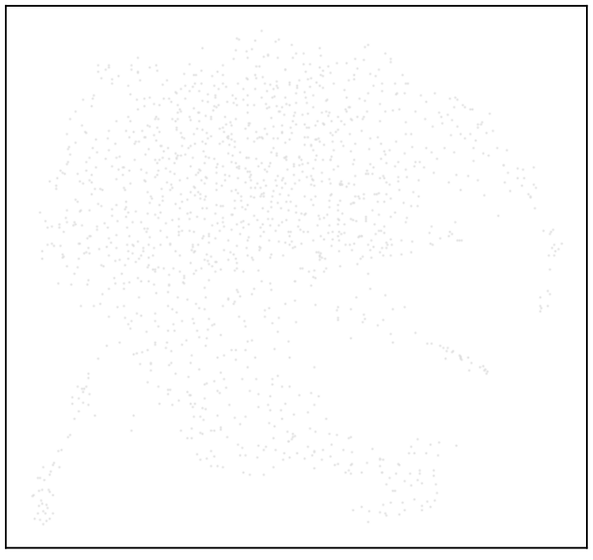
oblong: not defined



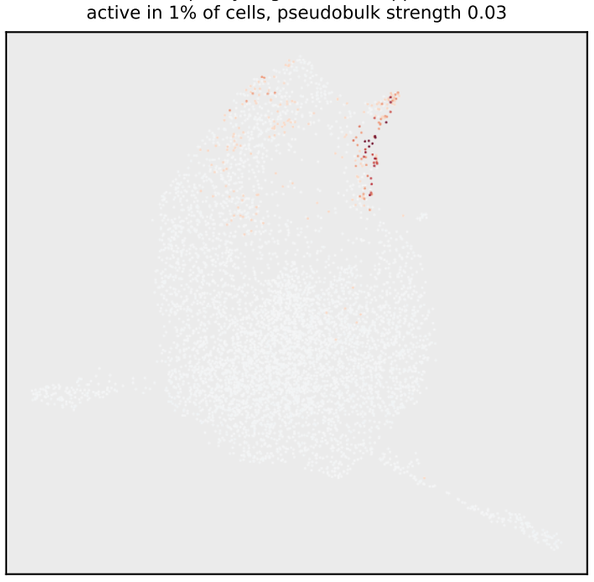
dome: not defined



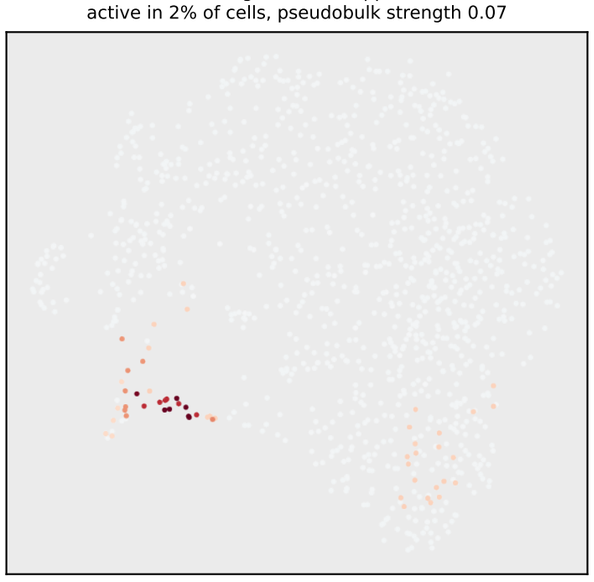
30% epiboly: not defined



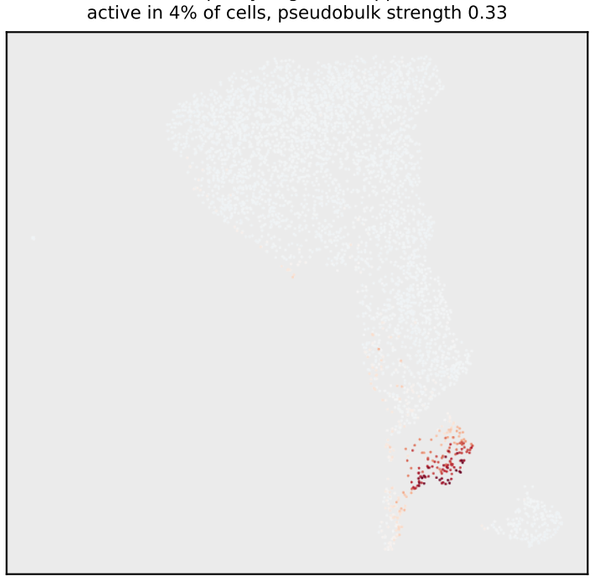
50% epiboly: 2 genes, not supported
active in 1% of cells, pseudobulk strength 0.03



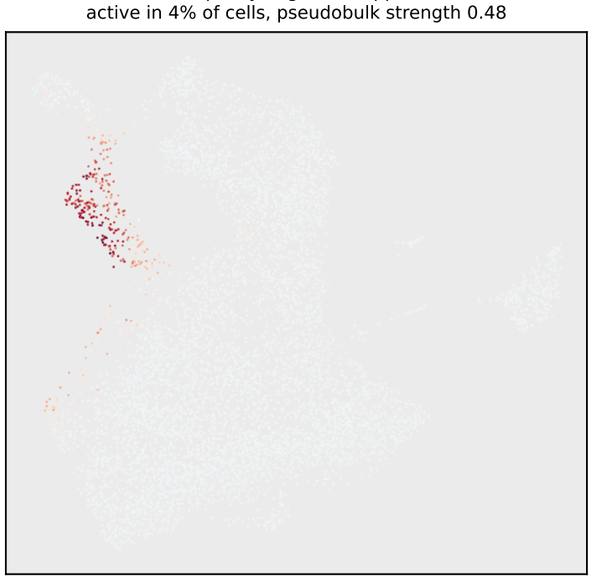
shield: 1 genes, not supported
active in 2% of cells, pseudobulk strength 0.07



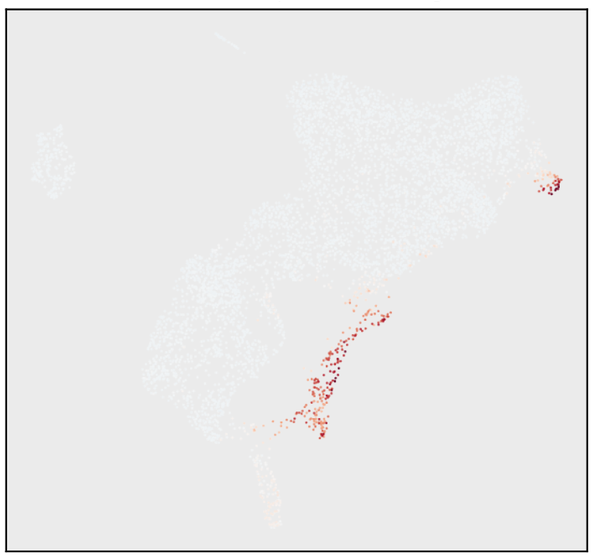
60% epiboly: 2 genes, supported
active in 4% of cells, pseudobulk strength 0.33



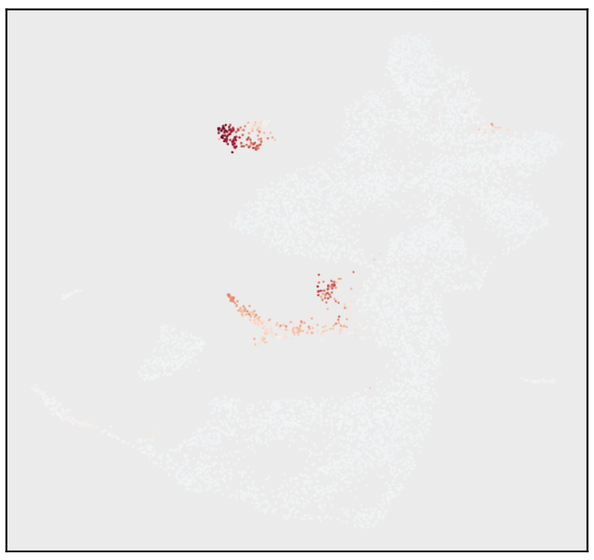
75% epiboly: 2 genes, supported
active in 4% of cells, pseudobulk strength 0.48



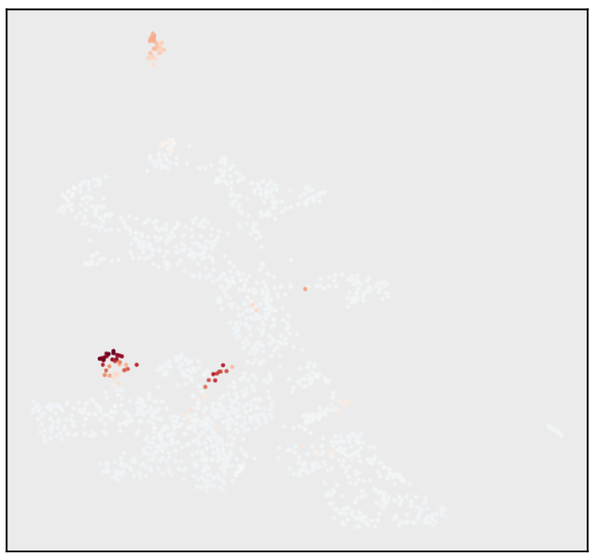
90% epiboly: 2 genes, supported
active in 4% of cells, pseudobulk strength 0.58



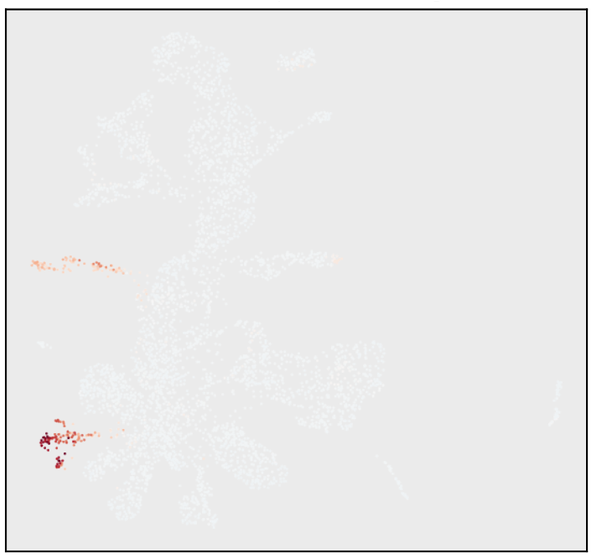
bud: 2 genes, supported
active in 5% of cells, pseudobulk strength 0.86



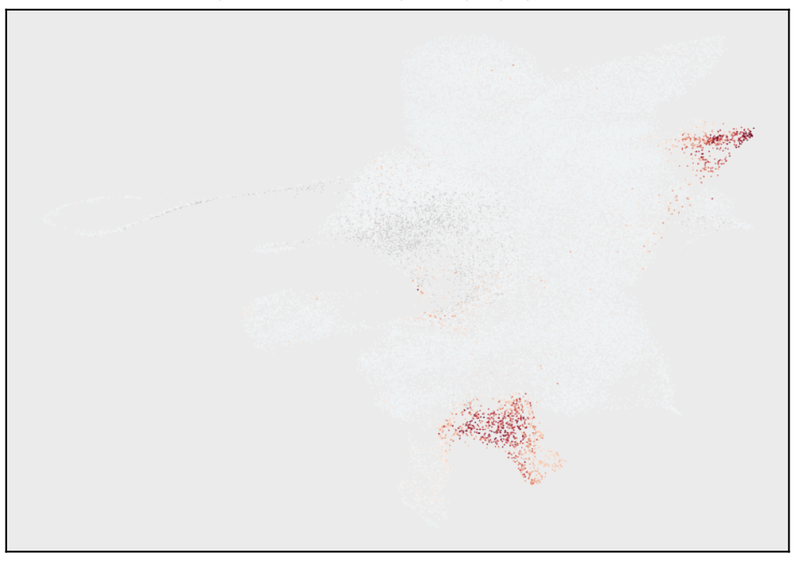
3-somite: 2 genes, supported
active in 4% of cells, pseudobulk strength 0.94



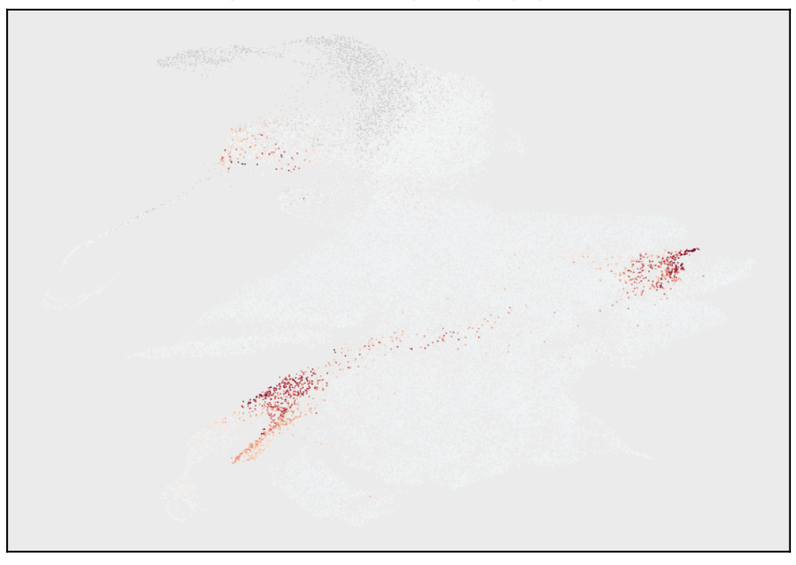
6-somite: 2 genes, supported
active in 4% of cells, pseudobulk strength 0.77



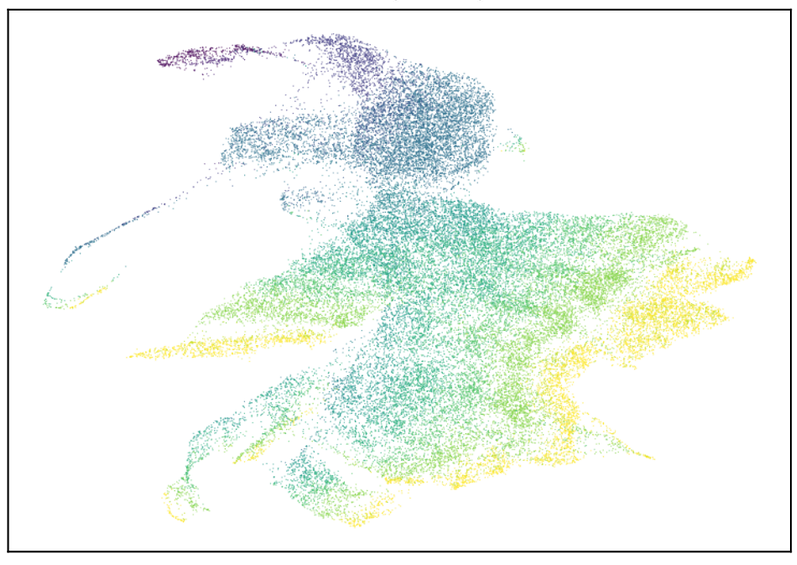
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



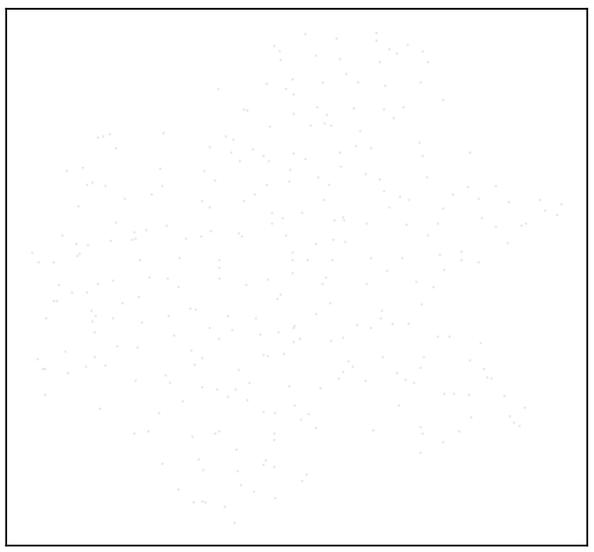
atlas, unaligned: stages



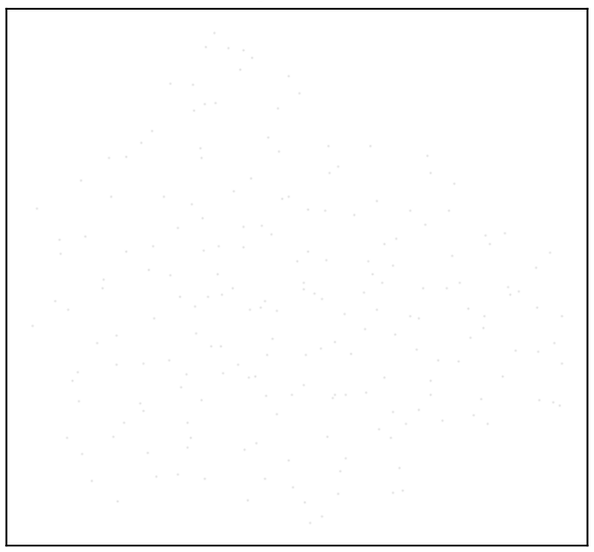
Genes (number of stages): shha (8), shhb (7)

sending:Notch

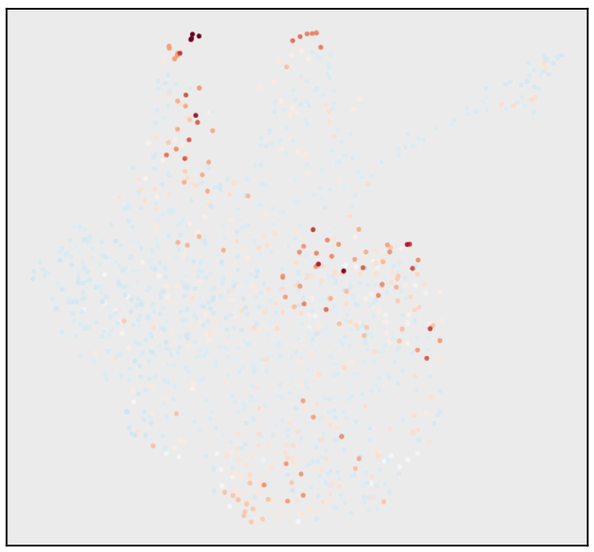
high: not defined



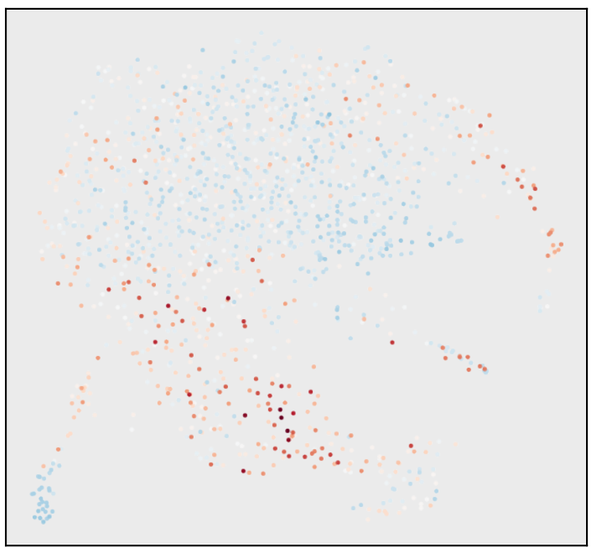
oblong: not defined



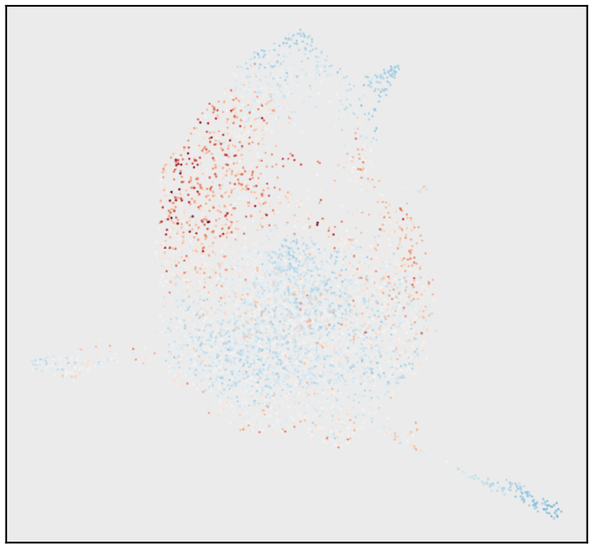
dome: 4 genes, not supported
active in 1% of cells, pseudobulk strength 0.06



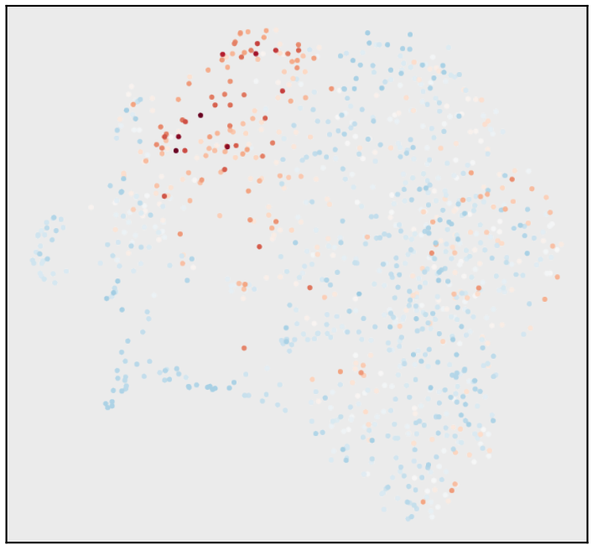
30% epiboly: 4 genes, not supported
active in 1% of cells, pseudobulk strength 0.35



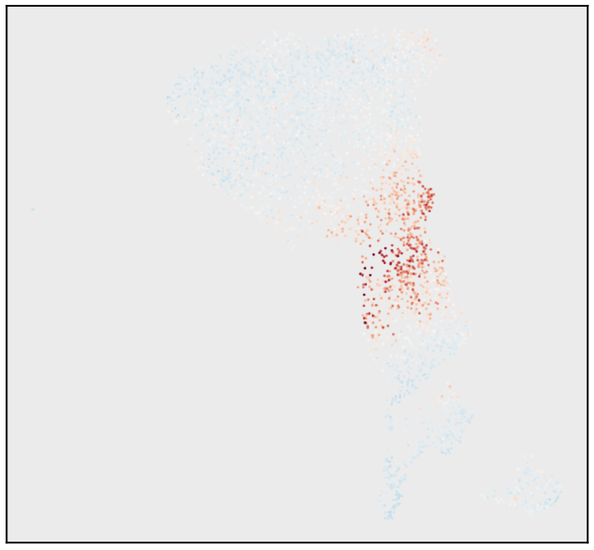
50% epiboly: 4 genes, not supported
active in 1% of cells, pseudobulk strength 0.41



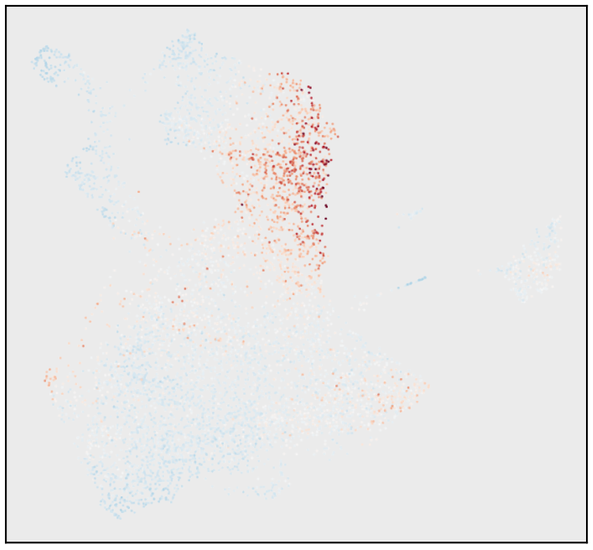
shield: 3 genes, not supported
active in 0% of cells, pseudobulk strength 0.32



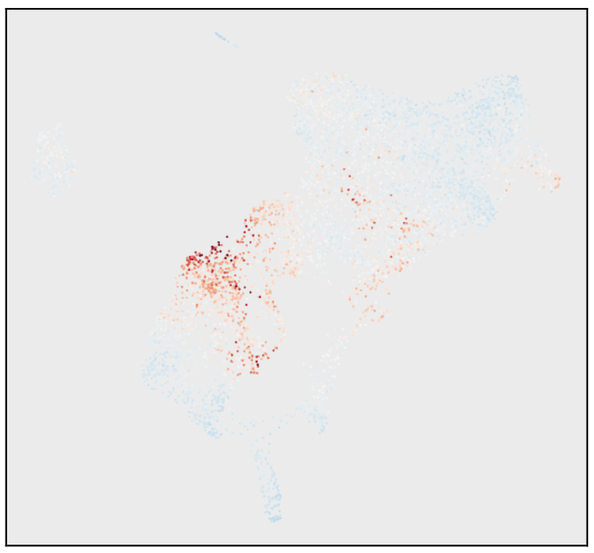
60% epiboly: 4 genes, not supported
active in 7% of cells, pseudobulk strength 0.56



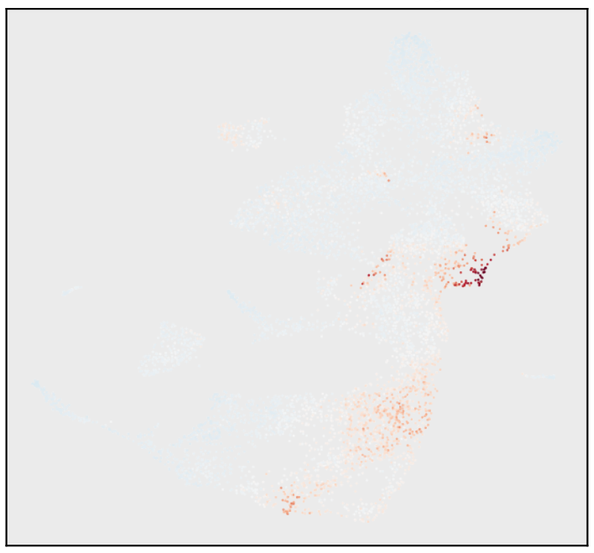
75% epiboly: 4 genes, not supported
active in 5% of cells, pseudobulk strength 0.59



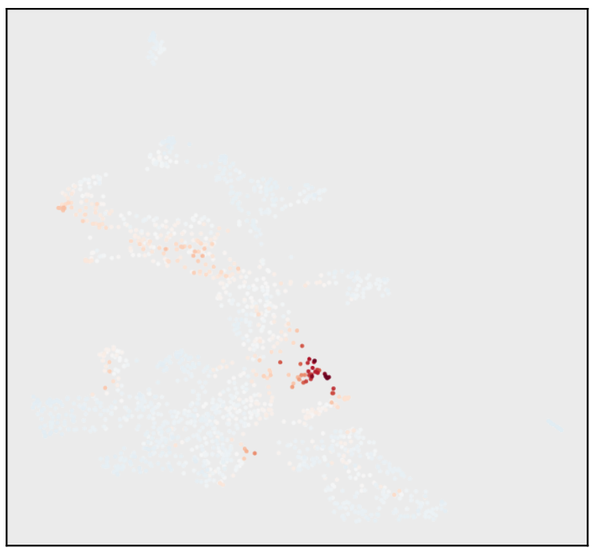
90% epiboly: 4 genes, not supported
active in 5% of cells, pseudobulk strength 0.64



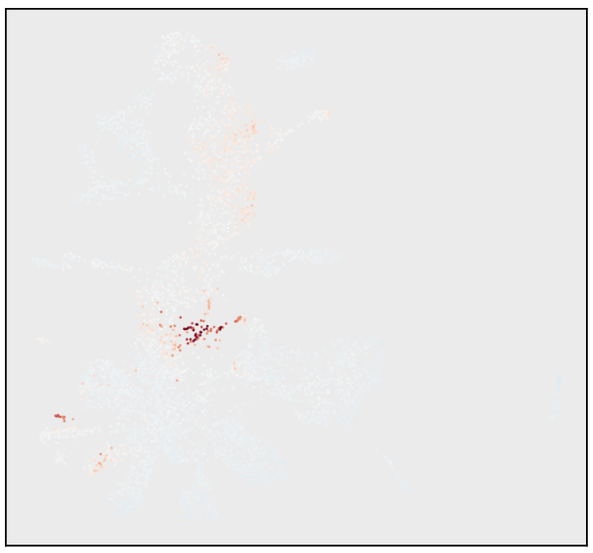
bud: 4 genes, supported
active in 6% of cells, pseudobulk strength 0.88



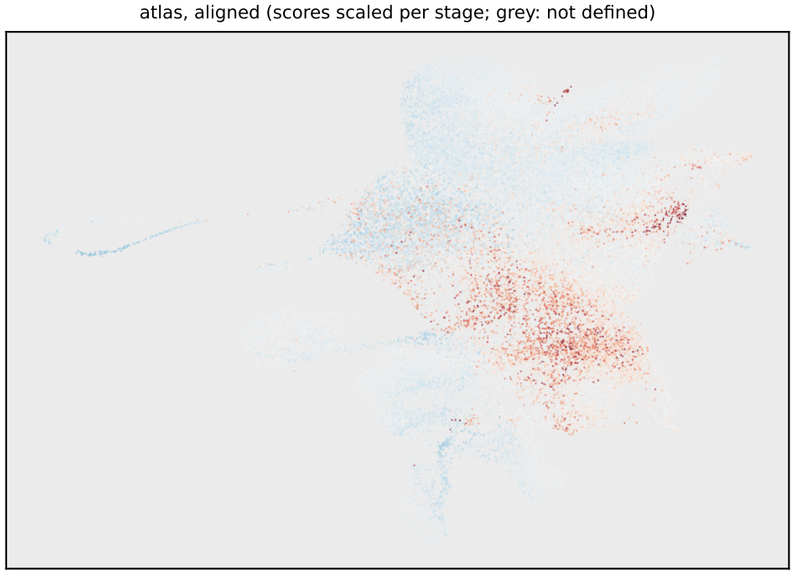
3-somite: 4 genes, supported
active in 4% of cells, pseudobulk strength 0.69



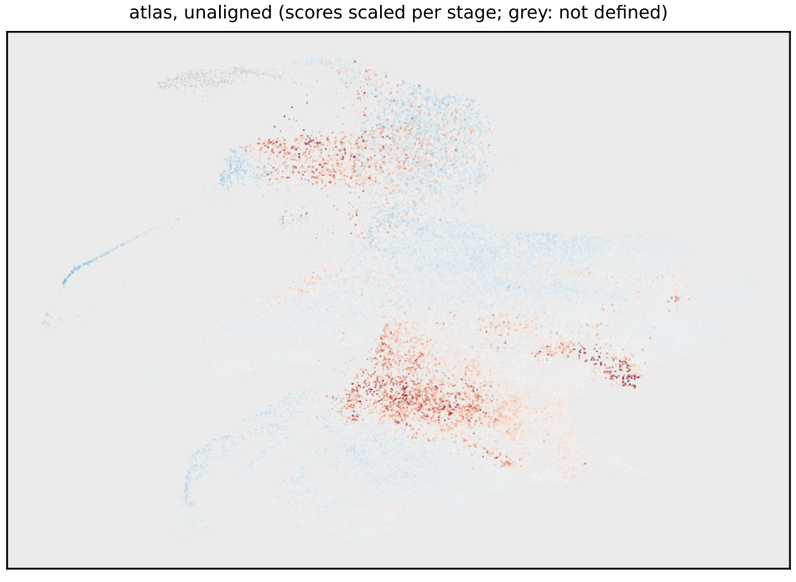
6-somite: 4 genes, supported
active in 4% of cells, pseudobulk strength 0.75



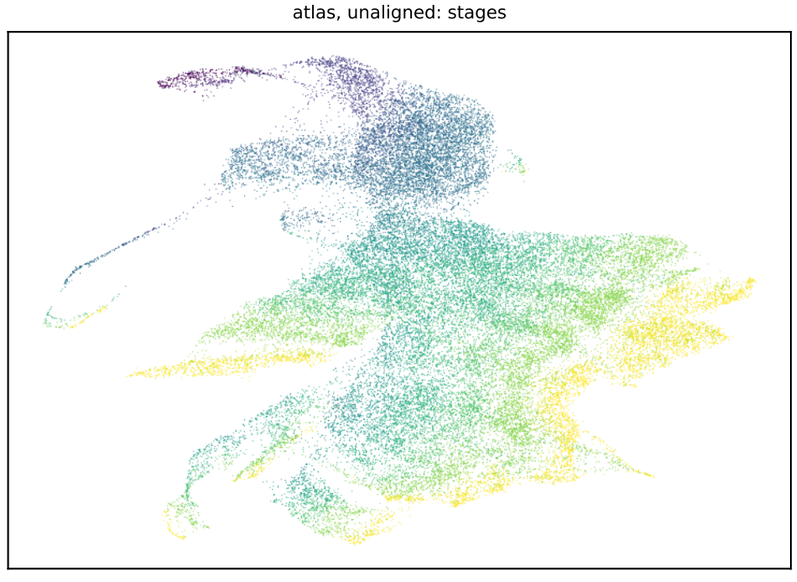
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



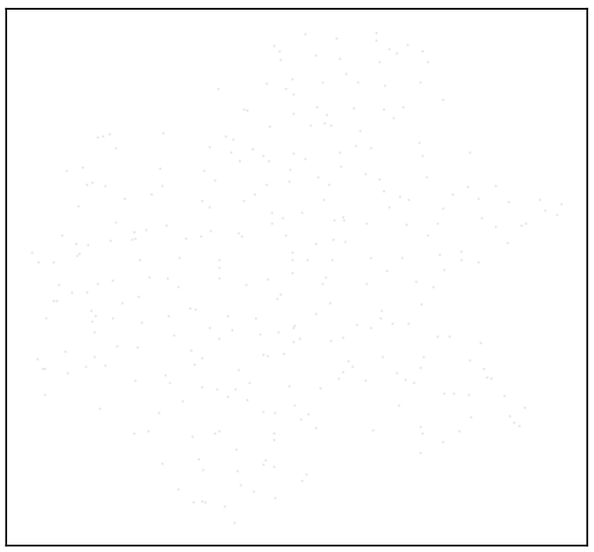
atlas, unaligned: stages



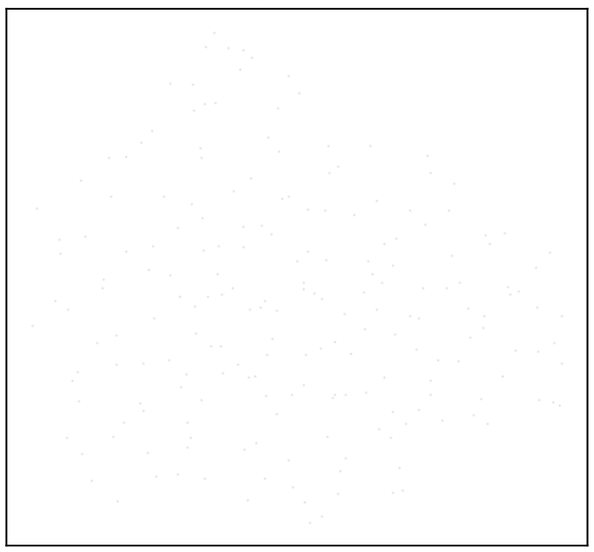
Genes (number of stages): dla (10), dlc (10), dld (10), dlb (9)

sending:RA

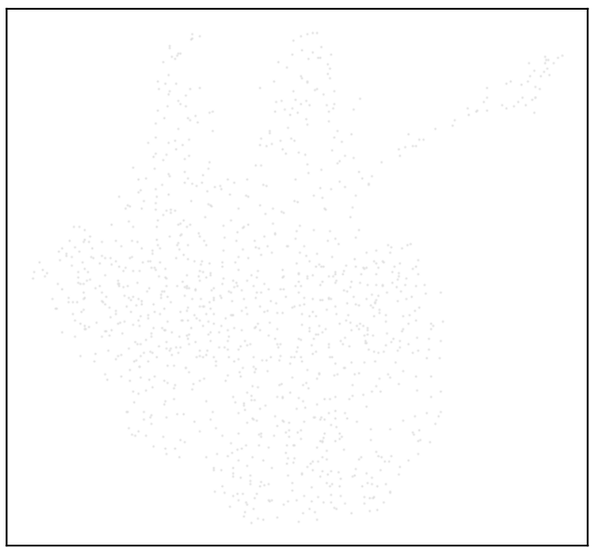
high: not defined



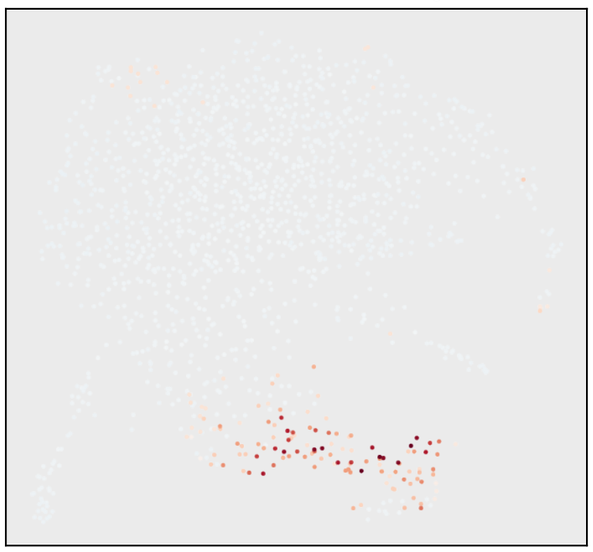
oblong: not defined



dome: not defined

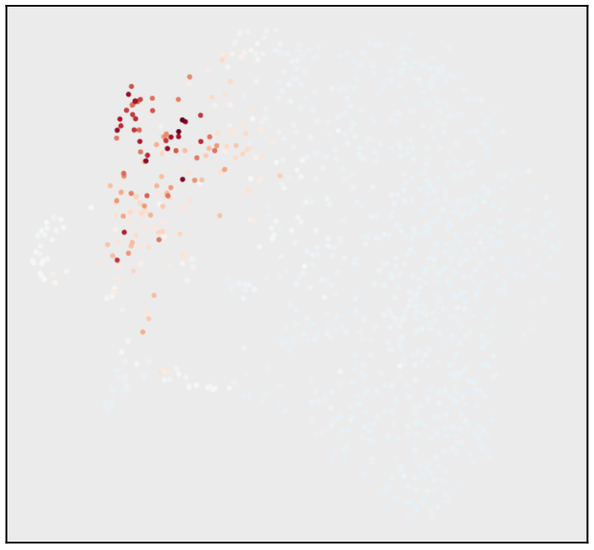


30% epiboly: 1 genes, not supported
active in 1% of cells, pseudobulk strength 0.07

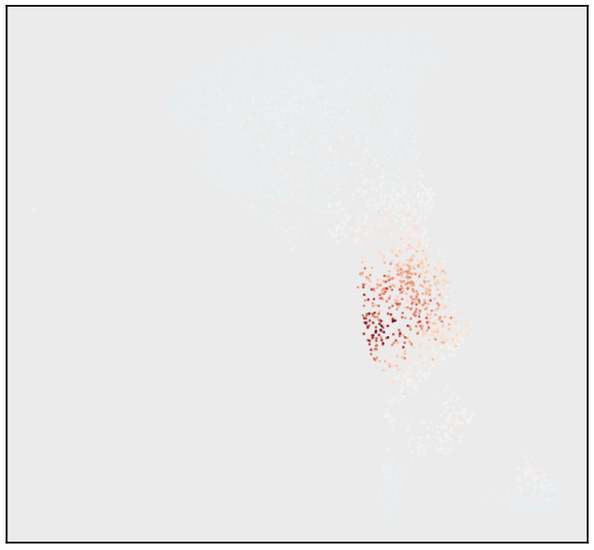


50% epiboly: 1 genes, not supported
active in 4% of cells, pseudobulk strength 0.23

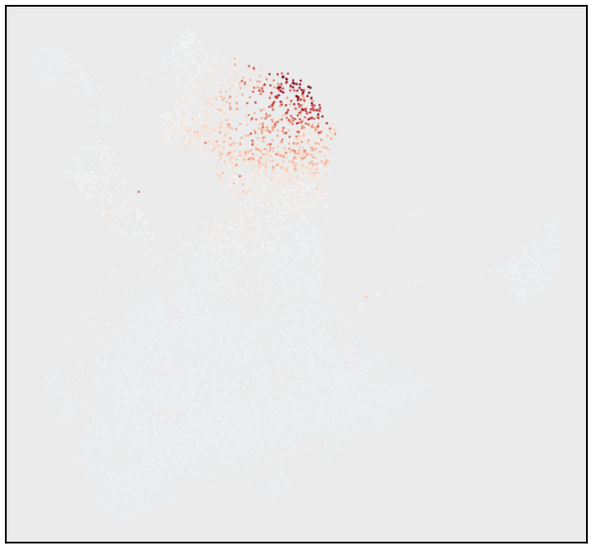
shield: 1 genes, not supported
active in 5% of cells, pseudobulk strength 0.64



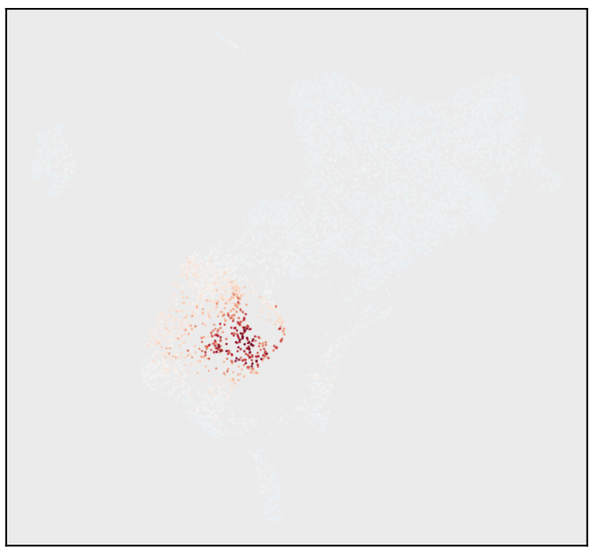
60% epiboly: 1 genes, not supported
active in 7% of cells, pseudobulk strength 1.00



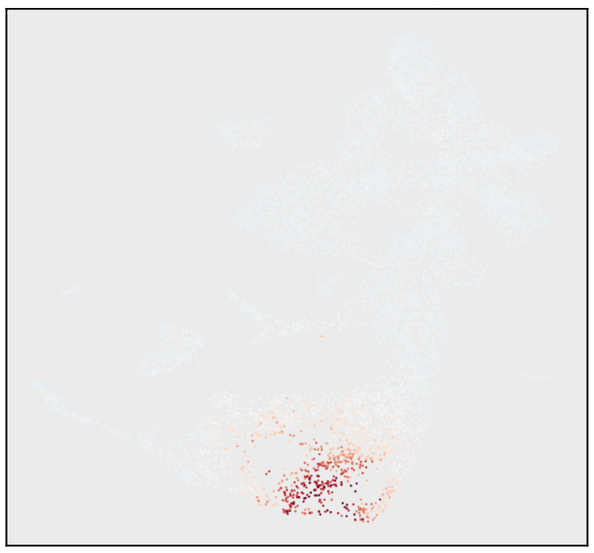
75% epiboly: 1 genes, not supported
active in 7% of cells, pseudobulk strength 0.90



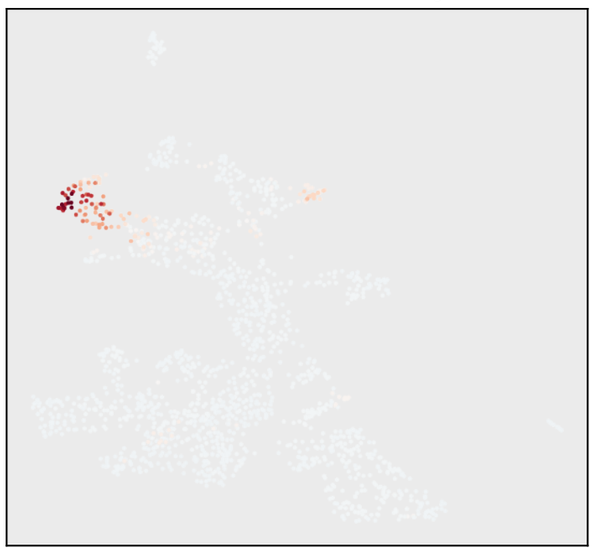
90% epiboly: 1 genes, not supported
active in 5% of cells, pseudobulk strength 0.80



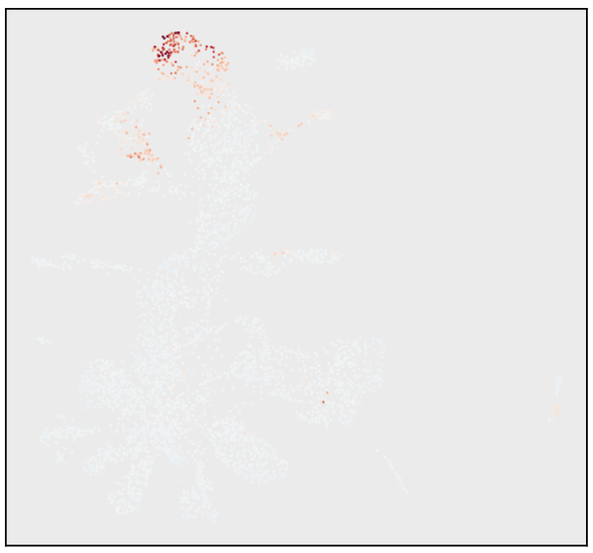
bud: 1 genes, not supported
active in 5% of cells, pseudobulk strength 0.84



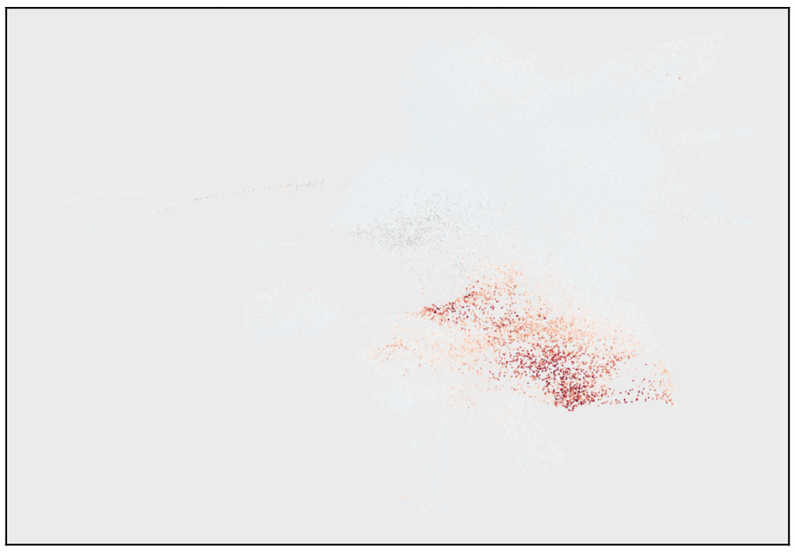
3-somite: 1 genes, not supported
active in 3% of cells, pseudobulk strength 0.18



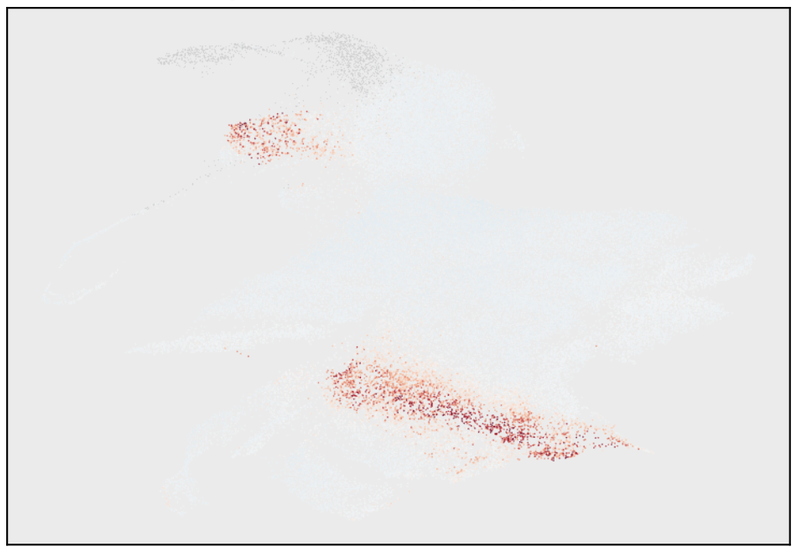
6-somite: 1 genes, not supported
active in 3% of cells, pseudobulk strength 0.18



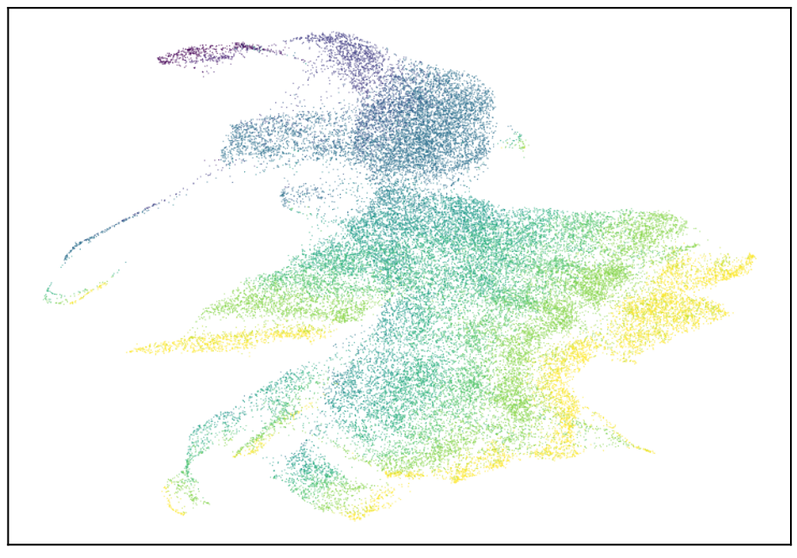
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



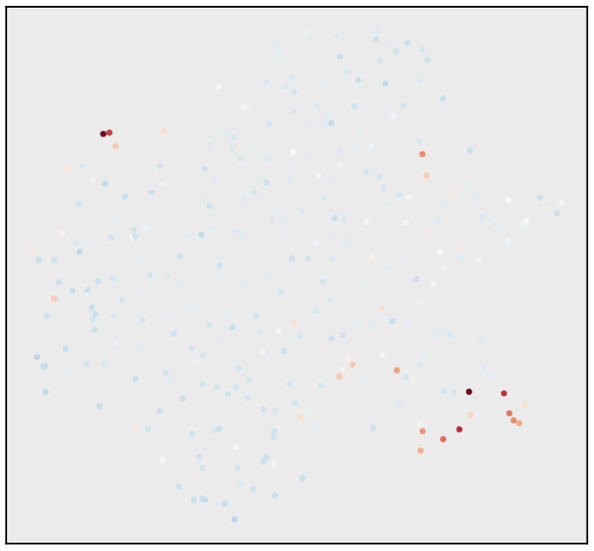
atlas, unaligned: stages



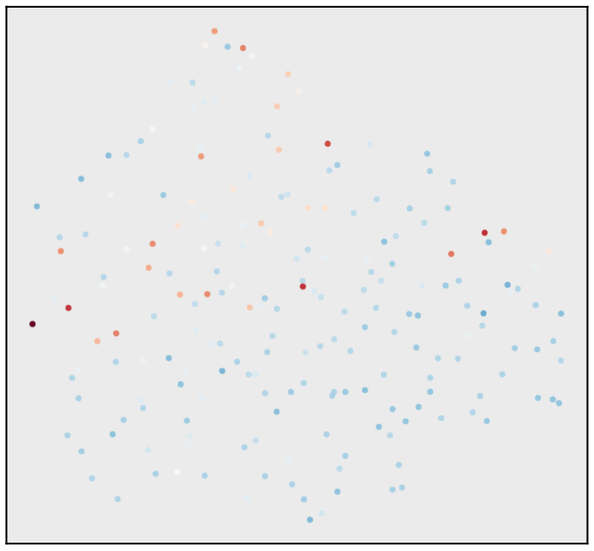
Genes (number of stages): aldh1a2 (9)

sending:Wnt/PCP

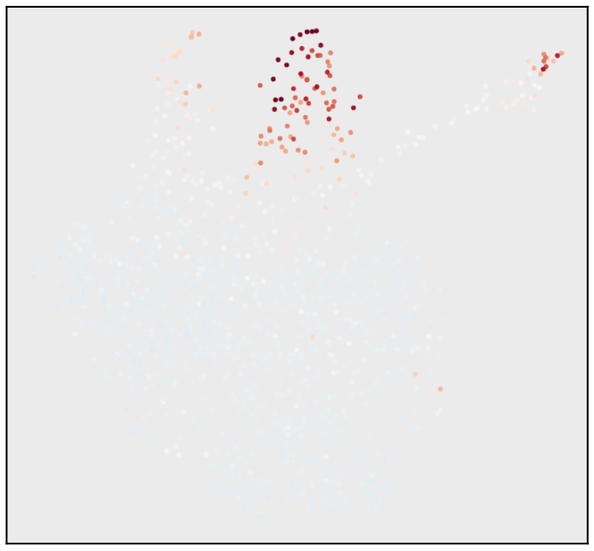
high: 1 genes, not supported
active in 0% of cells, pseudobulk strength 0.06



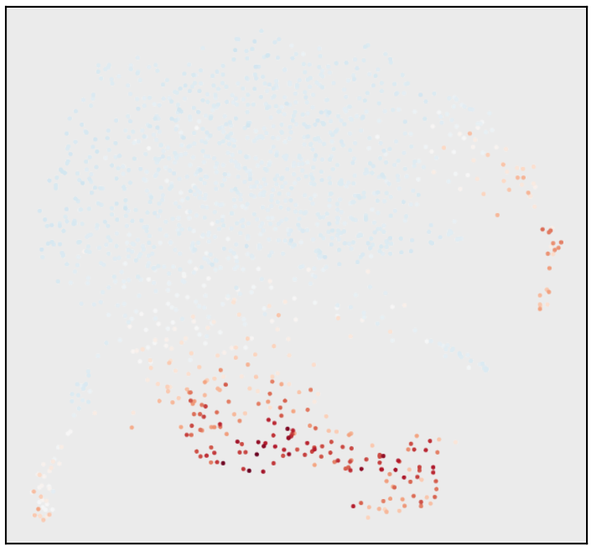
oblong: 1 genes, not supported
active in 1% of cells, pseudobulk strength 0.09



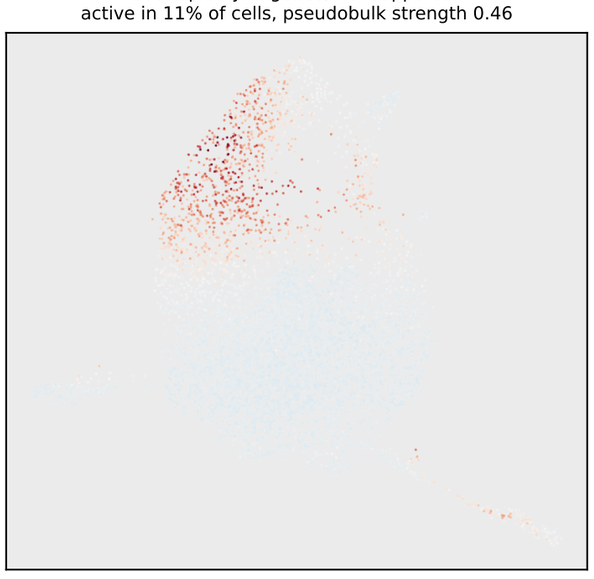
dome: 1 genes, not supported
active in 7% of cells, pseudobulk strength 0.28



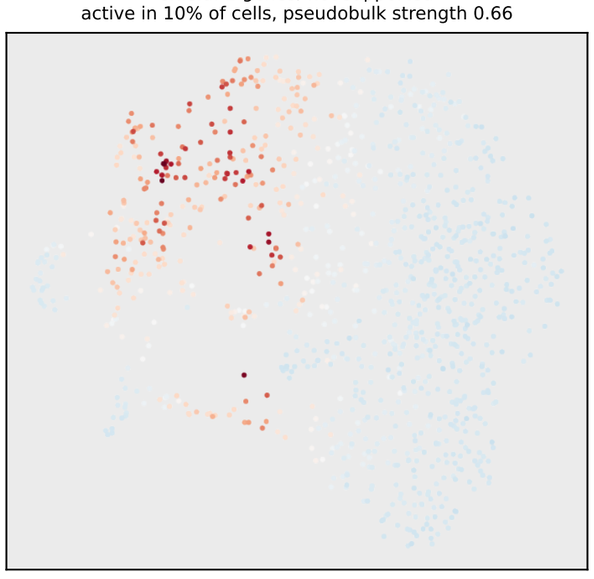
30% epiboly: 2 genes, not supported
active in 10% of cells, pseudobulk strength 0.57



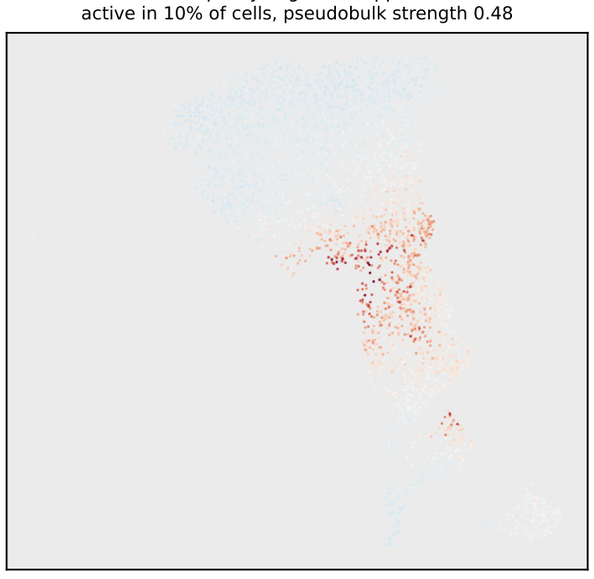
50% epiboly: 2 genes, not supported
active in 11% of cells, pseudobulk strength 0.46



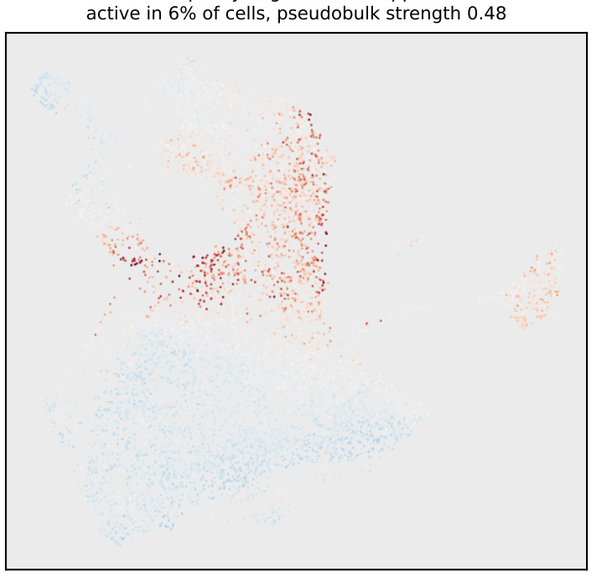
shield: 2 genes, not supported
active in 10% of cells, pseudobulk strength 0.66



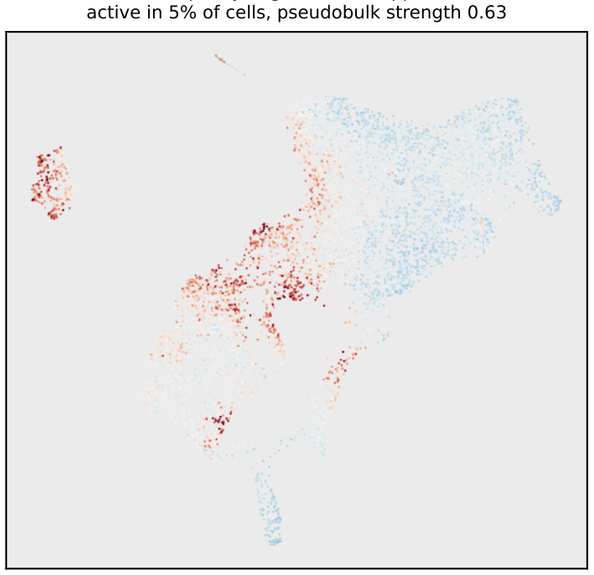
60% epiboly: 2 genes, supported
active in 10% of cells, pseudobulk strength 0.48



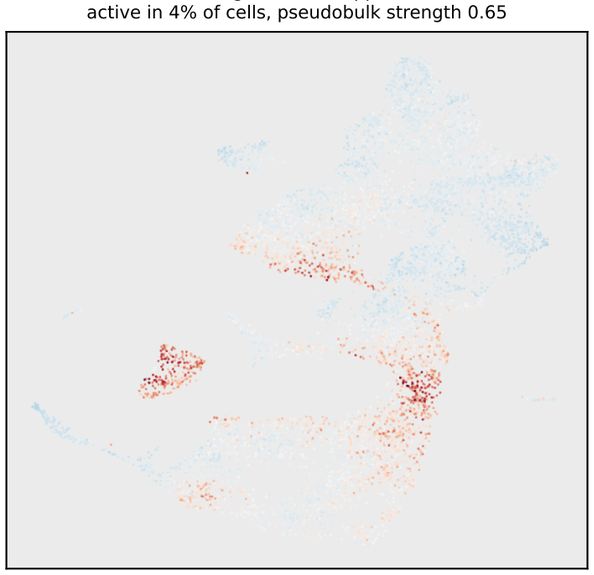
75% epiboly: 2 genes, not supported
active in 6% of cells, pseudobulk strength 0.48



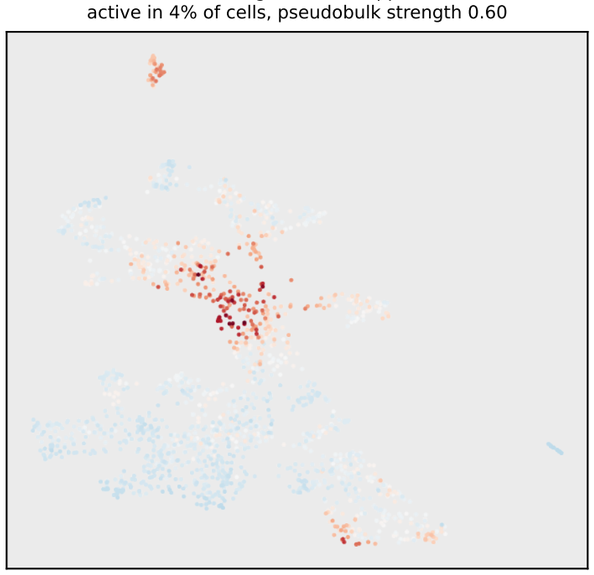
90% epiboly: 2 genes, not supported
active in 5% of cells, pseudobulk strength 0.63



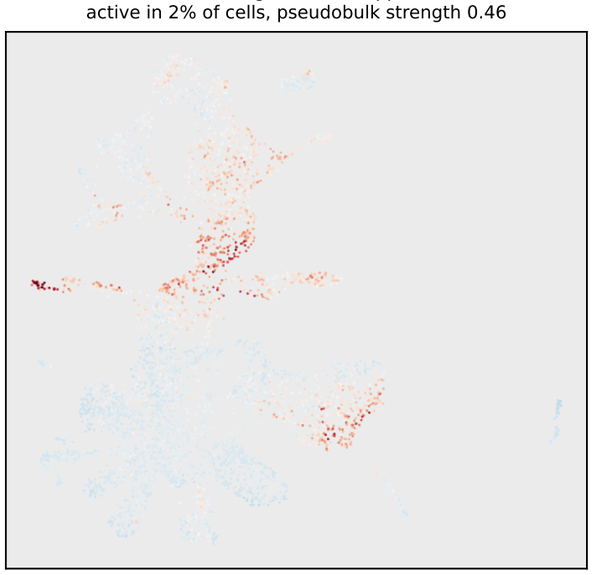
bud: 2 genes, not supported
active in 4% of cells, pseudobulk strength 0.65



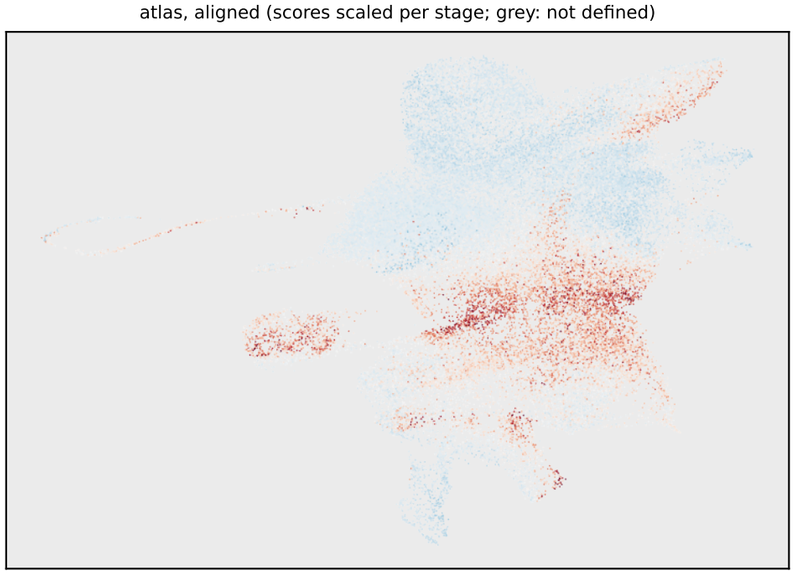
3-somite: 2 genes, not supported
active in 4% of cells, pseudobulk strength 0.60



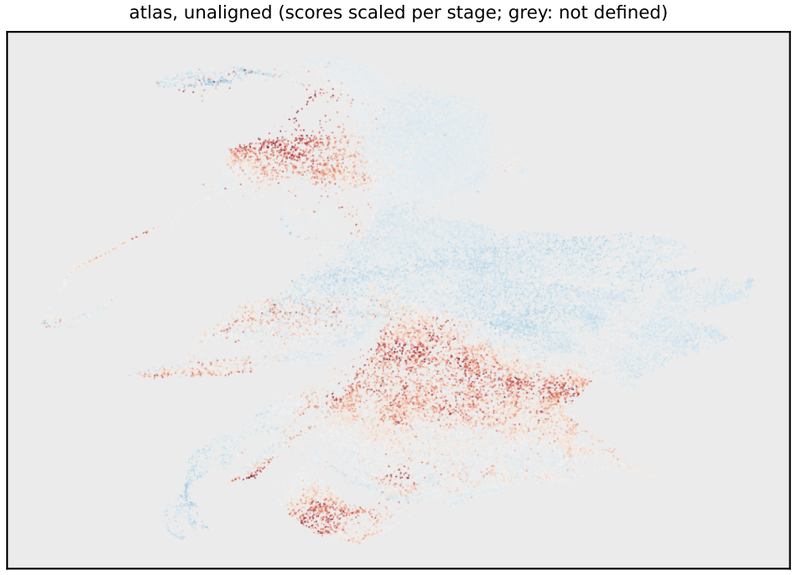
6-somite: 2 genes, not supported
active in 2% of cells, pseudobulk strength 0.46



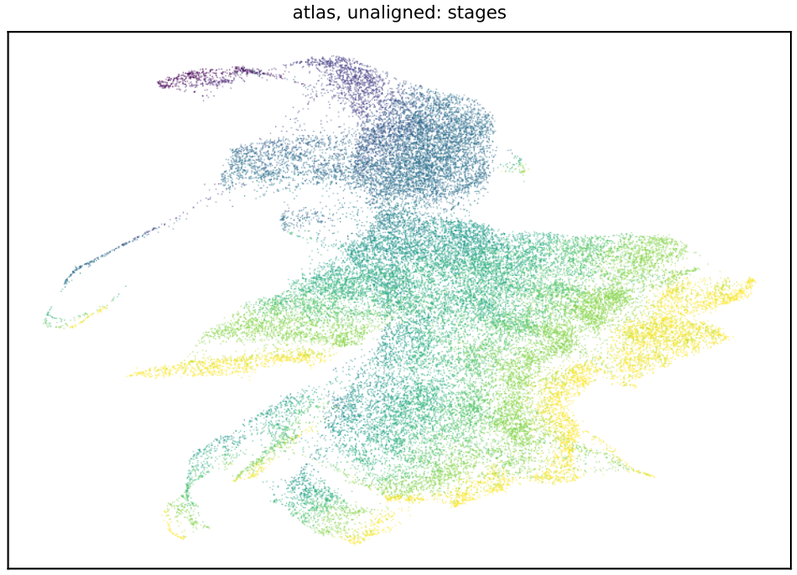
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



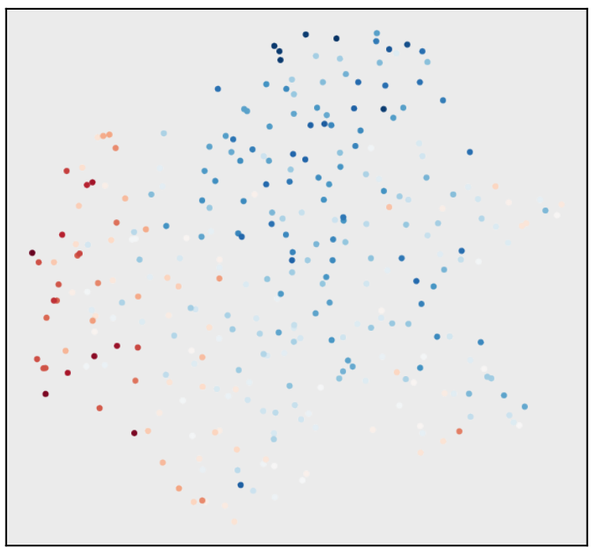
atlas, unaligned: stages



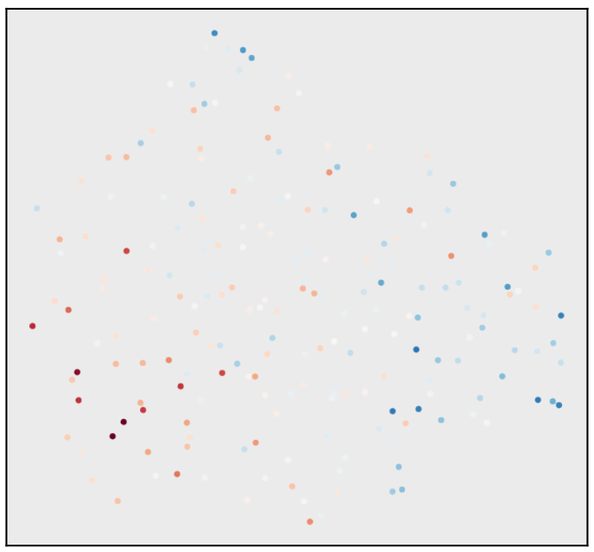
Genes (number of stages): wnt11 (12), wnt5b (9)

signal:BMP

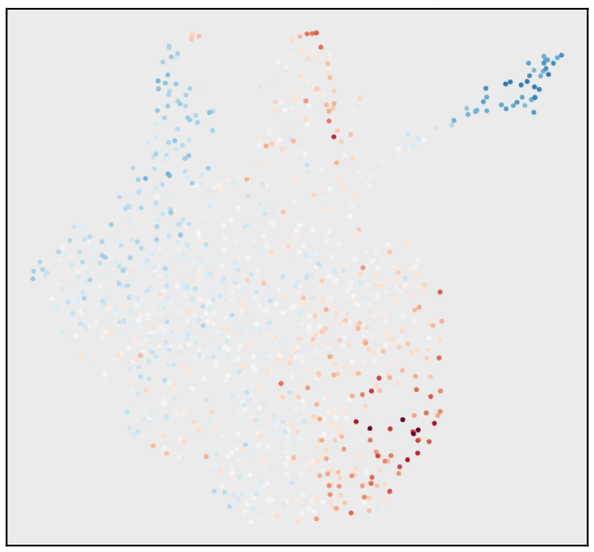
high: 6 genes, not supported
active in 0% of cells, pseudobulk strength 0.13



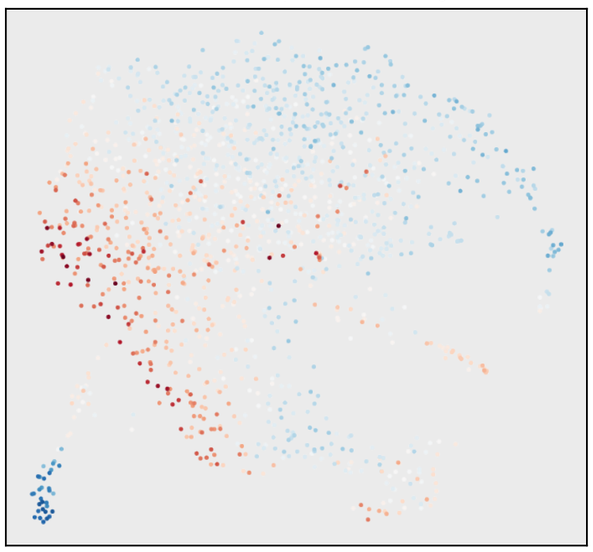
oblong: 5 genes, not supported
active in 2% of cells, pseudobulk strength 0.25



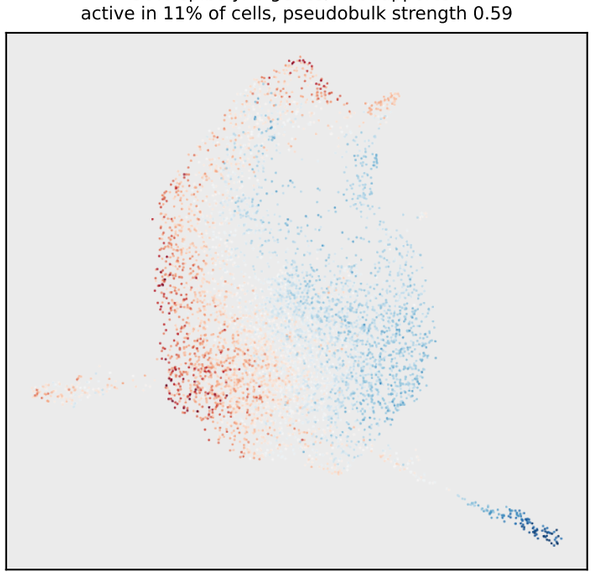
dome: 8 genes, not supported
active in 3% of cells, pseudobulk strength 0.43



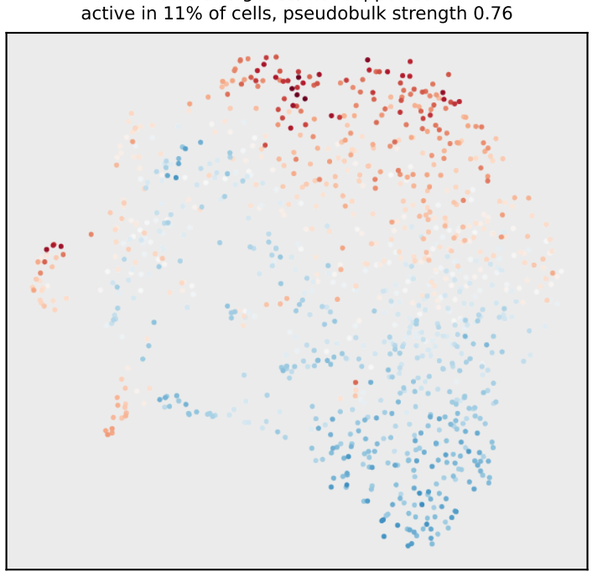
30% epiboly: 8 genes, not supported
active in 10% of cells, pseudobulk strength 0.68



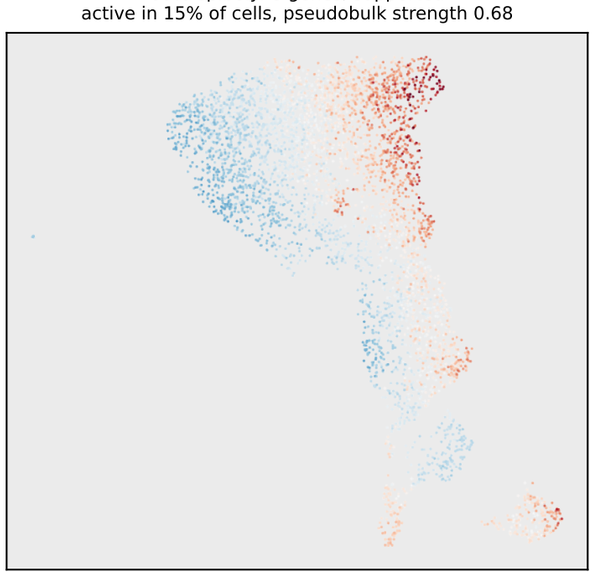
50% epiboly: 8 genes, not supported
active in 11% of cells, pseudobulk strength 0.59



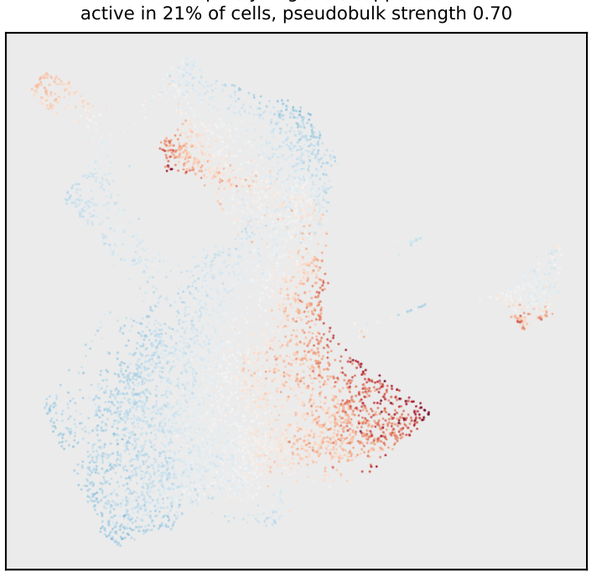
shield: 8 genes, not supported
active in 11% of cells, pseudobulk strength 0.76



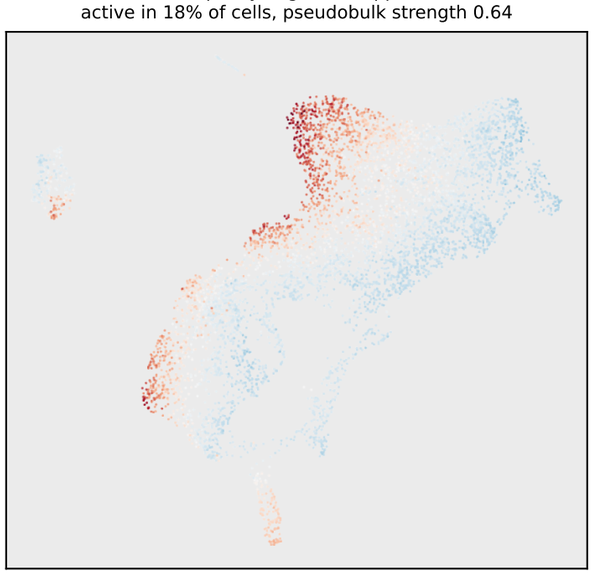
60% epiboly: 8 genes, supported
active in 15% of cells, pseudobulk strength 0.68



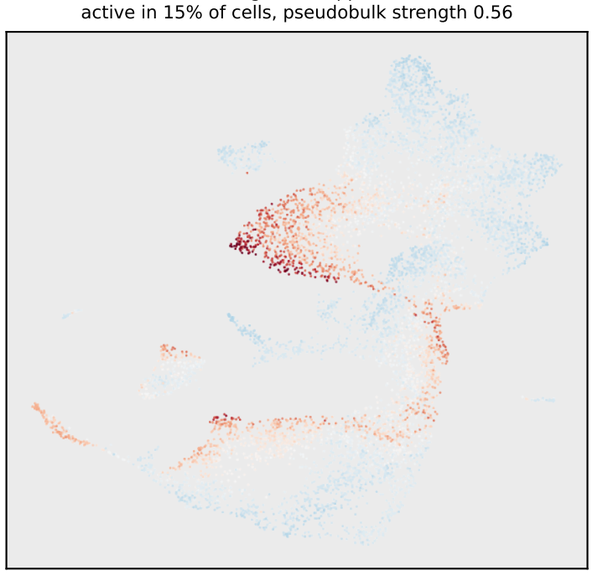
75% epiboly: 8 genes, supported
active in 21% of cells, pseudobulk strength 0.70



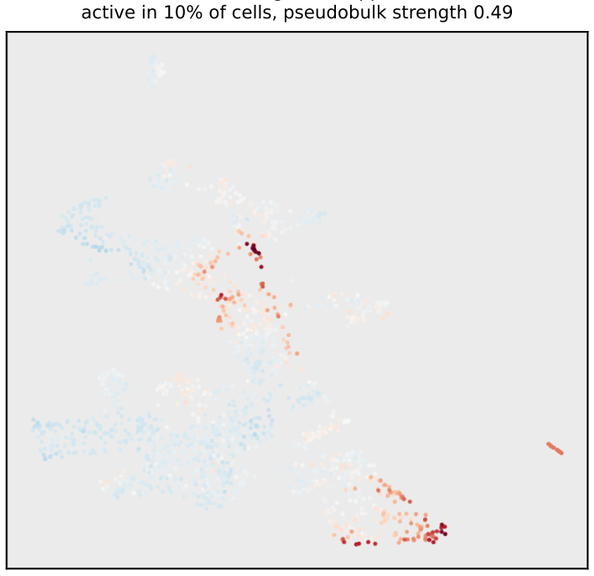
90% epiboly: 8 genes, supported
active in 18% of cells, pseudobulk strength 0.64



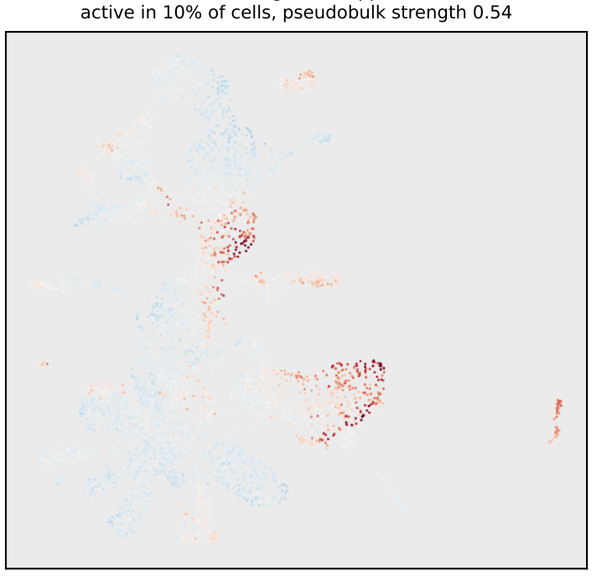
bud: 8 genes, supported
active in 15% of cells, pseudobulk strength 0.56



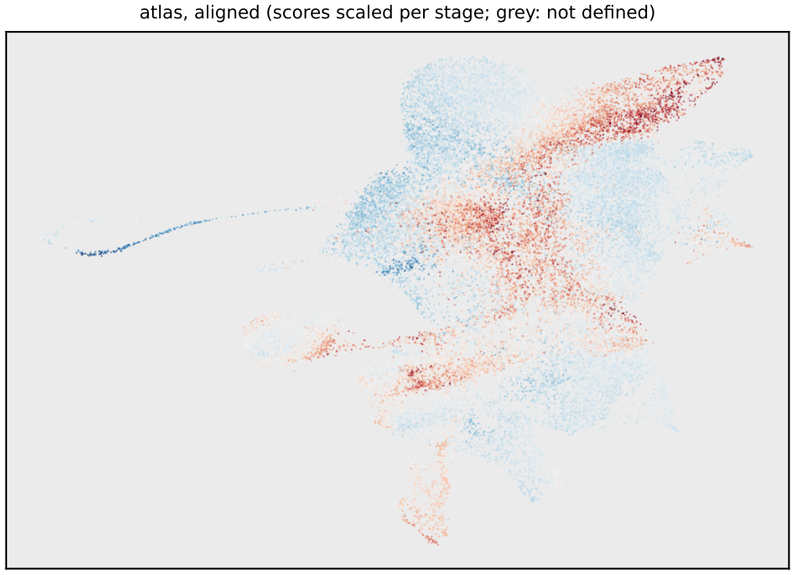
3-somite: 8 genes, supported
active in 10% of cells, pseudobulk strength 0.49



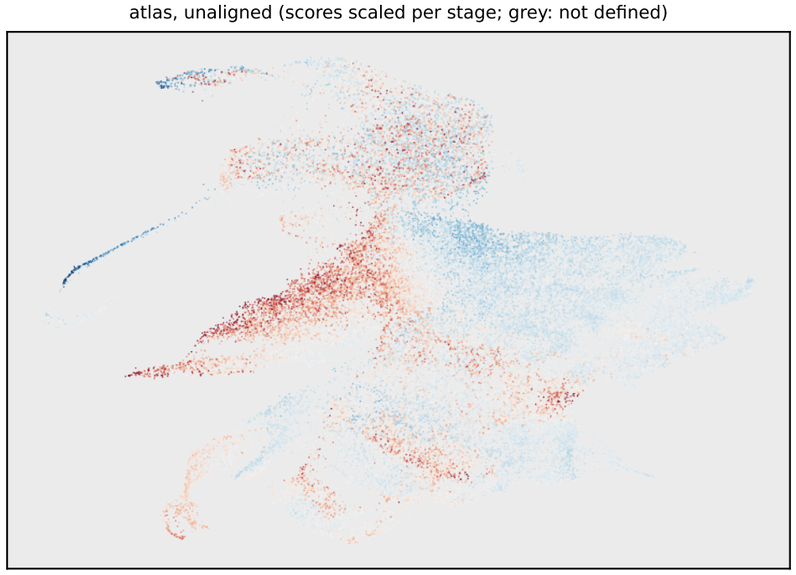
6-somite: 8 genes, supported
active in 10% of cells, pseudobulk strength 0.54



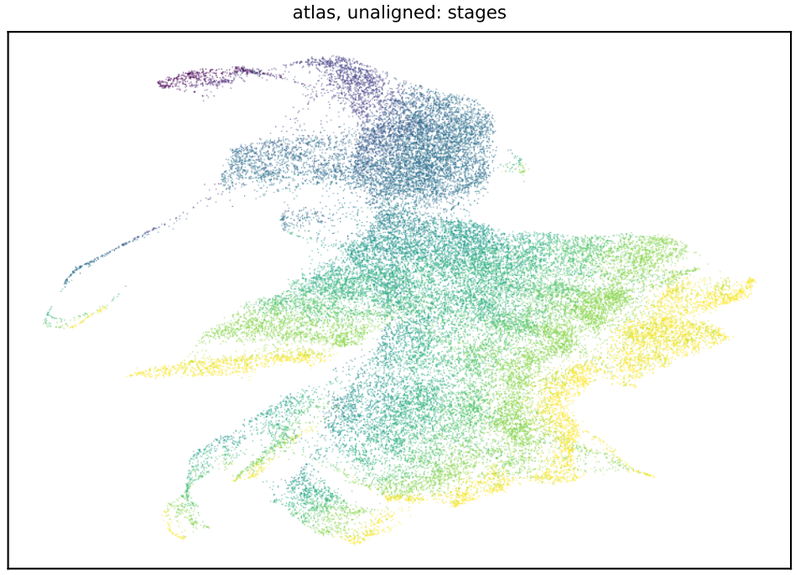
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



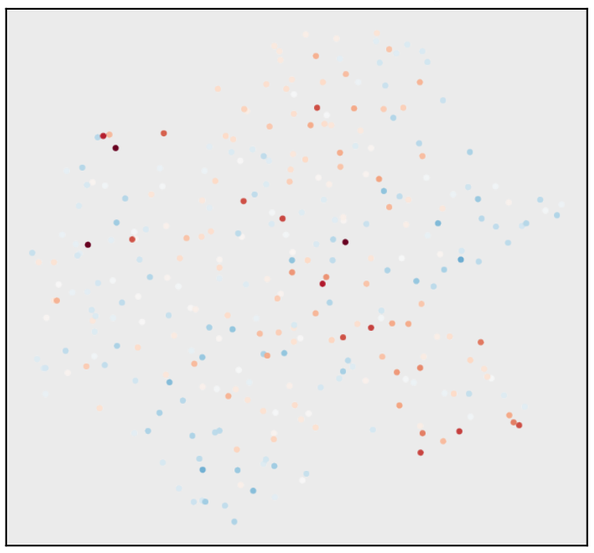
atlas, unaligned: stages



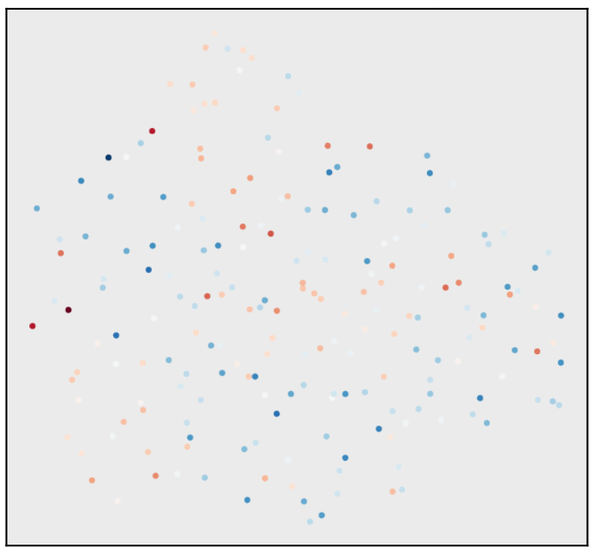
Genes (number of stages): id1 (12), id3 (12), bambia (12), bmp2b (12), bmp7a (12), msx1b (11), szl (10), bmp4 (10)

signal:FGF

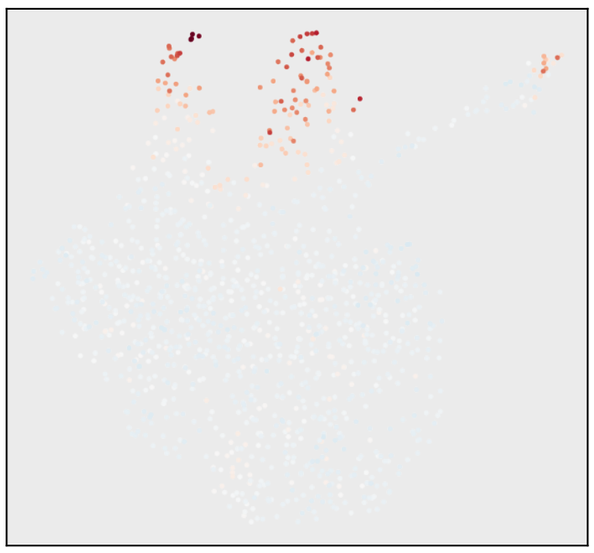
high: 4 genes, not supported
active in 0% of cells, pseudobulk strength 0.20



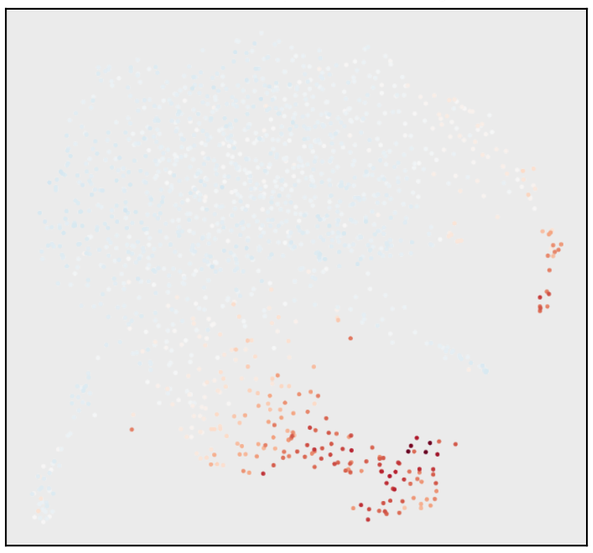
oblong: 3 genes, not supported
active in 0% of cells, pseudobulk strength 0.24



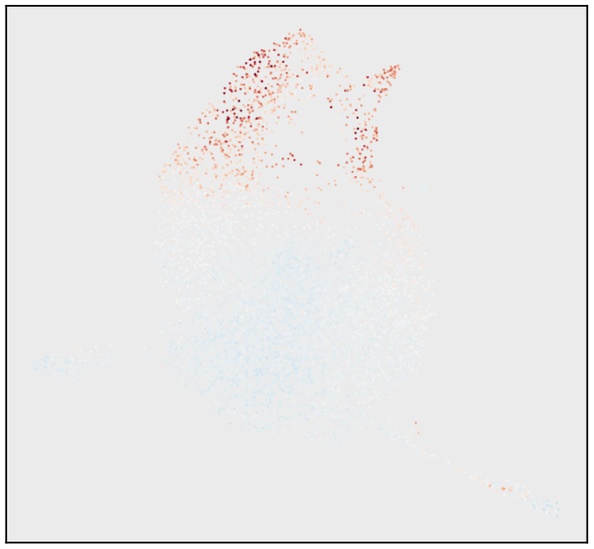
dome: 6 genes, not supported
active in 6% of cells, pseudobulk strength 0.43



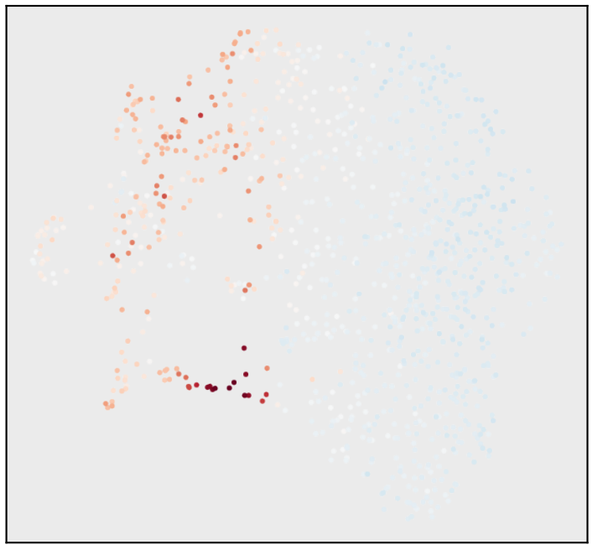
30% epiboly: 6 genes, not supported
active in 10% of cells, pseudobulk strength 0.68



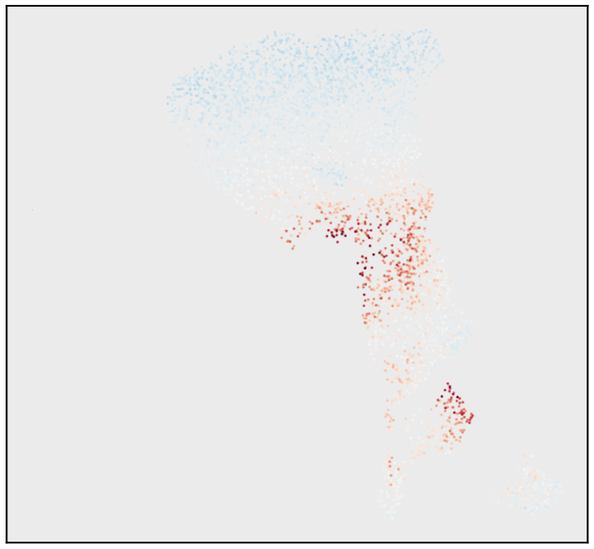
50% epiboly: 6 genes, not supported
active in 10% of cells, pseudobulk strength 0.51



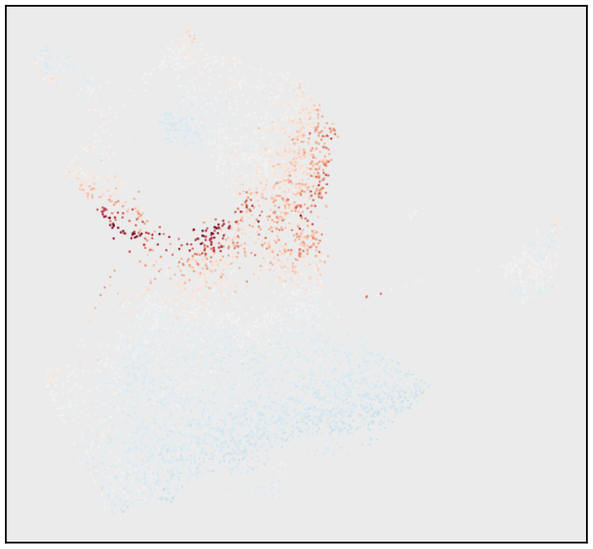
shield: 5 genes, not supported
active in 7% of cells, pseudobulk strength 0.57



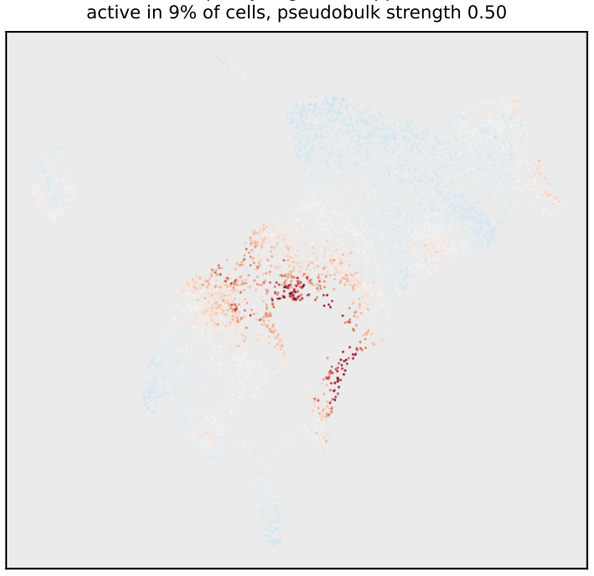
60% epiboly: 6 genes, supported
active in 11% of cells, pseudobulk strength 0.58



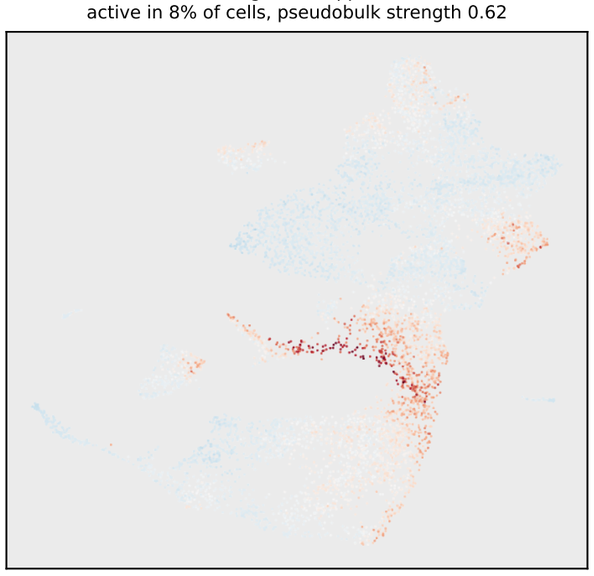
75% epiboly: 5 genes, supported
active in 9% of cells, pseudobulk strength 0.51



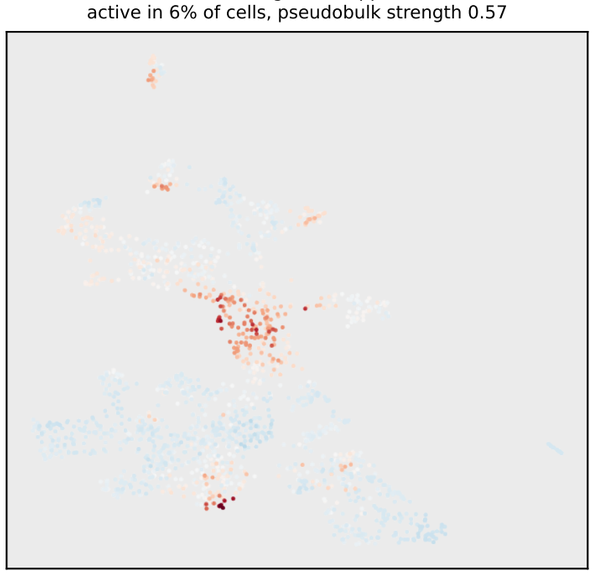
90% epiboly: 5 genes, supported
active in 9% of cells, pseudobulk strength 0.50



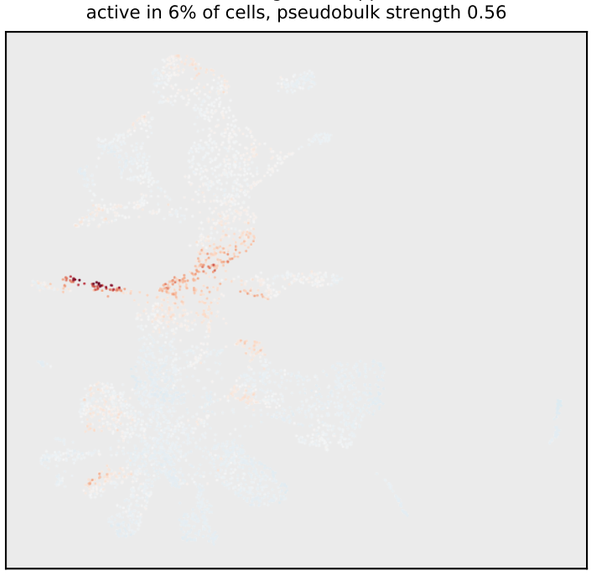
bud: 6 genes, supported
active in 8% of cells, pseudobulk strength 0.62



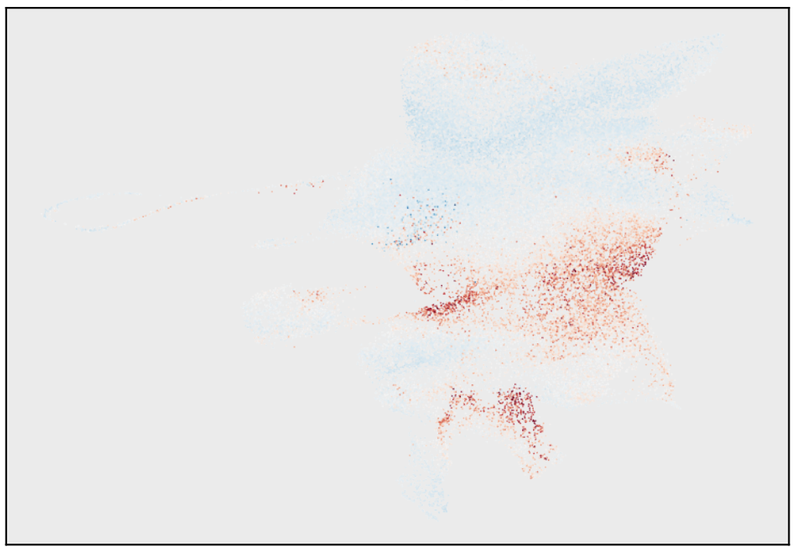
3-somite: 6 genes, supported
active in 6% of cells, pseudobulk strength 0.57



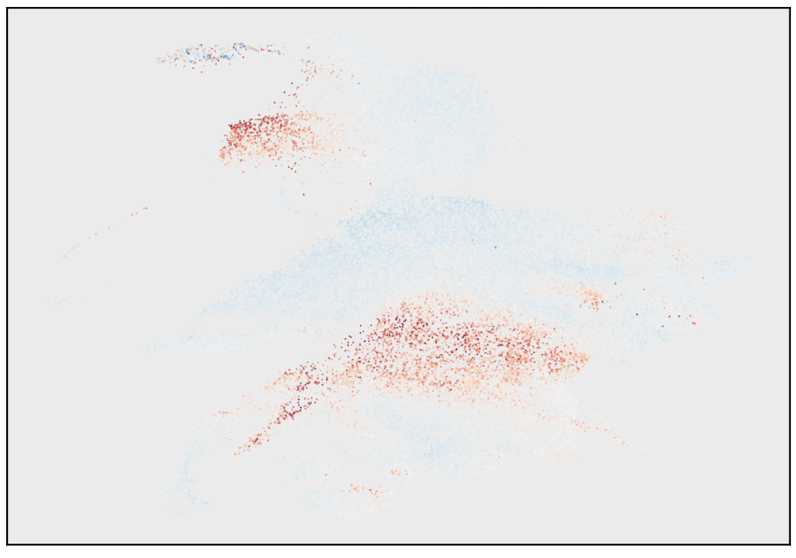
6-somite: 6 genes, supported
active in 6% of cells, pseudobulk strength 0.56



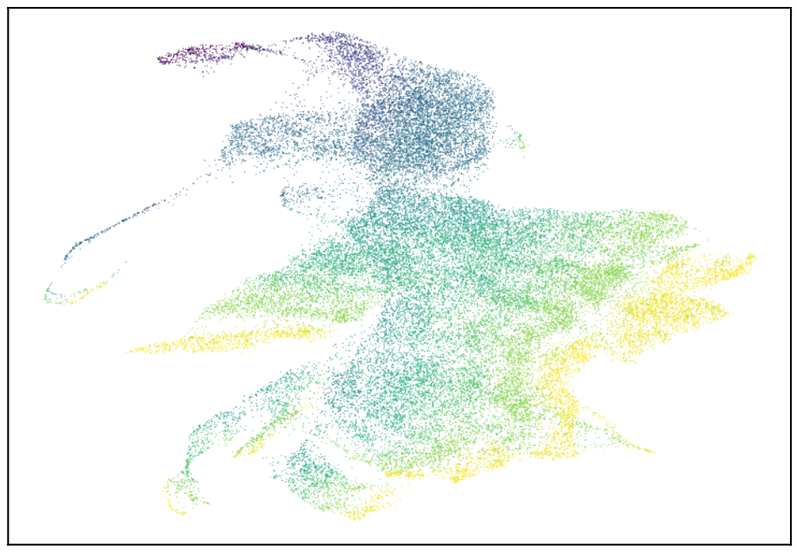
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



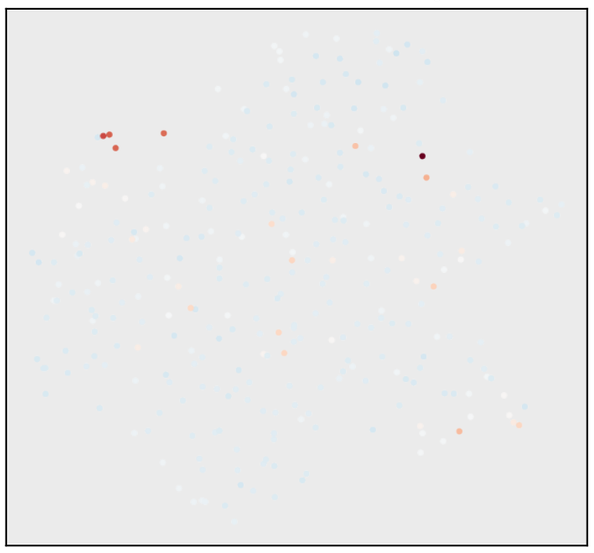
atlas, unaligned: stages



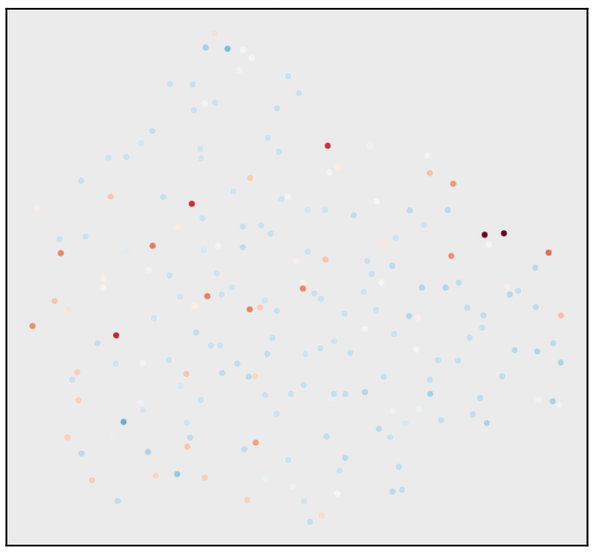
Genes (number of stages): etv4 (12), il17rd (12), fgf8a (12), fgf3 (11), fgf24 (10), fgf4 (7)

signal:Nodal

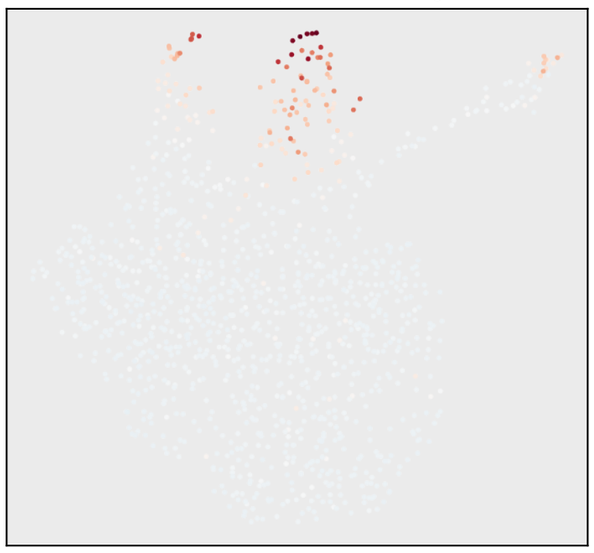
high: 3 genes, not supported
active in 1% of cells, pseudobulk strength 0.18



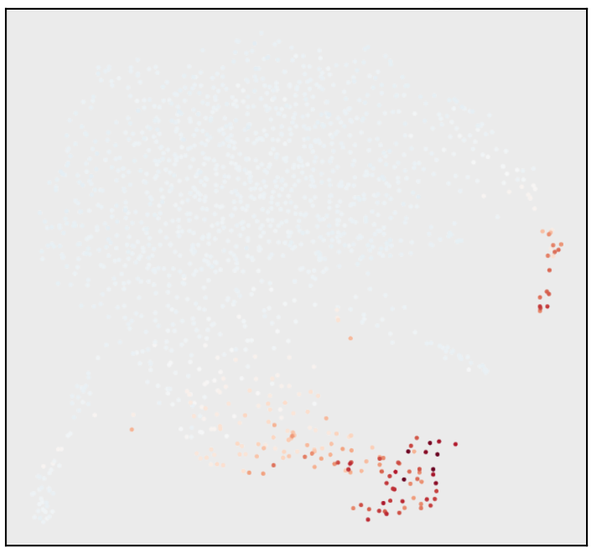
oblong: 3 genes, not supported
active in 0% of cells, pseudobulk strength 0.25



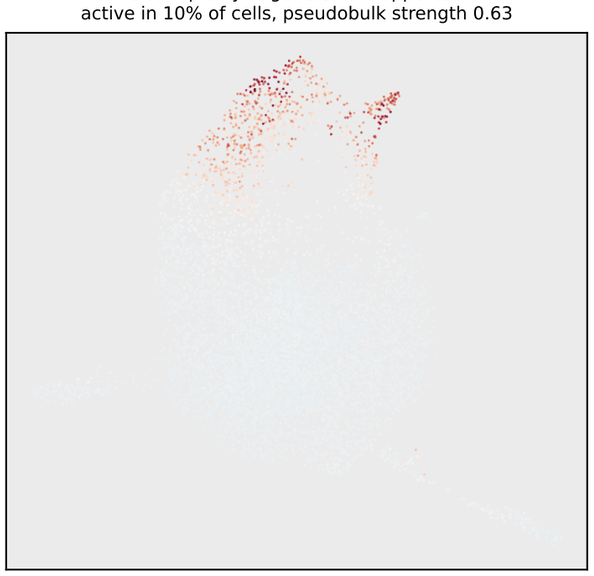
dome: 3 genes, supported
active in 6% of cells, pseudobulk strength 0.65



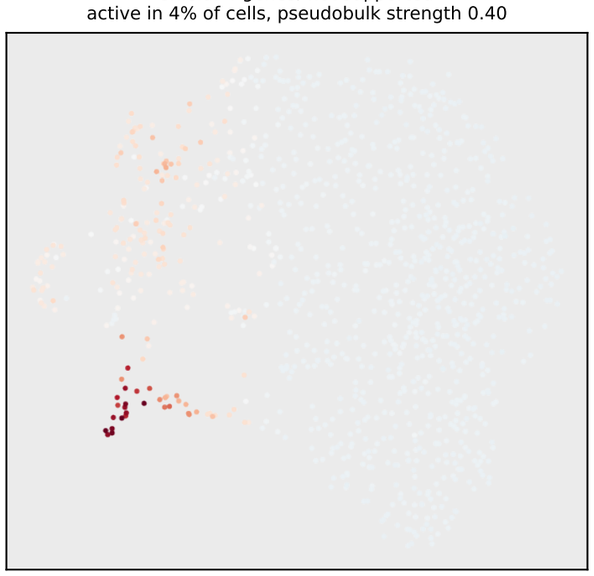
30% epiboly: 4 genes, supported
active in 7% of cells, pseudobulk strength 0.83



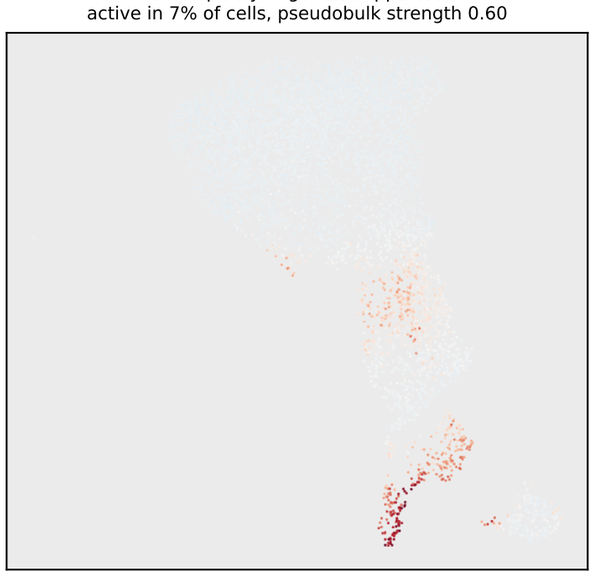
50% epiboly: 4 genes, not supported
active in 10% of cells, pseudobulk strength 0.63



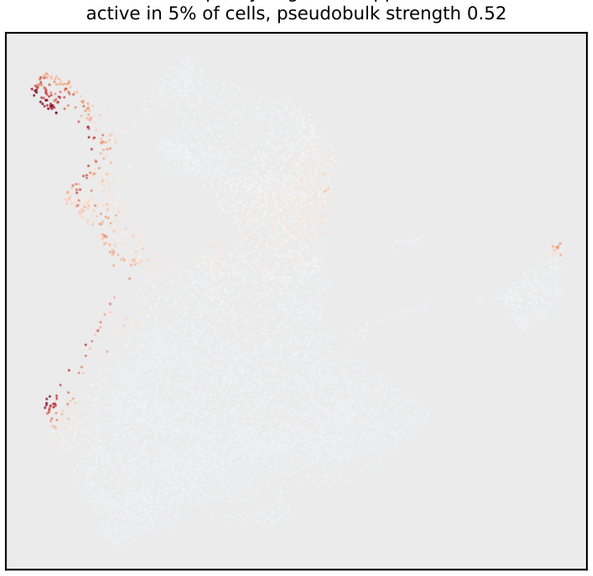
shield: 4 genes, not supported
active in 4% of cells, pseudobulk strength 0.40



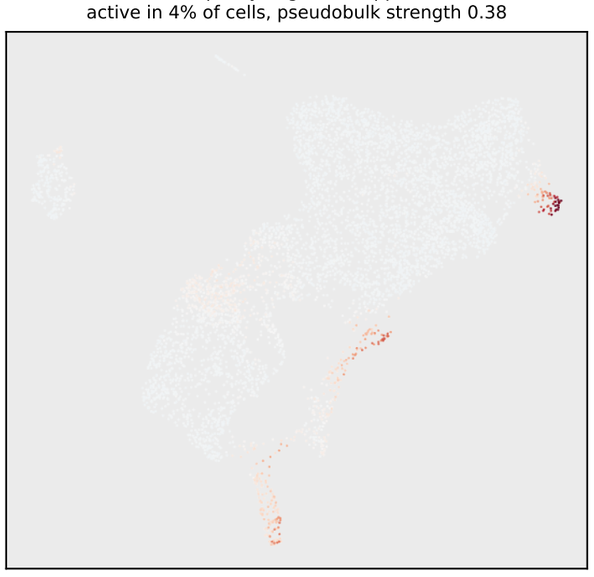
60% epiboly: 4 genes, supported
active in 7% of cells, pseudobulk strength 0.60



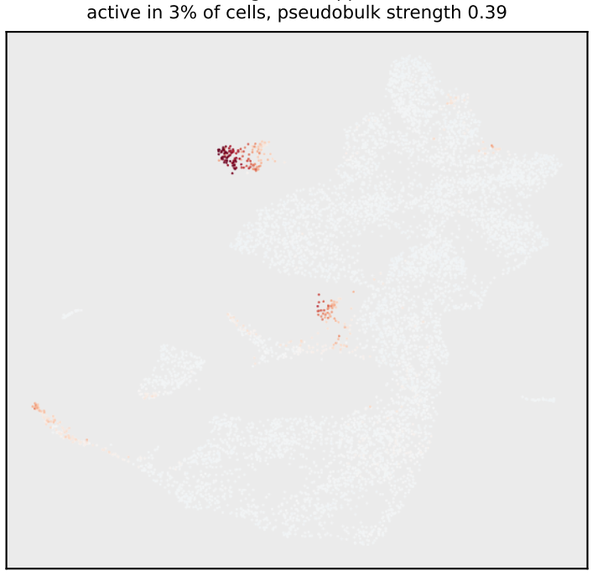
75% epiboly: 4 genes, supported
active in 5% of cells, pseudobulk strength 0.52



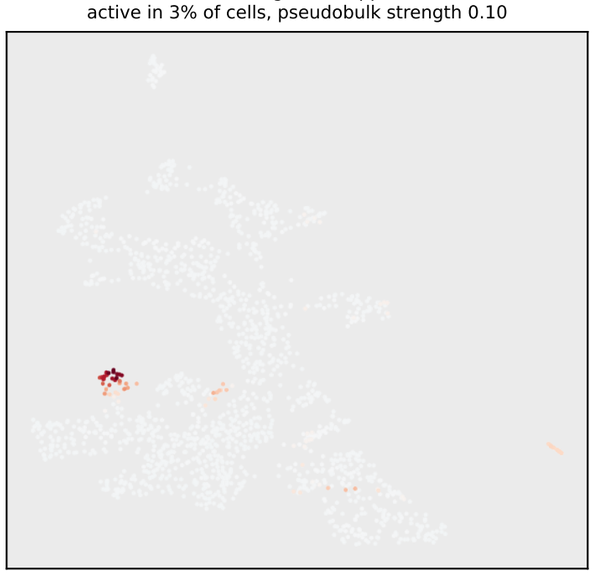
90% epiboly: 3 genes, supported
active in 4% of cells, pseudobulk strength 0.38



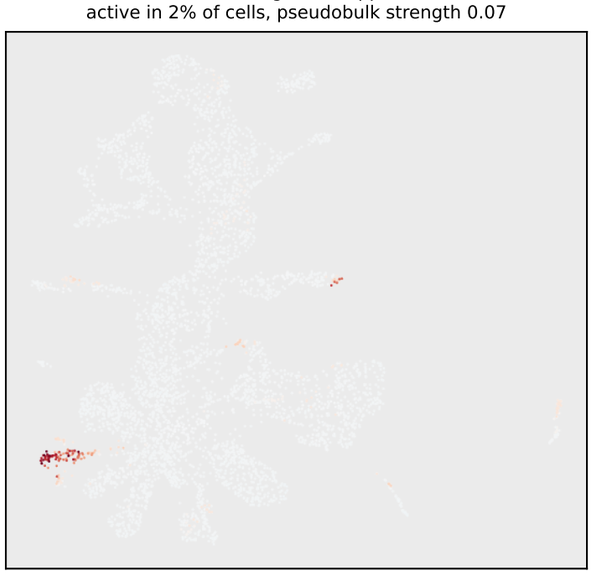
bud: 3 genes, supported
active in 3% of cells, pseudobulk strength 0.39



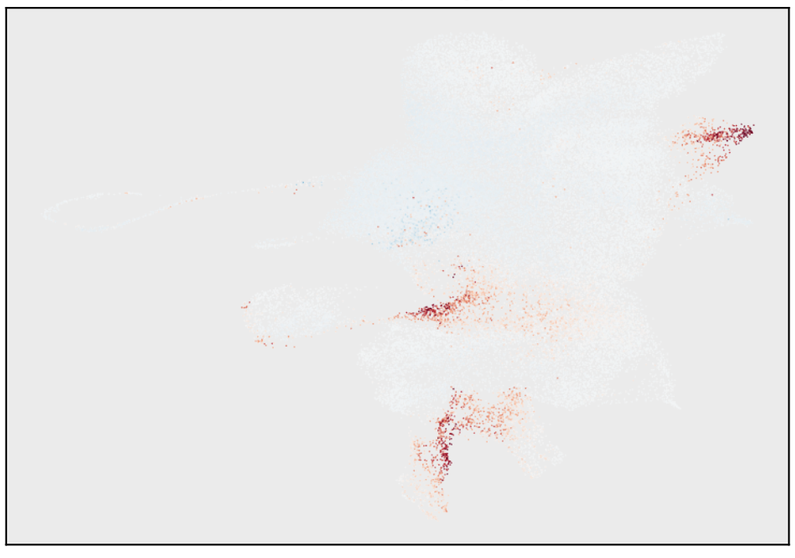
3-somite: 2 genes, supported
active in 3% of cells, pseudobulk strength 0.10



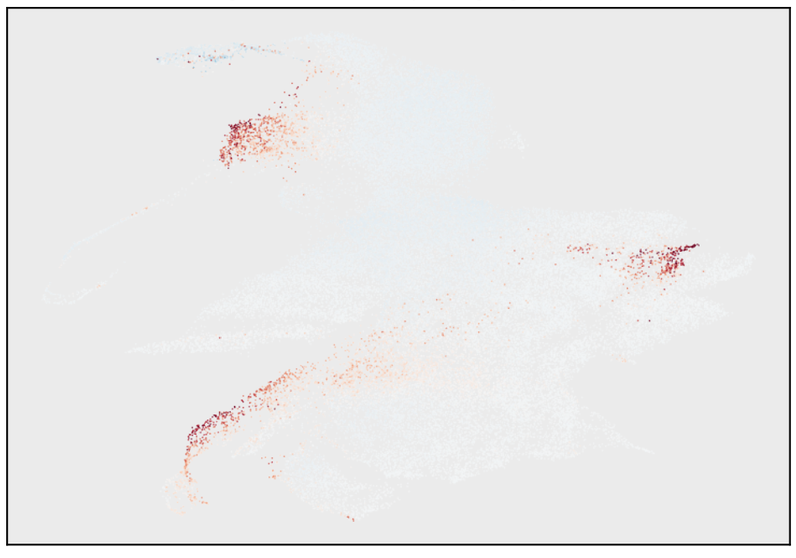
6-somite: 2 genes, supported
active in 2% of cells, pseudobulk strength 0.07



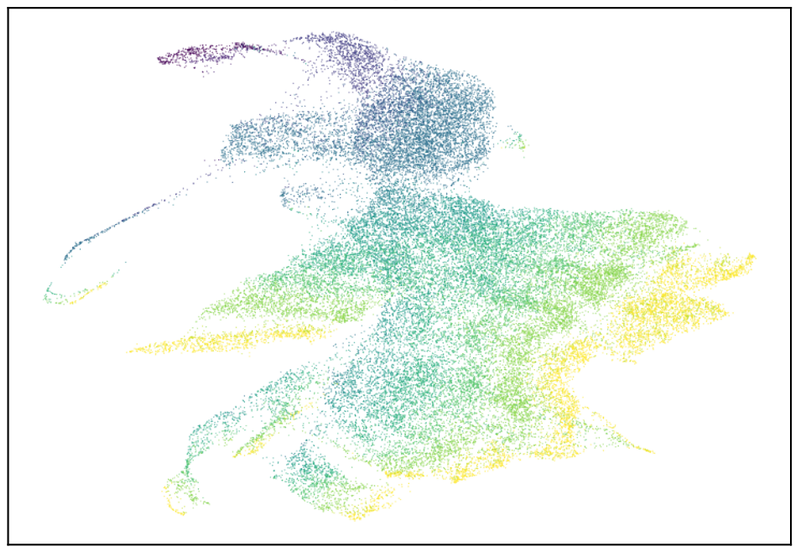
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



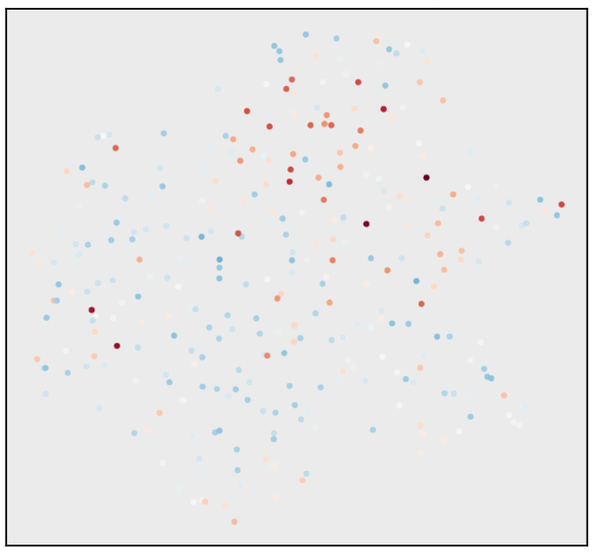
atlas, unaligned: stages



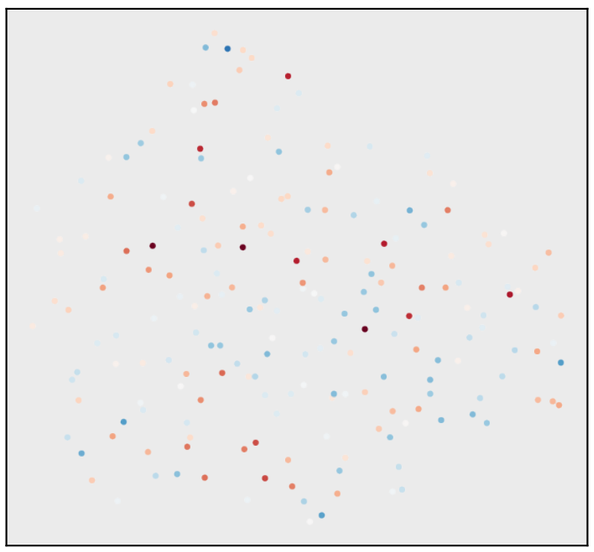
Genes (number of stages): lft1 (12), lft2 (12), ndr1 (8), ndr2 (7)

signal:Wnt

high: 3 genes, not supported
active in 0% of cells, pseudobulk strength 0.10

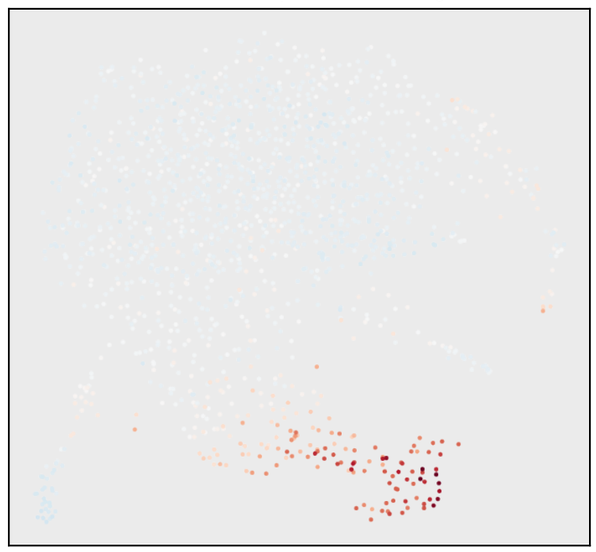


oblong: 3 genes, not supported
active in 0% of cells, pseudobulk strength 0.16

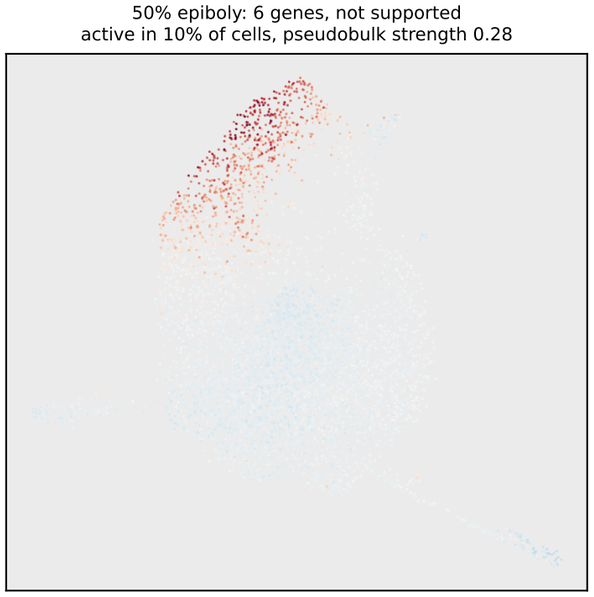


dome: 4 genes, not supported
active in 2% of cells, pseudobulk strength 0.06

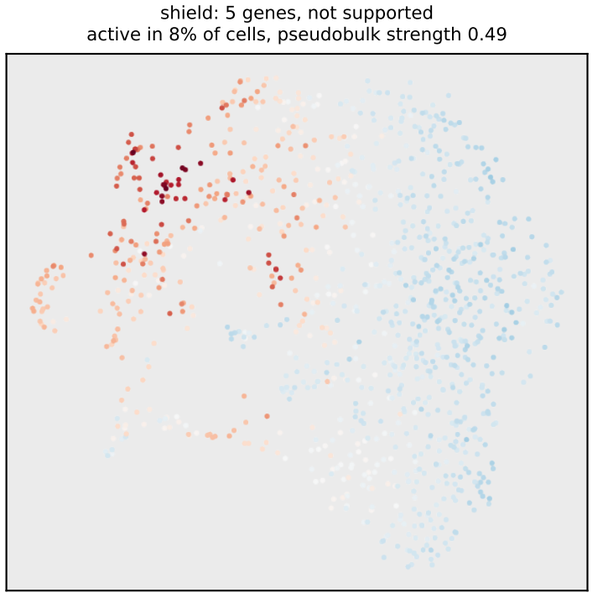
30% epiboly: 5 genes, not supported
active in 6% of cells, pseudobulk strength 0.28



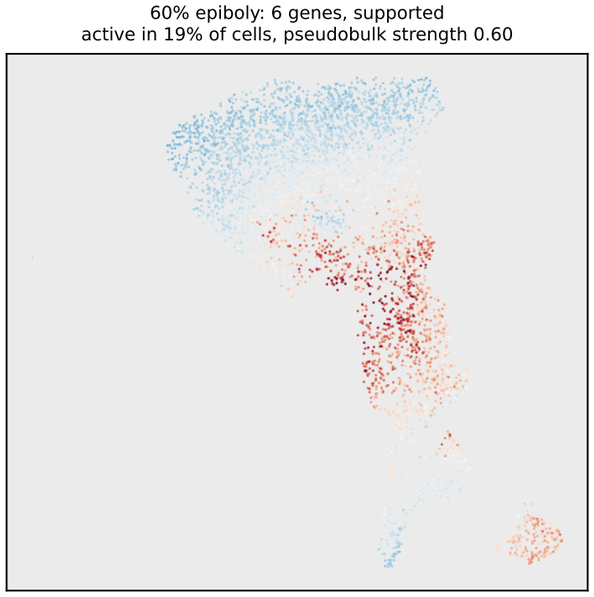
50% epiboly: 6 genes, not supported
active in 10% of cells, pseudobulk strength 0.28



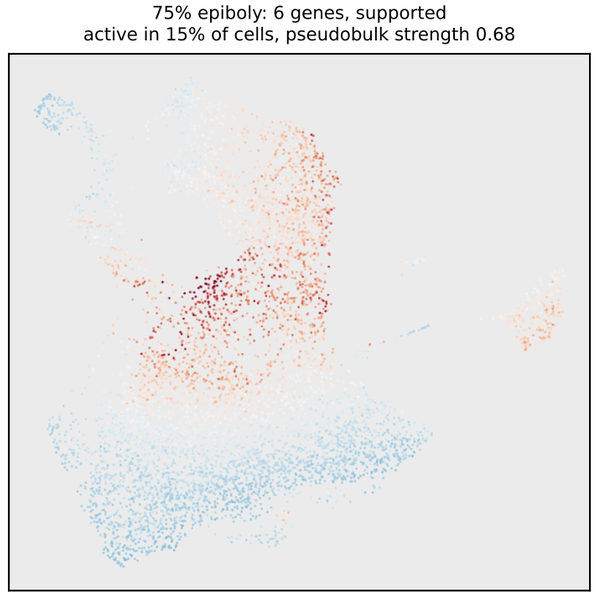
shield: 5 genes, not supported
active in 8% of cells, pseudobulk strength 0.49



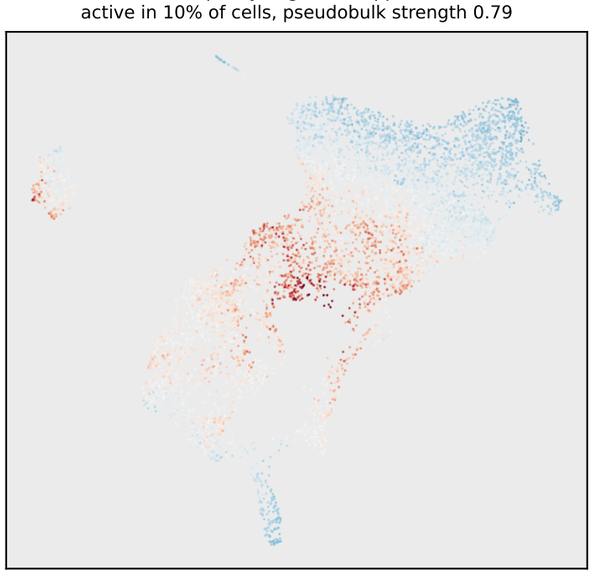
60% epiboly: 6 genes, supported
active in 19% of cells, pseudobulk strength 0.60



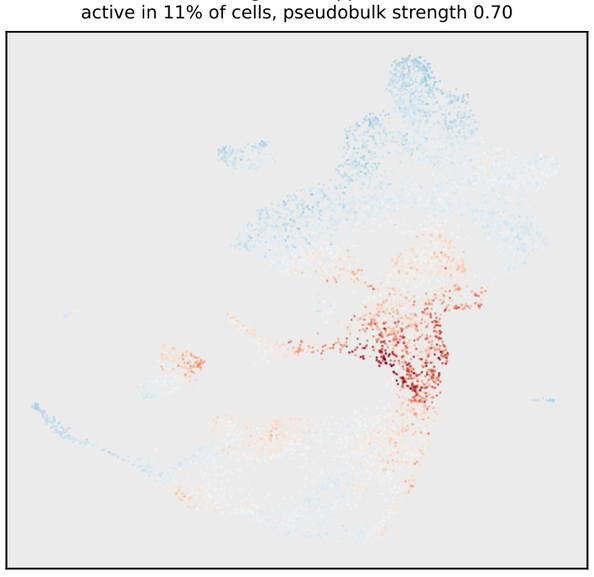
75% epiboly: 6 genes, supported
active in 15% of cells, pseudobulk strength 0.68



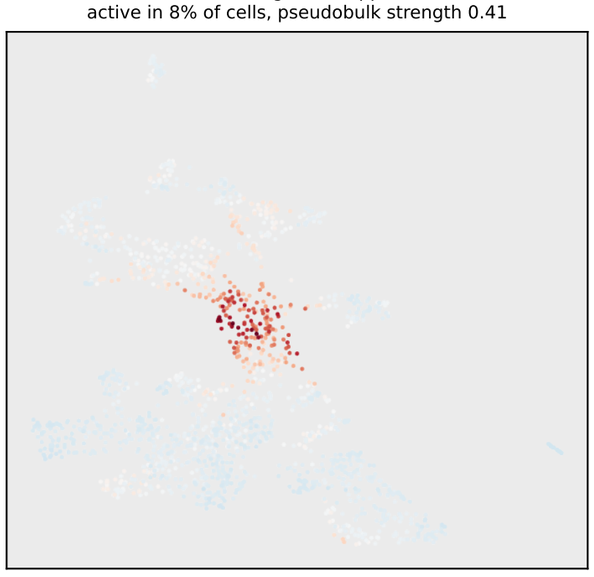
90% epiboly: 6 genes, supported
active in 10% of cells, pseudobulk strength 0.79



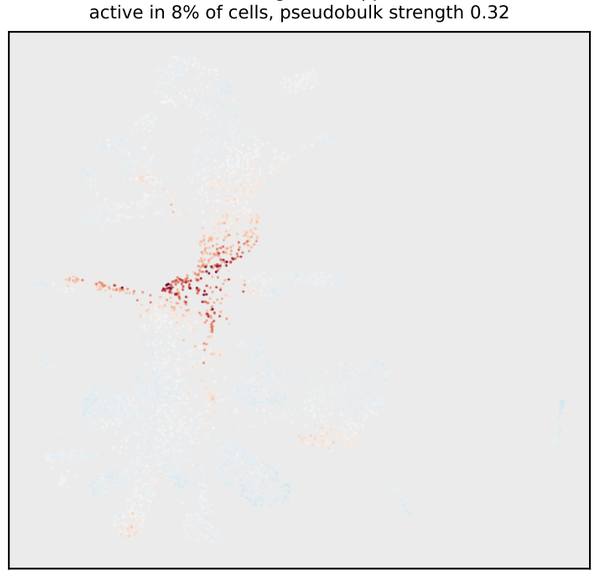
bud: 6 genes, supported
active in 11% of cells, pseudobulk strength 0.70



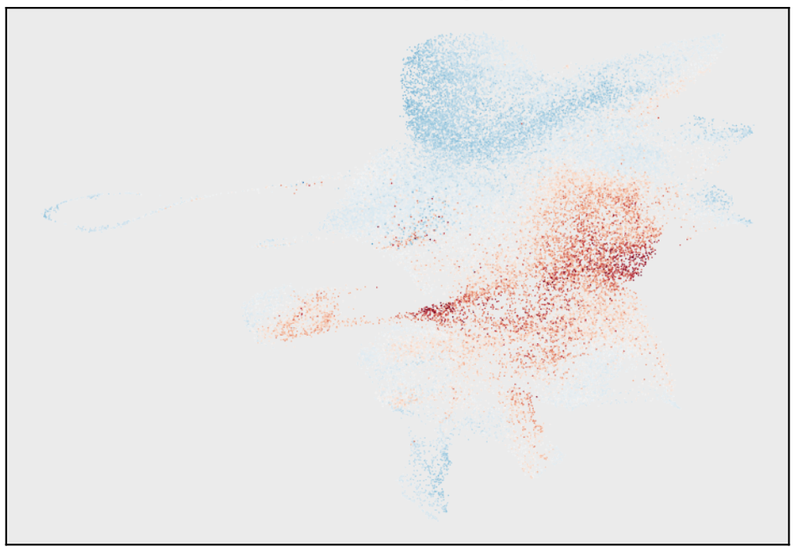
3-somite: 6 genes, supported
active in 8% of cells, pseudobulk strength 0.41



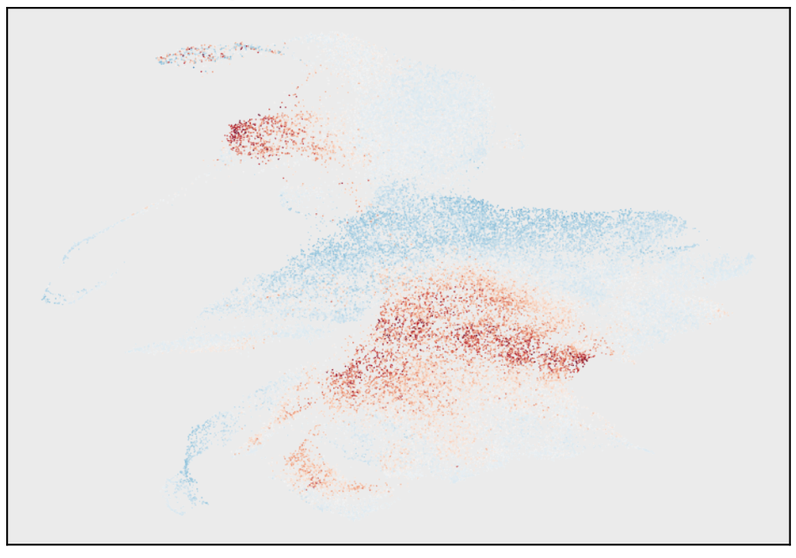
6-somite: 6 genes, supported
active in 8% of cells, pseudobulk strength 0.32



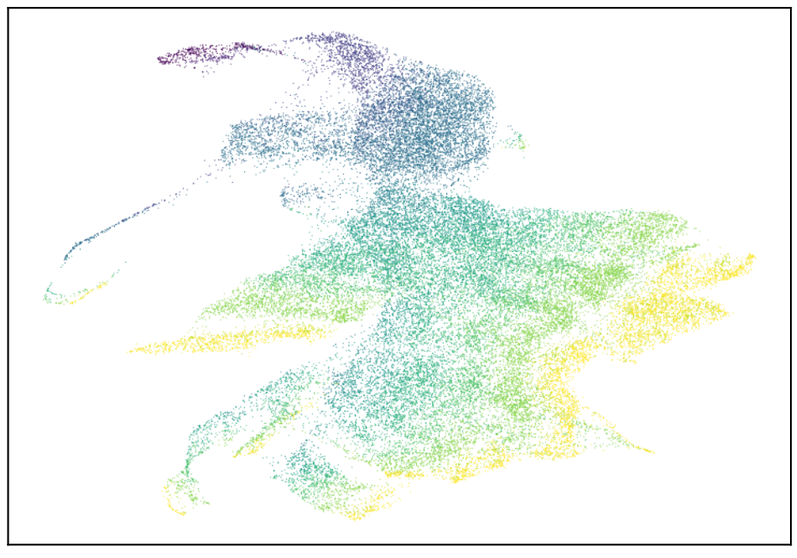
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



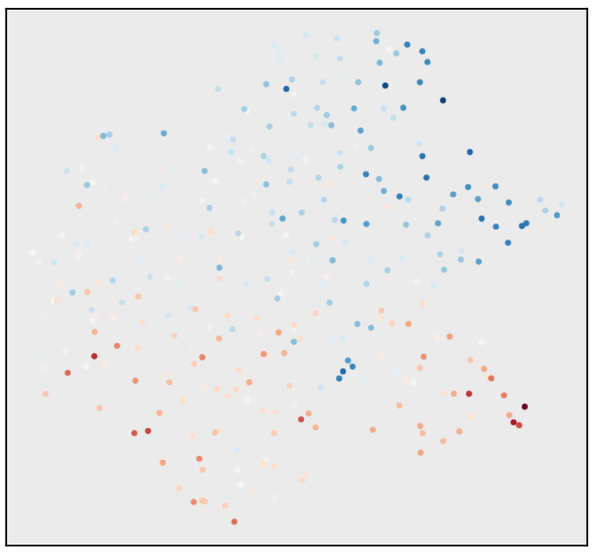
atlas, unaligned: stages



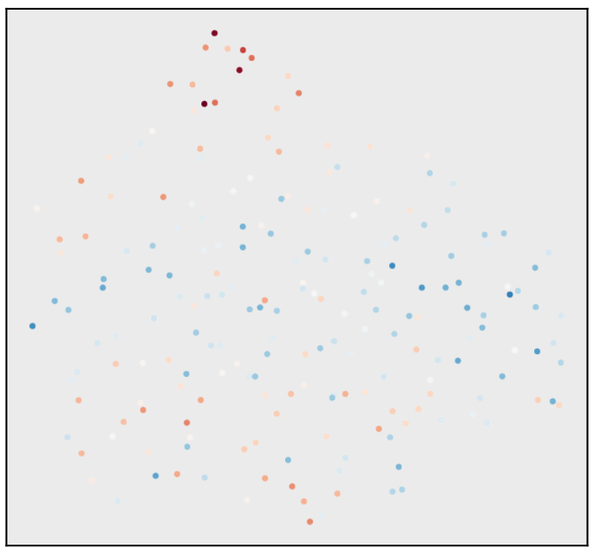
Genes (number of stages): sp5l (12), lef1 (12), wnt8a (12), sp5a (10), notum1a (9), wnt3a (7)

state:blastomere

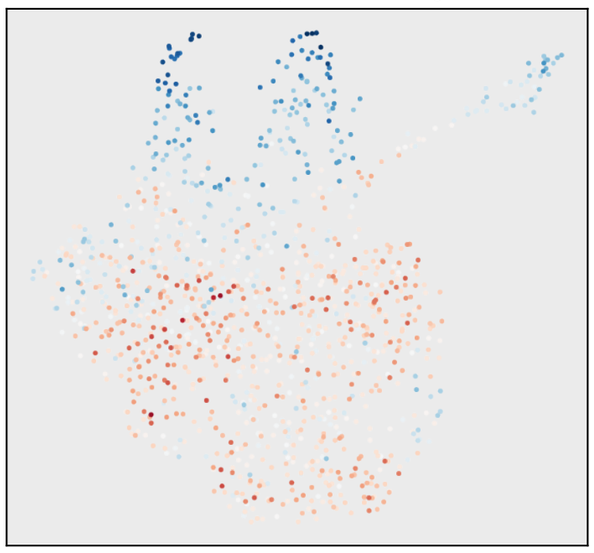
high: 6 genes, not supported
active in 7% of cells, pseudobulk strength 0.49



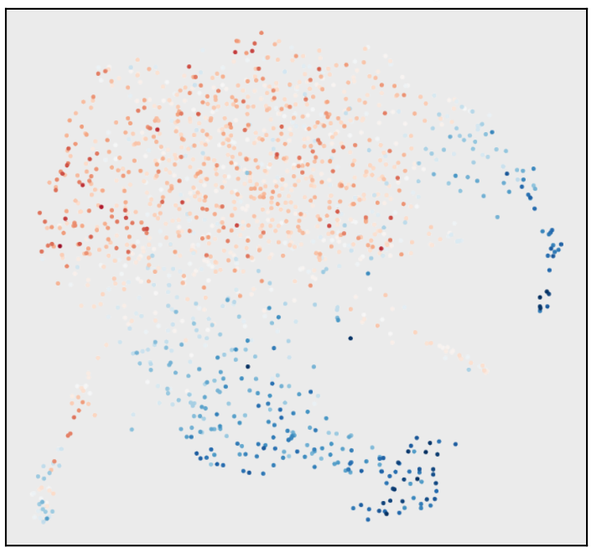
oblong: 6 genes, not supported
active in 1% of cells, pseudobulk strength 0.55



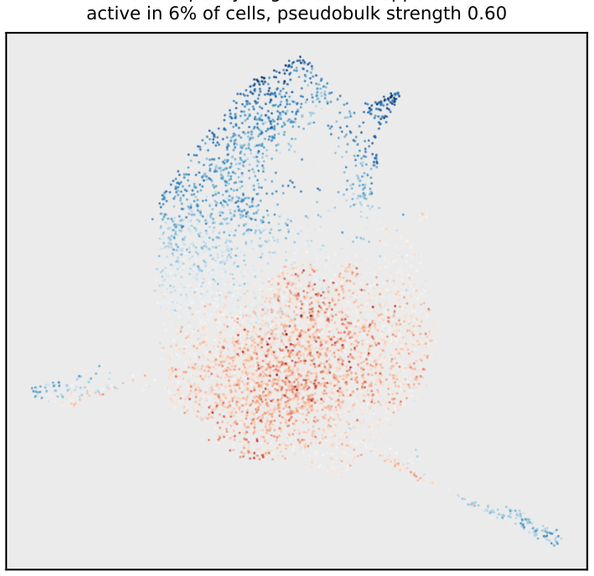
dome: 6 genes, not supported
active in 1% of cells, pseudobulk strength 0.57



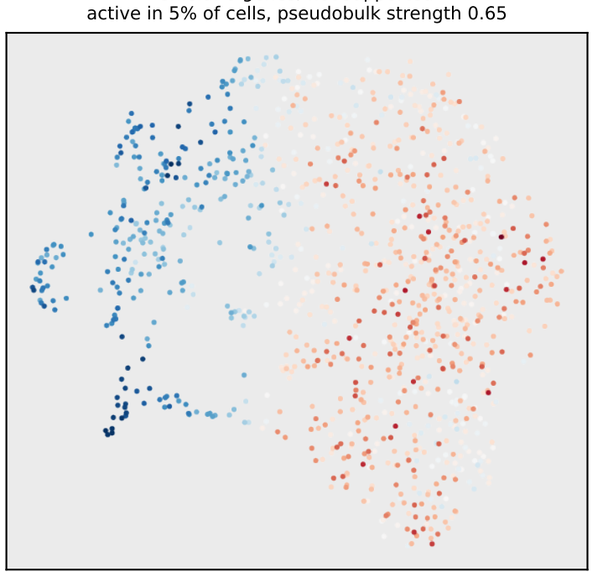
30% epiboly: 6 genes, not supported
active in 1% of cells, pseudobulk strength 0.65



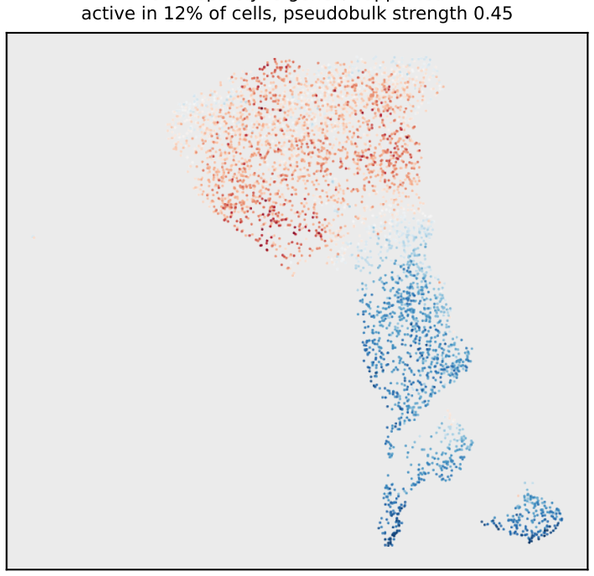
50% epiboly: 6 genes, not supported
active in 6% of cells, pseudobulk strength 0.60



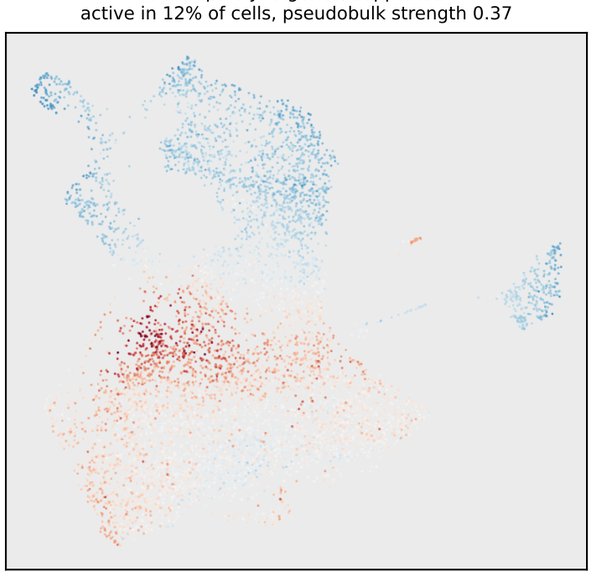
shield: 6 genes, not supported
active in 5% of cells, pseudobulk strength 0.65



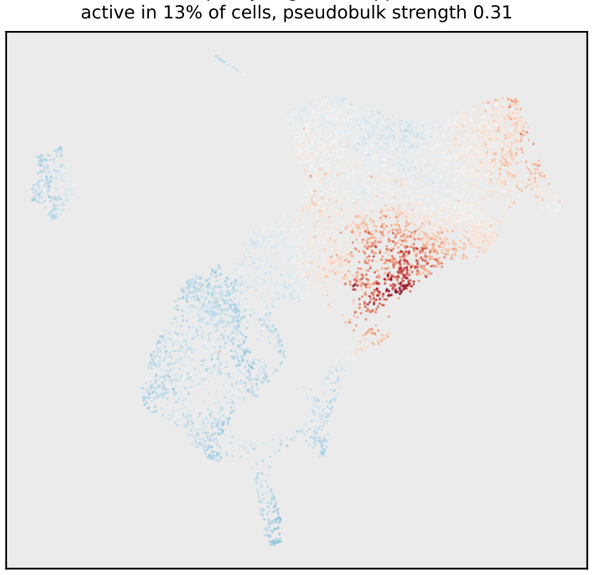
60% epiboly: 6 genes, supported
active in 12% of cells, pseudobulk strength 0.45



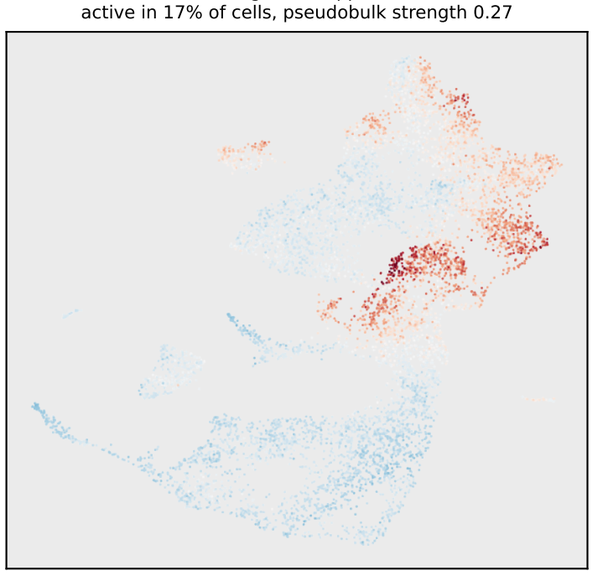
75% epiboly: 6 genes, supported
active in 12% of cells, pseudobulk strength 0.37



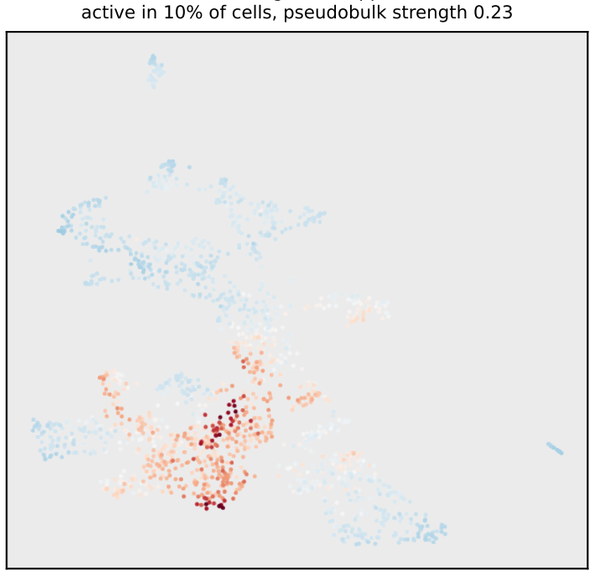
90% epiboly: 6 genes, supported
active in 13% of cells, pseudobulk strength 0.31



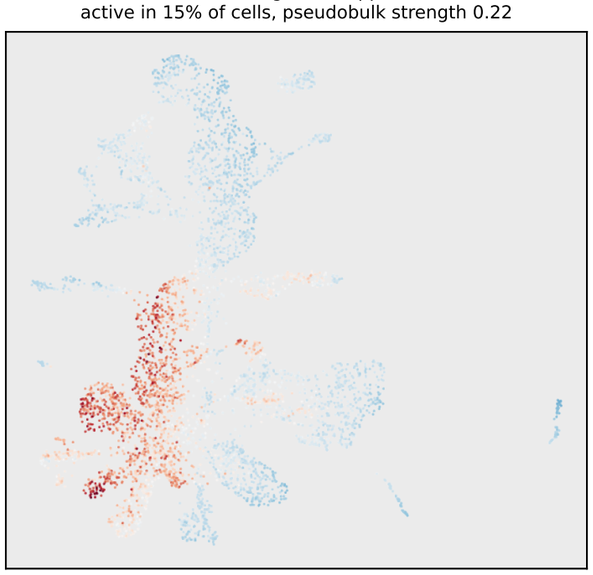
bud: 6 genes, supported
active in 17% of cells, pseudobulk strength 0.27



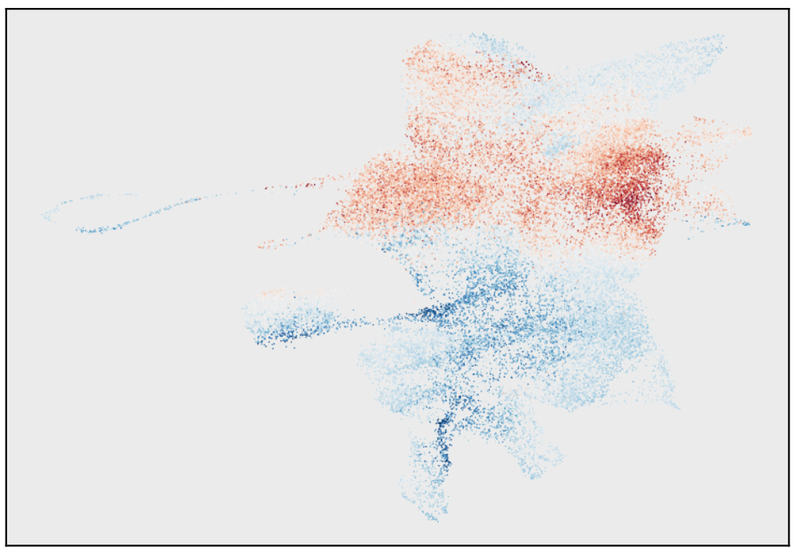
3-somite: 5 genes, supported
active in 10% of cells, pseudobulk strength 0.23



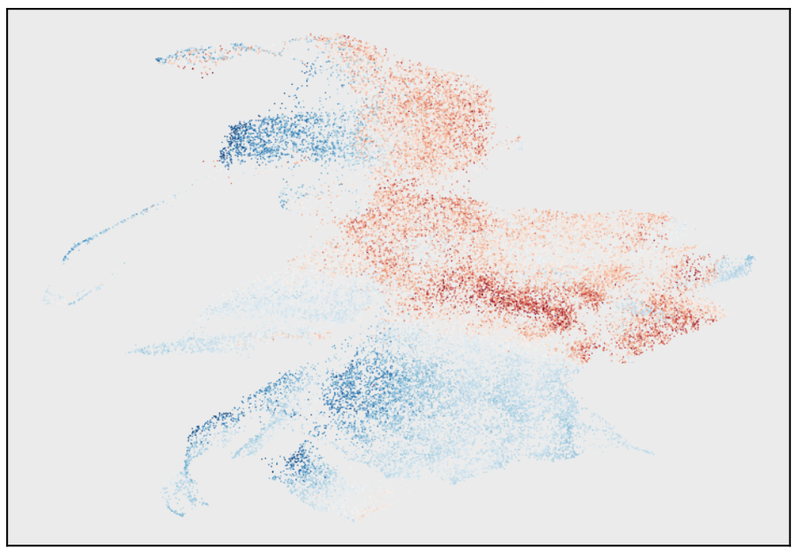
6-somite: 6 genes, supported
active in 15% of cells, pseudobulk strength 0.22



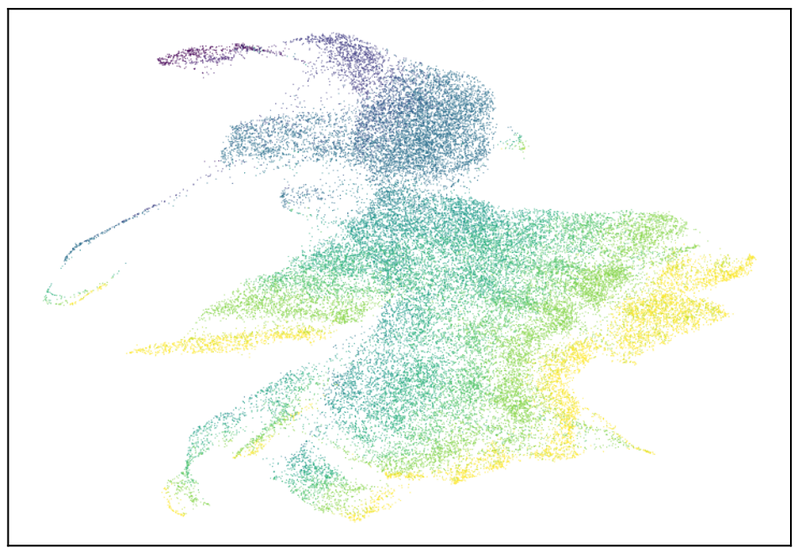
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



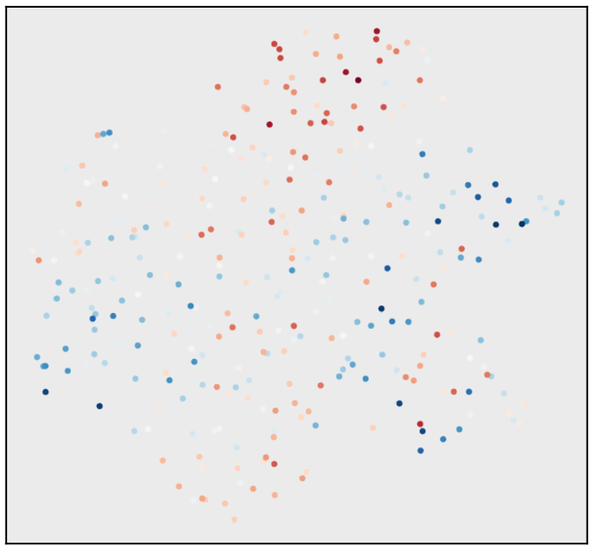
atlas, unaligned: stages



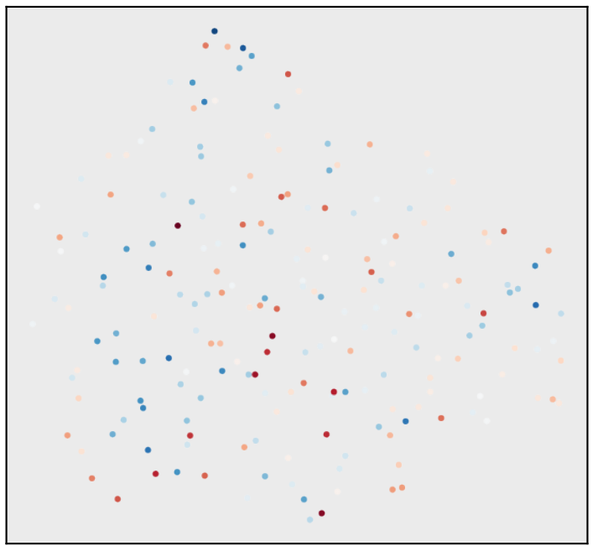
Genes (number of stages): pou5f3 (12), sox19b (12), lin28a (12), sox3 (12), sox19a (12), nanog (11)

state:cell_cycle_g2m

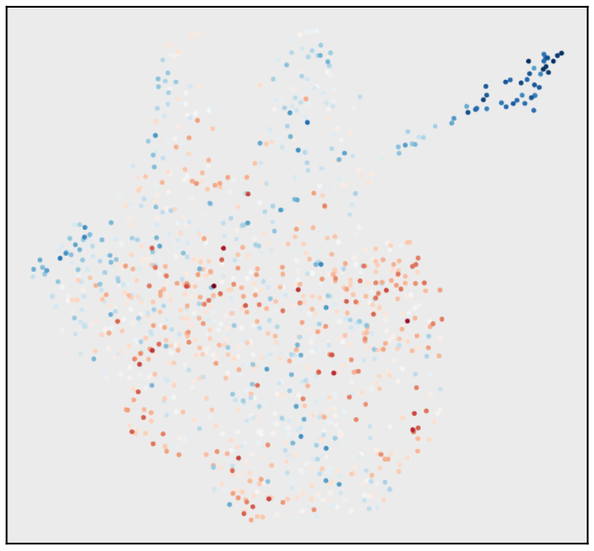
high: 10 genes, not supported
active in 0% of cells, pseudobulk strength 0.55



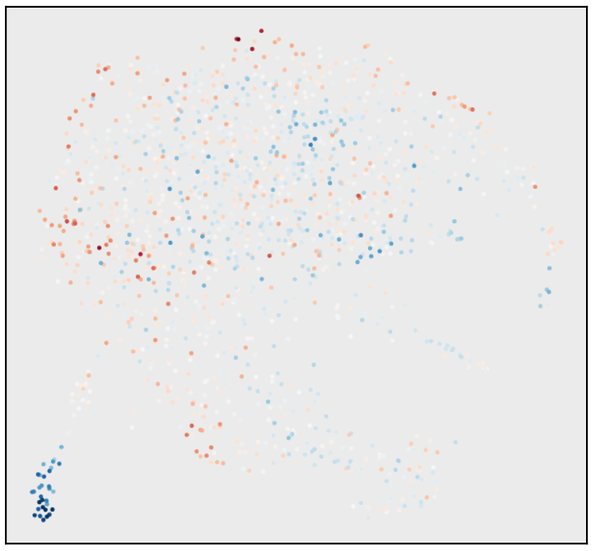
oblong: 10 genes, not supported
active in 0% of cells, pseudobulk strength 0.50



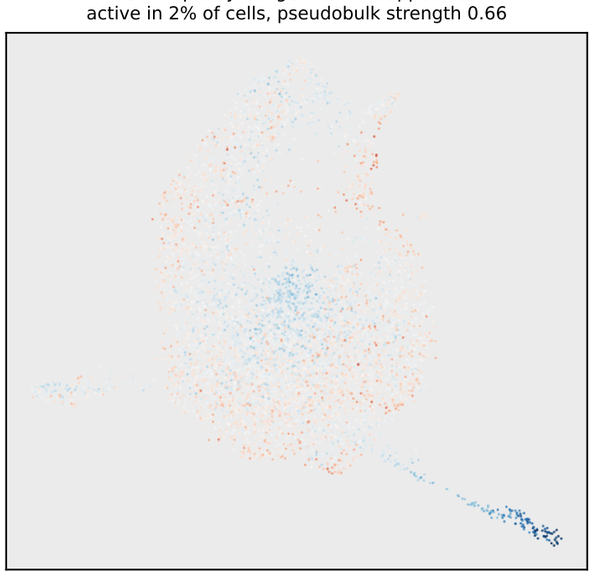
dome: 10 genes, not supported
active in 0% of cells, pseudobulk strength 0.43



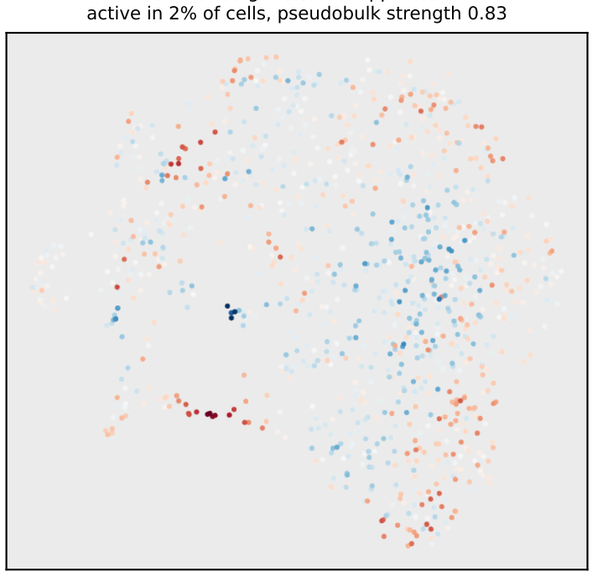
30% epiboly: 10 genes, not supported
active in 1% of cells, pseudobulk strength 0.54



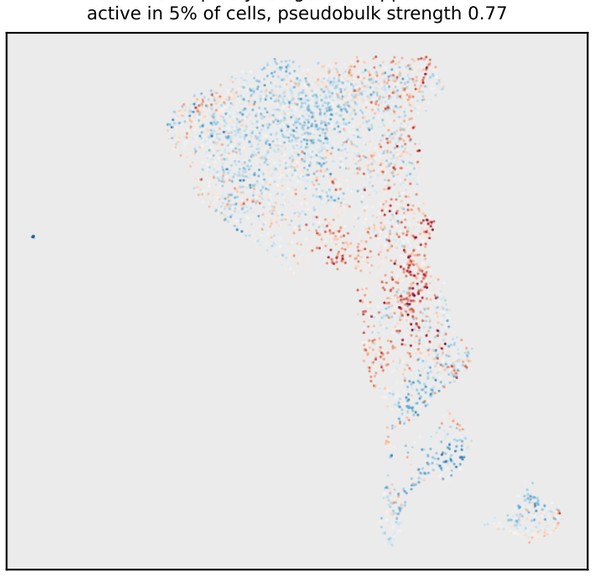
50% epiboly: 10 genes, not supported
active in 2% of cells, pseudobulk strength 0.66



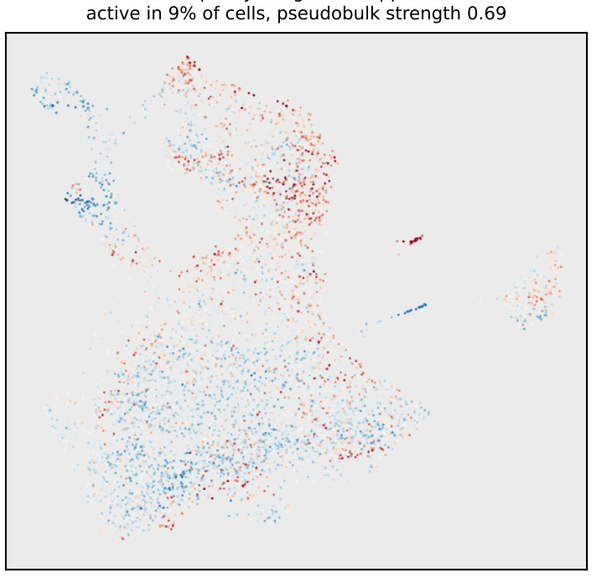
shield: 10 genes, not supported
active in 2% of cells, pseudobulk strength 0.83



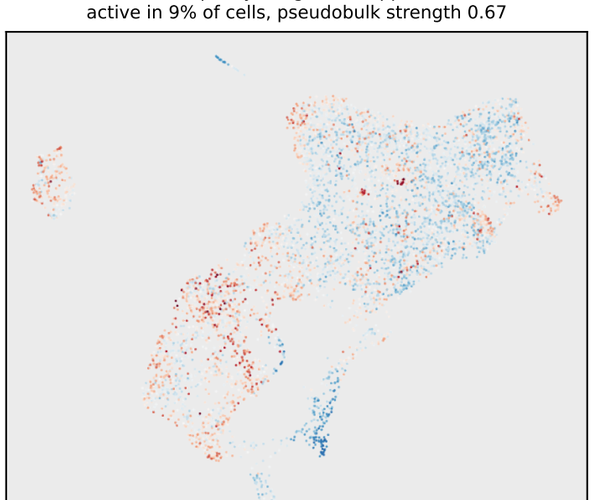
60% epiboly: 10 genes, supported
active in 5% of cells, pseudobulk strength 0.77



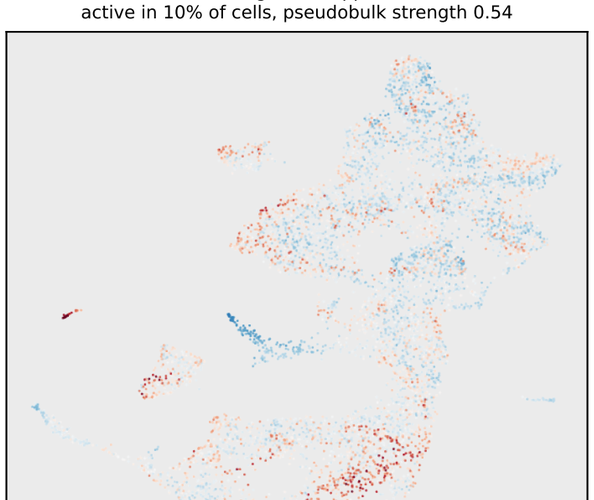
75% epiboly: 10 genes, supported
active in 9% of cells, pseudobulk strength 0.69



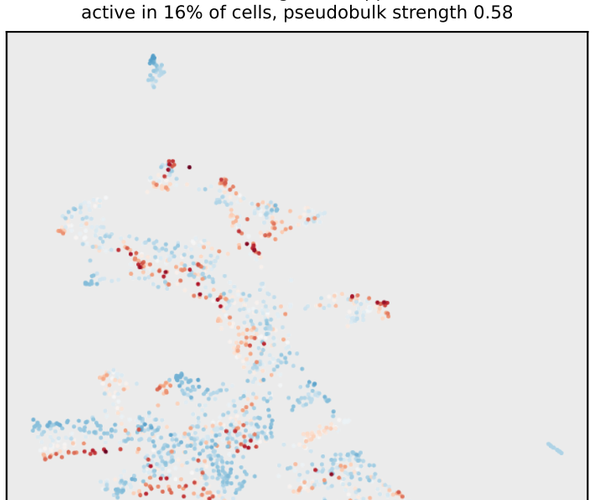
90% epiboly: 10 genes, supported
active in 9% of cells, pseudobulk strength 0.67



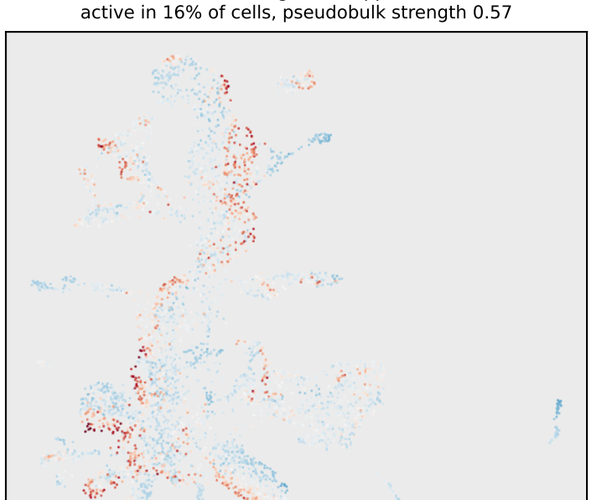
bud: 10 genes, supported
active in 10% of cells, pseudobulk strength 0.54



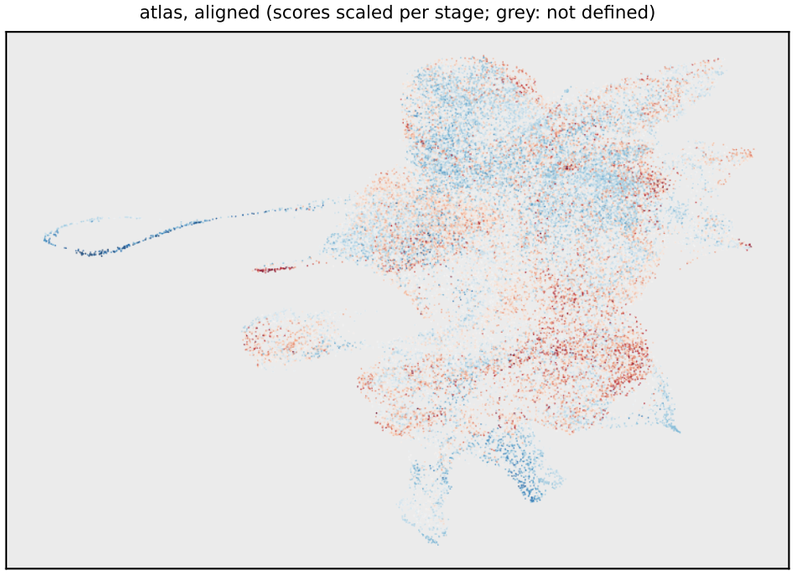
3-somite: 10 genes, supported
active in 16% of cells, pseudobulk strength 0.58



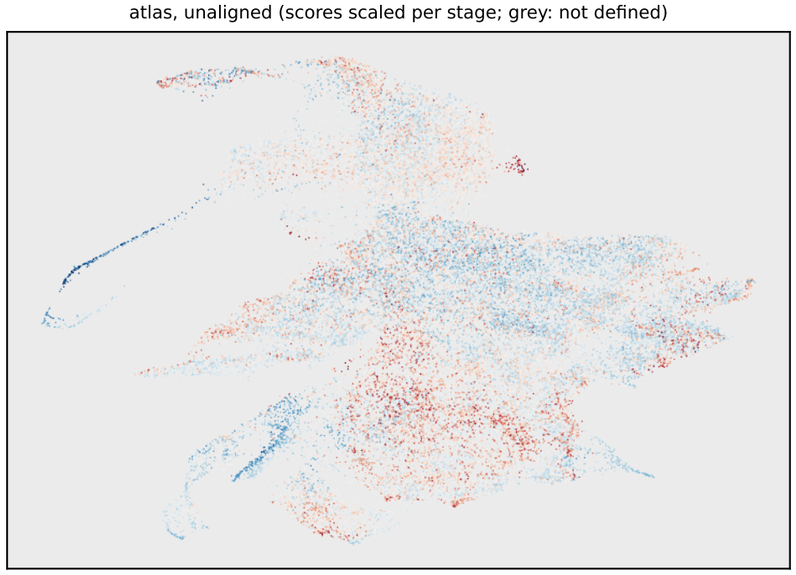
6-somite: 10 genes, supported
active in 16% of cells, pseudobulk strength 0.57



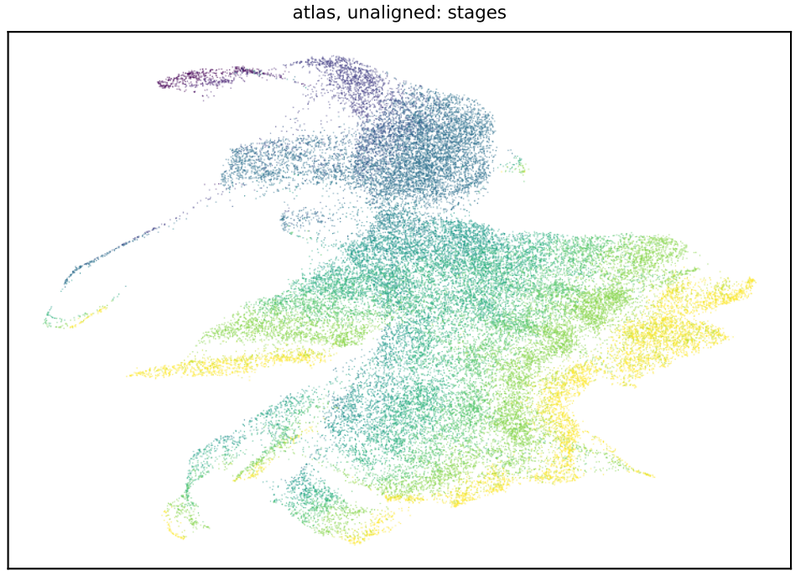
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



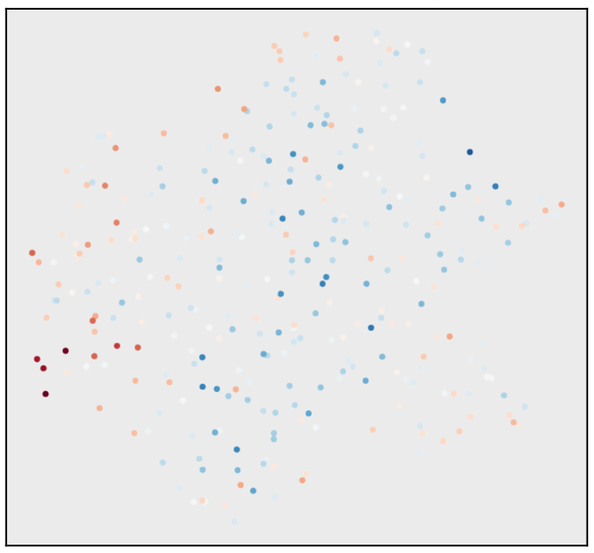
atlas, unaligned: stages



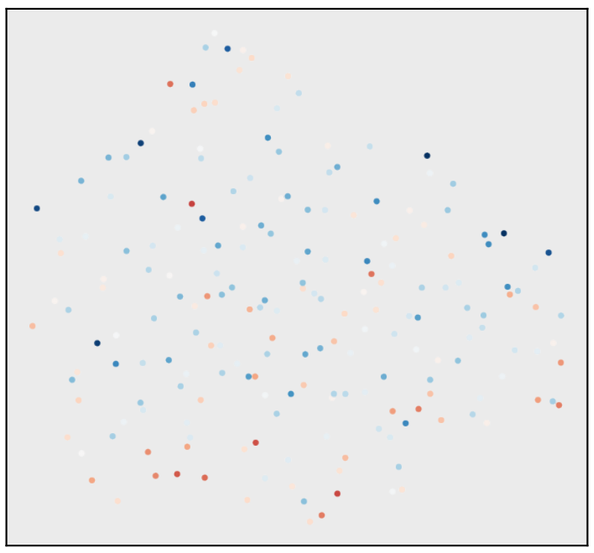
Genes (number of stages): cdk1 (12), ube2c (12), top2a (12), nusap1 (12), cenpf (12), tpx2 (12), aurkb (12), cdc20 (12), birc5a (12), kif11 (12)

state:cell_cycle_s

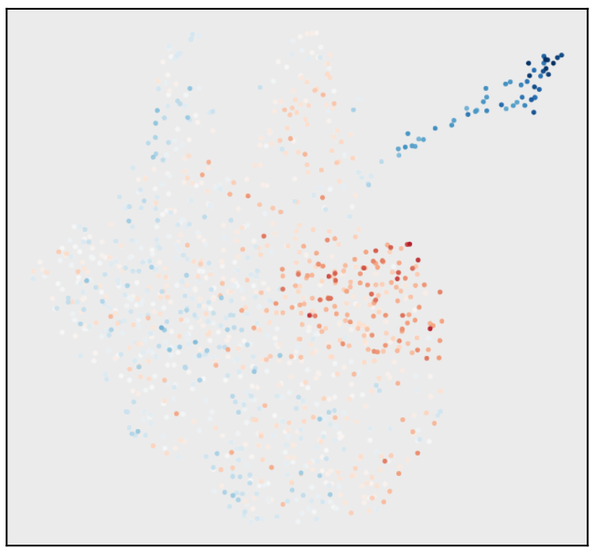
high: 10 genes, not supported
active in 1% of cells, pseudobulk strength 0.60



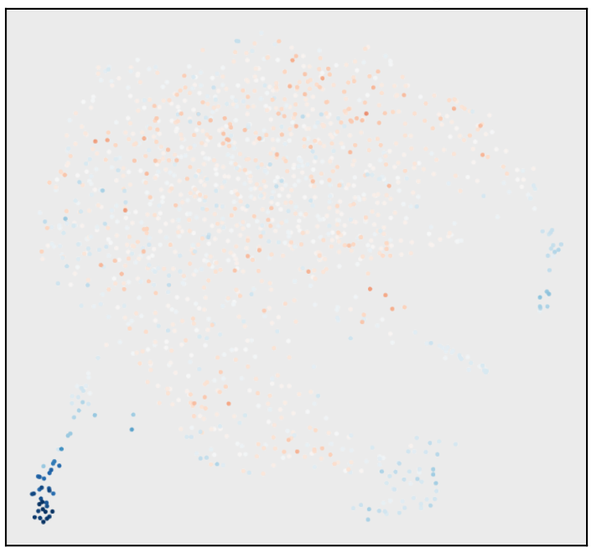
oblong: 10 genes, not supported
active in 0% of cells, pseudobulk strength 0.53



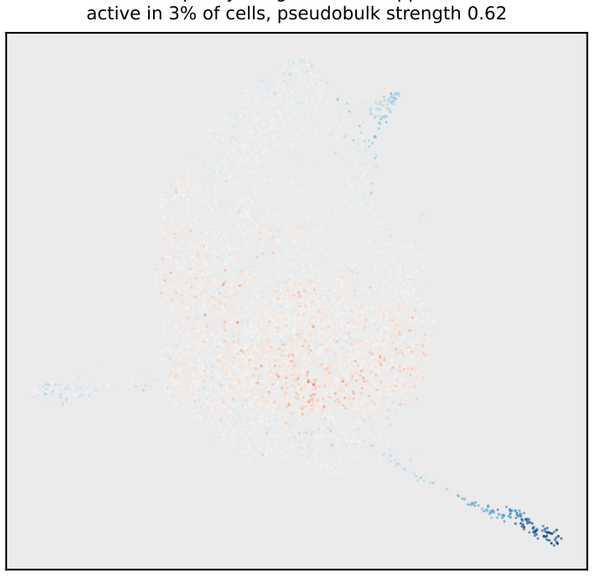
dome: 10 genes, not supported
active in 2% of cells, pseudobulk strength 0.64



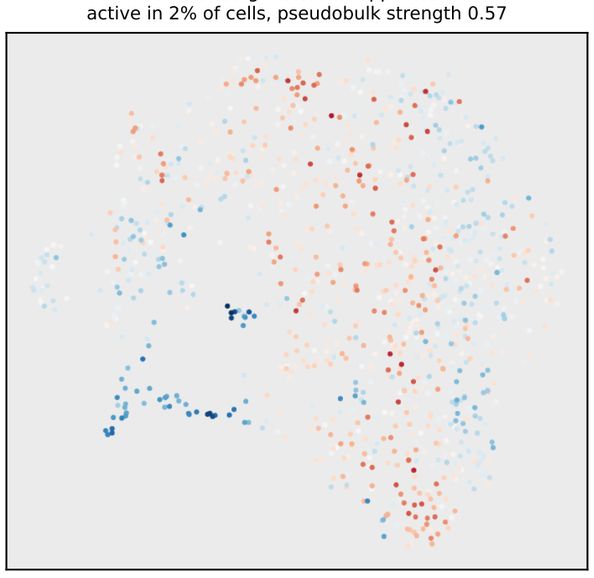
30% epiboly: 10 genes, not supported
active in 0% of cells, pseudobulk strength 0.58



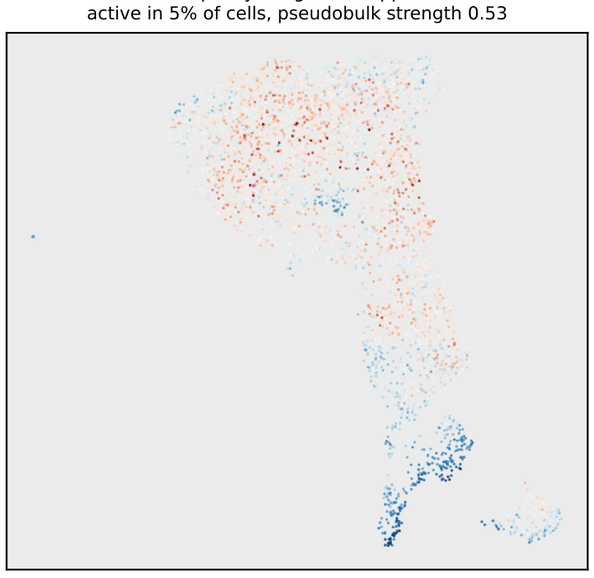
50% epiboly: 10 genes, not supported
active in 3% of cells, pseudobulk strength 0.62



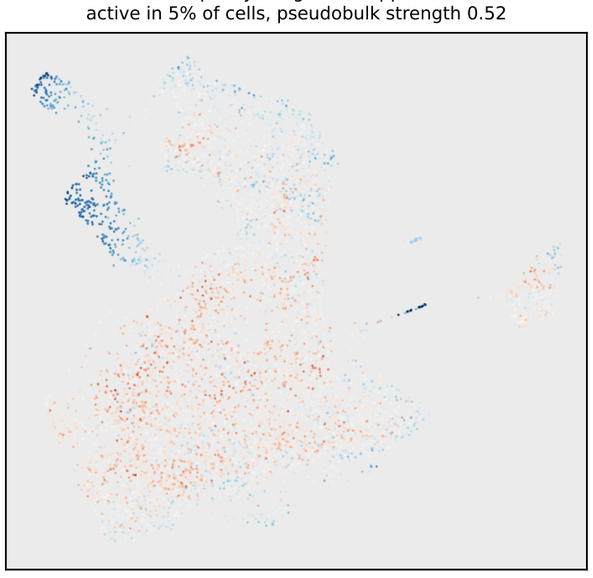
shield: 10 genes, not supported
active in 2% of cells, pseudobulk strength 0.57



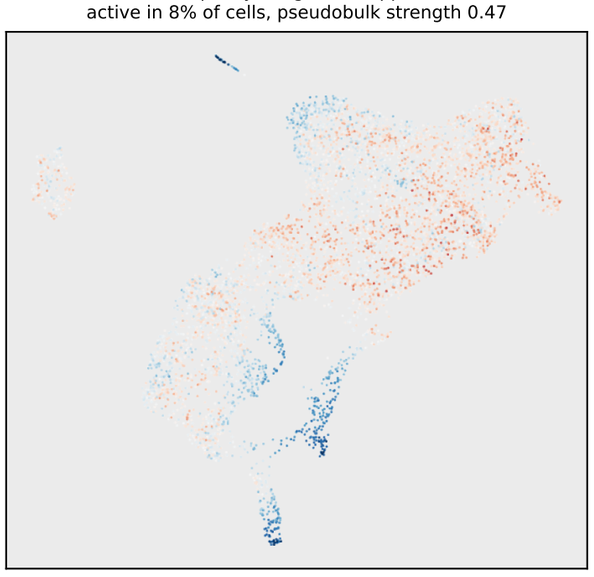
60% epiboly: 10 genes, supported
active in 5% of cells, pseudobulk strength 0.53



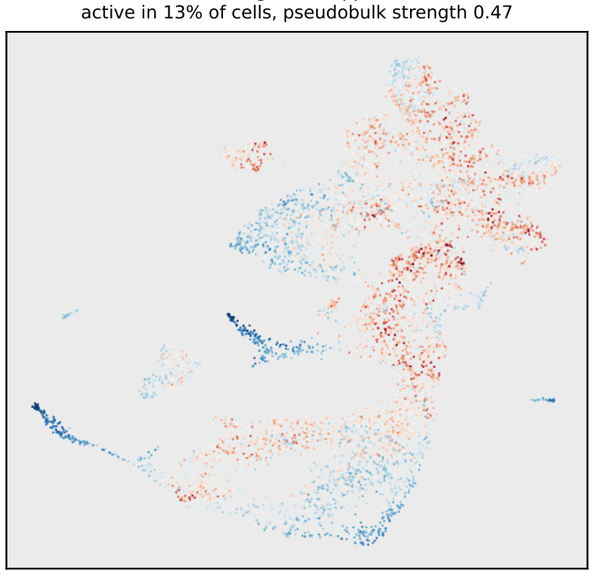
75% epiboly: 10 genes, supported
active in 5% of cells, pseudobulk strength 0.52



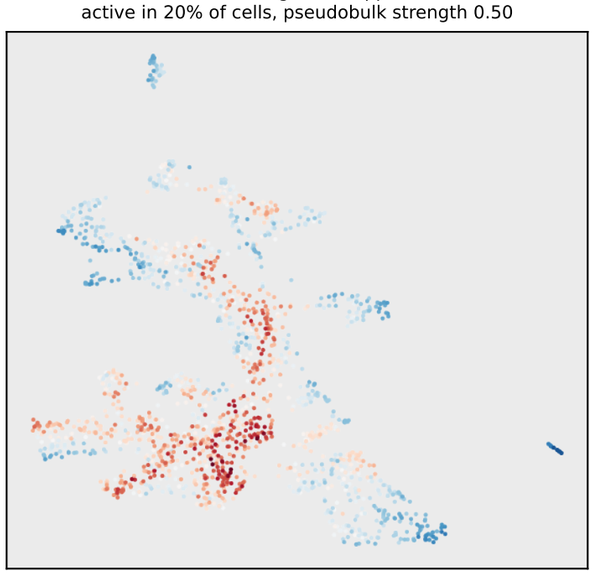
90% epiboly: 10 genes, supported
active in 8% of cells, pseudobulk strength 0.47



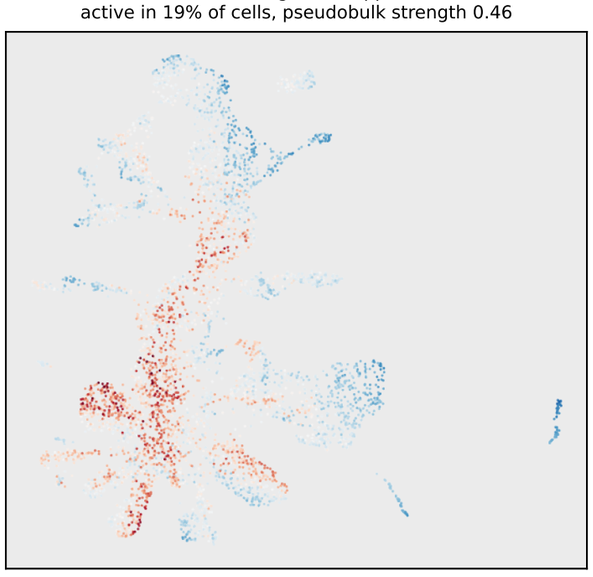
bud: 10 genes, supported
active in 13% of cells, pseudobulk strength 0.47



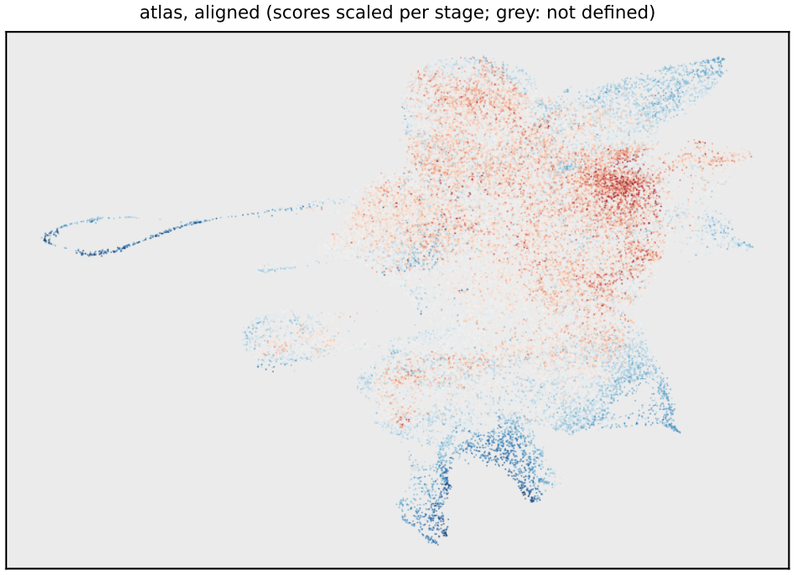
3-somite: 10 genes, supported
active in 20% of cells, pseudobulk strength 0.50



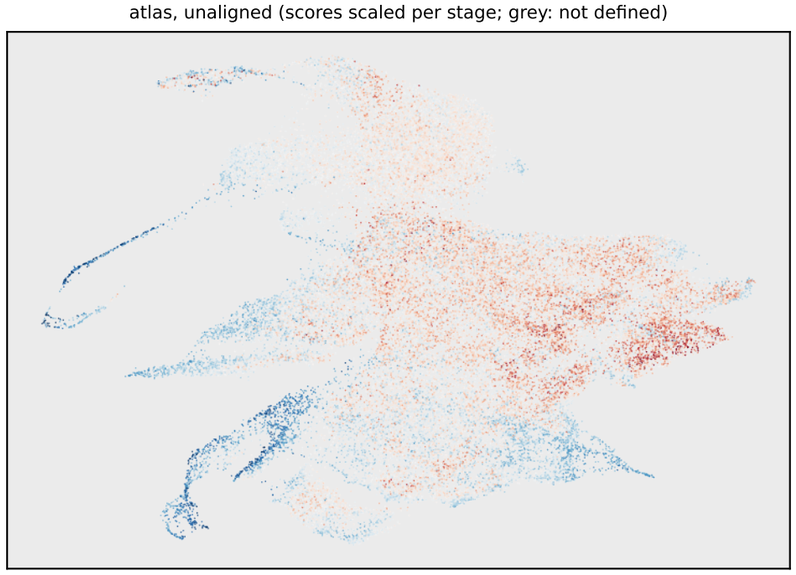
6-somite: 10 genes, supported
active in 19% of cells, pseudobulk strength 0.46



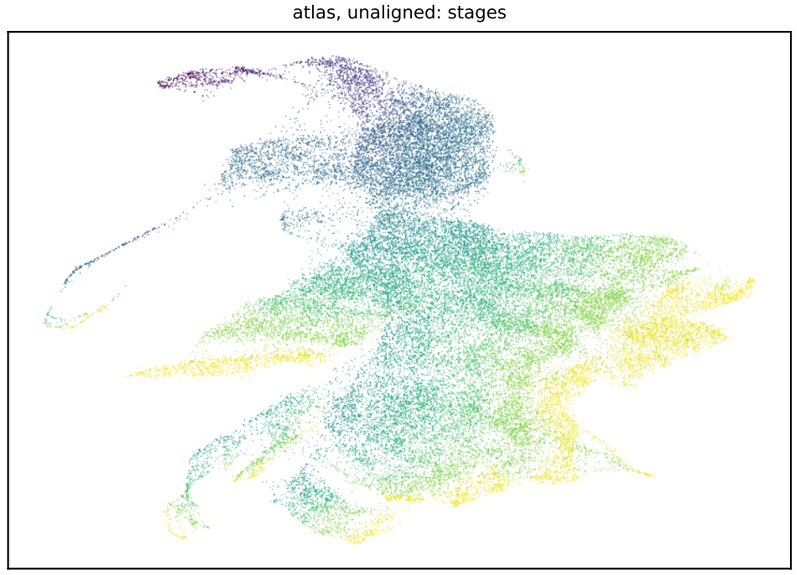
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



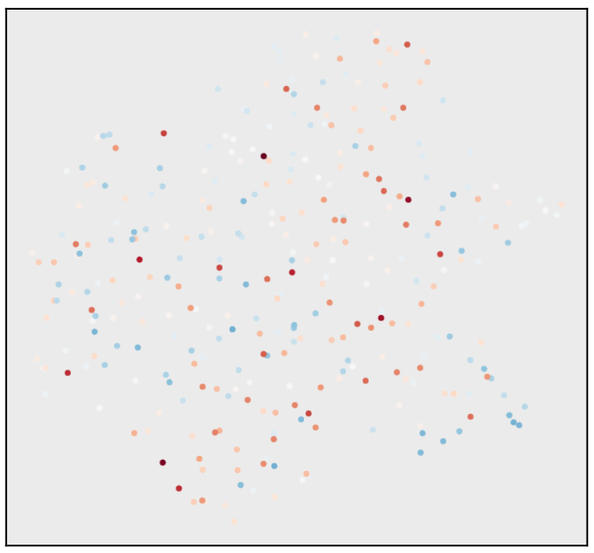
atlas, unaligned: stages



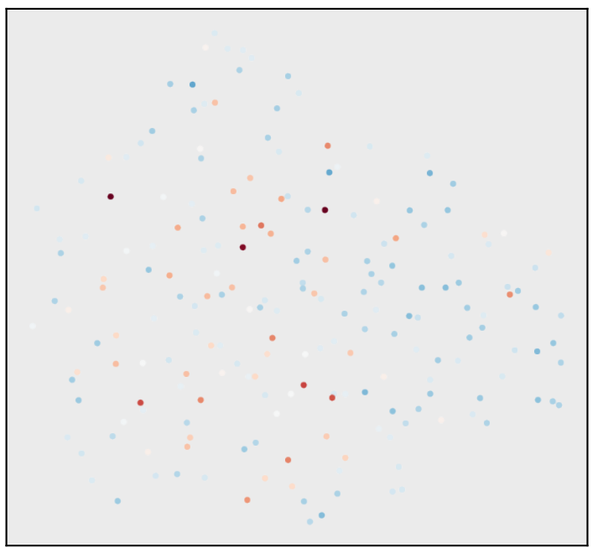
Genes (number of stages): pcna (12), mcm2 (12), mcm5 (12), mcm6 (12), cdc6 (12), slbp (12), tyms (12), mcm4 (12), uhrf1 (12), rrm2 (12)

state:heat_shock

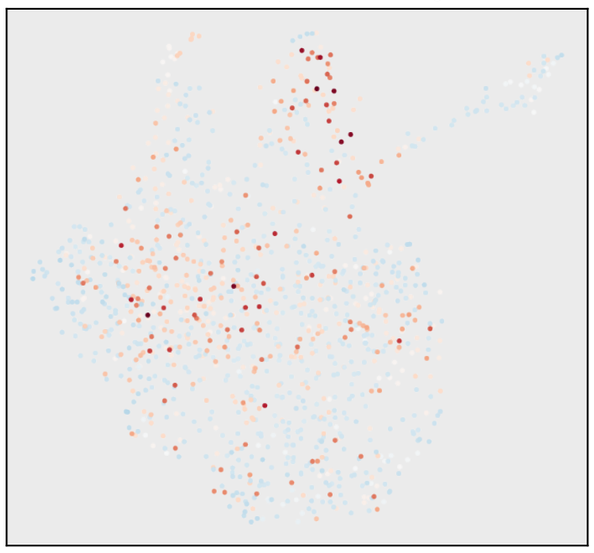
high: 3 genes, not supported
active in 0% of cells, pseudobulk strength 0.38



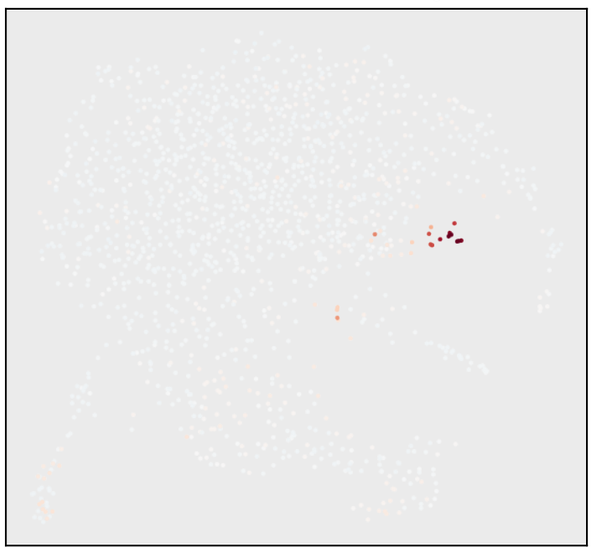
oblong: 2 genes, not supported
active in 0% of cells, pseudobulk strength 0.24



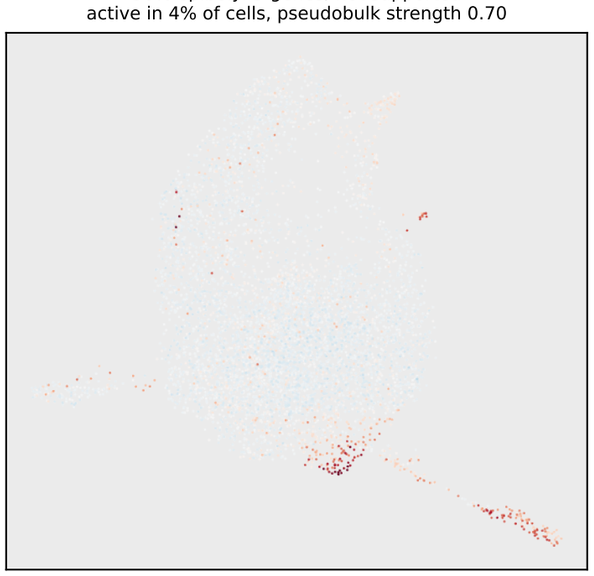
dome: 3 genes, not supported
active in 0% of cells, pseudobulk strength 0.07



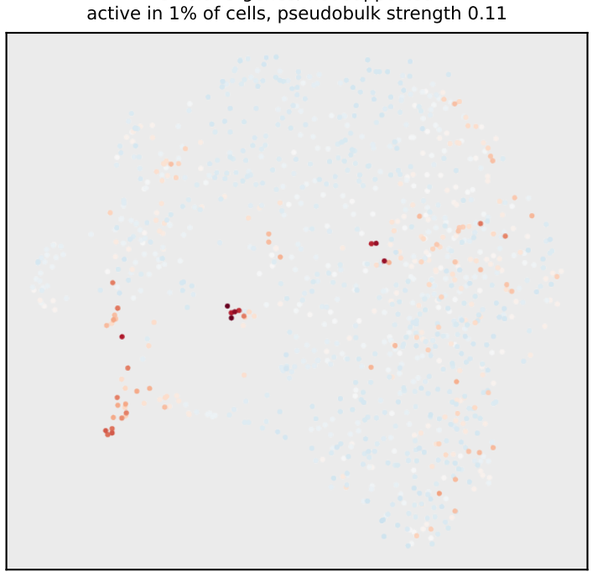
30% epiboly: 4 genes, not supported
active in 1% of cells, pseudobulk strength 0.10



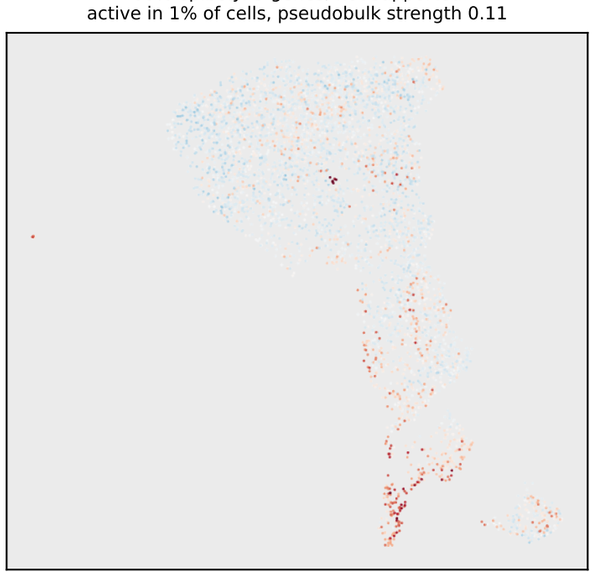
50% epiboly: 5 genes, not supported
active in 4% of cells, pseudobulk strength 0.70



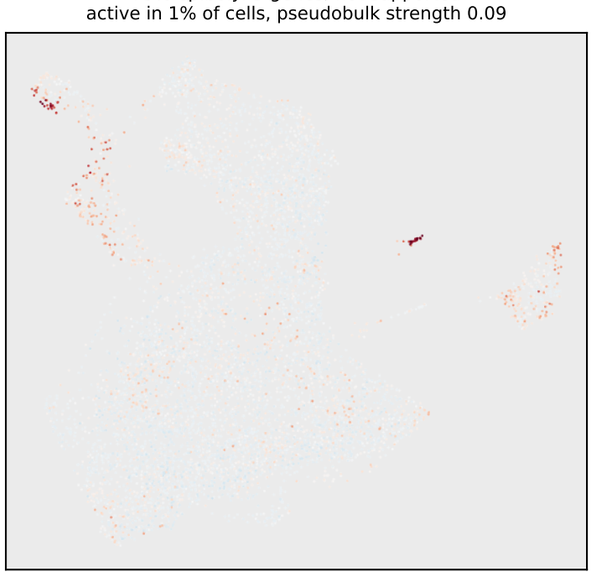
shield: 3 genes, not supported
active in 1% of cells, pseudobulk strength 0.11



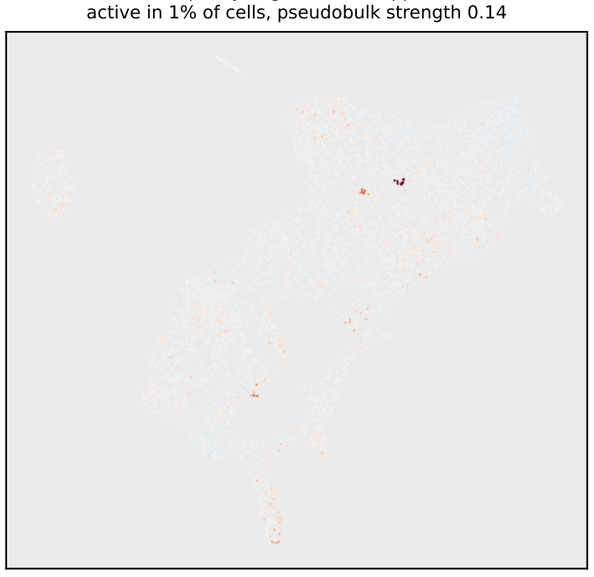
60% epiboly: 5 genes, not supported
active in 1% of cells, pseudobulk strength 0.11



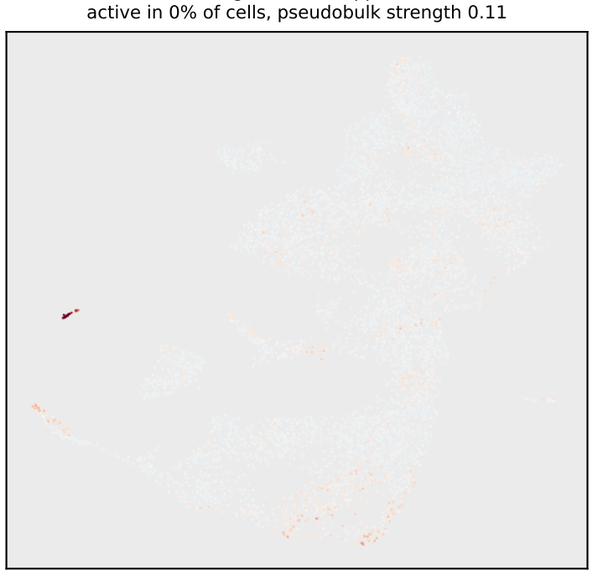
75% epiboly: 5 genes, not supported
active in 1% of cells, pseudobulk strength 0.09



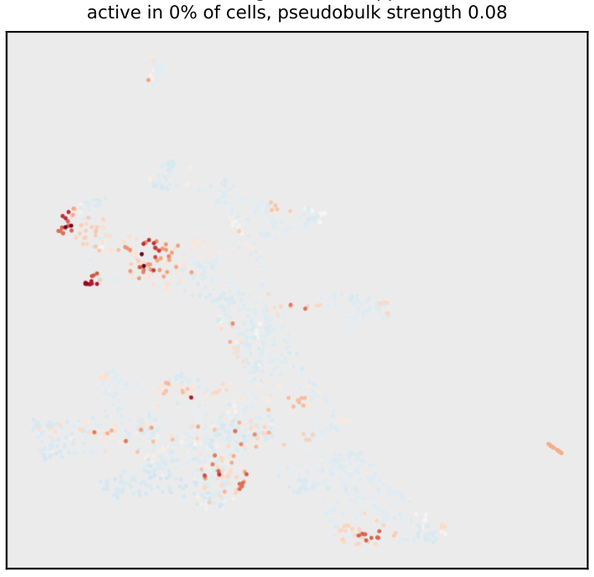
90% epiboly: 5 genes, not supported
active in 1% of cells, pseudobulk strength 0.14



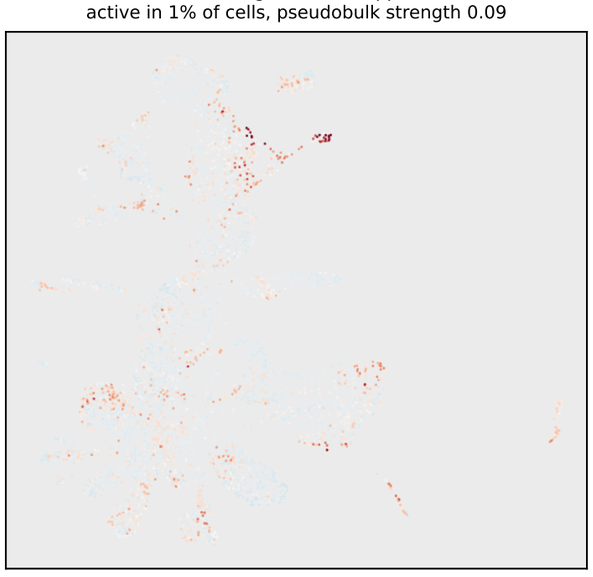
bud: 5 genes, not supported
active in 0% of cells, pseudobulk strength 0.11



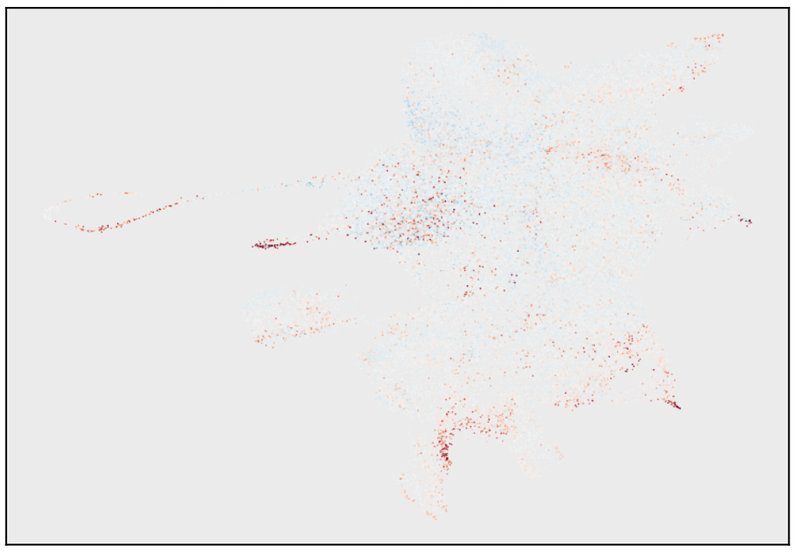
3-somite: 3 genes, not supported
active in 0% of cells, pseudobulk strength 0.08



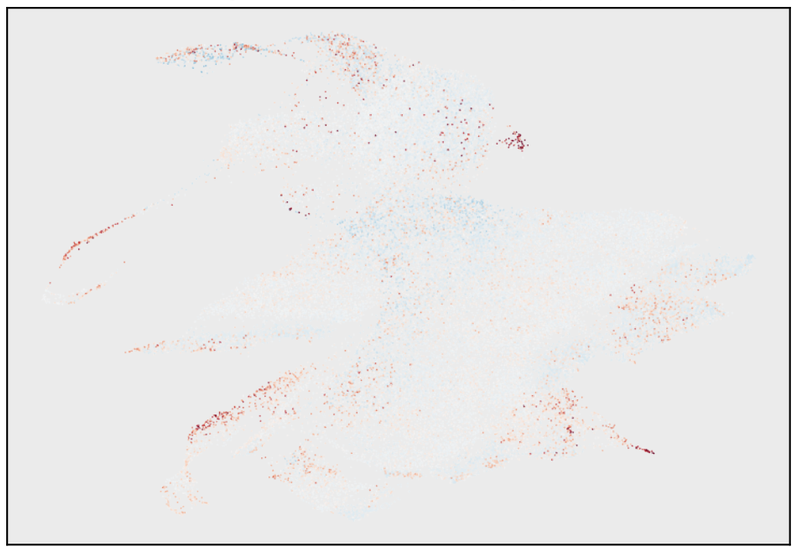
6-somite: 5 genes, not supported
active in 1% of cells, pseudobulk strength 0.09



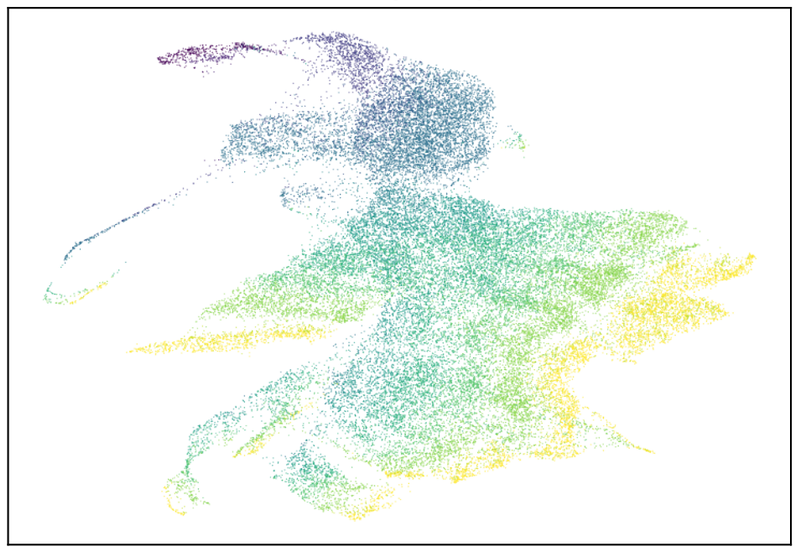
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



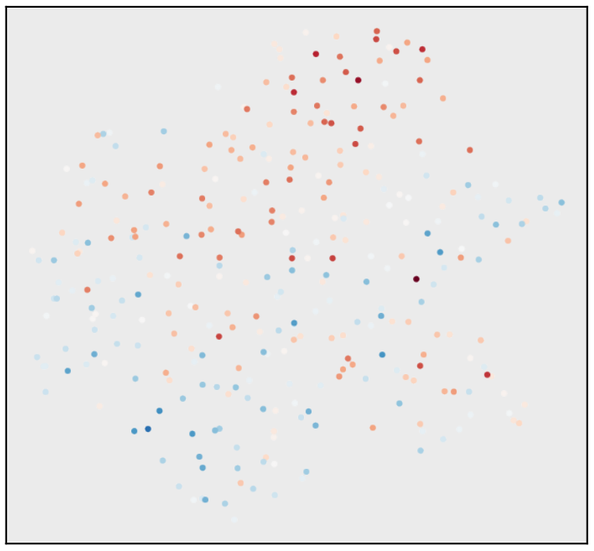
atlas, unaligned: stages



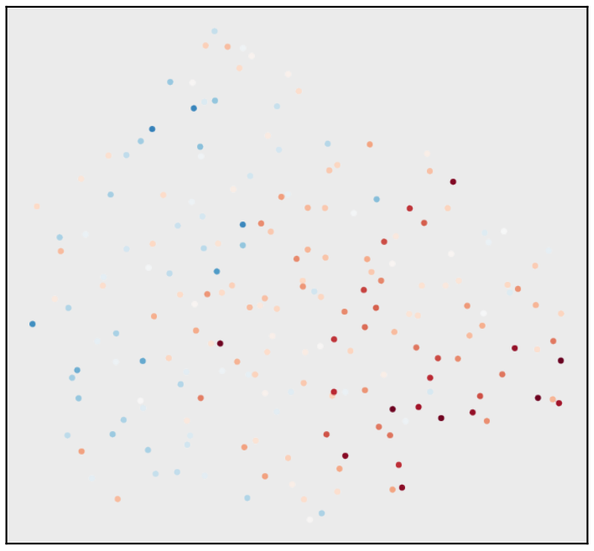
Genes (number of stages): hsp90aa1.2 (12), dnajb1a (11), hsp70.3 (10), bag3 (9), hsph1 (6)

state:maternal

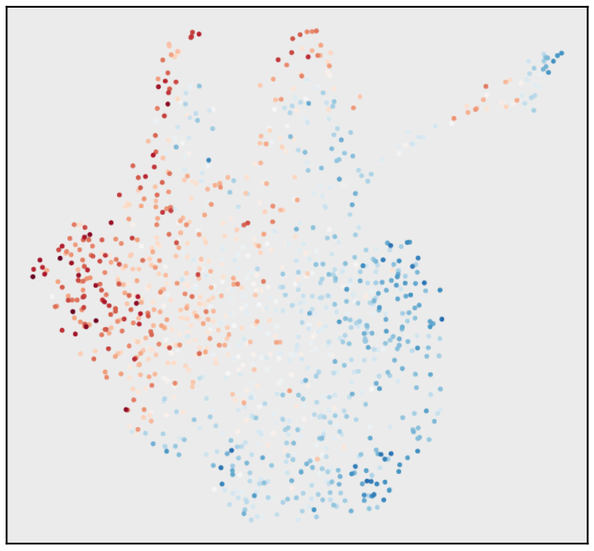
high: 10 genes, not supported
active in 14% of cells, pseudobulk strength 0.96



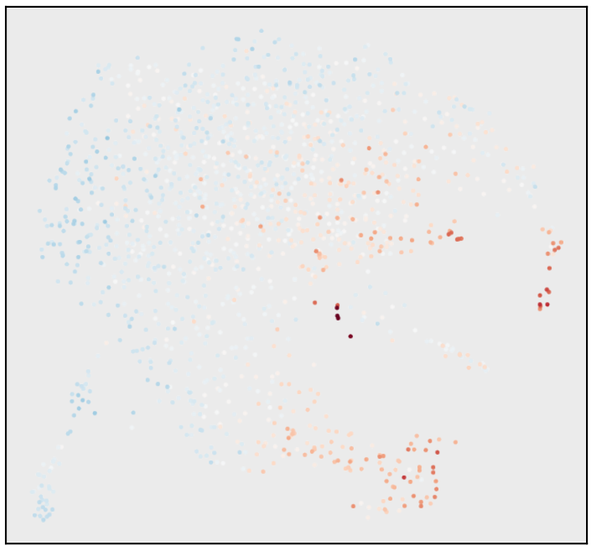
oblong: 10 genes, not supported
active in 9% of cells, pseudobulk strength 0.86



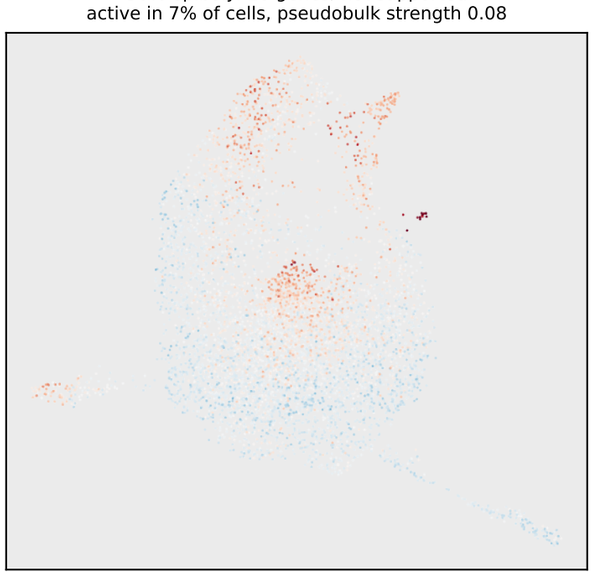
dome: 10 genes, not supported
active in 8% of cells, pseudobulk strength 0.25



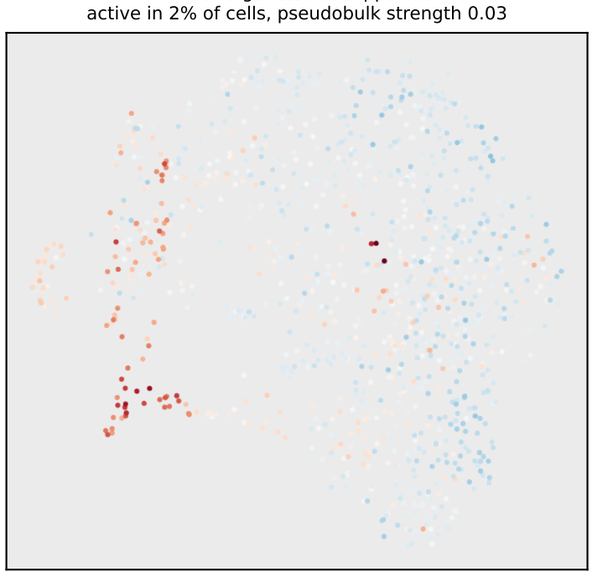
30% epiboly: 10 genes, not supported
active in 3% of cells, pseudobulk strength 0.12



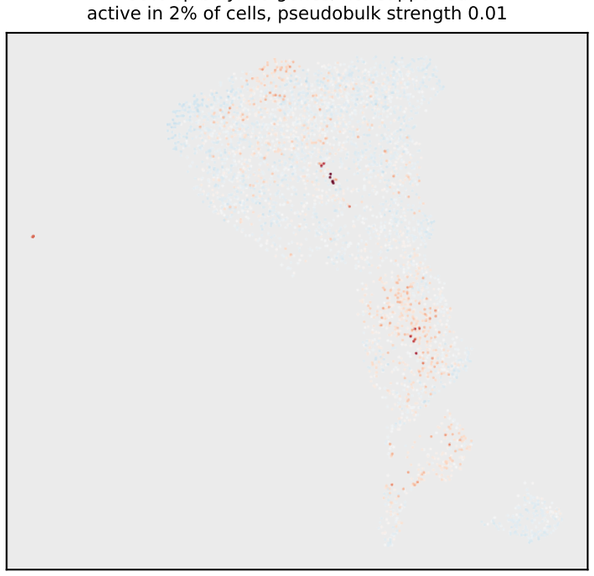
50% epiboly: 10 genes, not supported
active in 7% of cells, pseudobulk strength 0.08



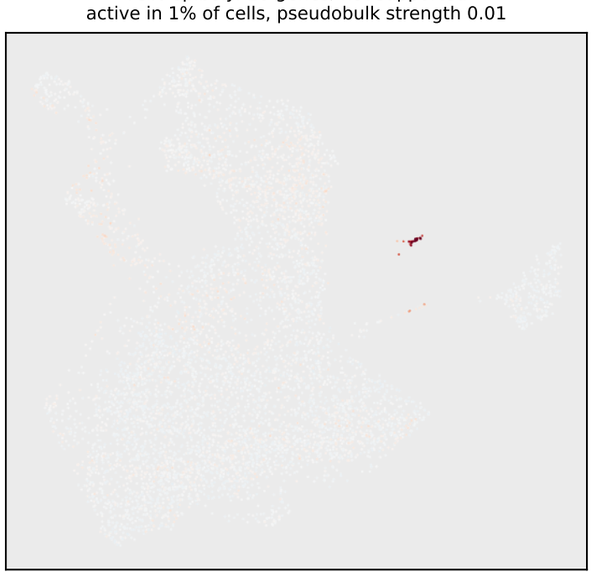
shield: 5 genes, not supported
active in 2% of cells, pseudobulk strength 0.03



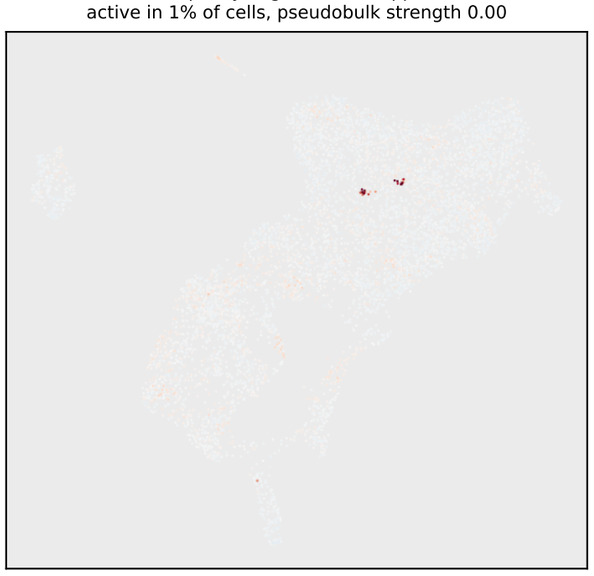
60% epiboly: 10 genes, not supported
active in 2% of cells, pseudobulk strength 0.01



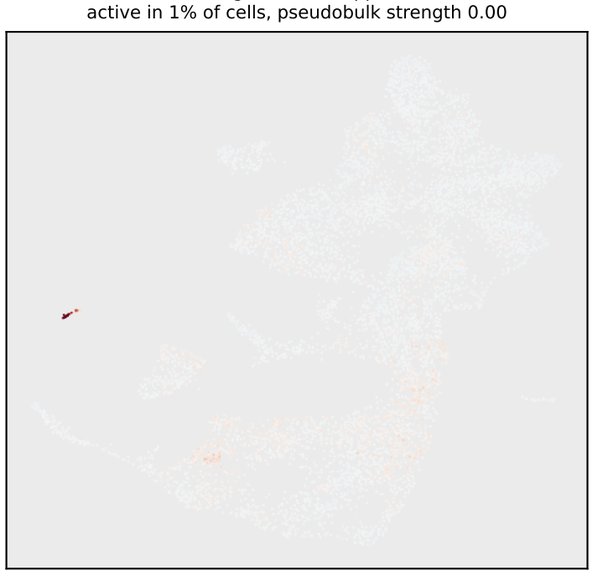
75% epiboly: 10 genes, not supported
active in 1% of cells, pseudobulk strength 0.01



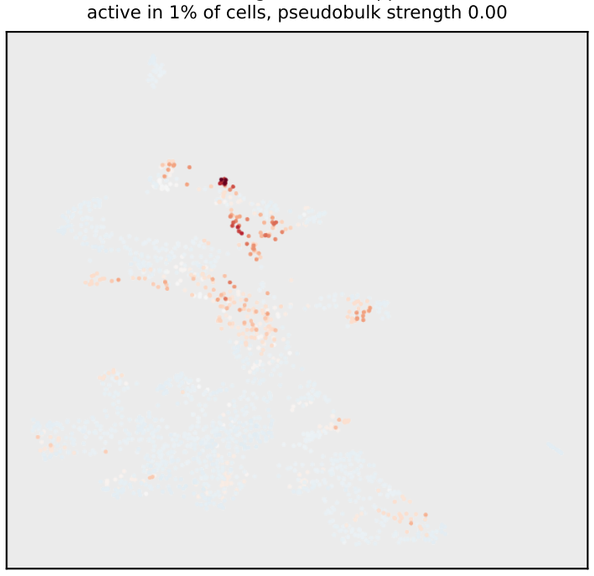
90% epiboly: 7 genes, not supported
active in 1% of cells, pseudobulk strength 0.00



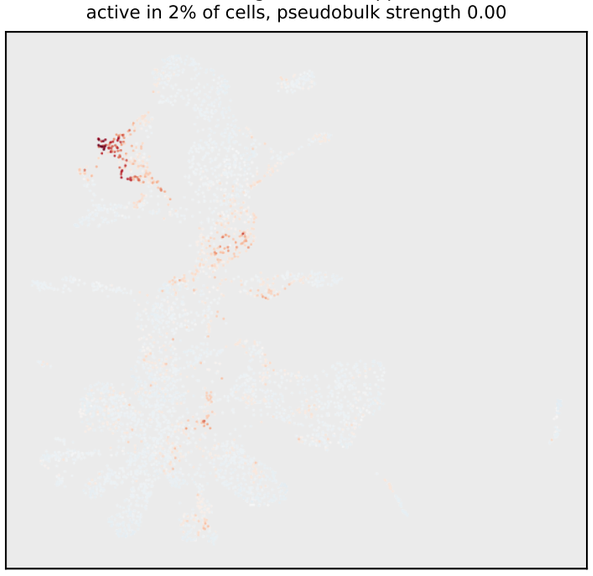
bud: 7 genes, not supported
active in 1% of cells, pseudobulk strength 0.00



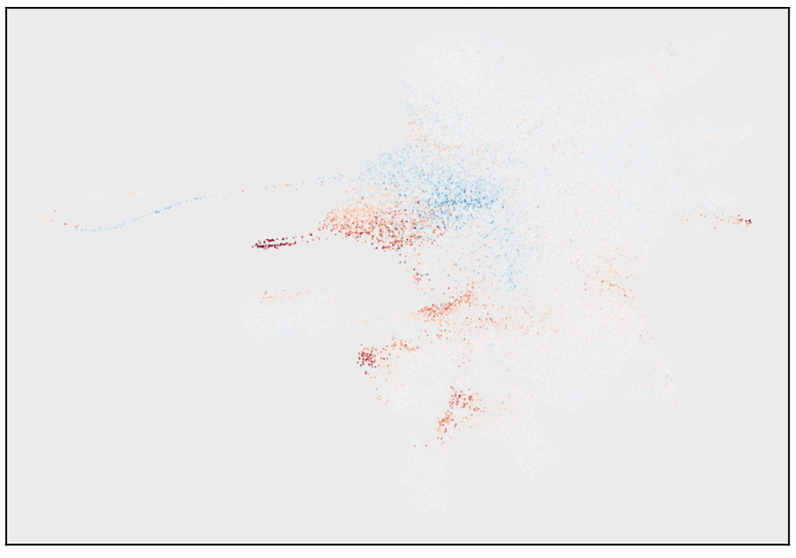
3-somite: 2 genes, not supported
active in 1% of cells, pseudobulk strength 0.00



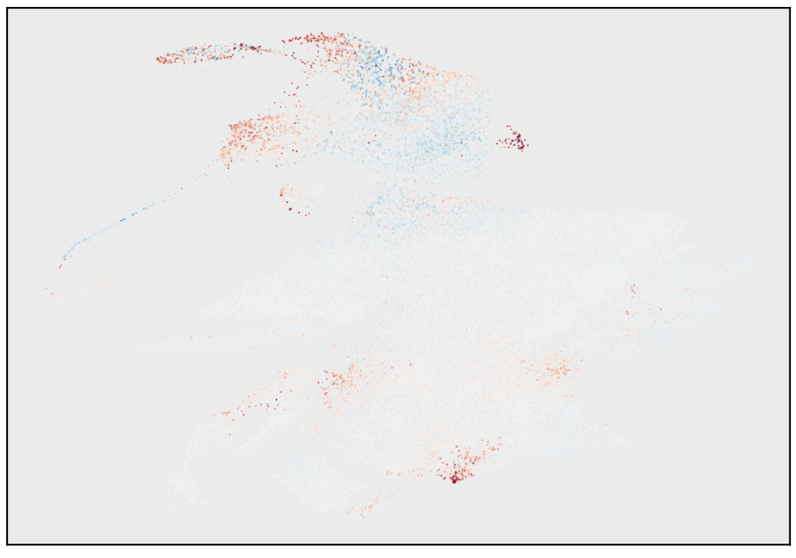
6-somite: 4 genes, not supported
active in 2% of cells, pseudobulk strength 0.00



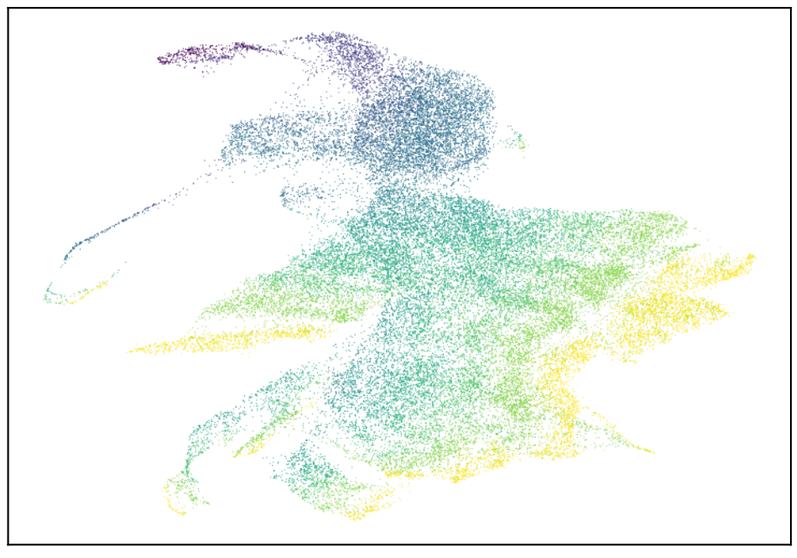
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



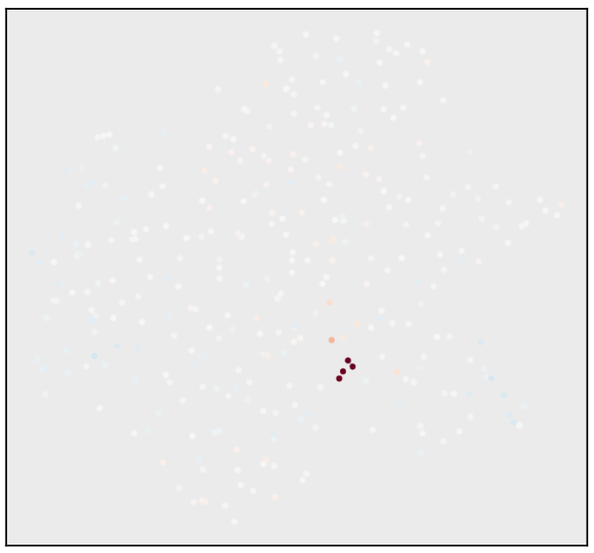
atlas, unaligned: stages



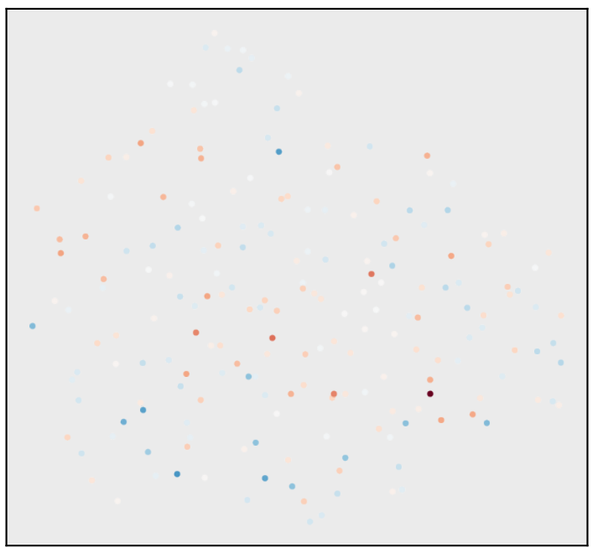
Genes (number of stages): h1m (12), gdf3 (12), cldnd (11), cth1 (10), mos (10), ccna1 (10), kpna7 (9), buc (7), btg4 (7), slbp2 (7)

state:p53_response

high: 10 genes, supported
active in 1% of cells, pseudobulk strength 0.74



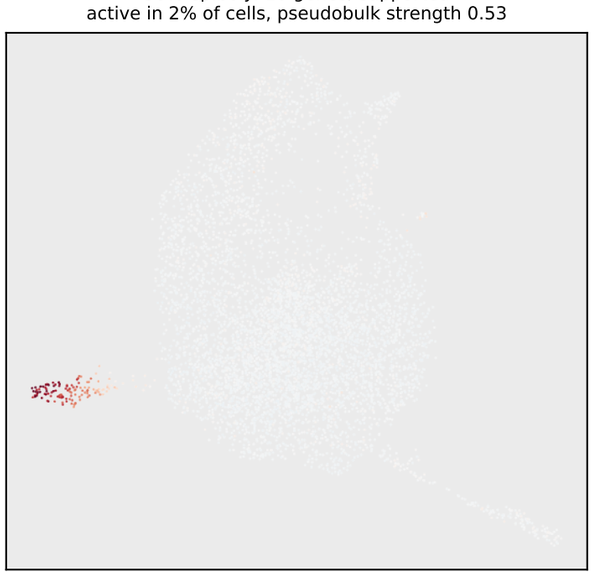
oblong: 9 genes, supported
active in 1% of cells, pseudobulk strength 0.70



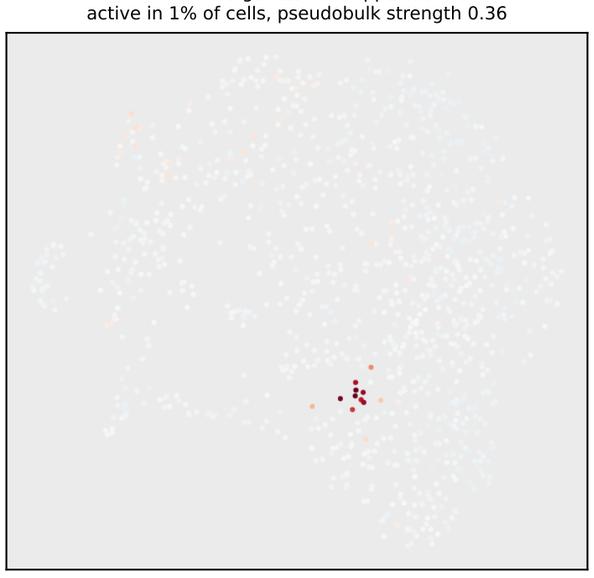
dome: 10 genes, not supported
active in 1% of cells, pseudobulk strength 0.38

30% epiboly: 10 genes, not supported
active in 2% of cells, pseudobulk strength 0.55

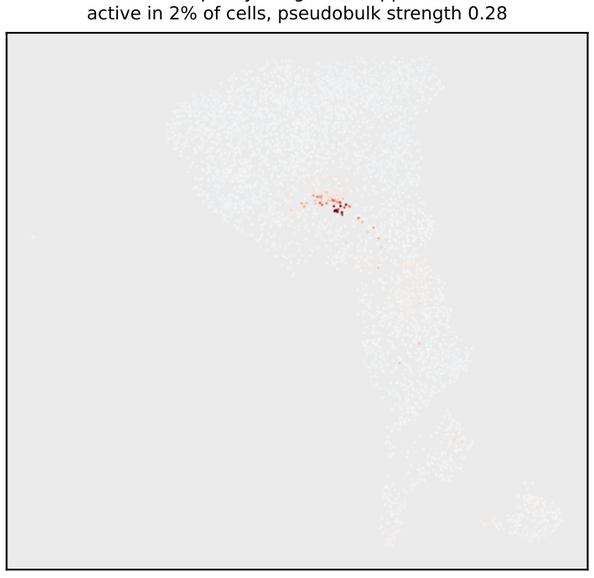
50% epiboly: 10 genes, supported
active in 2% of cells, pseudobulk strength 0.53



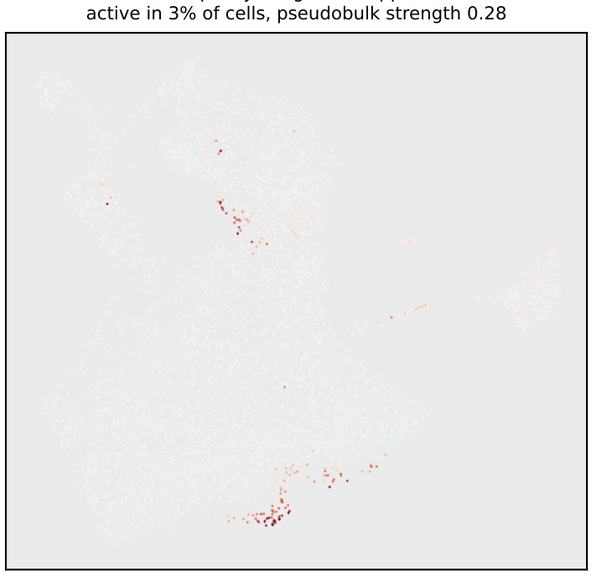
shield: 9 genes, not supported
active in 1% of cells, pseudobulk strength 0.36



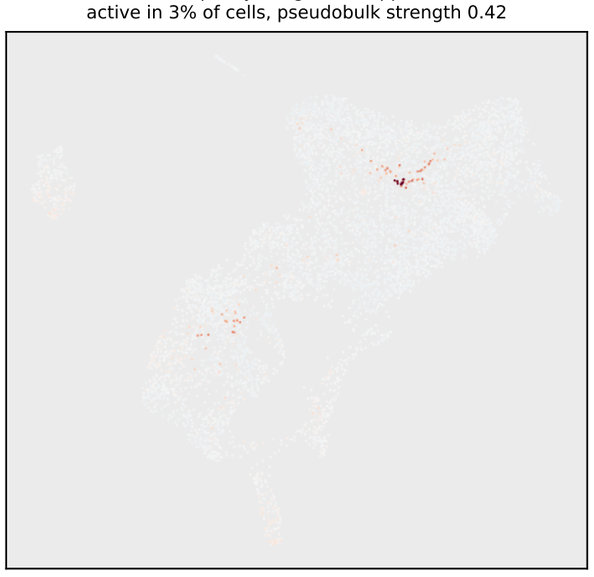
60% epiboly: 10 genes, supported
active in 2% of cells, pseudobulk strength 0.28



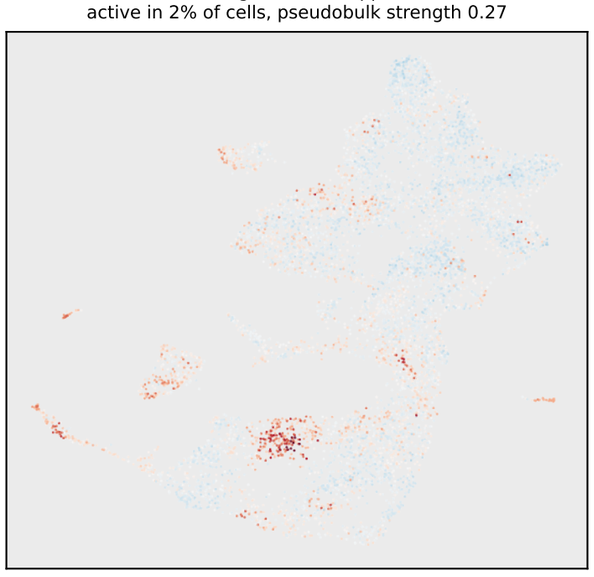
75% epiboly: 10 genes, supported
active in 3% of cells, pseudobulk strength 0.28



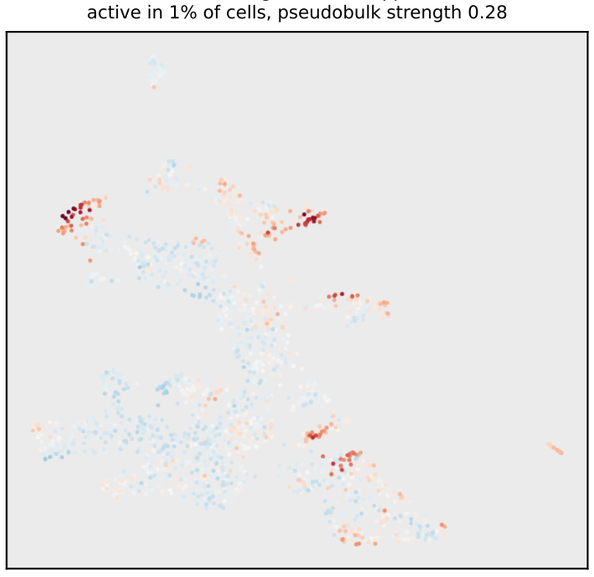
90% epiboly: 10 genes, supported
active in 3% of cells, pseudobulk strength 0.42



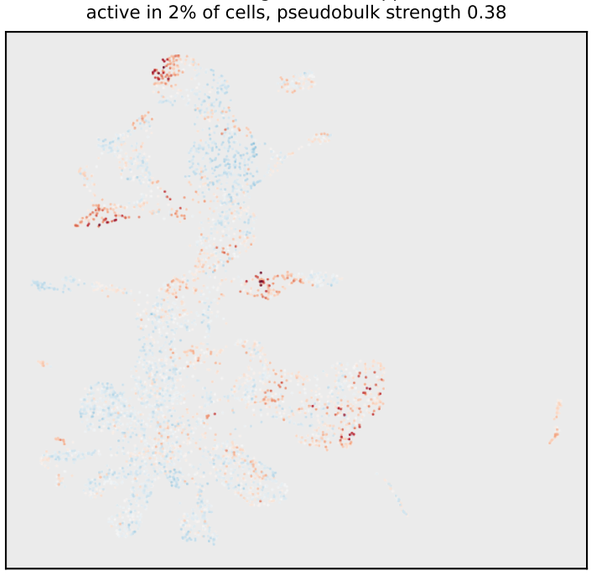
bud: 10 genes, not supported
active in 2% of cells, pseudobulk strength 0.27



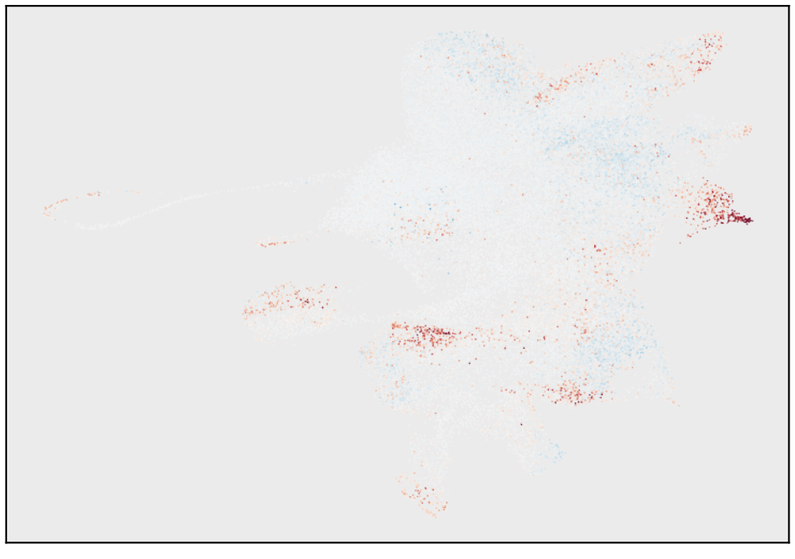
3-somite: 10 genes, not supported
active in 1% of cells, pseudobulk strength 0.28



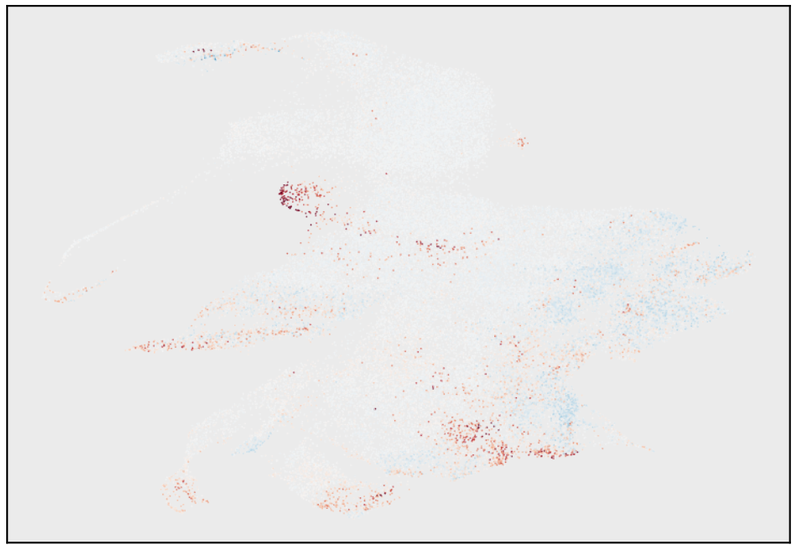
6-somite: 10 genes, not supported
active in 2% of cells, pseudobulk strength 0.38



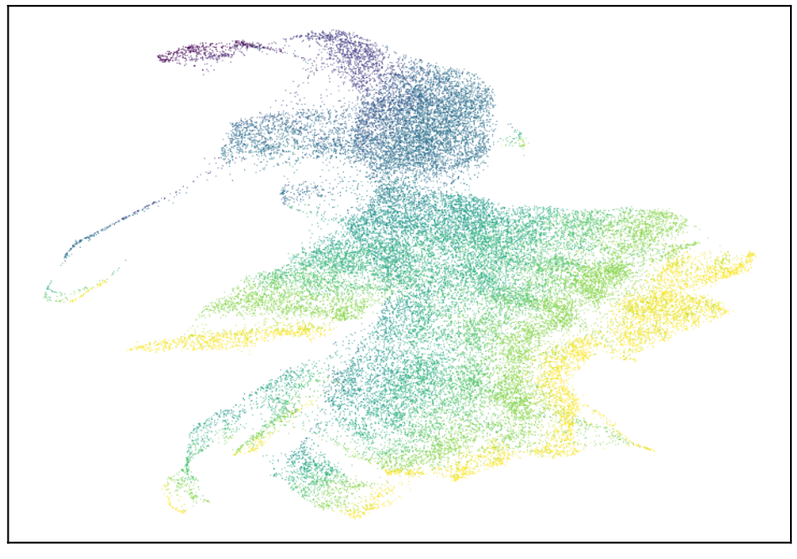
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



atlas, unaligned: stages



Genes (number of stages): mdm2 (12), cdkn1a (12), phlda3 (12), tp53inp1 (12), rrm2b (12), sesn1 (12), ddb2 (12), plk3 (12), gadd45aa (11), fdxr (11)